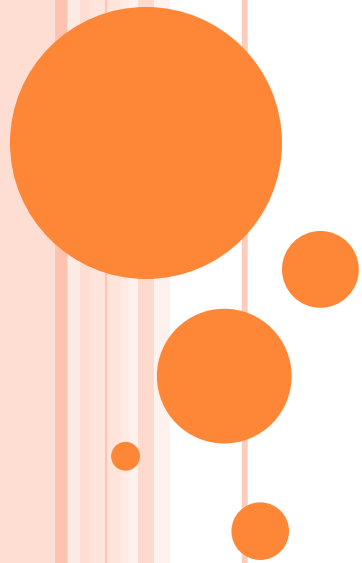


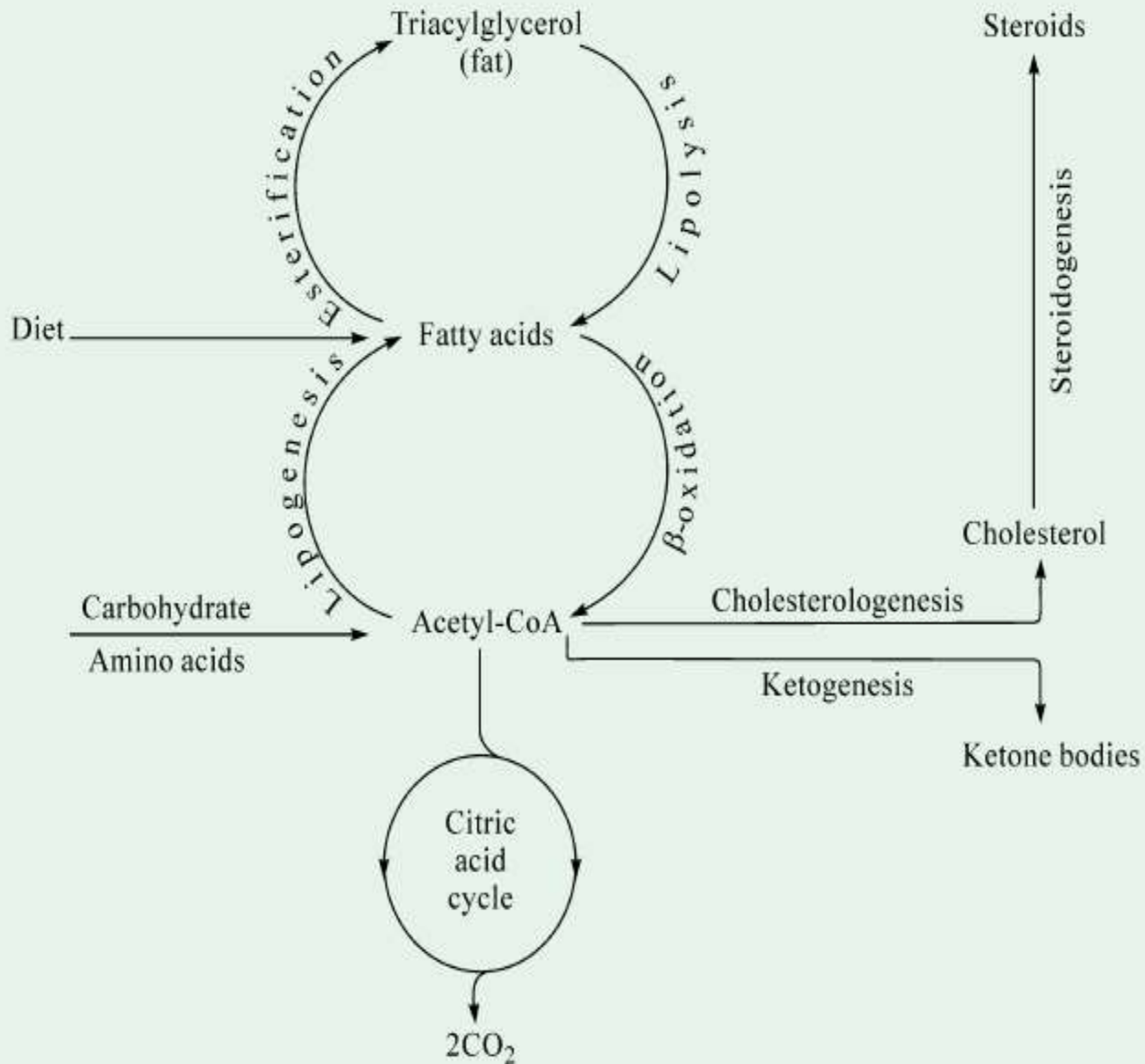
METABOLISM OF LIPIDS



DIGESTION AND ABSORPTION OF LIPIDS



OVERVIEW OF LIPID METABOLISM

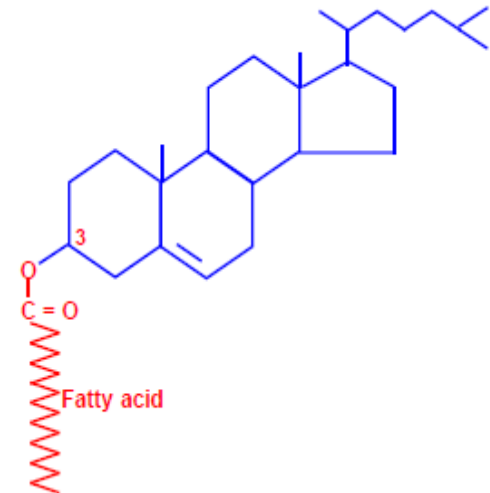
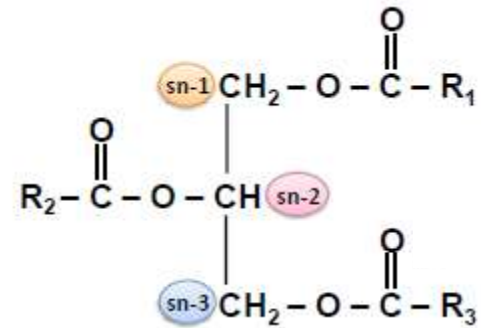
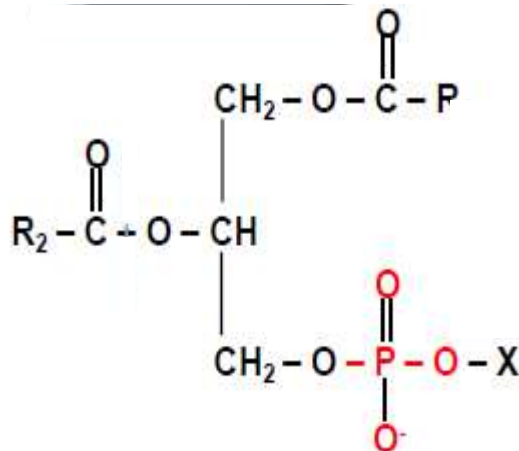


COMMON DIETARY LIPIDS

□ Mainly Triacylglycerol (TAG) (~95%)

Remaining 5%:

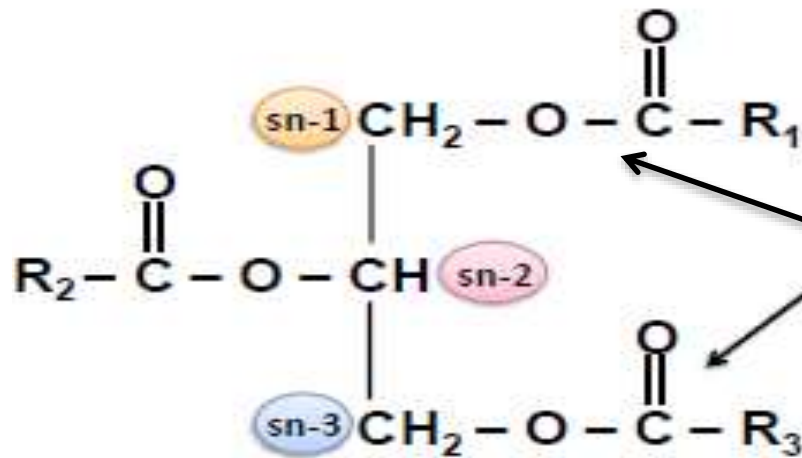
- Sterols(mainly cholesterol)
- Mostly as free cholesterol
- 10-15% are cholesterol esters
- Phospholipids



DIGESTION OF LIPIDS

Lingual lipase *released in mouth*

- **Primary target:** fatty acids of *short or medium chain* length (milk fat).
- Activity decreases with age and highly active in neonates



Removes **sn-1** and **sn-3** on TAG containing short and medium chain fatty acid

Product of lingual lipase: fatty acids+ 2-monoacylglycerol

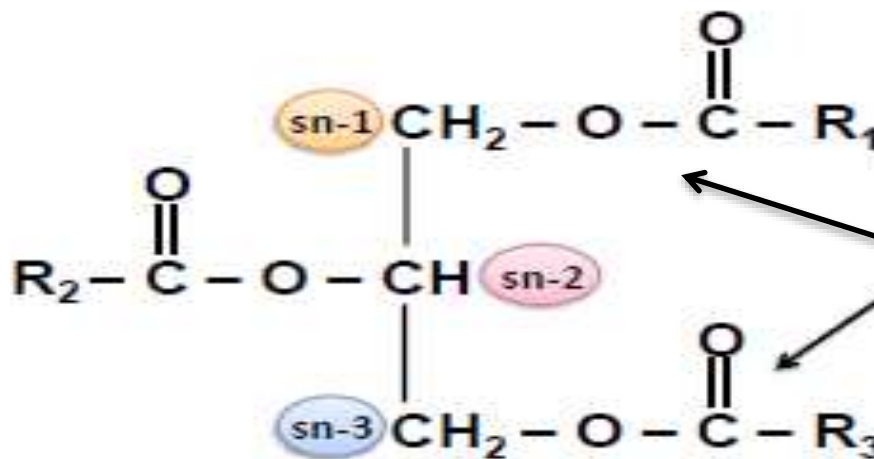
- Optimal work at pH: 4 to 6 ,it is acid stable

DIGESTION OF LIPIDS

Gastric lipase *released in stomach*

Lipids (non polar) –need to be emulsified in order to allow digestion in stomach

- Polysaccharides, phospholipids, digested proteins in **chyme** act as emulsifiers
- Optimal pH: 4 to 6



Removes sn-1 and sn-3 on TAG containing short and medium chain fatty acid

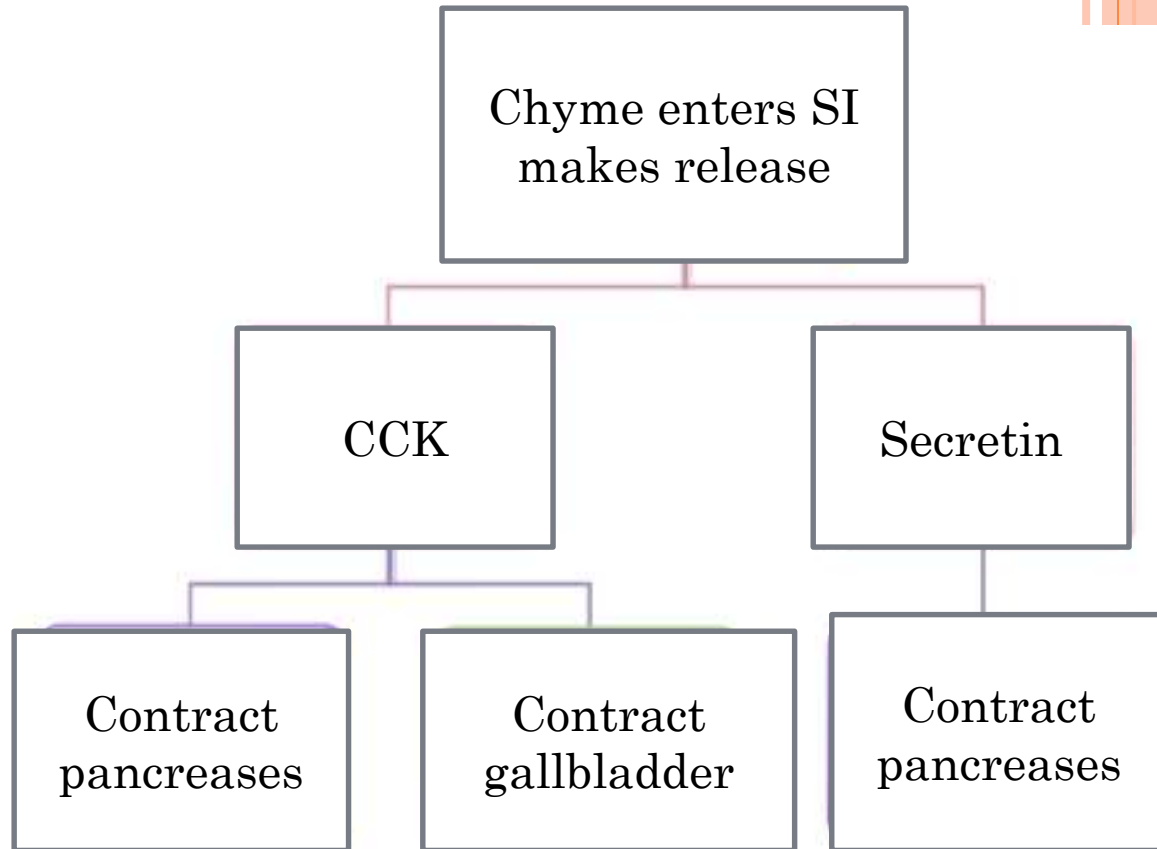
Product of gastric lipase is: fatty acids+ 2-monoacylglycerol

DIGESTION OF LIPIDS

❖ Presence of fat in stomach causes release of hormones

- CCK
- Secretin
- GIP
- GLP-1

❖ Lipids have a high satiety value



EMULSIFICATION OF LIPIDS IN SMALL INTESTINE

Emulsification increases the surface area of the hydrophobic fat droplets so that digestive enzymes can act effectively.

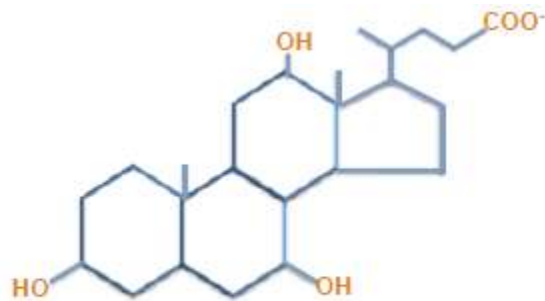
Done by

- **Peristalsis:** Mechanical mixing
- **Bile salts:** Detergent properties as they decrease the surface tension and cause fat emulsification.
- **Phospholipids**

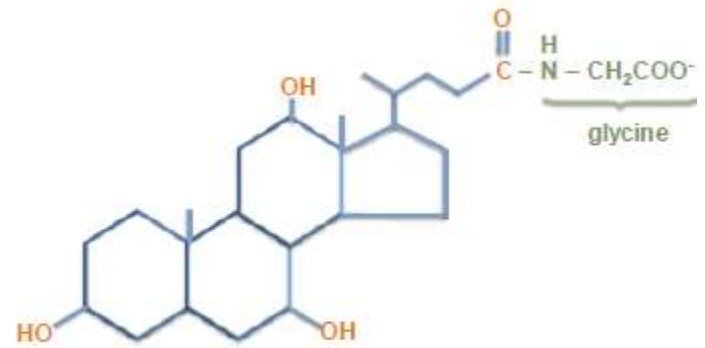


BILE ACIDS/BILE SALTS

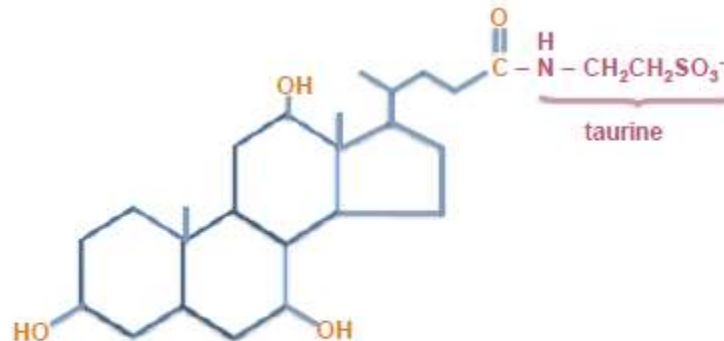
- ❖ Made in **liver** from **cholesterol**
- ❖ Bile acids are oxygenated derivatives of cholesterol that have several hydroxyl groups on the sterol rings
- ❖ Stored in the **gallbladder**



Cholic acid (cholate)

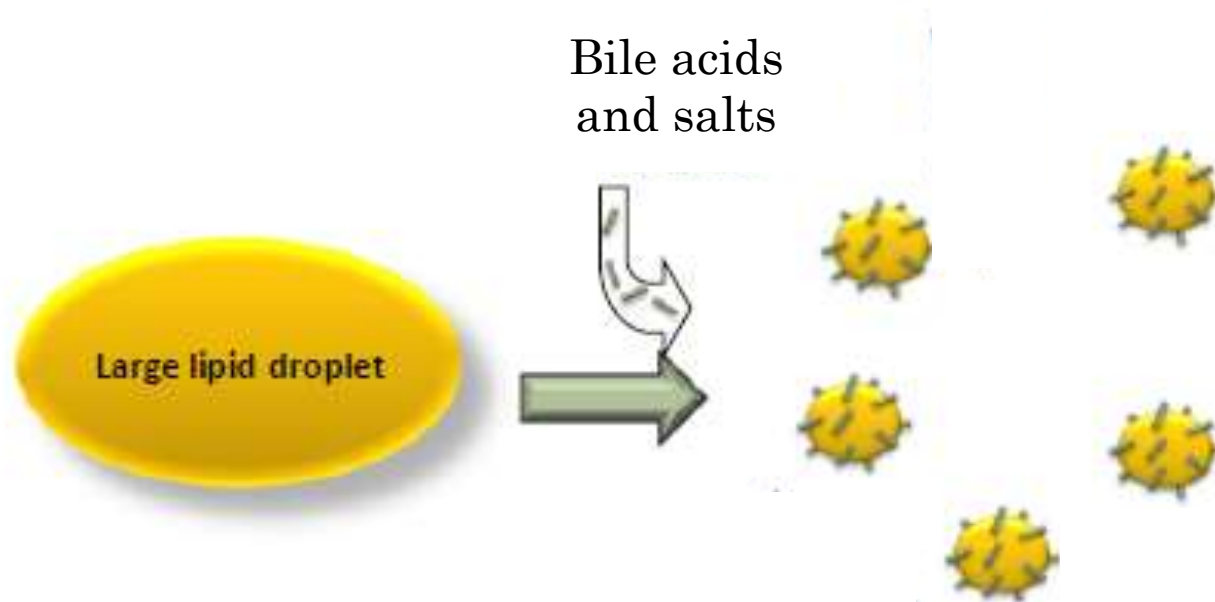


Glycocholate



Taurocholate

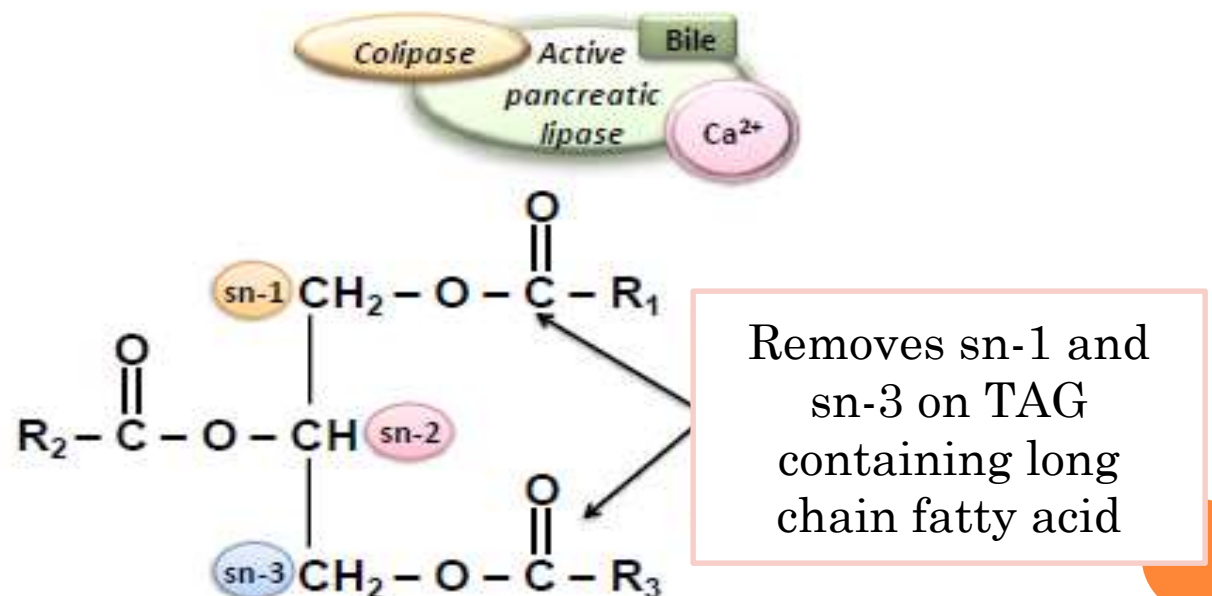
EMULSIFICATION BY BILE ACIDS/SALTS



DIGESTION OF LIPIDS

1- Degradation of triacylglycerol (TAG):-

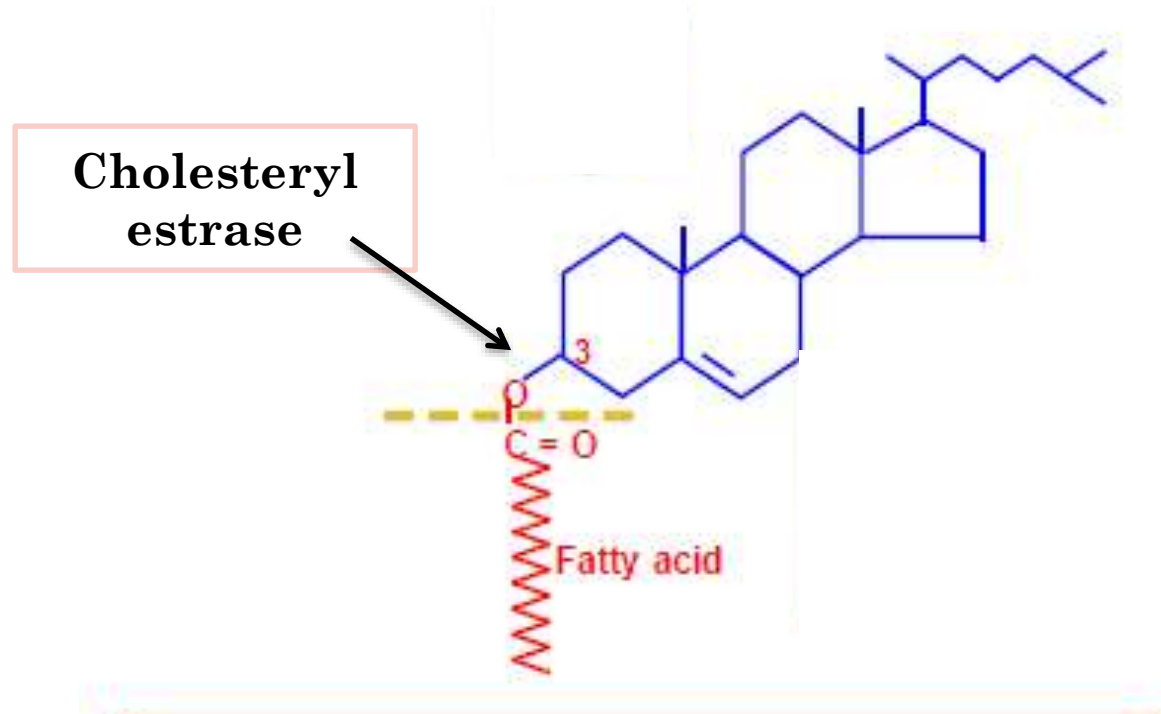
- **Pancreatic Lipase:** Released from pancreases into duodenum
- Activated by calcium and Colipase, bile salt inhibit PL



Product of pancreatic lipase is: fatty acids+ 2-monoacylglycerol

2- DEGRADATION OF CHOLESTERYL ESTERS

- ❖ **Cholesteryl esterase** produces cholesterol and free fatty acids.
- ❖ Activity is increased in presence of Bile salts.



Products of esterase digestion: free fatty acids + cholesterol + Vit. A, E

3- DEGRADATION OF PHOSPHOLIPIDS

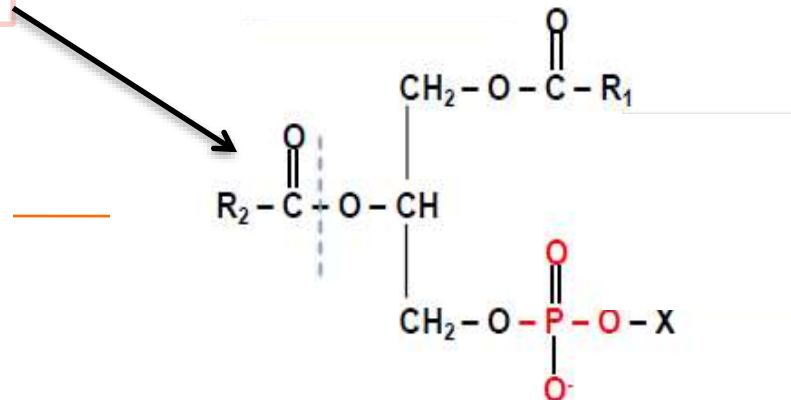
Phospholipase A₂ in proenzyme form,

- Activated by Trypsin,
- Requires bile salts for activity.
- Removes one fatty acid from carbon 2 of a phospholipid **lysophospholipid**.

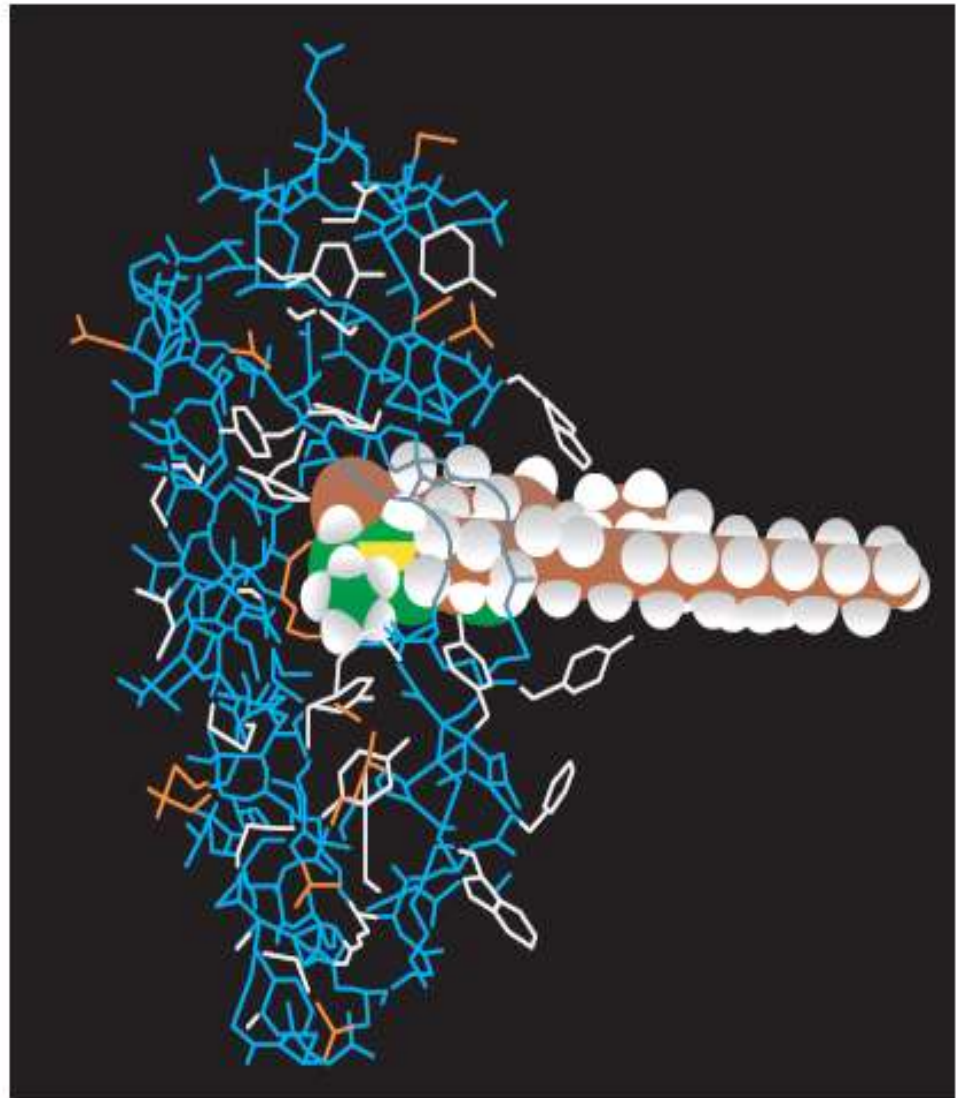
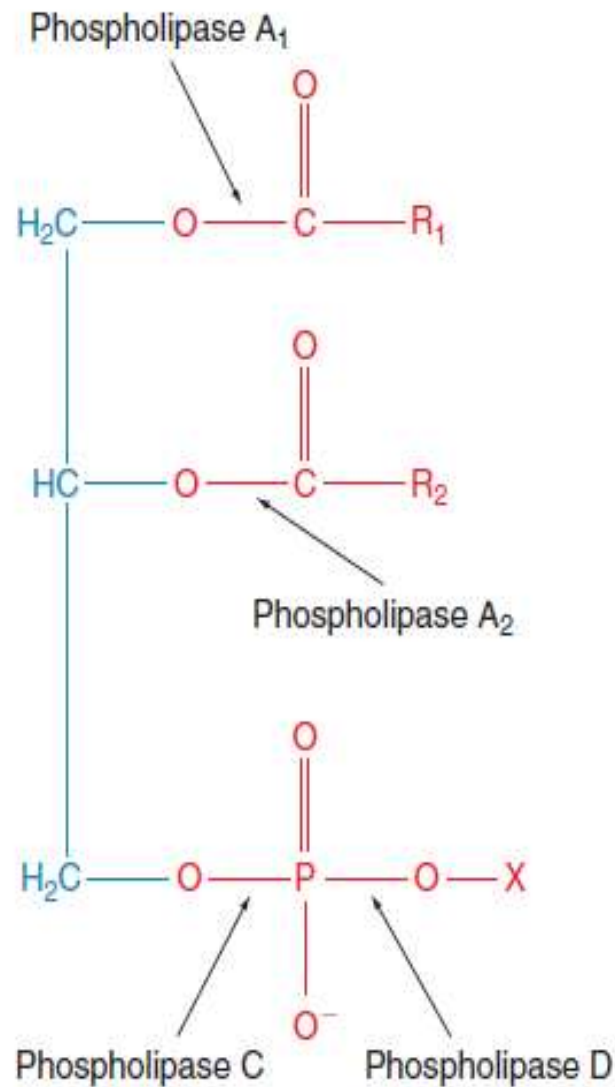
Lysophospholipase A₁

- Removes fatty acid at carbon 1 and forms glycerylphosphoryl base that is excreted, degraded or absorbed.

Phospholipase A₂



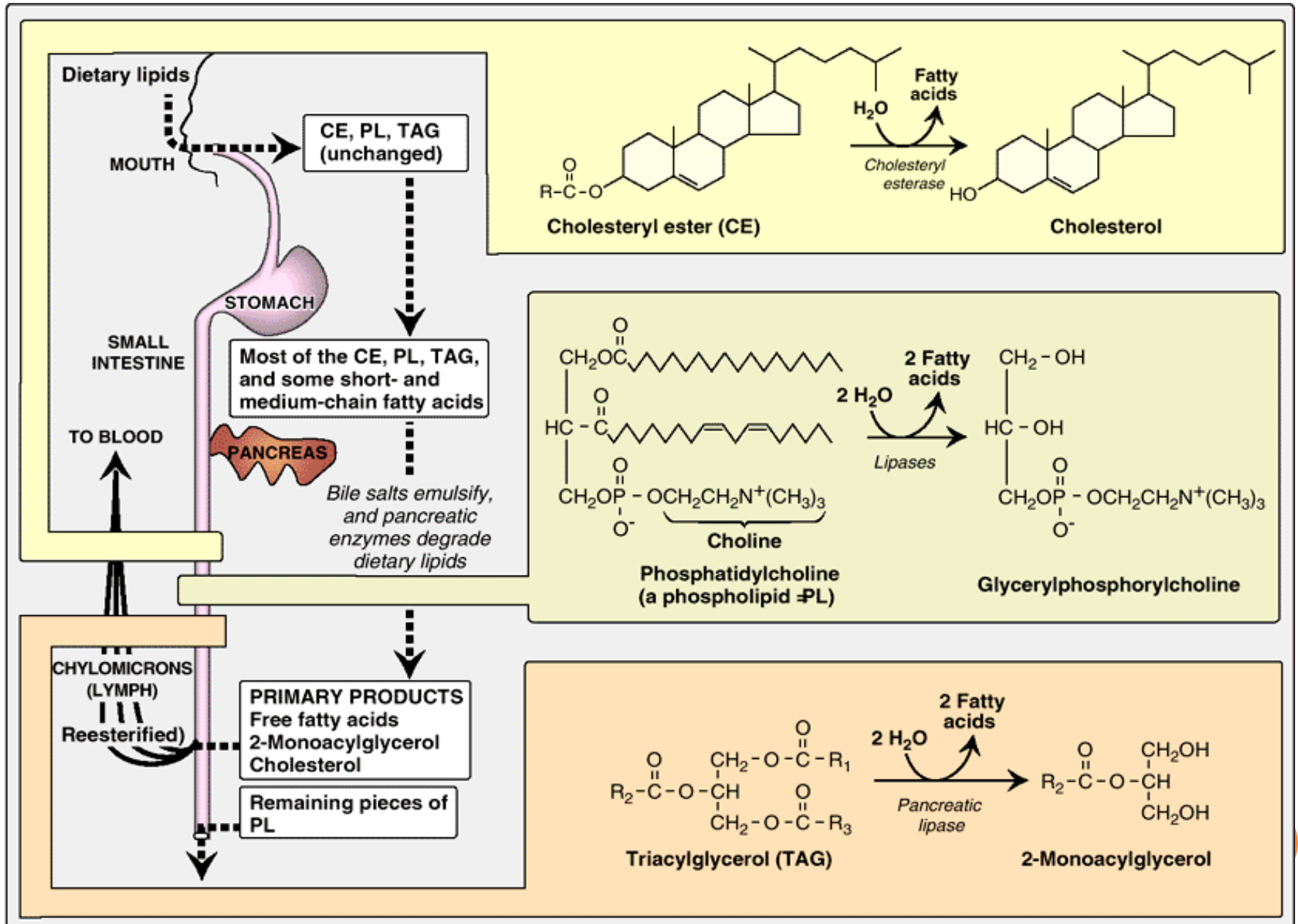
Product of phospholipase A₂ is free fatty acid and lysophospholipid



- Digestion of a phosphoglyceride by pancreatic phospholipase.



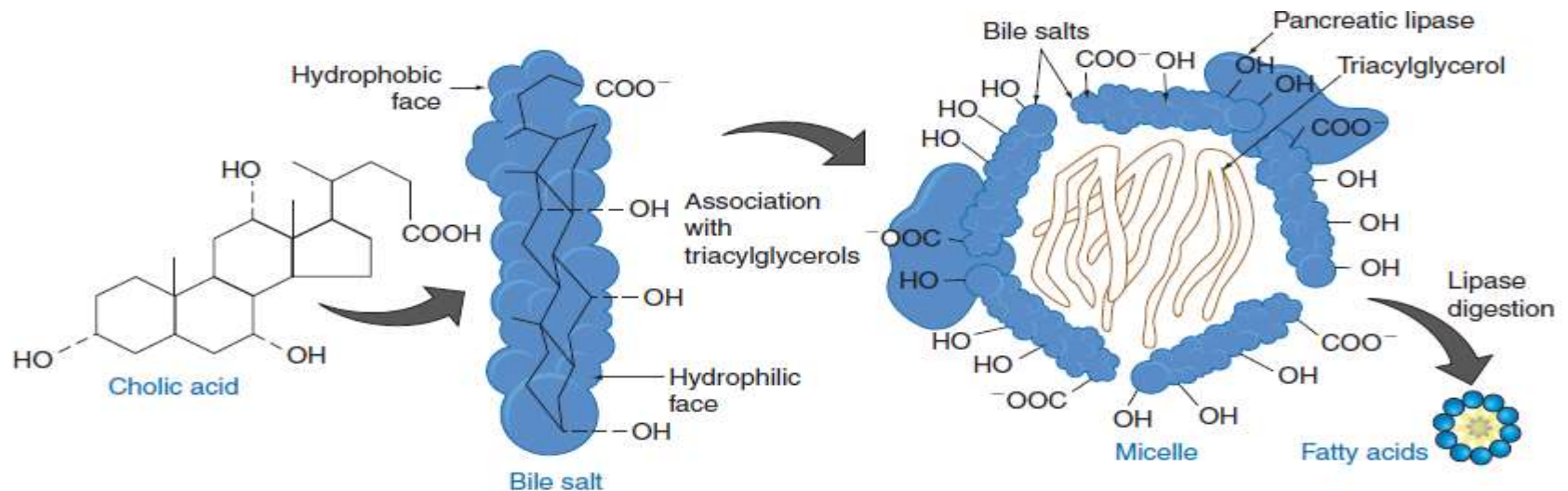
Summary of digestion of lipids



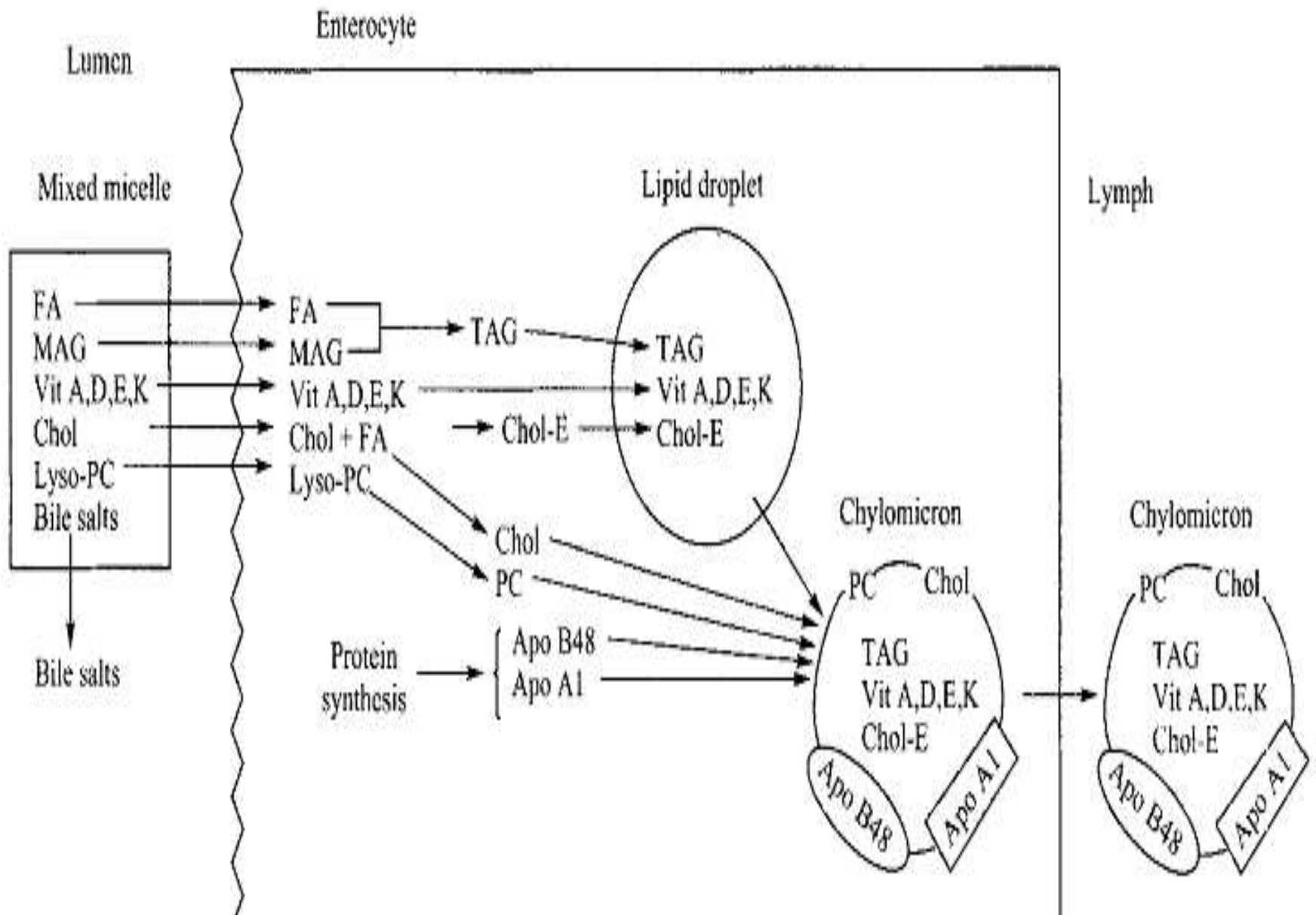
ABSORPTION OF LIPIDS BY INTESTINAL MUCOSAL CELLS

Jejunum gets : **Free fatty acids**
Free cholesterol
2-monoacylglycerol
Lysophospholipid } Combine with bile and fat soluble vitamins

Form micelles : Soluble in aqueous intestinal environment, absorbed at the brush border of enterocytes .



Notes: Fatty acids with short and medium chain length do not need micelles for absorption.



Absorption of fat into enterocytes is mainly done through diffusion and by FAT/CD36

RESYNTHESIS OF TRIACYLGLYCEROL

The conversion of free fatty acids to fatty acyl-CoA in enterocytes utilizes the ubiquitous **fatty acyl-CoA synthetase** reaction



- In enterocytes, synthesis of triacylglycerol occurs through the sequential action of **monoacylglycerol acyltransferase** and **diacylglycerol acyltransferase**, for a net reaction



This pathway is distinct from the triacylglycerol synthesis pathway in other cells, such as **hepatocytes** and **adipocytes**, which utilizes **glycerol 3-phosphate** as **the** acceptor of acyl groups.

RESYNTHESIS OF PHOSPHOLIPIDS/ CHOLESTERYL ESTER

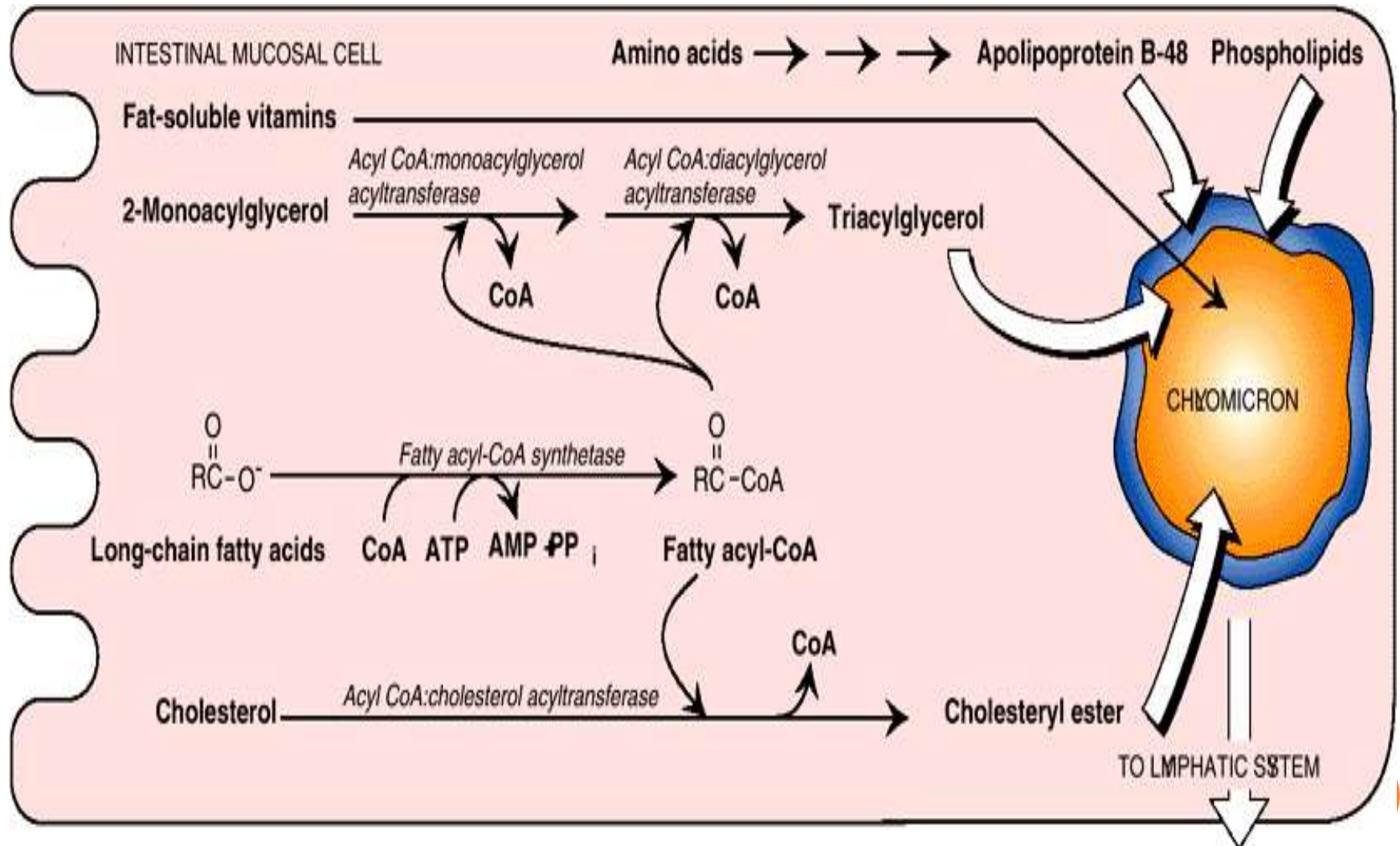
- ❑ Reacylation of lysophospholipids caused by acyltransferases forms phospholipids.
- ❑ Cholesteryl ester formation by Acyl CoA: cholesterol acyltransferase (ACAT).
- ❑ ACAT₁ and ACAT₂, each with different expression patterns and unique physiological functions. ACAT₂ is expressed exclusively in the **liver and intestine**, where it participates in lipoprotein assembly. ACAT₁ is present in most tissues, and is the primary ACAT isoform in *macrophages and steroidogenic tissues*.

Note: Free fatty acids with short and medium chains are released into portal circulation.

Glycerol: diffuses to portal vein and then reach liver.



SUMMARY OF RESYNTHESIS OF LIPIDS IN ENTEROCYTES



microsomal triglyceride transfer protein(MTP)

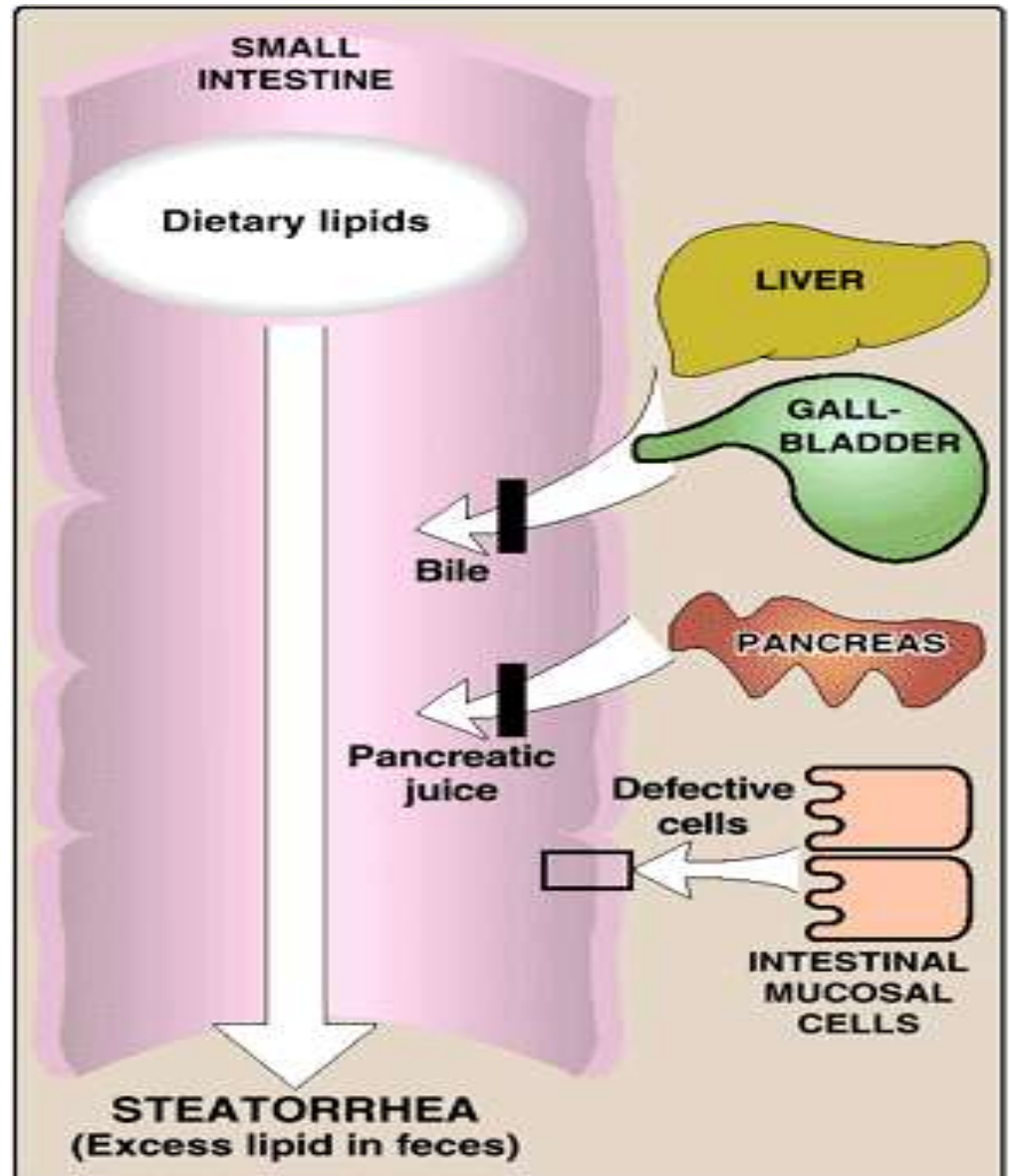
ABSORPTION OF BILE SALTS

- Bile salts are not absorbed together with the products of hydrolysis of dietary triglycerides, phospholipids, and cholesteryl esters, and they are not incorporated into chylomicrons.
- Instead, bile salts remain in the intestinal lumen until they reach the **distal ileum**, where most are **absorbed by an active transport mechanism that utilizes a Na^+ -bile salt co-transport**



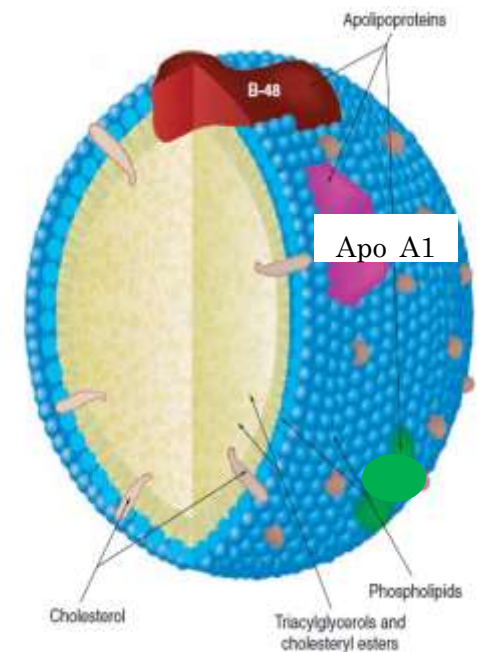
LIPID MALABSORPTION

- ❑ **Steatorrhea:**
increased lipid and fat soluble vitamin excretion in feces ($> 5\text{g/day}$).
- Caused by defects in lipid digestion and/or lipid absorption



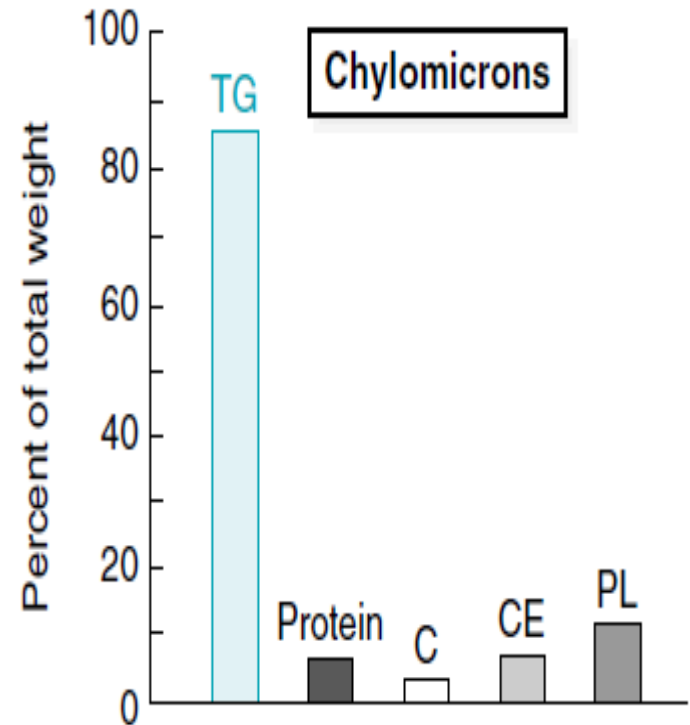
SECRETION OF LIPIDS FROM ENTEROCYTES

- ❑ **Formation of Chylomicrons:-**
Aggregates of TAG and cholesteryl esters are formed, surrounded by a thin layer phospholipids, free cholesterol and special protein Apolipoprotein B-48 .
- ❑ Chylomicrons are only synthesized in small intestine released into lacteals by exocytosis.
- ❑ After a lipid rich meal , lymph is called chyle.
- ❑ From lymph, chylomicrons finally enter blood.



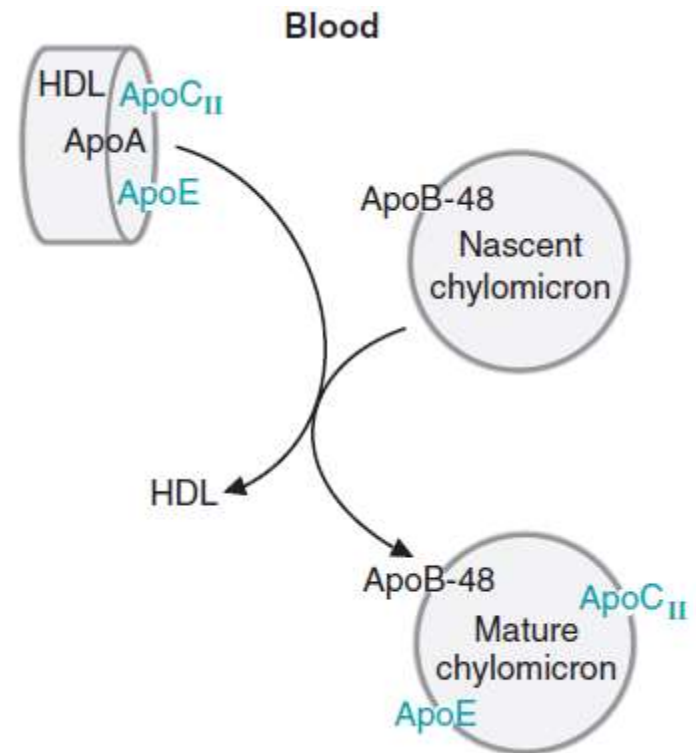
CHYLOMICRONS

- ❑ Assembled in intestinal mucosal cells
- ❑ Lowest density
- ❑ Largest size
- ❑ Highest % of lipids and lowest % proteins
- ❑ Highest triacylglycerol (dietary origin)
- ❑ Carry dietary lipids to peripheral tissues
- ❑ Responsible for physiological milky appearance of plasma (up to 2 hours after meal)



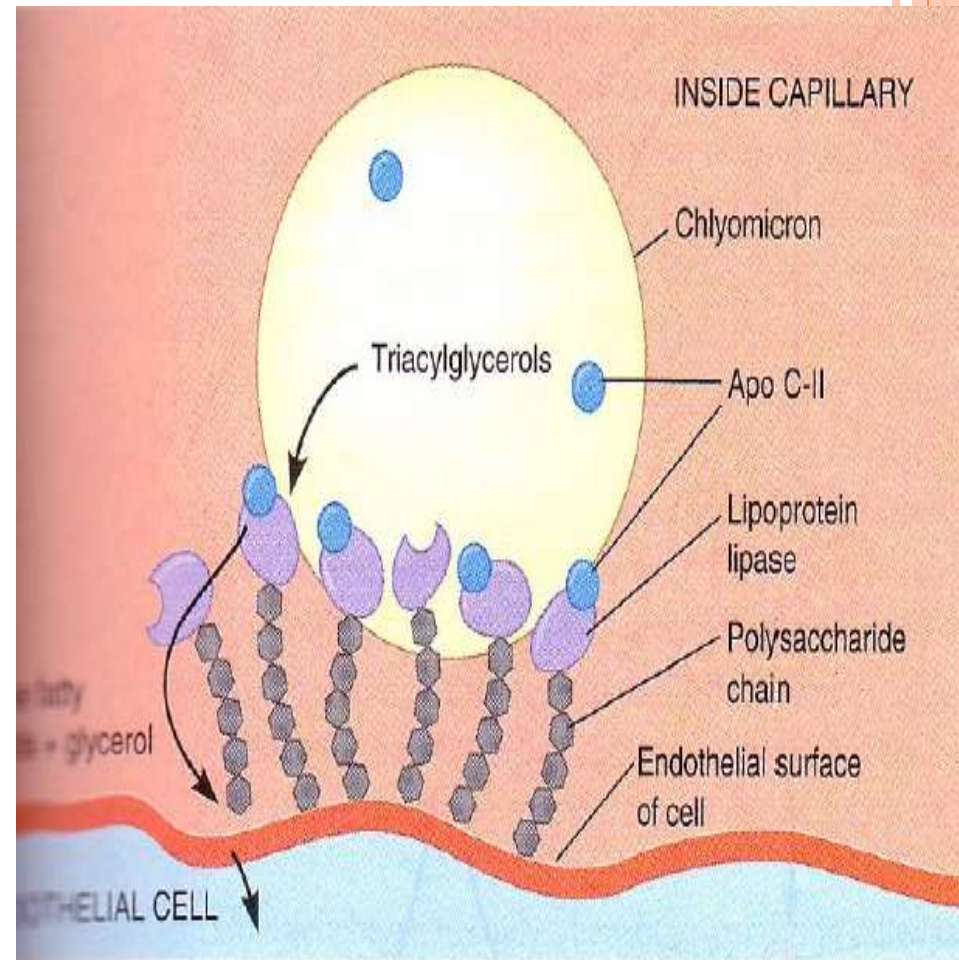
METABOLISM OF CHYLOMICRONS

- ❑ These nascent chylomicrons are released in blood through lymphatic system.
- ❑ In plasma it receives from HDL two more apoproteins i.e. **apo C-II** and **apo-E**.
- ❑ Apo C-II activates the enzyme Lipoprotein lipase, located on the capillary walls.
- ❑ Lipoprotein lipase degrades TAG in chylomicrons and forms free fatty acids and glycerol.



LIPOPROTEIN LIPASE

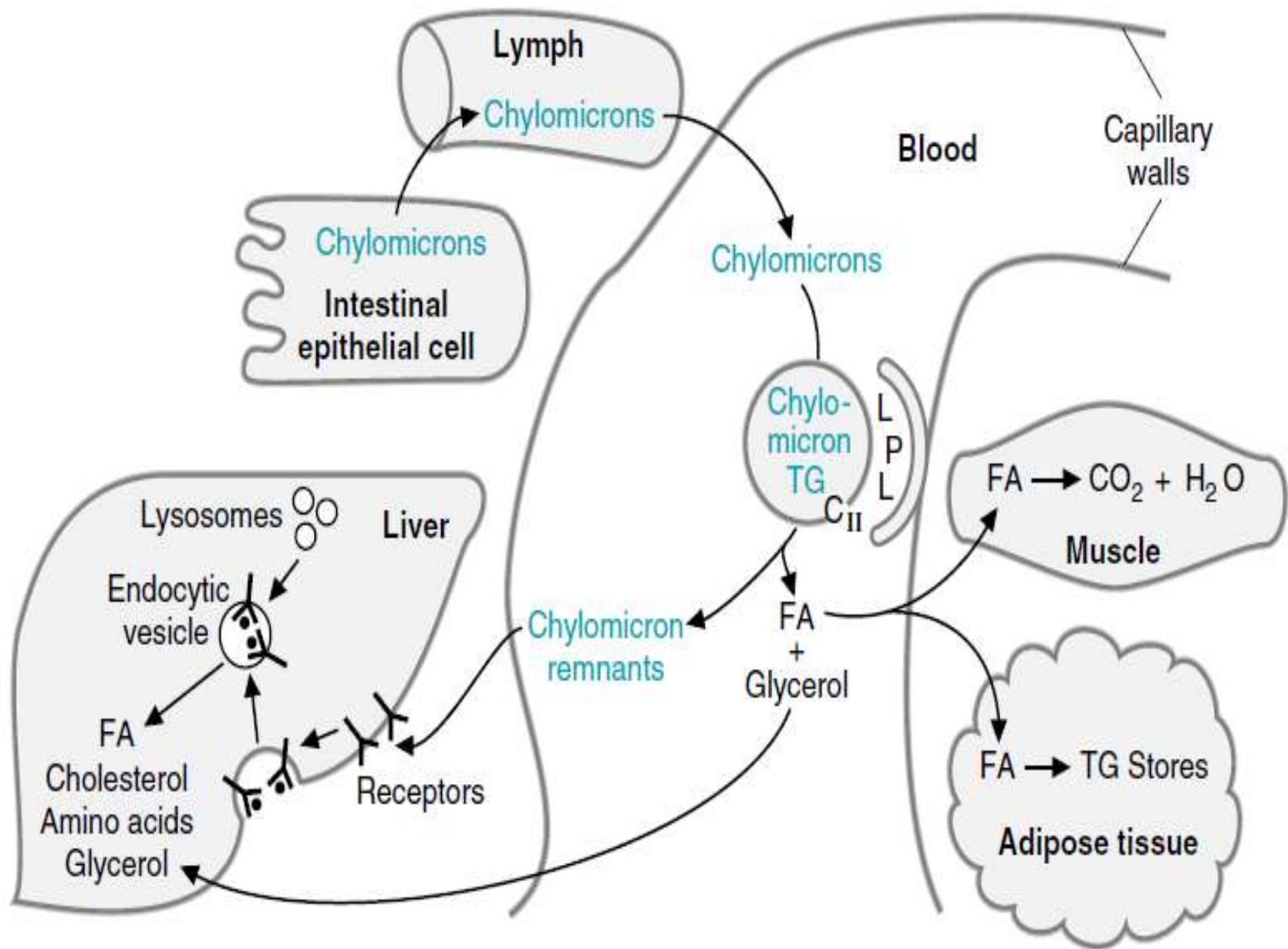
- ❑ Extracellular enzyme, anchored by heparan sulfate to the capillary walls of most tissues
- ❑ Predominantly present in **adipose tissue, cardiac & skeletal muscle**
- ❑ Requires ApoC-II for activation
- ❑ Degrades **TG into glycerol and free fatty acids**
- ❑ Insulin stimulates its synthesis and transfer to the luminal surface of the capillary
- ❑ During starvation activity declines in adipocytes while increases in cardiac muscles.



METABOLISM OF CHYLOMICRONS

- ❑ Degradation of TAG leads to decrease in the size of chylomicron particles and increases its density.
- ❑ apoC-II is returned to HDL leaving behind chylomicron remnant which has apoE and apoB48.
- ❑ Liver cells recognize apoE and rapidly take up chylomicron remnants.
- ❑ In liver cells they are acted upon by lysosomal enzymes and degradation of all the components take place , releasing amino acids, free cholesterol and fatty acids.
- ❑ The receptor is recycled.



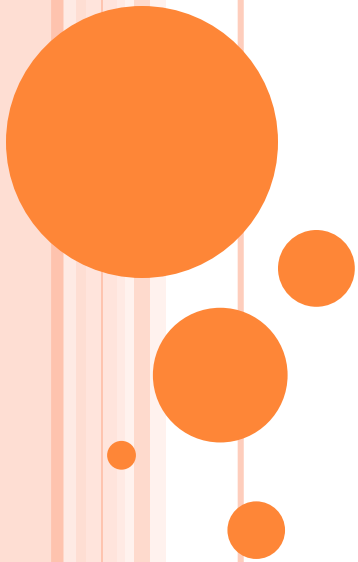


USE OF DIETARY LIPIDS BY THE TISSUES

- ❑ **Fate of fatty acids:** oxidized by various tissues for energy production and
- ❑ **Fate of Glycerol :** used by liver to form glycerol 3-phosphate which can enter glycolysis or gluconeogenesis.



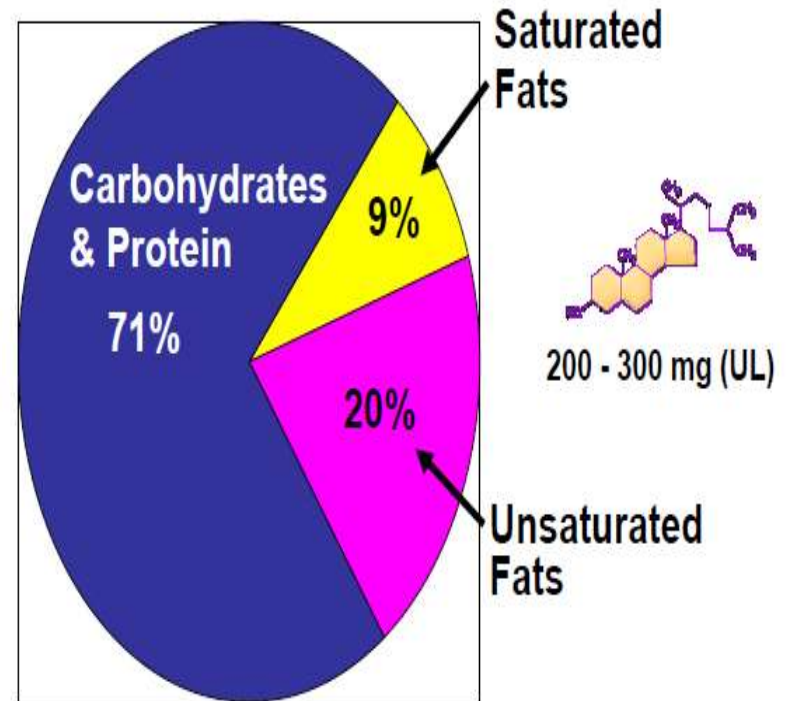
OXIDATION OF FATTY ACIDS



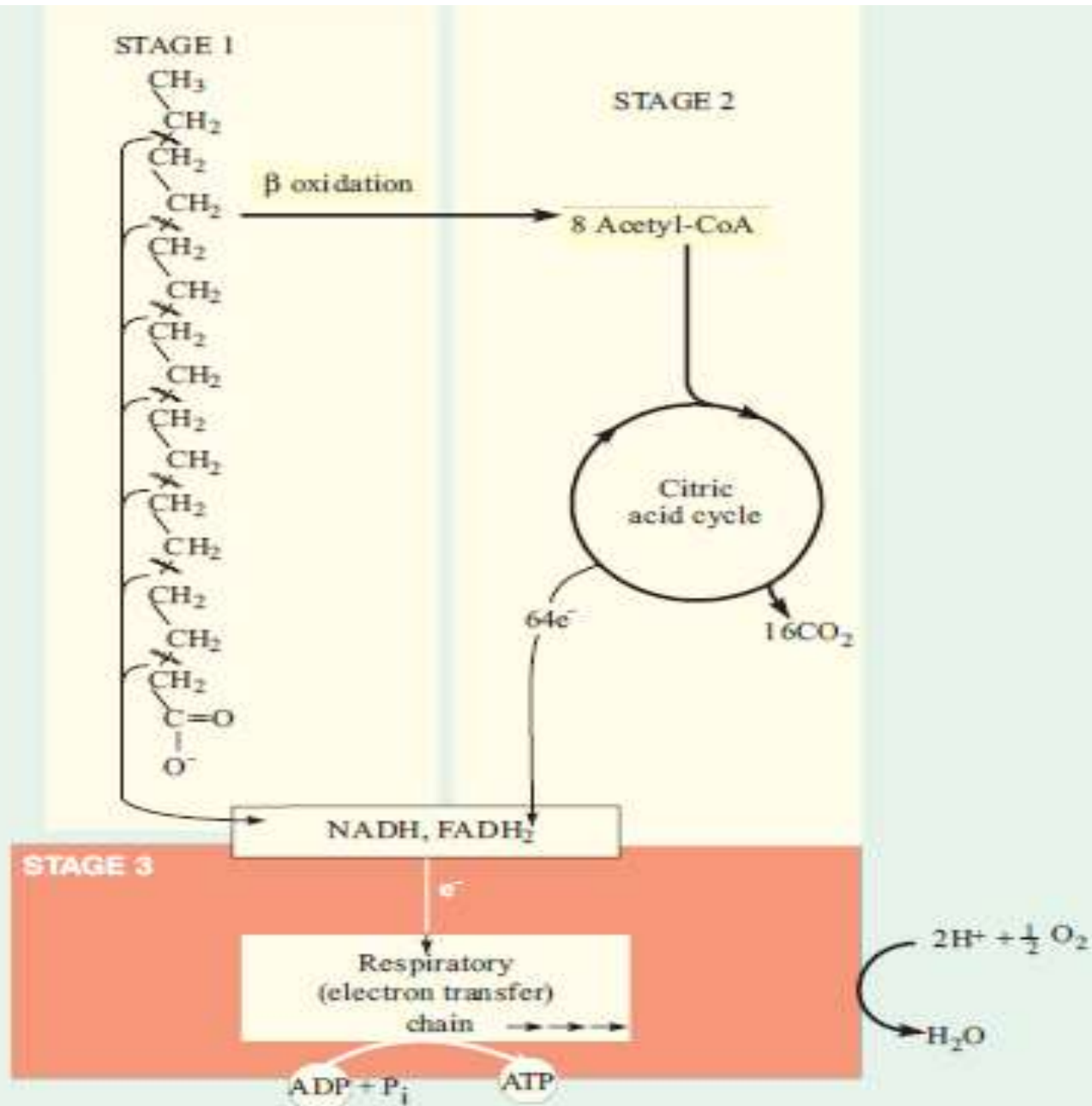
FUNCTIONS OF FATTY ACID OXIDATION

1. *Fatty Acid Oxidation Provides Energy for Cellular and Metabolic Work*

- ❑ Generating 9 kcal/g compared to 4 kcal/g from glucose
- ❑ Recommended fat in diet (29-31%)
- ❑ Current , American diets typically contain 35 to 50% of calories as fat
- ❑ Ethiopian diet ?



Oxidation of fatty acids takes place in 3 stages



FUNCTIONS OF FATTY ACID OXIDATION

2. Fatty Acid Oxidation Provides Fuel to the Brain during Starvation

- ❑ Brain preferably use glucose.
- ❑ During starvation, where is scarcity of glucose. it can also use ketone bodies as source of energy
- ❑ Oxidation of ketones by the brain reduces the brain's dependence on *glucose and thus decreases the body's need to catabolize muscle proteins to provide amino acid carbon skeletons for gluconeogenesis.*



FUNCTIONS OF FATTY ACID OXIDATION

3. *Fatty Acid Oxidation Generates Heat*

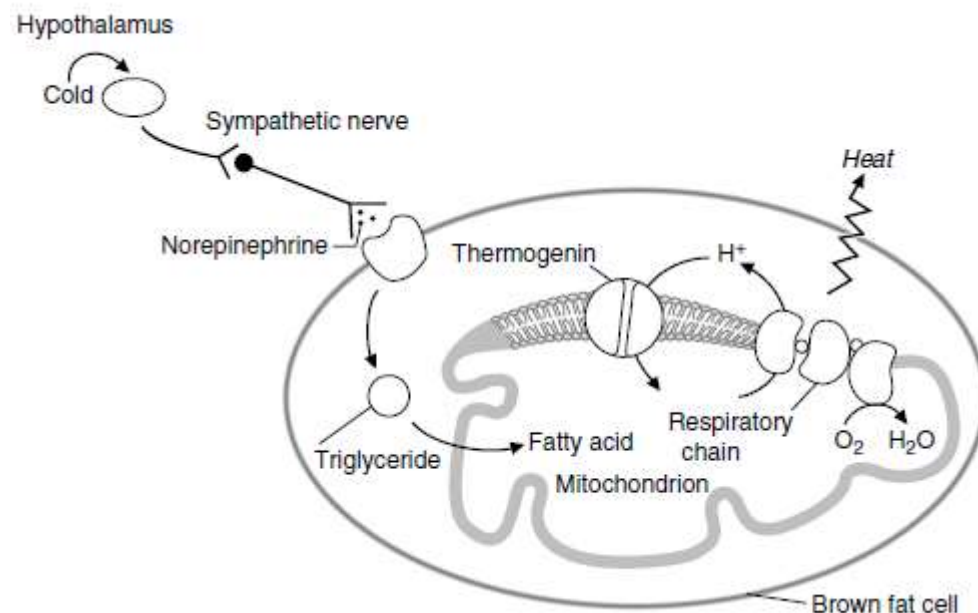
- Brown fat is a specialized tissue that has a high metabolic rate although it does not produce very much ATP from the NADH and FADH₂ generated during the oxidation of fatty acids.



Fig. 24-9. A grizzly bear preparing its hibernation nest, near the McNeil River in Canada.

FUNCTIONS OF FATTY ACID OXIDATION

- The presence of thermogenin or “uncoupling protein” in the inner membrane of the mitochondria of brown fat results in the generation of heat rather than ATP from electron flow through the electron-transport chain



FUNCTIONS OF FATTY ACID OXIDATION

4. Fatty Acid Oxidation Generates Intermediates in Synthetic processes

- ❑ Examples include: Phospholipids (in membranes).
- ❑ Eicosanoids, including prostaglandins and leucotrienes, which play a role in physiological regulation.
- ❑ Acetyl CoA
- ❑ Propionyl CoA



TISSUES IN WHICH FATTY ACID OXIDATION IS ACTIVE

- ❑ All cells and tissues **except red blood cells** and **the brain** oxidize fatty acids to generate ATP.
- ❑ Fatty acid oxidation is most active in tissues that are highly active metabolically.
- ❑ Thus, **skeletal and heart muscle** in particular have a large capacity for oxidizing fatty acids.
- ❑ Normally, 60 to 90% of the energy required for contraction of the heart is derived from the oxidation of fatty acids



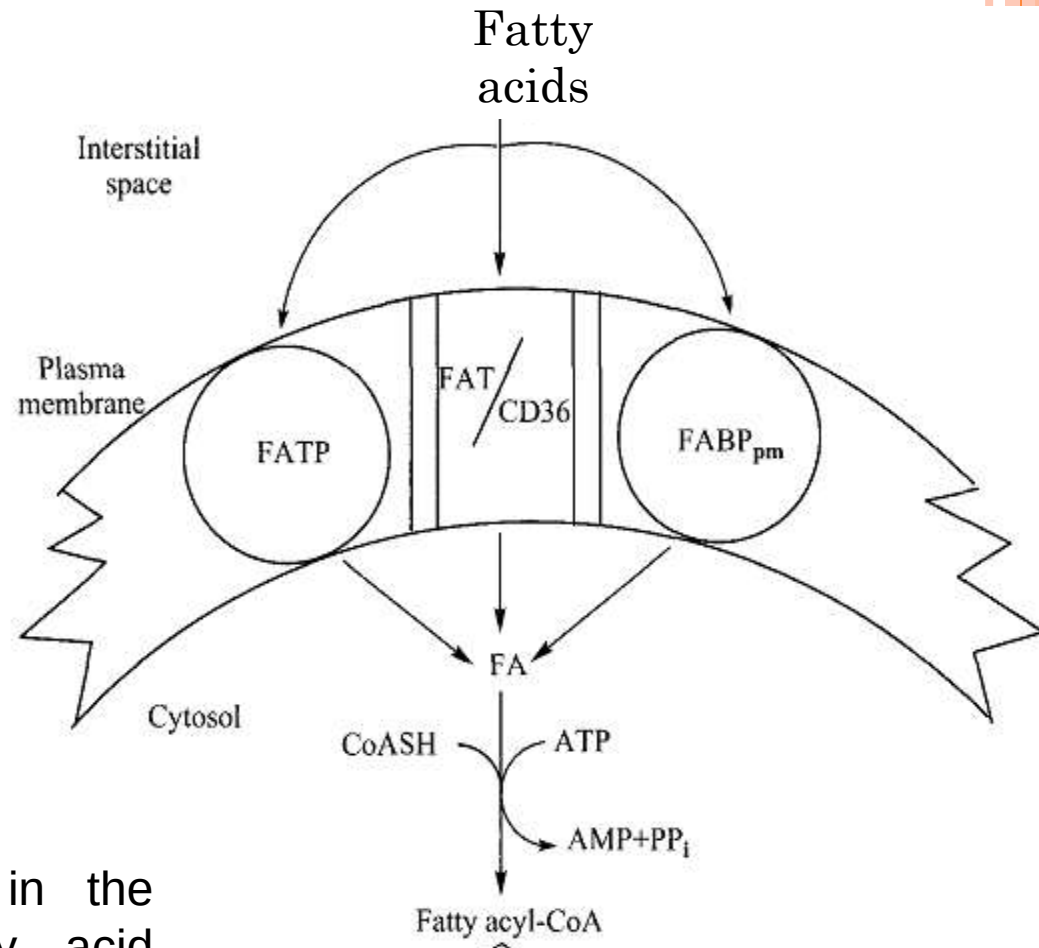
PHYSIOLOGICAL CONDITIONS IN WHICH FATTY ACID OXIDATION IS MOST ACTIVE

- Fatty Acid Oxidation Increases in the Fasted State
 - Depletion of glycogen in the liver/gluconeogenesis
- Fatty Acid Oxidation Increases During Exercise



TRANSPORT OF FATTY ACIDS

FAT/CD36 is the major fatty acid transporter in heart muscle, skeletal muscle, adipocytes, and intestine



FAT/CD36, Fatty acid translocase in the plasma membrane, MM. FATP, fatty acid transport protein; FABP, fatty acid-binding protein of the plasma membrane.

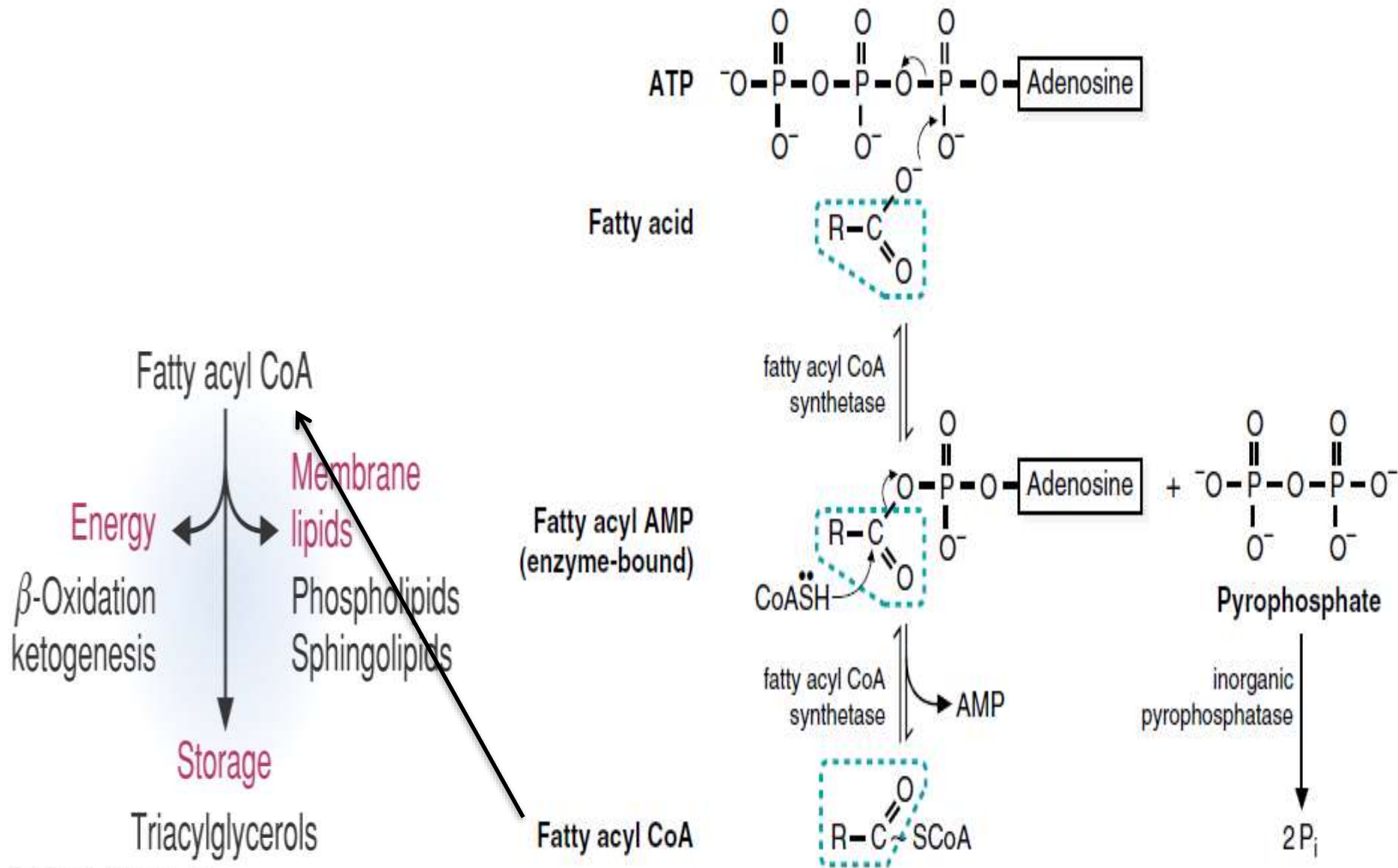
ACTIVATION OF FREE FATTY ACIDS

- Once inside cells, fatty acids must be activated before they can be metabolized.



- **Acyl-CoA synthetases** are **localized** to three different sites in cells:
 - The cytosolic face of the endoplasmic reticulum
 - The outer mitochondrial membrane
 - The peroxisomal membrane.
- There are at least five genetically distinct acyl-CoA synthetase (ACS) isoforms, each having its own specificity with regard to the fatty acid substrate.
- *For example, ACS4 prefers polyunsaturated fatty acids such as arachidonic acid*





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Activation of a fatty acid by a fatty acyl CoA synthetase. The fatty acid is activated by reacting with ATP to form a high-energy fatty acyl AMP and pyrophosphate. The AMP is then exchanged for CoA. Pyrophosphate is cleaved by a pyrophosphatase.

TYPES OF FATTY ACID OXIDATION

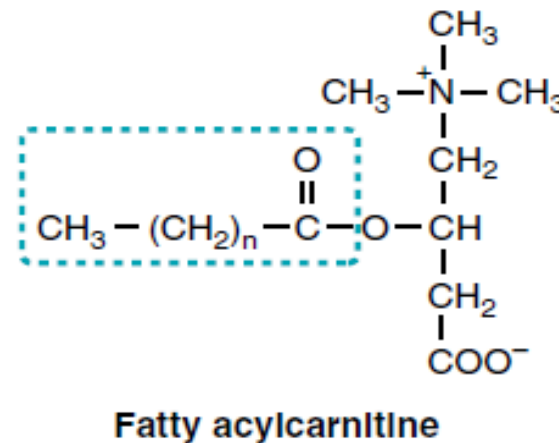
Fatty acid oxidation in the cell can be done in three ways

- ❑ α -oxidation
- ❑ β -oxidation
- ❑ ω -oxidation



1-TRANSPORT OF LONG CHAIN FATTY ACIDS INTO MITOCHONDRIA

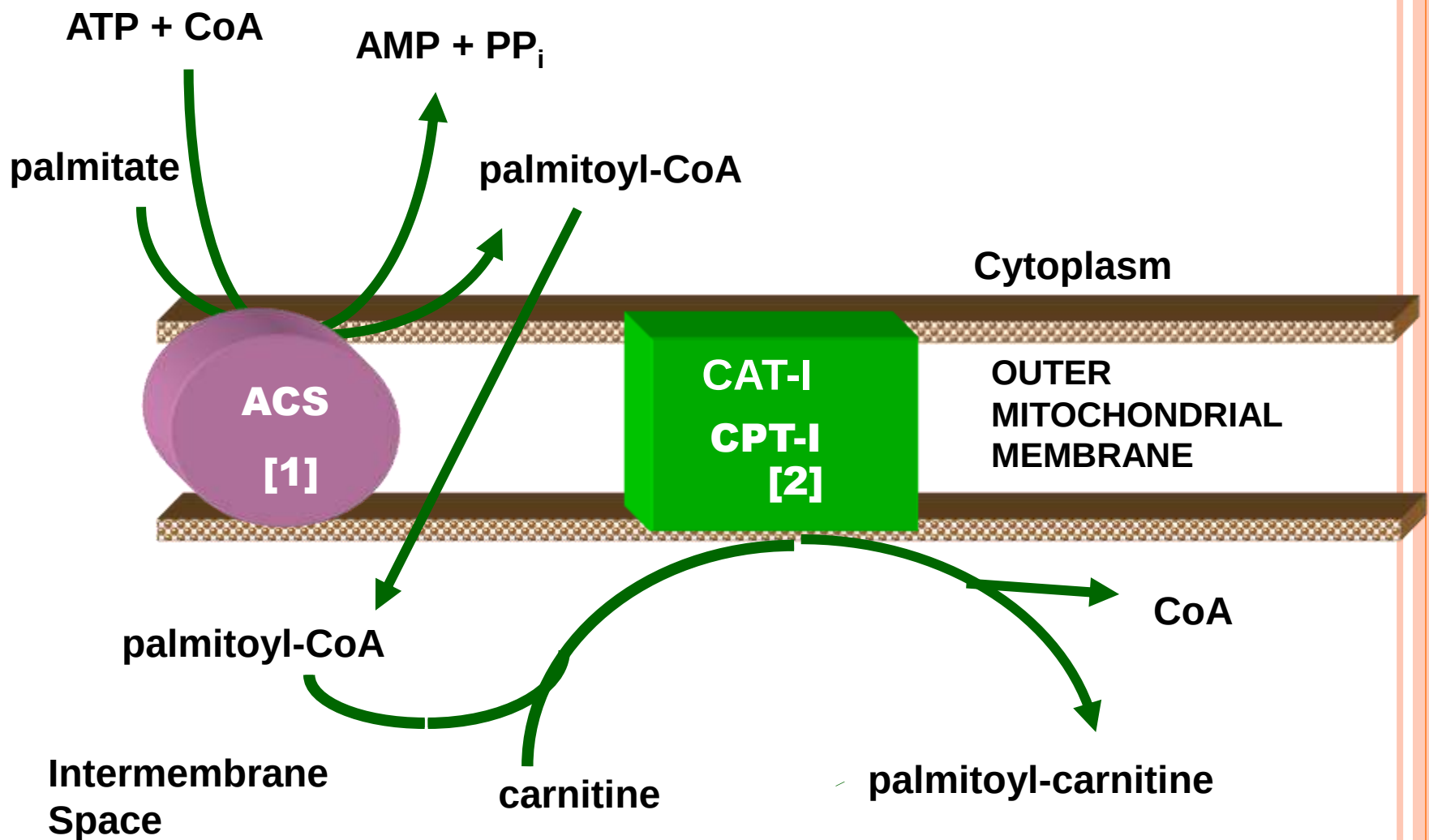
- ❑ β - oxidation occurs in mitochondrial matrix
- ❑ Inner mitochondrial membrane is impermeable to CoA
- ❑ Specialized carrier is required to transport long chain acyl groups from cytosol to mitochondria
- ❑ This carrier is CARNITINE
- ❑ It is a rate-limiting transport process and is called CARNITINE SHUTTLE.



STEPS IN LCFA TRANSPORT

- 1- In the inter-membrane space of the mitochondria, fatty acyl CoA reacts with carnitine in a reaction catalyzed by **carnitine acyltransferase I (CAT-I)**, yielding CoA and fatty acyl carnitine.
- The resulting acyl carnitine crosses the inner mitochondrial membrane.



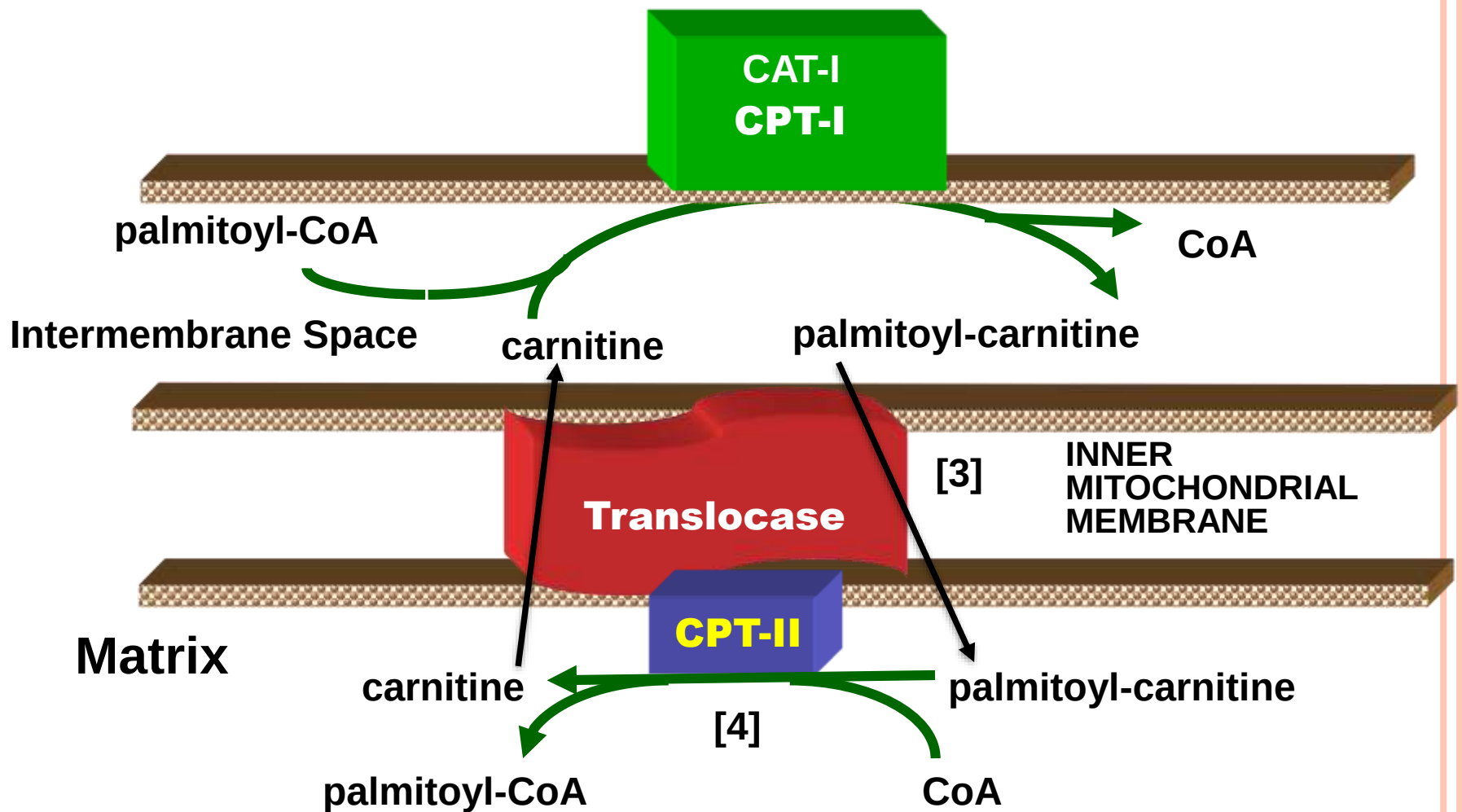


CPT-I defects cause severe muscle weakness because fatty acids are an important muscle fuel during exercise.

REGULATION OF CAT-I

- ❑ CAT-I is associated with the outer mitochondrial membrane.
- ❑ CAT-I reaction is rate-limiting;
- ❑ The enzyme is allosterically inhibited by **malonyl CoA**.
- ❑ Malonyl CoA concentration would be high during fatty acid synthesis.
- ❑ Inhibition of CAT-I by malonyl CoA **prevents simultaneous synthesis and degradation of fatty acids**.





Mitochondrial uptake via of palmitoyl-carnitine via the carnitine-acylcarnitine translocase.

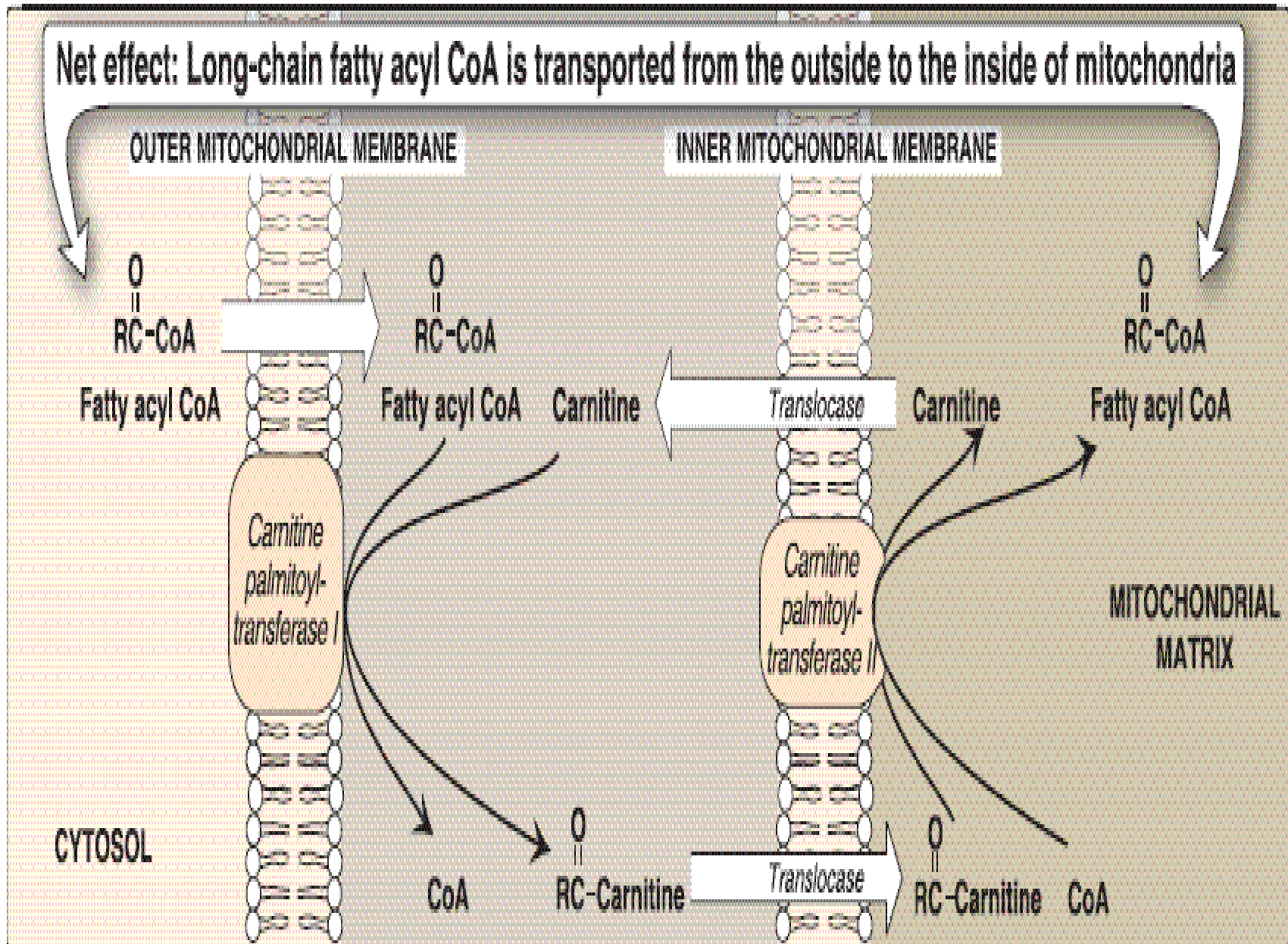


STEPS IN LCFA TRANSPORT

- ❑ 2-Fatty acyl carnitine is transported across the inner mitochondrial membrane in exchange for carnitine by **carnitine-acylcarnitine translocase**.
- ❑ In the mitochondrial matrix fatty acyl carnitine reacts with CoA in a reaction catalyzed by **carnitine acyltransferase II (CAT-II)**, yielding fatty acyl CoA and carnitine.
- ❑ The fatty acyl CoA is now ready to undergo beta-oxidation.



CARNITINE SHUTTLE



SOURCES OF CARNITINE

- Carnitine is obtained from the diet or can be synthesized in liver and kidney from amino acids **lysine and methionine**.
- Skeletal and heart muscles cannot synthesize carnitine and depend on diet or endogenous synthesis.
- The reactions use **S-adenosylmethionine (SAM)** to donate methyl groups, and **vitamin C** (ascorbic acid) is also required for these reactions



CARNITINE DEFICIENCIES

PRIMARY CAUSES:-

- ❑ Genetic CAT-I deficiency: Mainly affects liver. Liver cannot synthesize glucose in a fast, results in hypoglycemia, coma and death.
- ❑ CAT-II deficiency: Mainly affects skeletal and cardiac muscles.
- ❑ Defect in renal tubular reabsorption of carnitine.
- ❑ Defect in carnitine uptake by cells.



CARNITINE DEFICIENCIES

SECONDARY CAUSES :

- ❑ Liver diseases; decreased endogenous synthesis.
- ❑ Malnutrition or strict vegetarian diet
- ❑ Increased metabolic demands
- ❑ Hemodialysis



ENTRY OF SHORT AND MEDIUM CHAIN F.A INTO MITOCHONDRIA

- ❑ Carnitine and CAT system not required for fatty acids shorter than 12 carbon length.
- ❑ They are activated to their CoA form inside the mitochondrial matrix.
- ❑ Not inhibited by malonyl CoA.



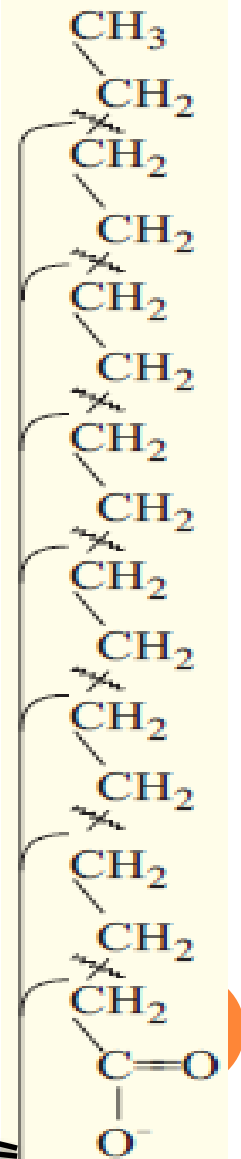
REACTIONS OF BETA -OXIDATION

β -oxidation is the process by which the fatty acids undergo oxidative removal of successive two-carbon units in the form of acetyl-CoA, starting from the carboxyl end of the fatty acyl chain.

The products of β -oxidation are:

□ Acetyl CoA

□ FADH_2 , NADH and H^+



FOUR STEPS- FOUR ENZYMES

There are four individual reactions of β -oxidation, each catalyzed by a separate enzyme.

- 1-Dehydrogenation between carbon 2 and 3 in a FAD-linked reaction. The enzyme is **acyl CoA dehydrogenase**.
- 2-Hydration of the double bond by **enoyl CoA hydratase**.
- 3-A second dehydrogenation in a NAD-linked reaction. Enzyme is **3-hydroxyacyl CoA dehydrogenase**.
- 4-Thiolytic cleavage of the thioester by **beta-ketoacyl CoA thiolase**.

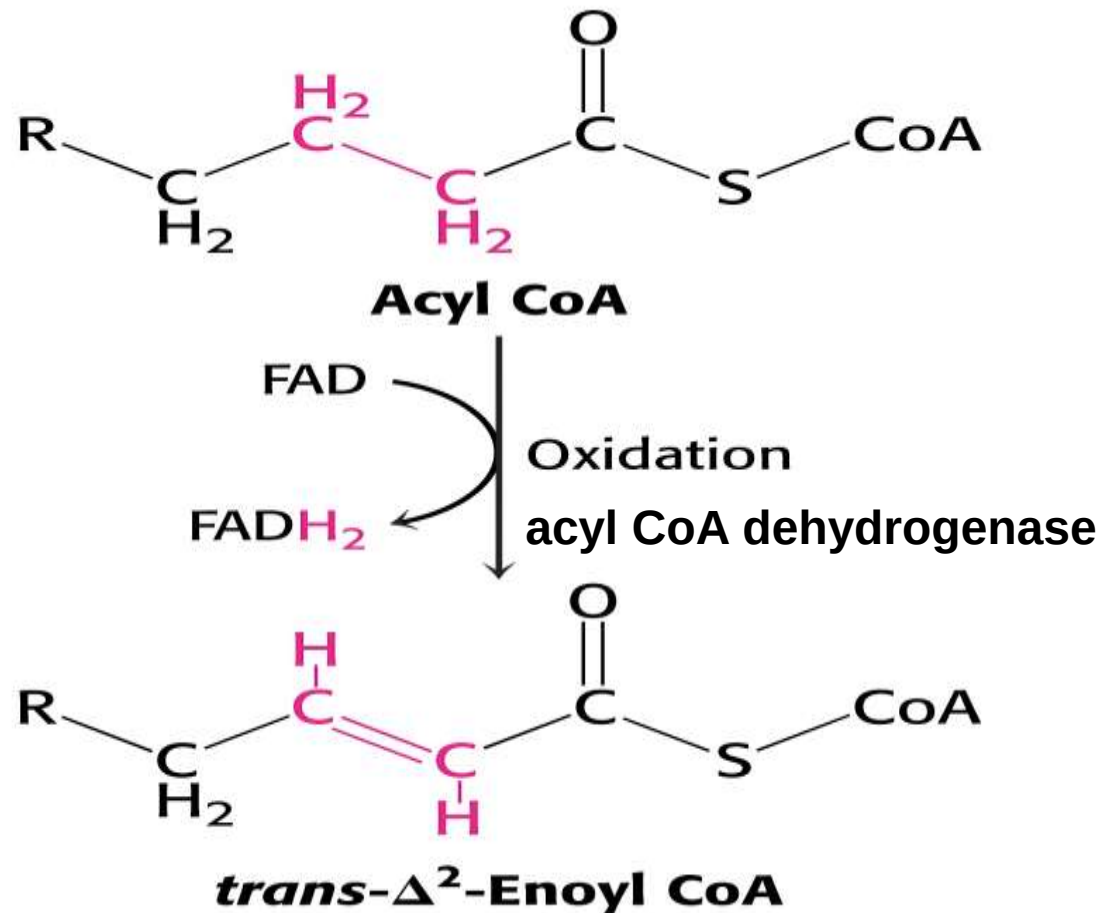
This sequence of four steps is repeated until the fatty acyl chain is completely degraded to acetyl CoA



FOUR STEPS- FOUR ENZYMES

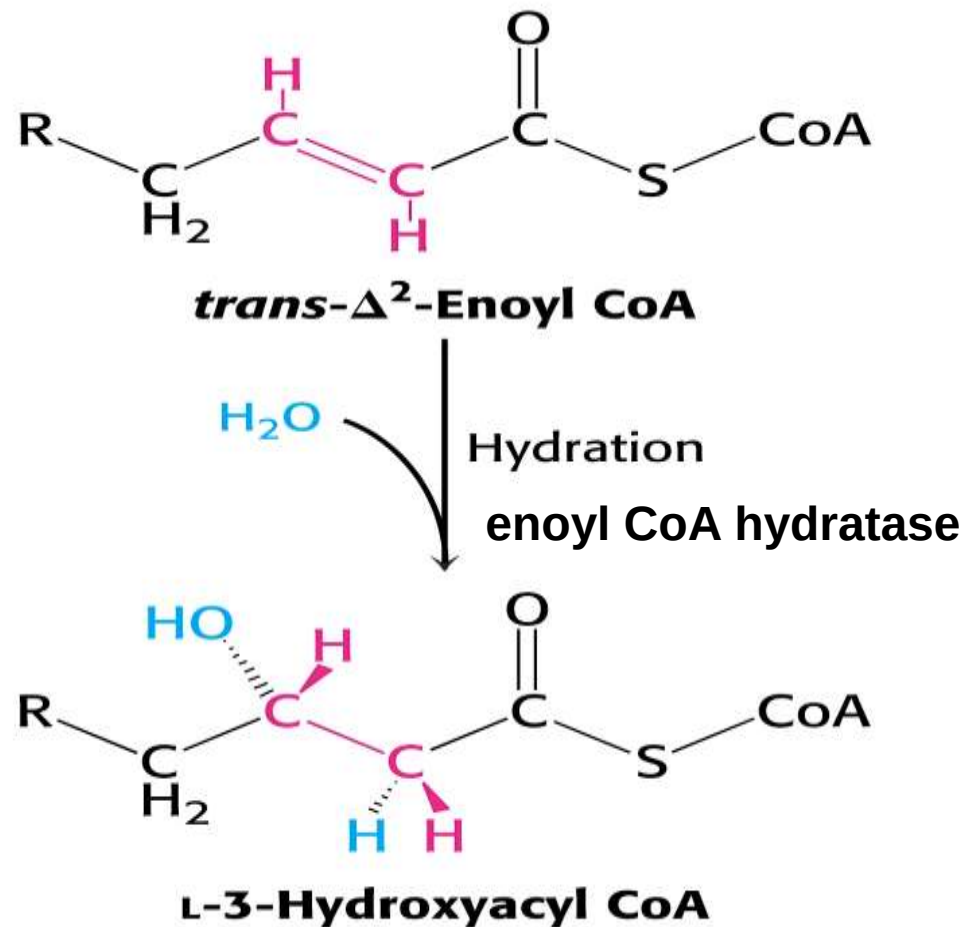
Each round in fatty acid degradation involves four reactions

1. Oxidation to *trans*- Δ^2 -Enoyl-CoA by



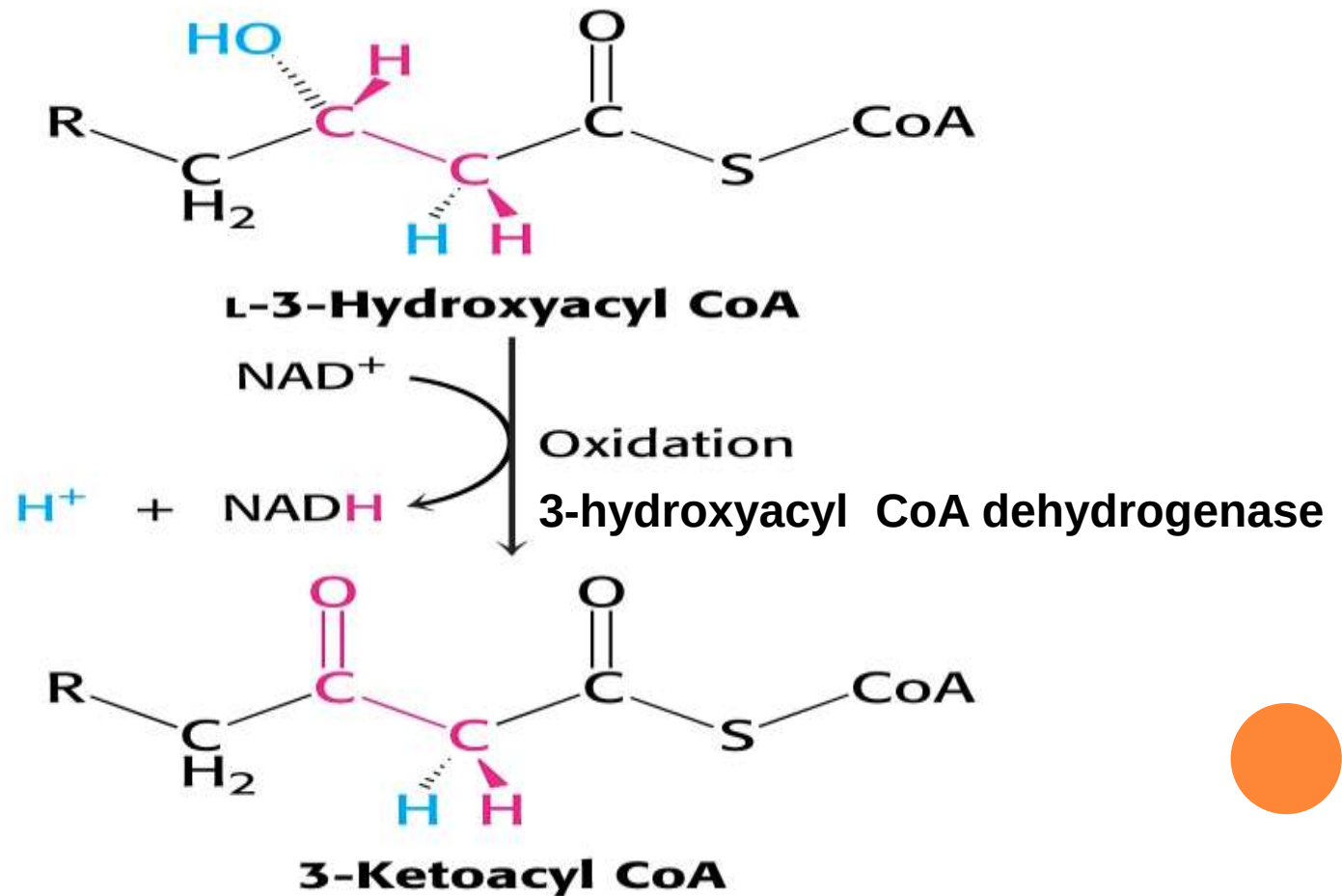
FOUR STEPS- FOUR ENZYMES

2. Hydration to L-3-Hydroxyacyl CoA



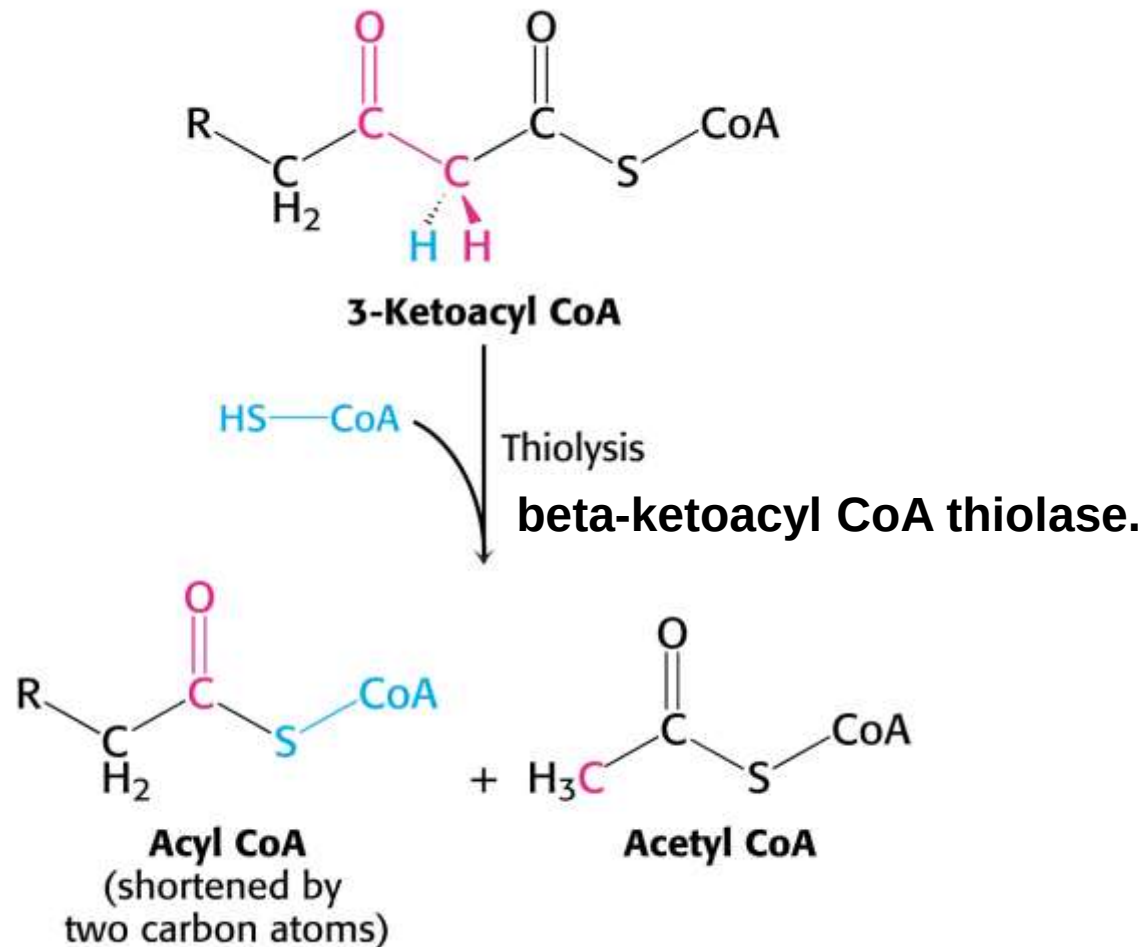
FOUR STEPS- FOUR ENZYMES

3. Oxidation to 3-Ketoacyl CoA

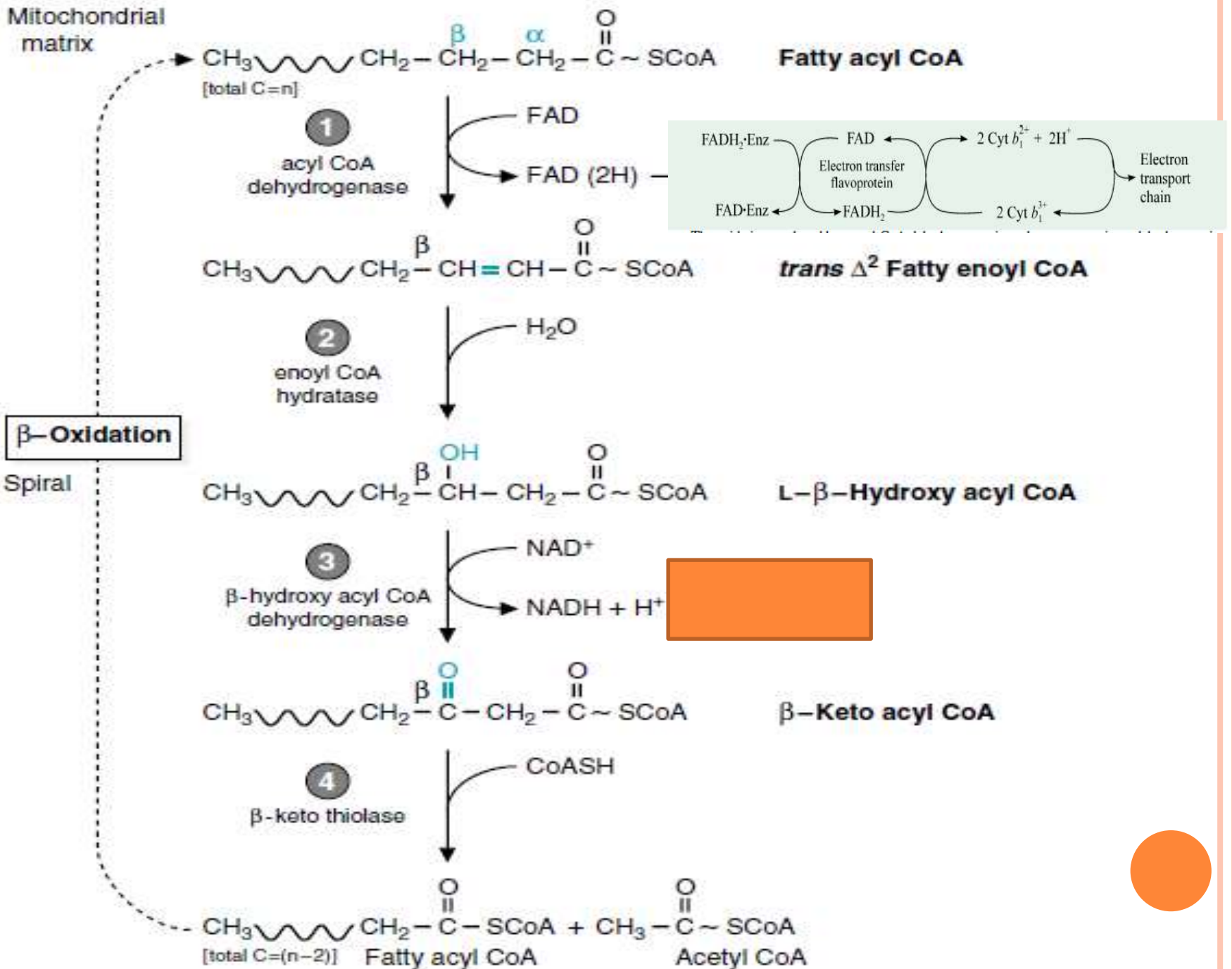


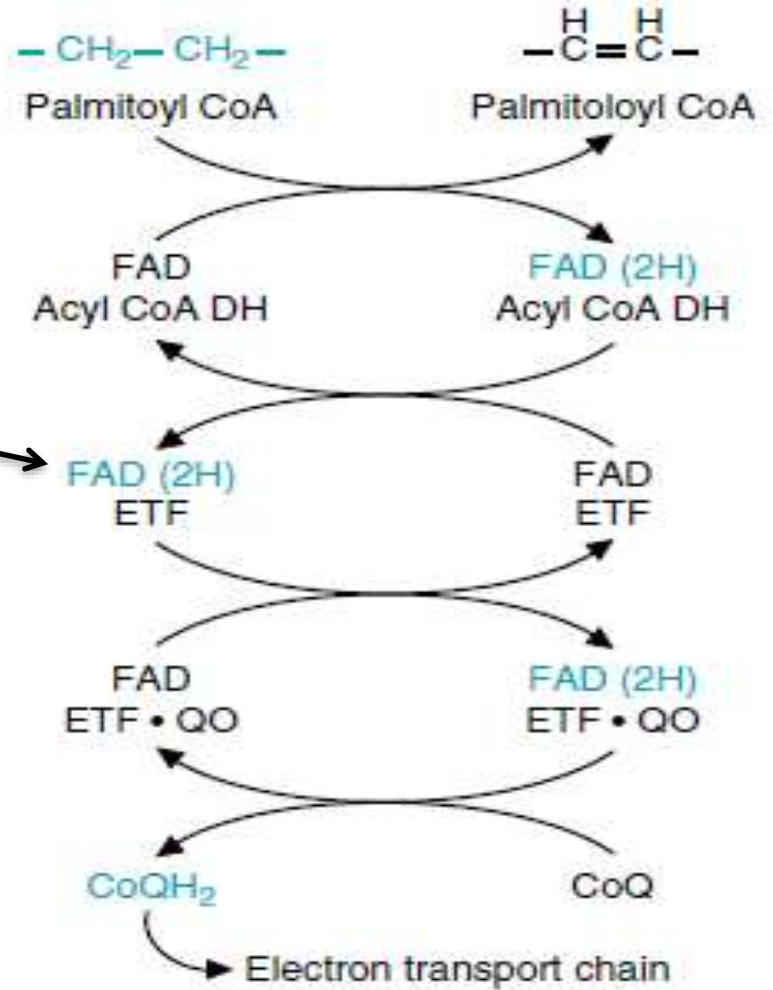
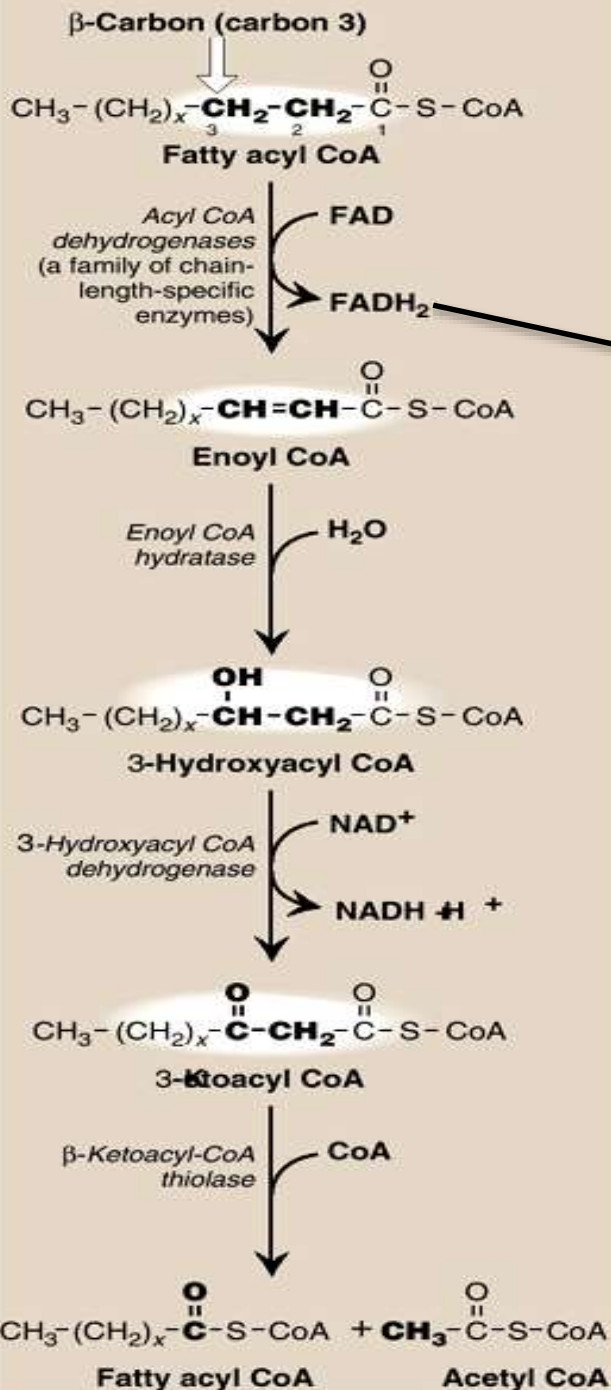
FOUR STEPS- FOUR ENZYMES

4. Thiolysis to produce Acetyl-CoA



Mitochondrial matrix





Transfer of electrons from acyl CoA dehydrogenase to the electron transport chain. Abbreviations: ETF, electron-transferring flavoprotein; ETF-QO, electron-transferring flavoprotein-Coenzyme Q oxidoreductase.

TYPES OF FATTY ACYL CoA DEHYDROGENASES

- ❑ Long chain fatty **acyl CoA dehydrogenase** (LCAD) acts on chains greater than C12.
- ❑ Medium chain fatty acyl CoA dehydrogenase (MCAD) acts on chains of C6 to C12.
- ❑ Short chain fatty acyl CoA dehydrogenase (SCAD) acts on chains of C4 to C6.

MCAD deficiency is thought to be one of the most common inborn errors of metabolism.



REACTION PRODUCTS

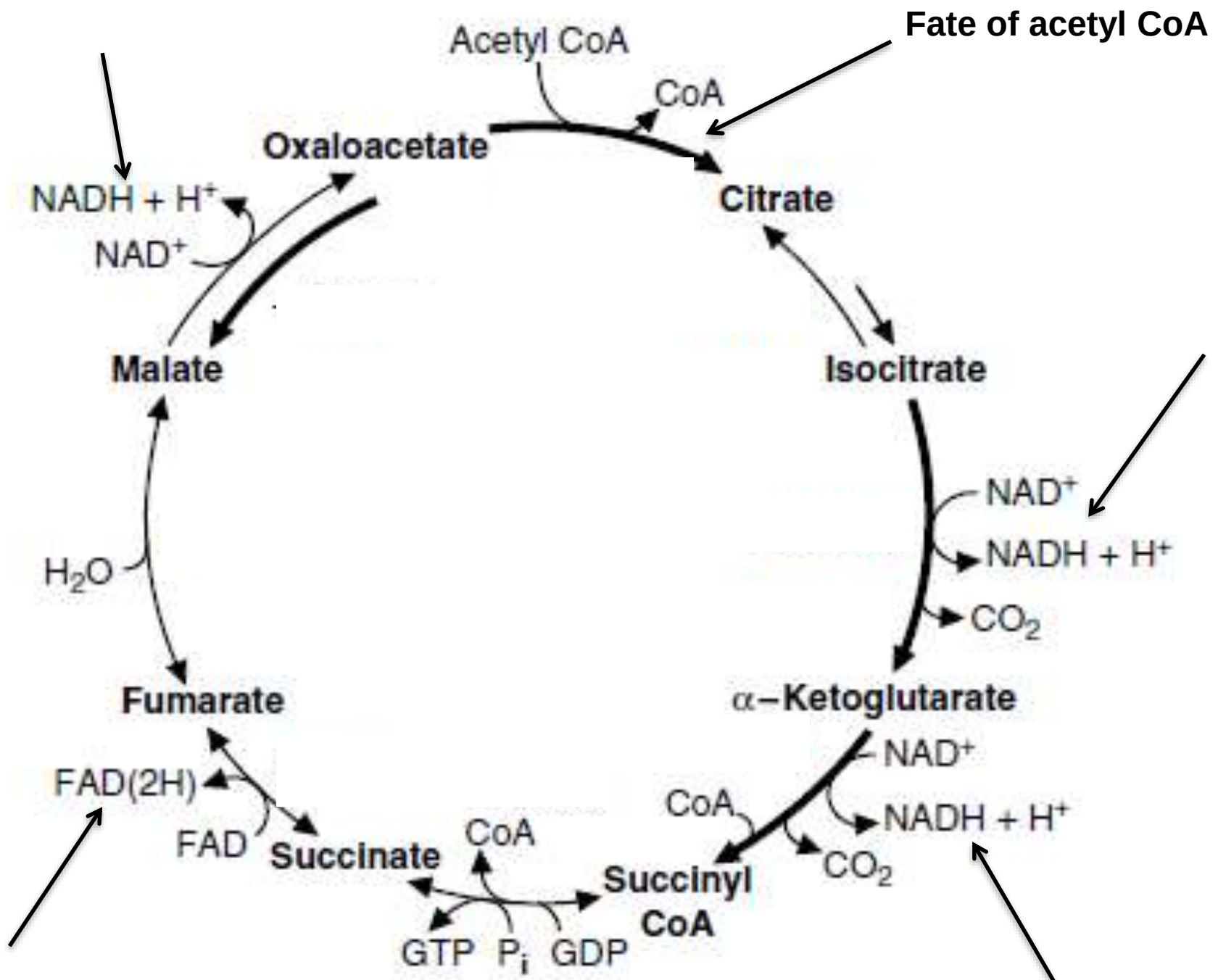
Fate of acetyl CoA

- Oxidation by the citric acid cycle to CO_2 and H_2O .
- In the liver only, acetyl CoA may be used for ketone body synthesis.

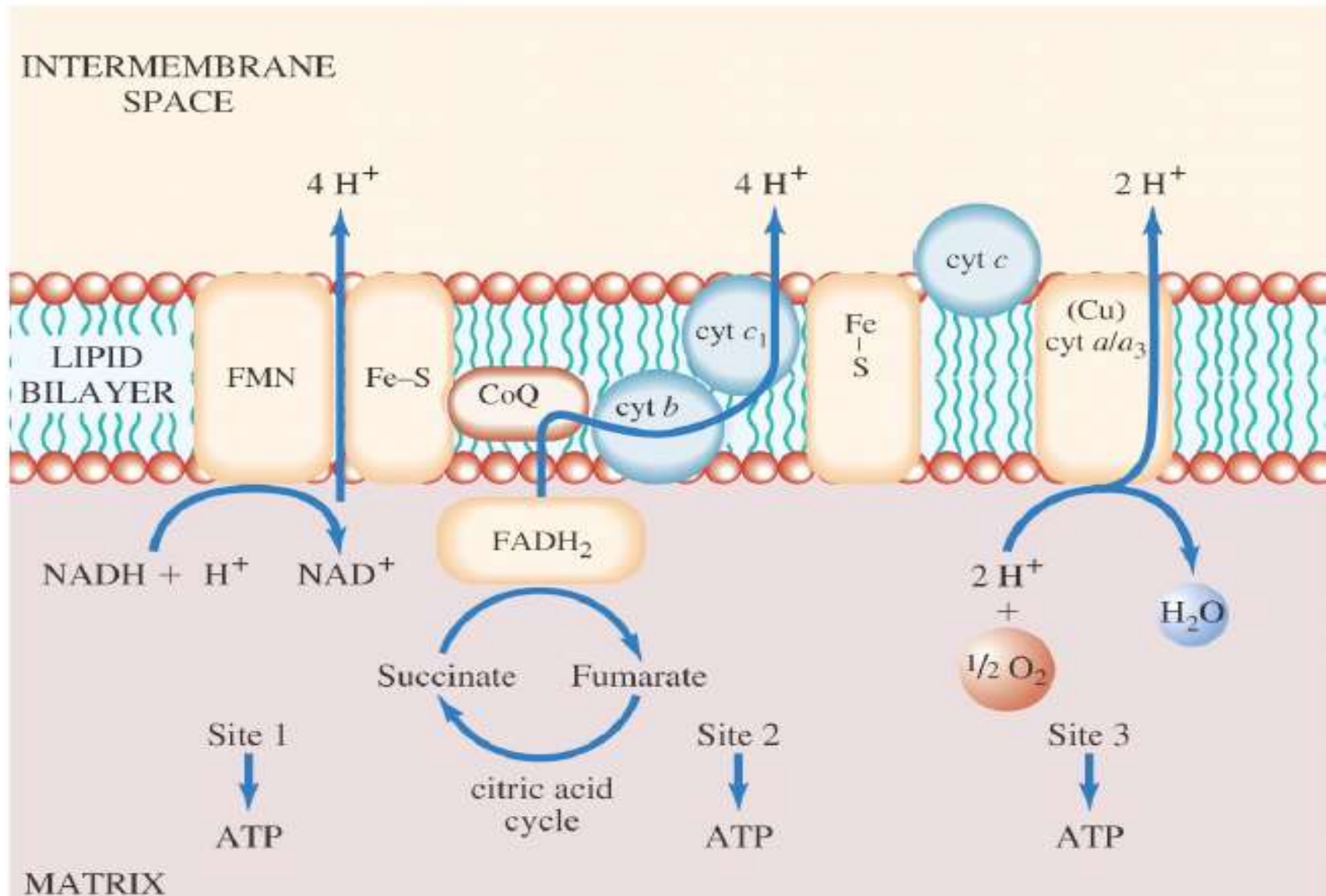
Fate of the FADH_2 and $\text{NADH} + \text{H}^+$

- FADH_2 and $\text{NADH} + \text{H}^+$ are oxidized by the mitochondrial electron transport system, yielding ATP.





FATE OF THE FADH_2 AND $\text{NADH} + \text{H}^+$



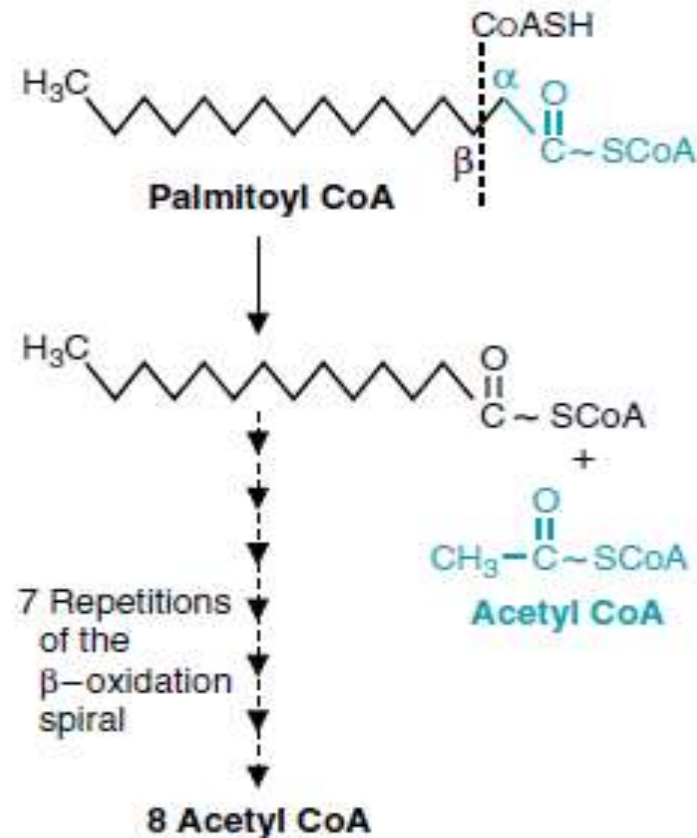
What is the total ATP yield for the oxidation of 1 mole of palmitic acid to carbon dioxide and water?

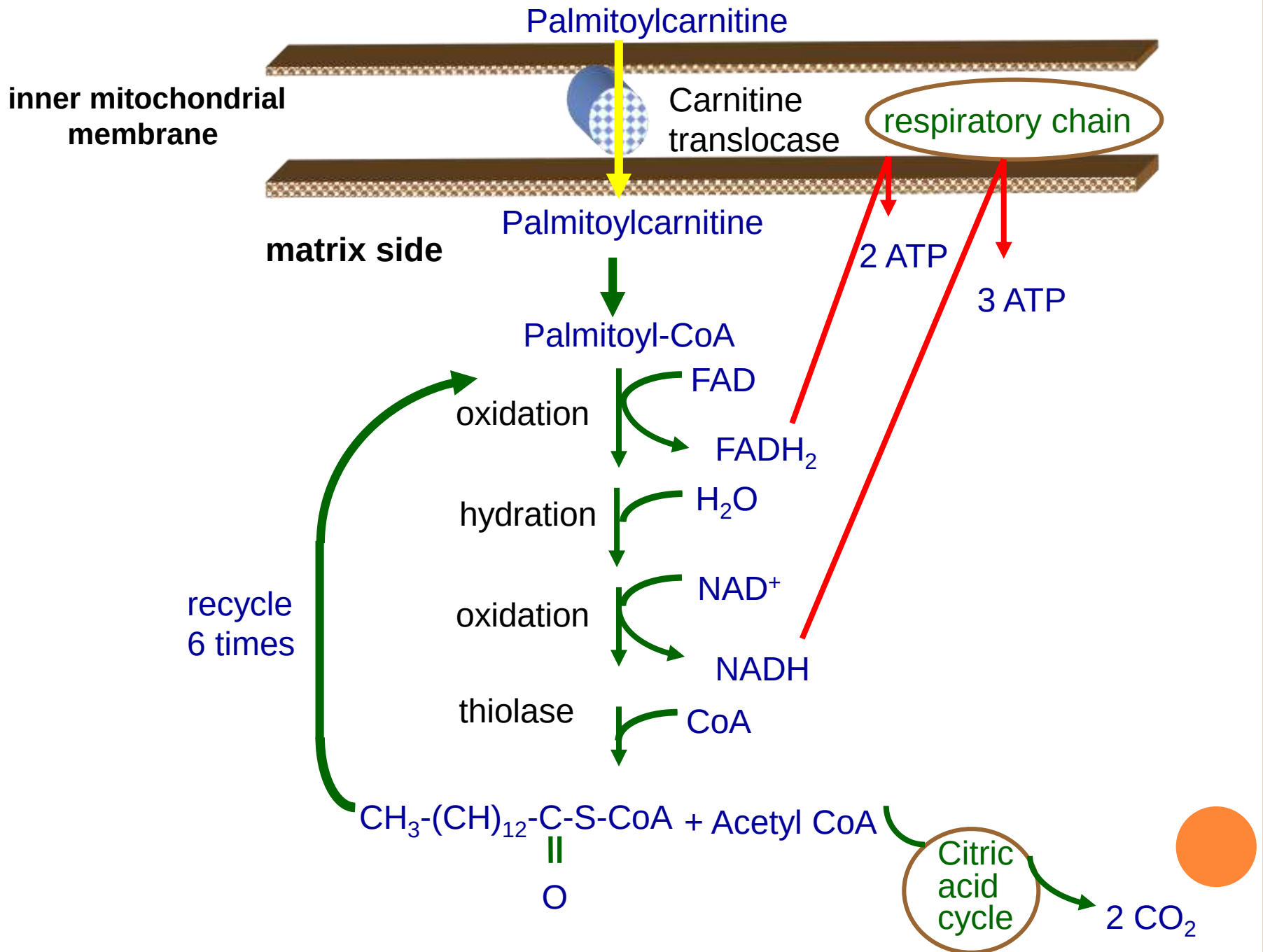


ENERGY YIELD FROM BETA OXIDATION OF PALMITIC ACID

Oxidation of one molecule of palmitoyl CoA to CO_2 and water produces

- 8 acetyl CoA
- 7 NADH
- 7 FADH_2





NET ENERGY PRODUCTION

$$\square 7 \text{ FADH}_2 = 2 \times 7 = 14 \text{ ATP}$$

$$\square 7 \text{ NADH} = 3 \times 7 = 21 \text{ ATP}$$

$$\square 8 \text{ Acetyl CoA} = 12 \times 8 = 96 \text{ ATP}$$

$$\text{Total ATP} = 131 \text{ ATP}$$

$\square$ 2 ATP are utilized during the formation of acyl CoA . Therefore net yield is 129 ATP.

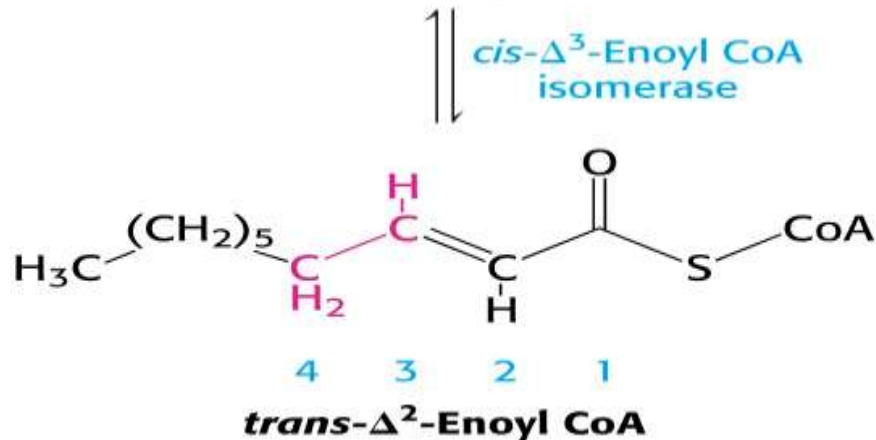
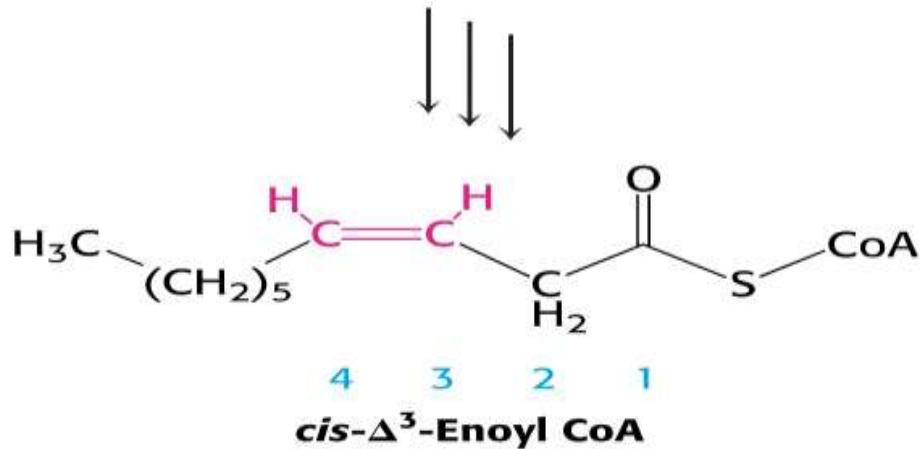
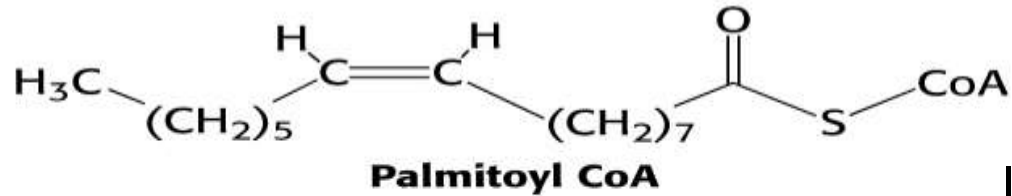


OXIDATION OF UNSATURATED FATTY ACIDS

- ❑ Undergoes beta oxidation
- ❑ Less energy yield
- ❑ Less formation of reducing equivalents as unsaturated F.A are not highly reduced.
- ❑ One or two additional enzymes are required based on the number of double bonds in the fatty acid backbone.



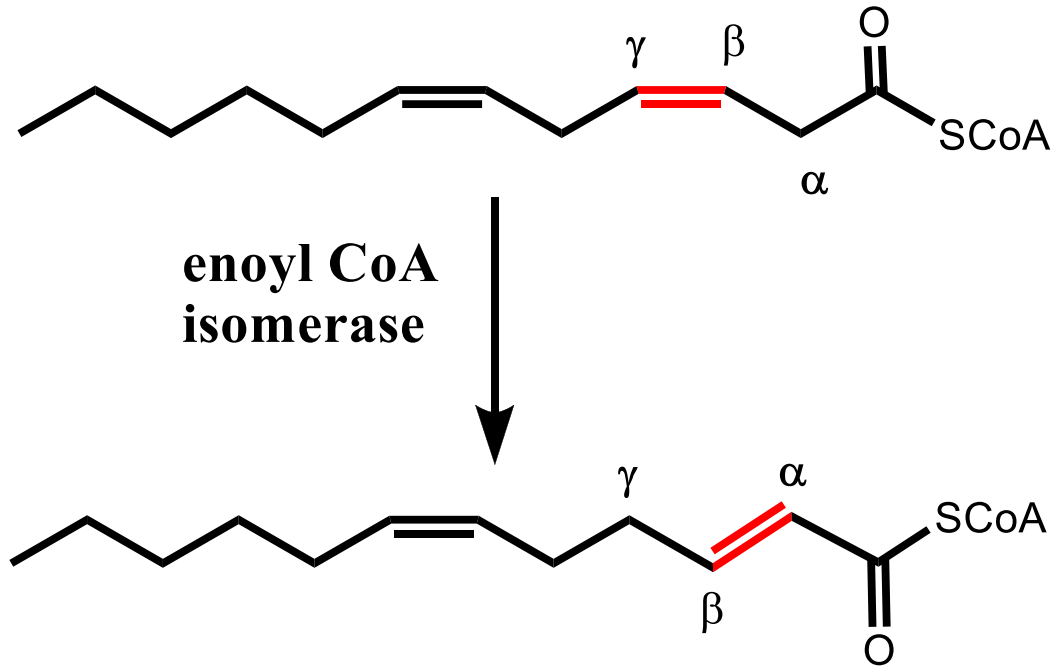
UNSATURATED FATTY ACIDS (MONOUNSATURATED)



If there is a double bond at an odd-numbered carbon (e.g., 16:1⁹), the action of **enoyl CoA isomerase** is required to move the naturally occurring *cis*- bond and convert it to the *trans*-bond used in beta-oxidation.

The product, with a *trans*-double bond, is a substrate for enoyl CoA hydratase, the second enzyme of beta-oxidation.

Special Case 1: Odd Unsaturation



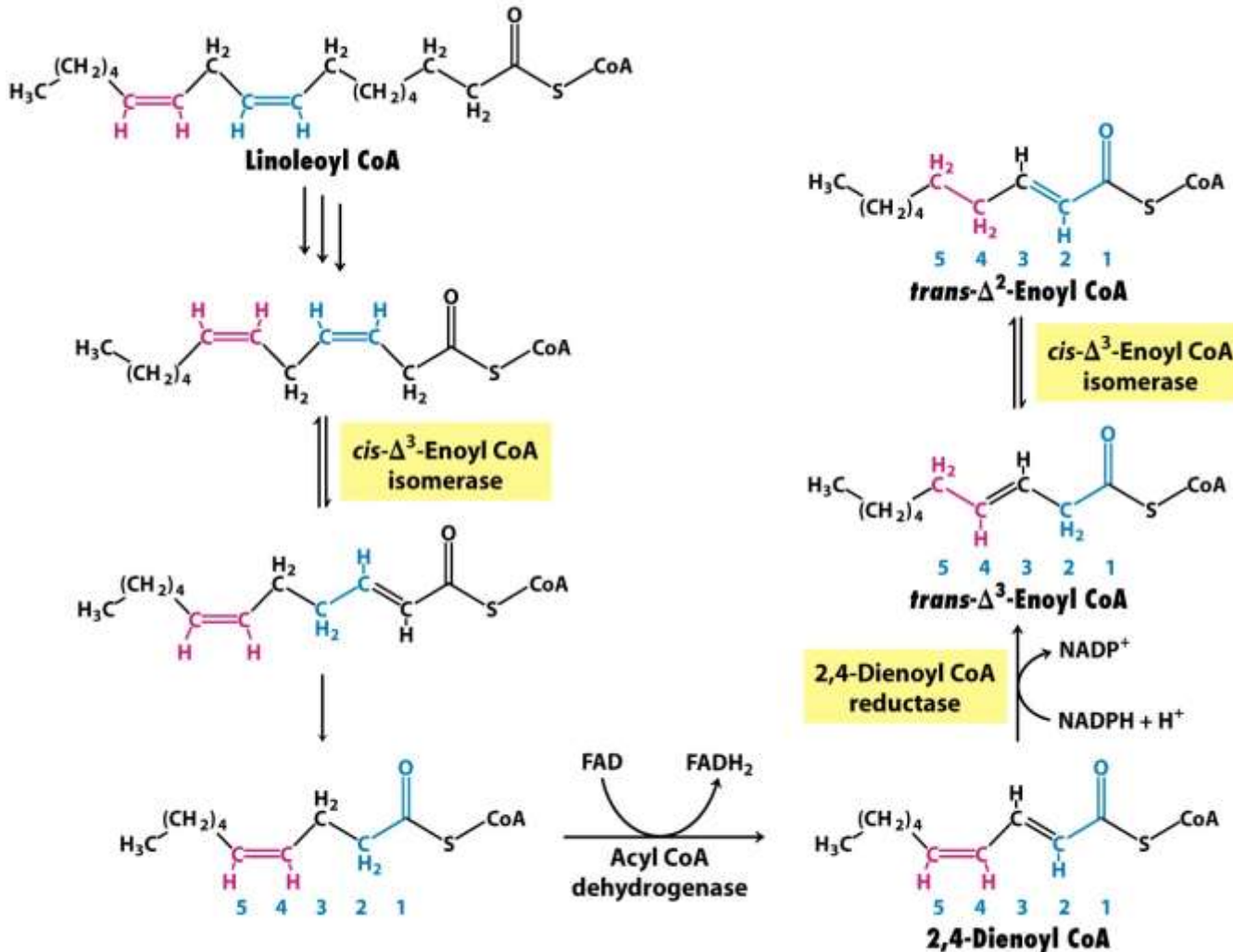
The GOOD News:

- Enzyme **enoyl-CoA isomerase** moves the double bond over 1 position.
- Oxidation process resumes

The BAD News:

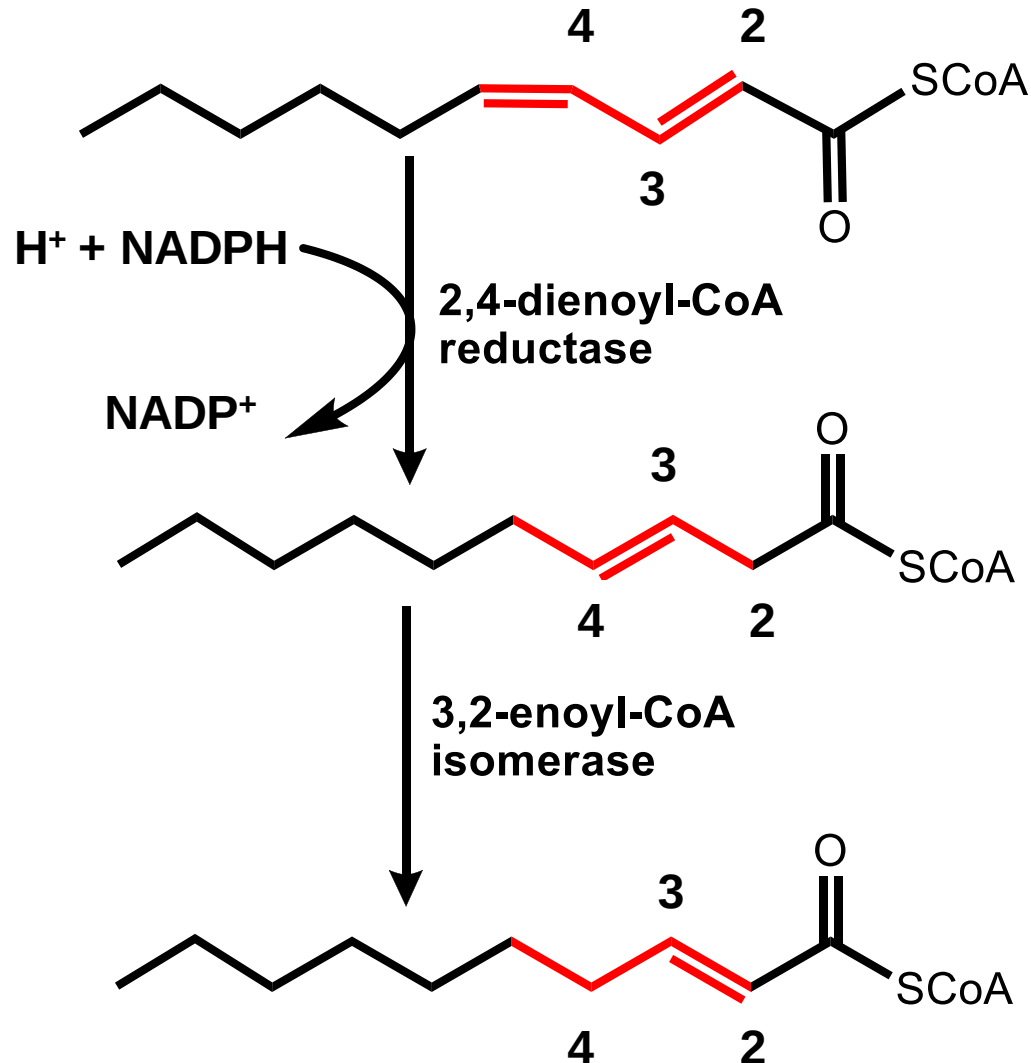
- The first oxidation step is skipped
- One less FADH_2 is made
 - 2 fewer ATPs

UNSATURATED FATTY ACIDS (POLYUNSATURATED)



In case of polyunsaturated fatty acids, e.g. linoleic acid that is 18:2(9,12), NADPH- dependent **Dienoyl CoA Reductase** is required in addition to isomerase

Special Case 2: Even Unsaturation

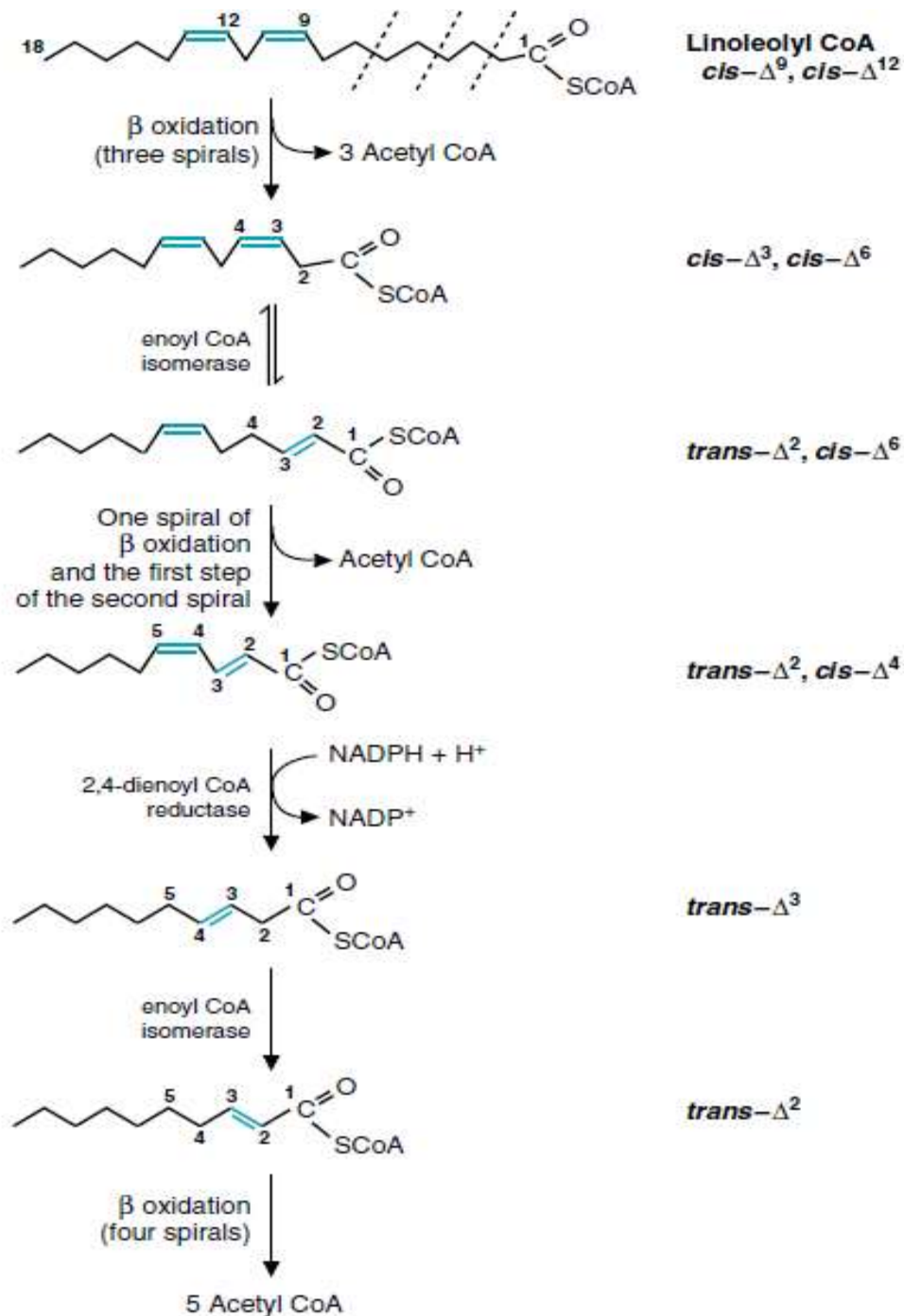


The GOOD News:

- Enzymes **2,4 dienoyl-CoA reductase** and **3,2 dienoyl-CoA isomerase** convert the γ - δ and α - β double bonds into a single α - β unsaturation.
- Oxidation process resumes

The BAD News:

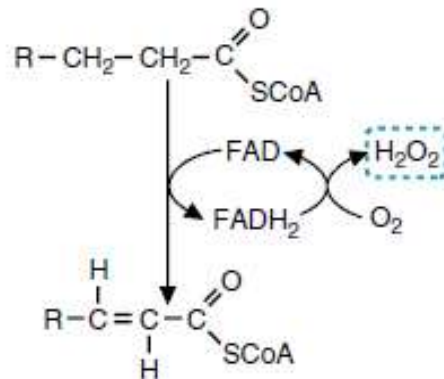
- The conversion costs one NADPH directly.
- Net effect is the loss of one NADH
 - 3 fewer ATPs

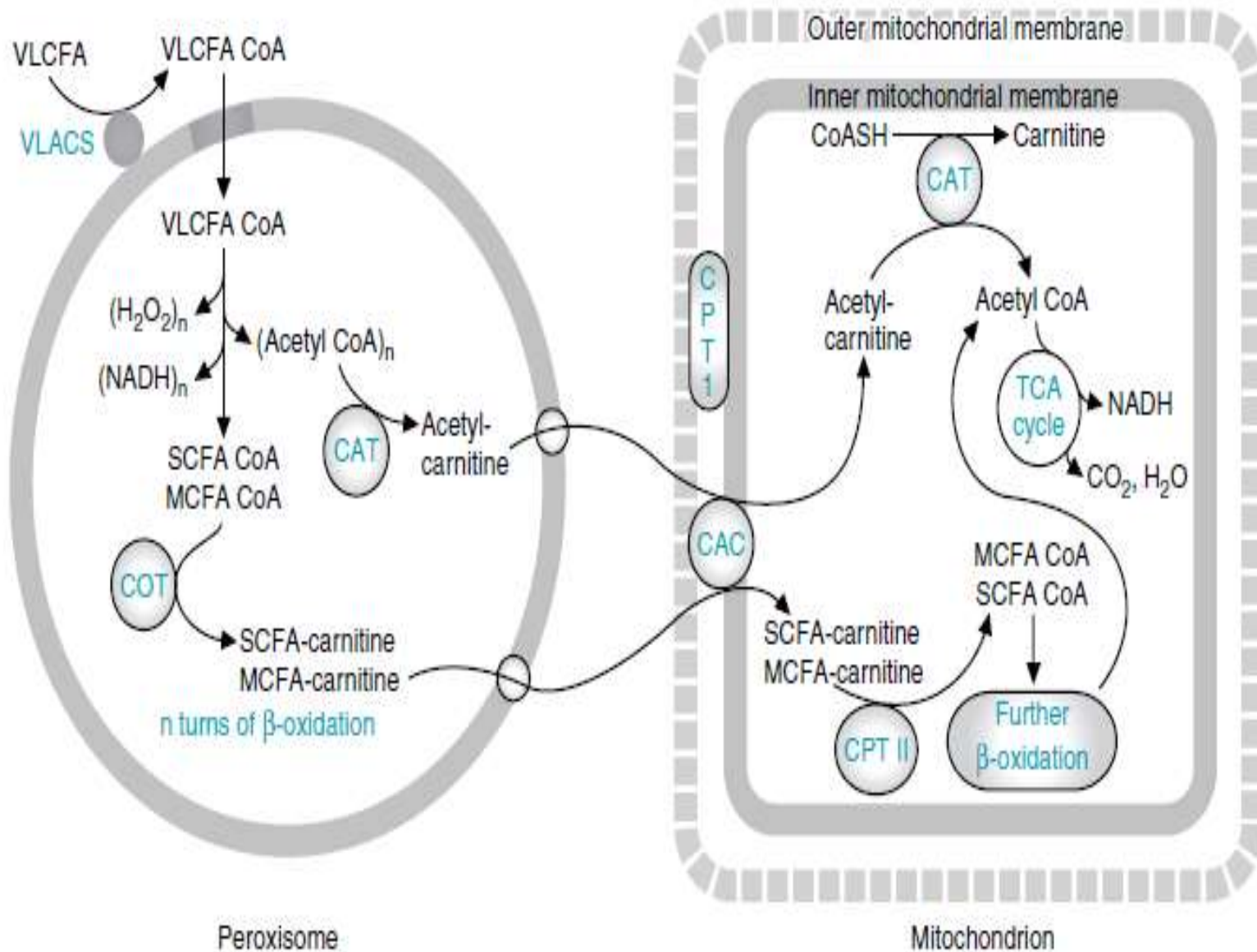


Oxidation of linoleate. After three spirals of β -oxidation (dashed lines), there is now a 3,4 *cis* double bond and a 6,7 *cis* double bond. The 3,4 *cis* double bond is isomerized to a 2,3-*trans* double bond, which is in the proper configuration for the normal enzymes to act. One spiral of β -oxidation occurs, plus the first step of a second spiral. A reductase that uses NADPH now converts these two double bonds (between carbons 2 and 3 and carbons 4 and 5) to one double bond between carbons 3 and 4 in a *trans* configuration. The isomerase (which can act on double bonds that are in either the *cis* or the *trans* configuration) moves this double bond to the 2,3-*trans* position, and β -oxidation can resume.

BETA OXIDATION IN PEROXISOME

- ❑ Fatty acids with 20 or more carbons (VLCFA) are first oxidized in the peroxisomes.
- ❑ The shortened fatty acid then goes to the mitochondria.
- ❑ The enzyme for initial dehydrogenation is FAD containing **Acyl CoA oxidase**.
- ❑ H_2O_2 is produced during the process which is toxic to cells and is therefore converted to H_2O by Catalase.





Very-long-chain fatty acids of 20 to 26 carbons are oxidized exclusively in peroxisomes by a sequence of reactions similar to mitochondrial oxidation in that they generate acetyl CoA and NADH

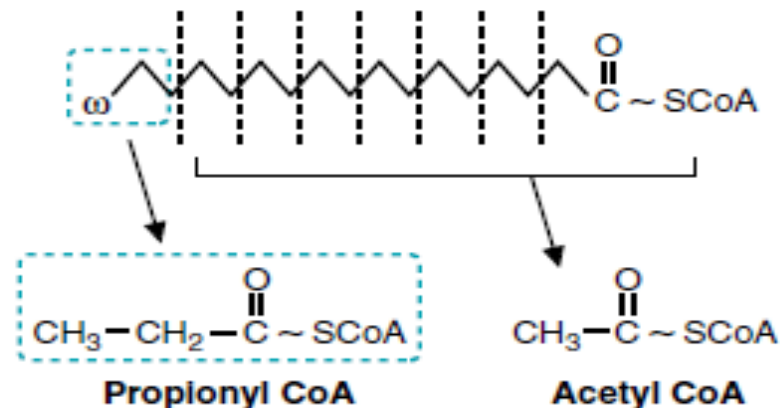
ZELLWEGER SYNDROME

- ❑ Rare inherited disorder.
- ❑ Absence of peroxisomes.
- ❑ VLCFA cannot be oxidized
- ❑ Accumulation of VLCFA in brain, blood and other tissues like liver and kidney.

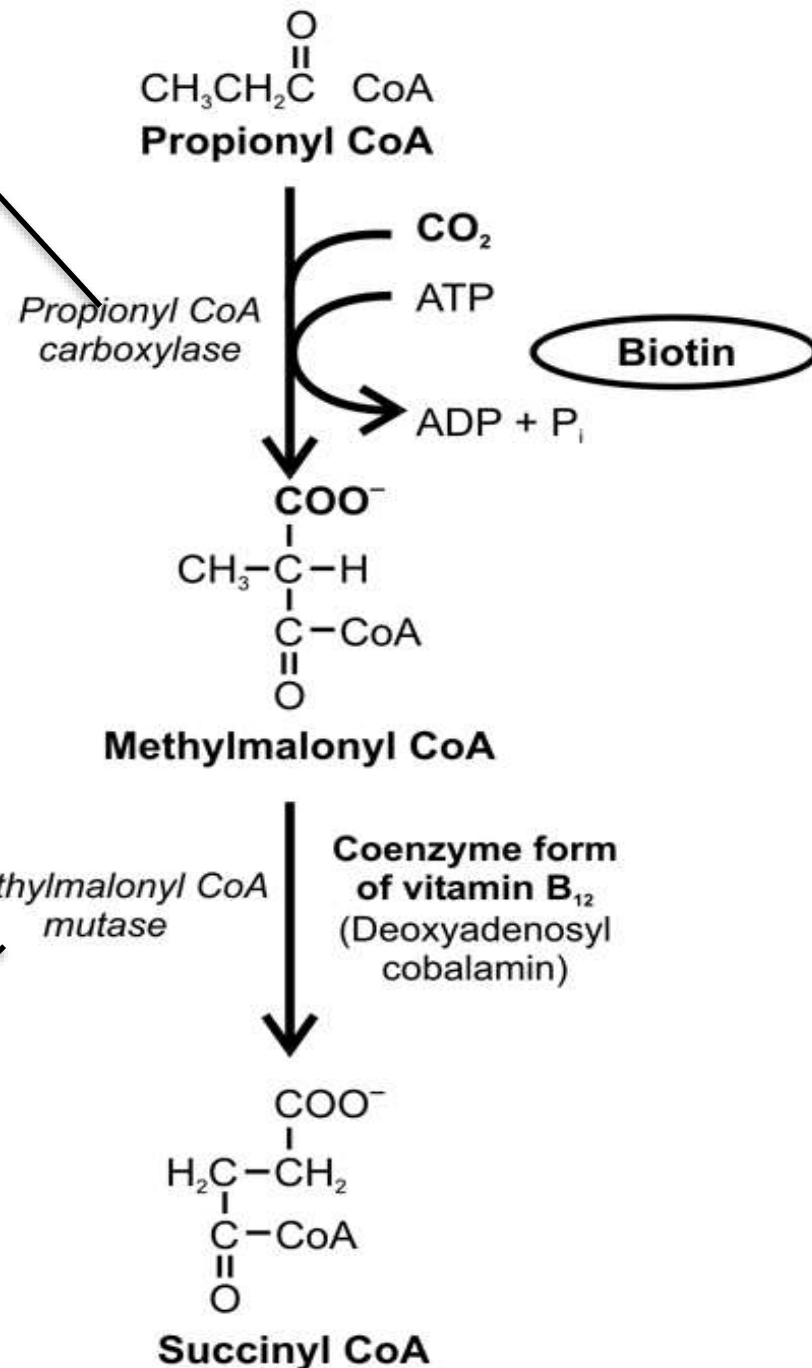


BETA OXIDATION OF ODD CHAIN FATTY ACIDS

- ❑ Oxidation of fatty acids with an odd number of carbons yield acetyl CoA and one molecule of **propionyl CoA**.
- ❑ Propionyl CoA is converted to methyl malonyl CoA by **carboxylase** (a biotin-requiring enzyme).
- ❑ MMCoA is moved within the molecule by MMCoA mutase (vit.B12 coenzyme) to form **succinyl CoA**... gluconeogenic.



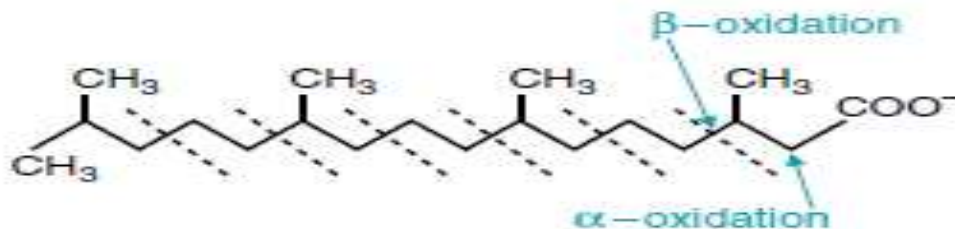
Inheritable disorder of propionate metabolism is due to a defect in propionyl-CoA carboxylase, resulting in **propionic acidemia** (and **aciduria**). Such individuals, as well as those with methylmalonic acidemia, are capable of oxidizing some propionate to CO₂, even in the absence of propionyl-CoA carboxylase.



Two inheritable types of **methylmalonic acidemia** (and **aciduria**) are associated in young children with failure to grow and mental retardness.

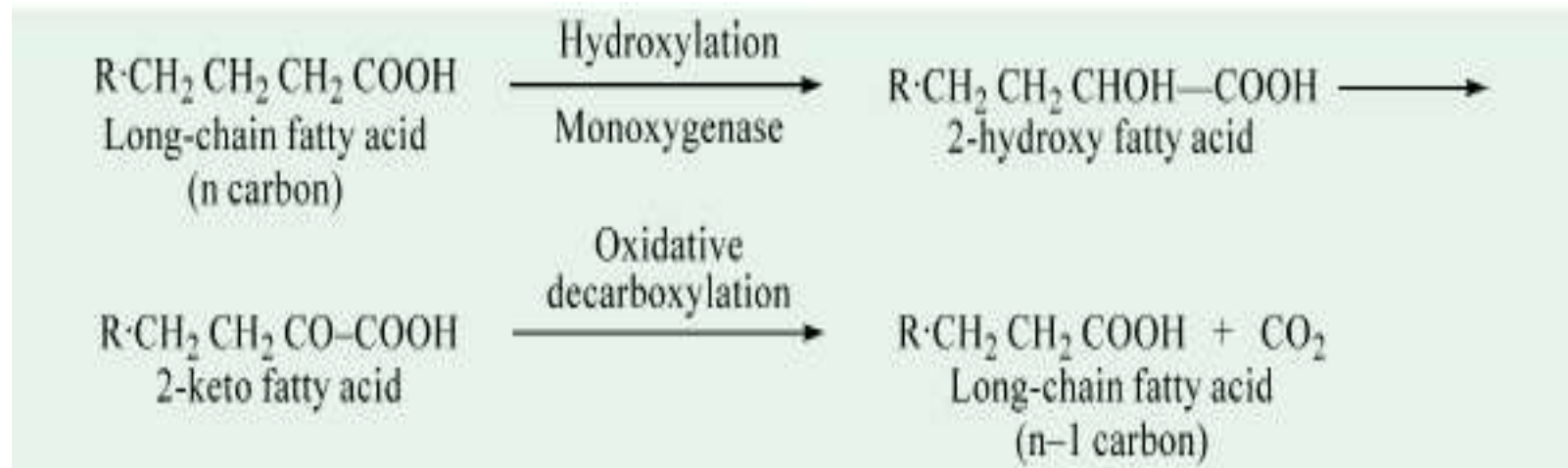
ALPHA OXIDATION

- α oxidation is the removal of one carbon atom (*i.e.*, a carbon) at a time from the carboxyl end of the BRANCHED FATTY ACIDS molecule.
- α oxidation of long-chain fatty acids to 2-hydroxy acids and then to fatty acids with one carbon atom less than the original substrate have been demonstrated in the **microsomes** of brain and other tissues also.
- Long-chain α hydroxy fatty acids are constituents of brain lipids, *e.g.*, the C24 cerebronic acid (= 2-hydroxylignoceric acid), $\text{CH}_3(\text{CH}_2)_{21}\text{CH}(\text{OH})\text{COOH}$.
- These hydroxy fatty acids can be converted to the 2-keto acids, followed by oxidative decarboxylation, resulting in the formation of long-chain fatty acids with an odd number of carbon atoms :



ALPHA OXIDATION

- The initial hydroxylation reaction is catalyzed by a mitochondrial enzyme, monooxygenase which requires O_2 , Mg^{2+} , NADPH.

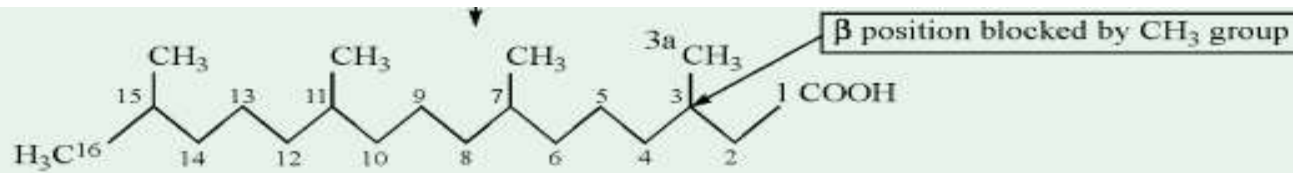


- Conversion of the α hydroxy fatty acid to CO_2 and the next lower unsubstituted acid appears to occur in the endoplasmic reticulum and to require O_2 , Fe^{2+} and ascorbate.

PHYTANIC ACID

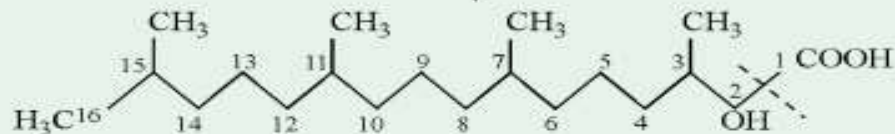
- ❑ The α -oxidation system plays a key role in the capacity of mammalian tissues to oxidize **phytanic acid**.
- ❑ Phytanic acid is an oxidation product of phytol and is present in animal fat, cow's milk and foods derived from milk.
- ❑ The phytol presumably originates from plant sources, as it is a substituent of chlorophyll and the side chain of vitamin K2.
- ❑ Normally, phytanic acid is rarely found in serum lipids because of the ability of normal tissue to degrade (or oxidize) the acid very rapidly.





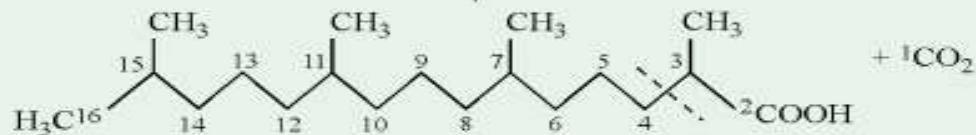
Phytanic acid

α oxidation *Phytanate α hydroxylase*



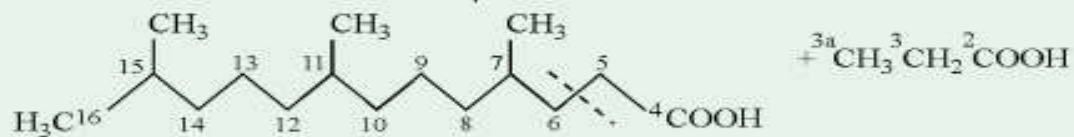
α hydroxyphytanic acid

α oxidation *Phytanate α oxidase*



Pristanic acid

β oxidation



4,8,12-trimethyldecanoic acid

β oxidation



Successive β oxidations

Products of Degradation



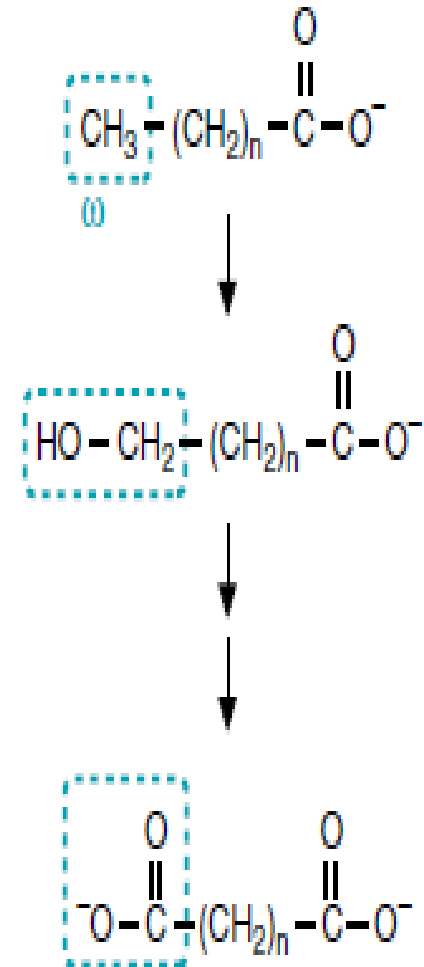
REFSUM'S DISEASE

- ❑ Large amounts of phytanic acid accumulate in the tissues and serum of individuals with **Refsum's disease**, a rare inheritable autosomal recessive disorder affecting the nervous system because of an inability to oxidize this acid.
- ❑ Diets low in animal fat and milk products appear to relieve some of the symptoms of Refsum's disease.
- ❑ *In Refsum's disease, there is a lack of the enzyme, **phytanate α hydroxylase***



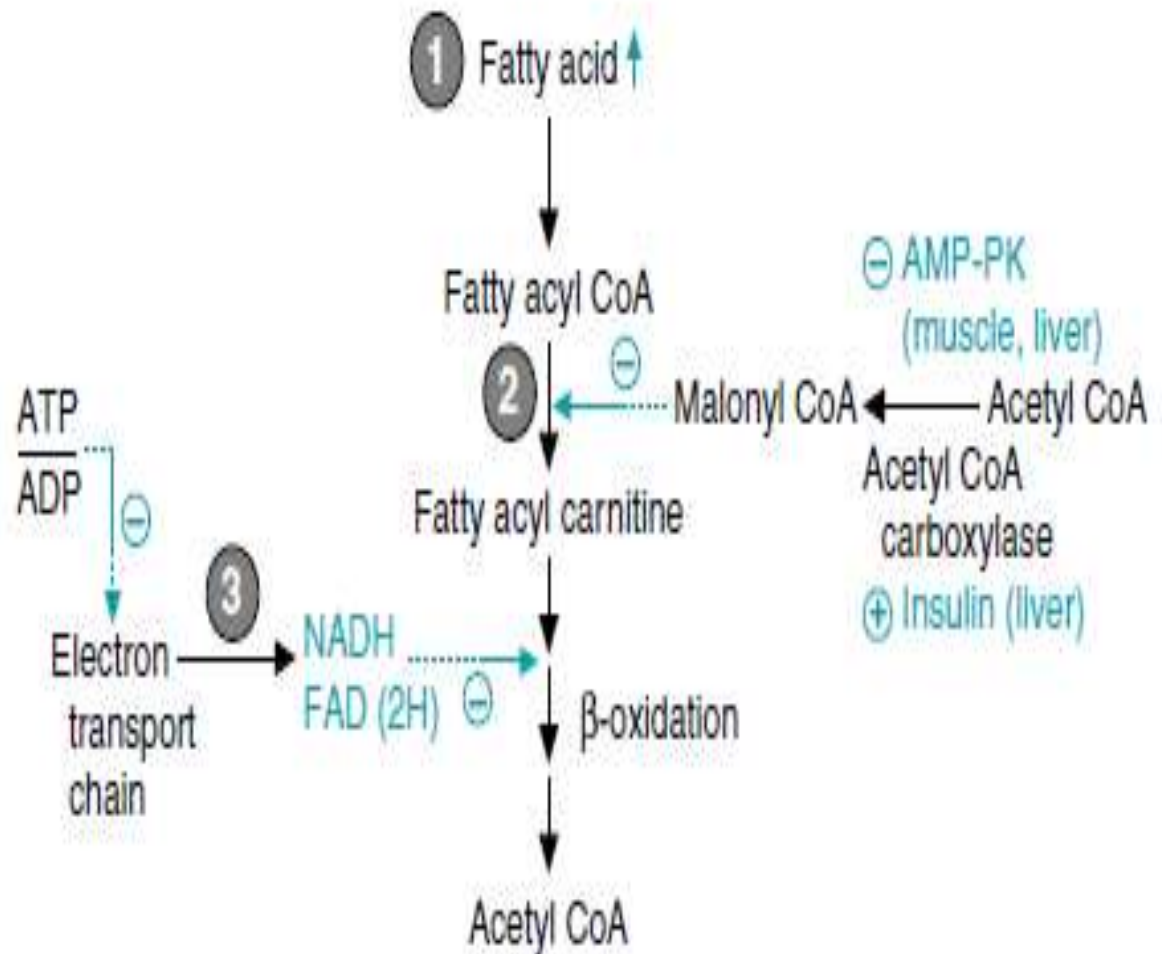
OMEGA -OXIDATION OF FATTY ACIDS

- ❑ Fatty acids also may be oxidized at the **ω -carbon** of the chain by enzymes in **the endoplasmic reticulum**.
- ❑ The terminal-methyl group is first oxidized to alcohol by an enzyme that **uses cytochrome P450, molecular oxygen, and NADPH**. Dehydrogenases convert the alcohol group to carboxylic acid.
- ❑ The dicarboxylic acids produced by **ω -oxidation** can undergo β -oxidation, forming compounds with 6 to 10 carbons that are water-soluble.
- ❑ Such compounds may then enter the blood, be oxidized as medium-chain fatty acids, or be excreted in urine as medium-chain dicarboxylic acids.



REGULATION OF BETA OXIDATION

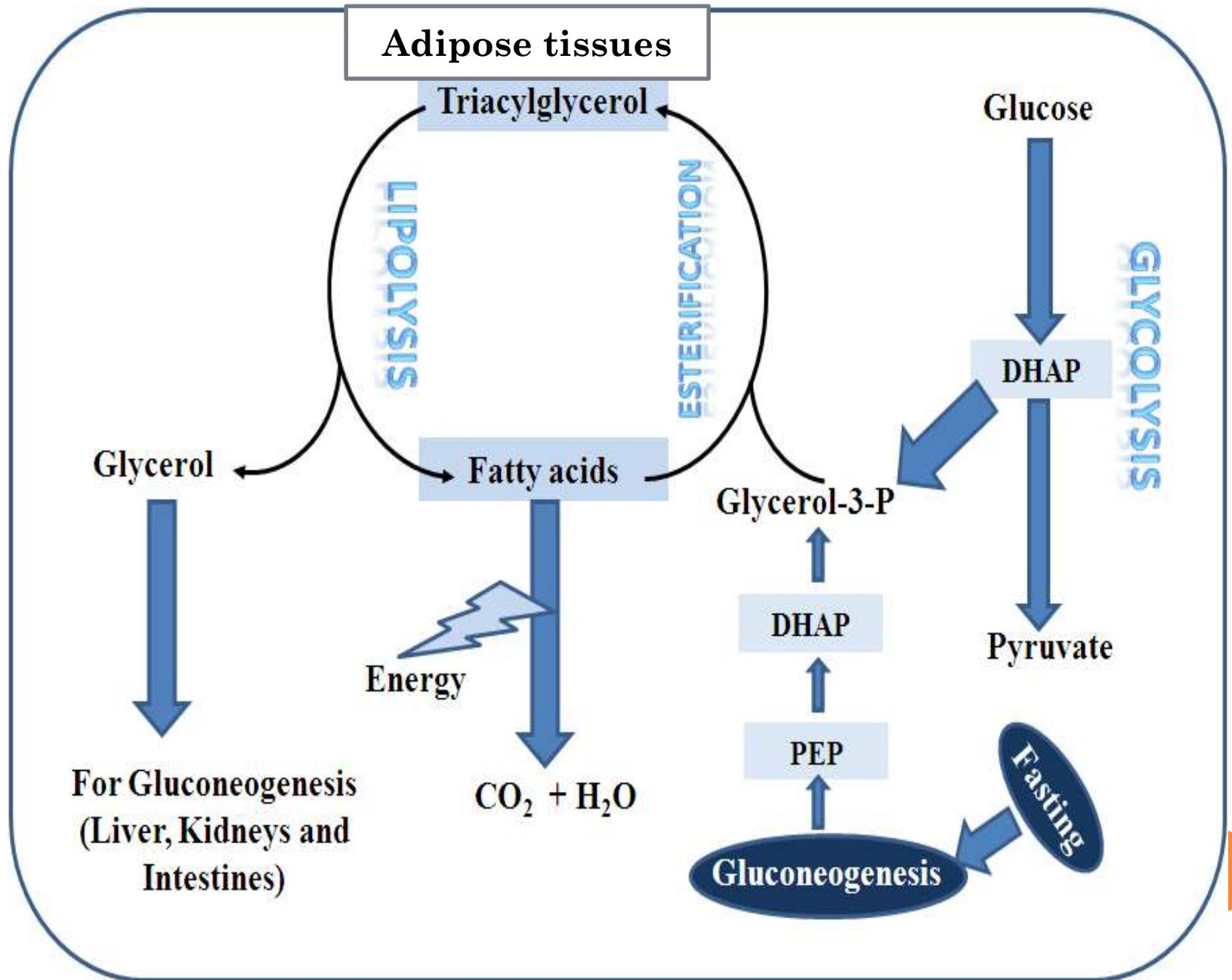
- Beta-oxidation is regulated as a whole primarily by fatty acid availability; once fatty acids are in the mitochondria they are oxidized as long as there is adequate NAD^+ and CoA.



FATTY ACID OXIDATION DURING FASTING

Mobilization of lipids from storage sites

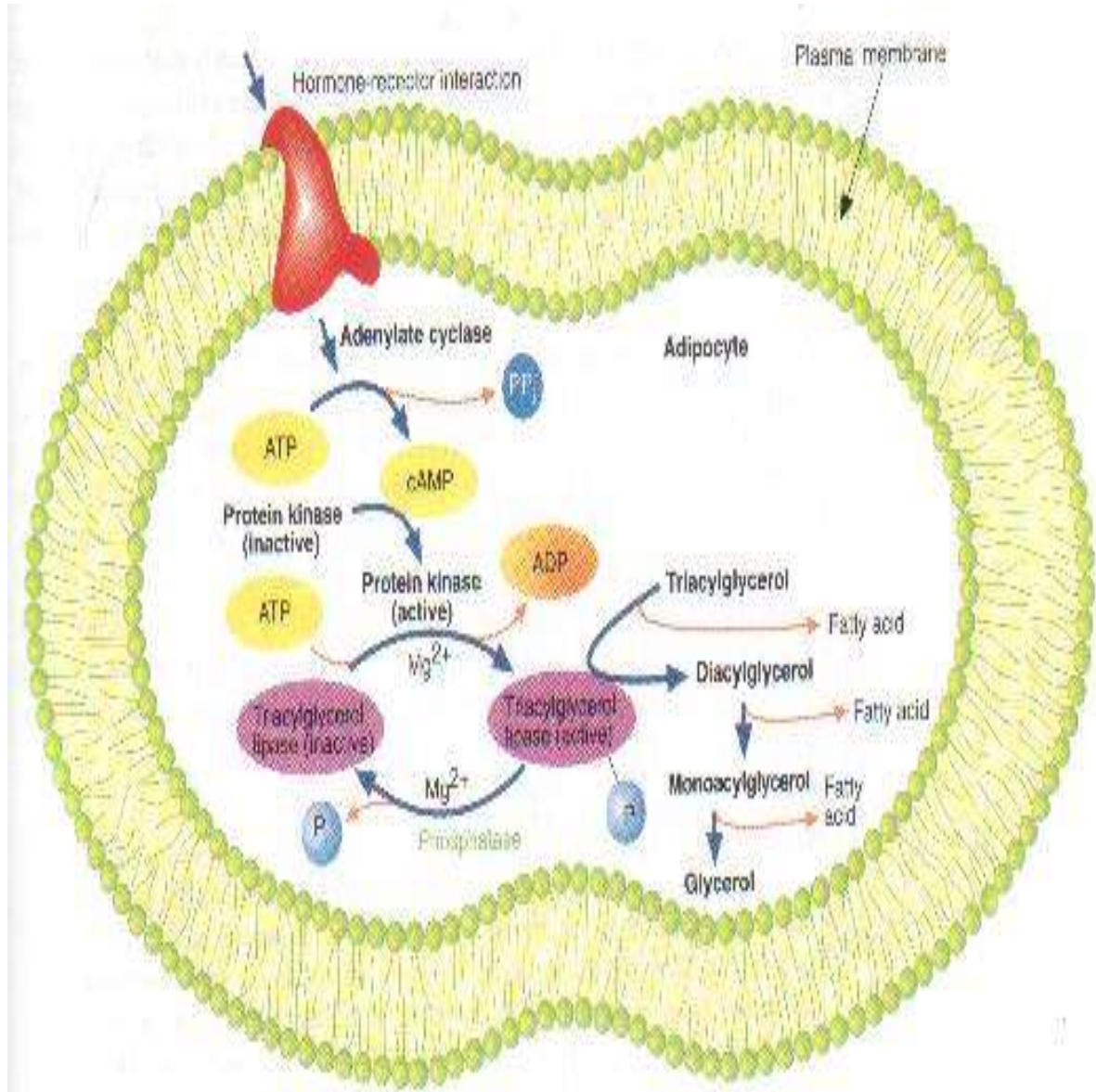
- ❑ Fatty acids are stored primarily in adipocytes as triacylglycerol.
 - ❑ Triacylglycerol must be hydrolyzed to release the fatty acids.
 - ❑ Adipocytes are found mostly in the abdominal cavity and subcutaneous tissue.
 - ❑ Adipocytes are metabolically very active; their stored triacylglycerol is constantly hydrolyzed and resynthesized.
- 



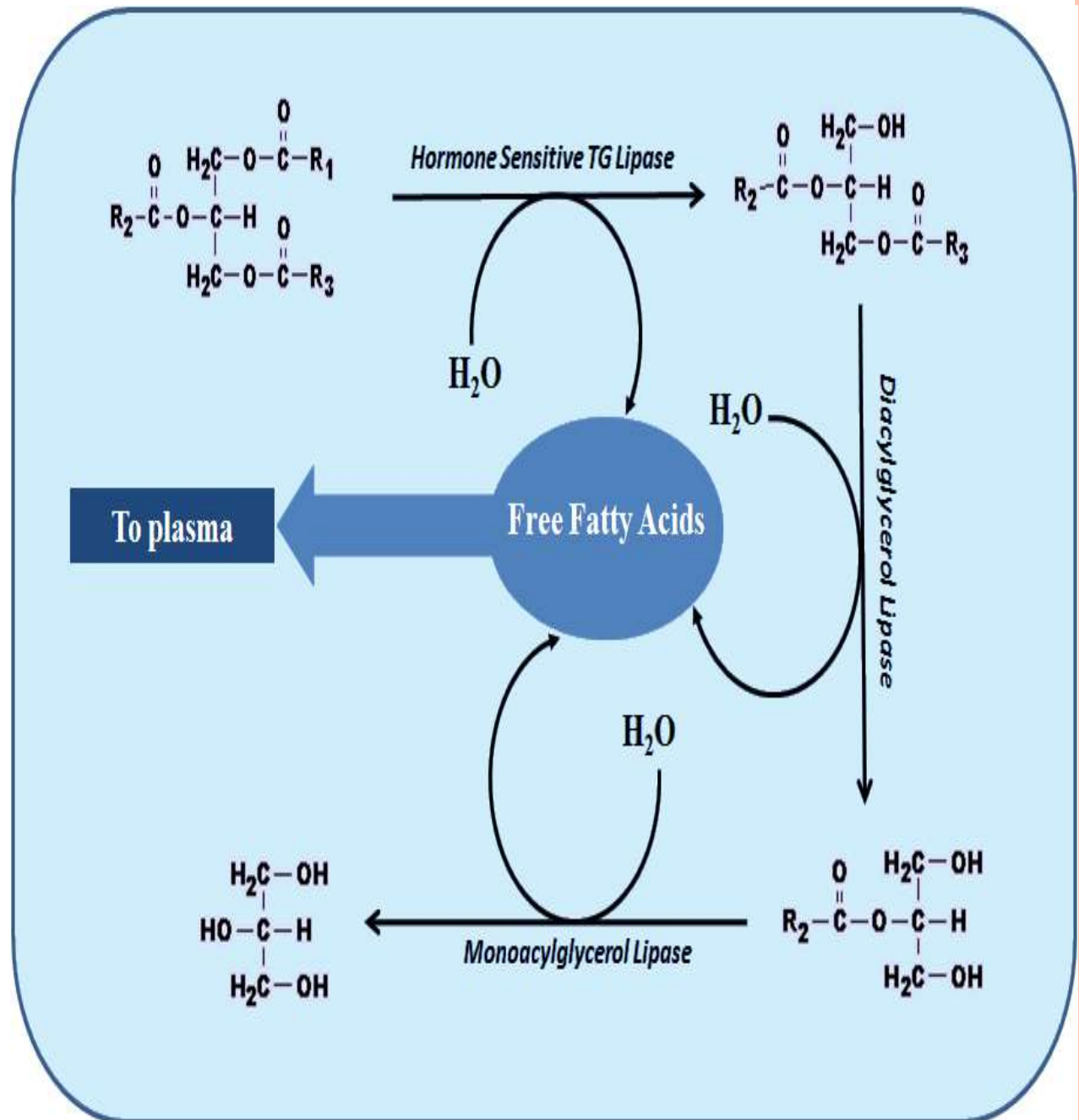
MOBILIZATION OF STORED FATS: LIPOLYSIS

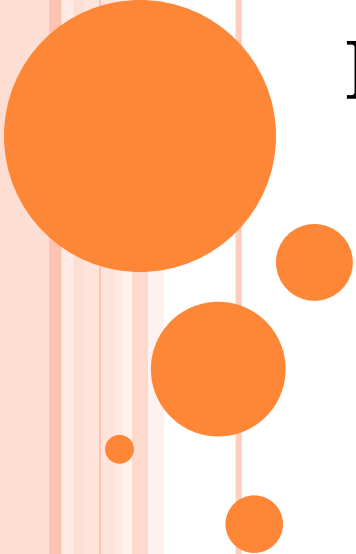
❑ Non-esterified fatty acid release from the adipocytes is initiated by the action of **hormone sensitive lipase (HSL)**, which begins to hydrolyze the stored triglyceride.

❑ The final products of triacylglycerol hydrolysis are glycerol and nonesterified fatty acids.



- ❑ HSL is activated by **epinephrine, norepinephrine, ACTH and glucagon**, acting via **phosphorylation of the enzyme**.
- ❑ It is inhibited by **insulin**.
- ❑ Non-esterified fatty acids are bound to **serum albumin** for transport to other tissues, where they are used.

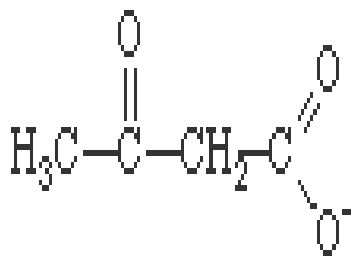




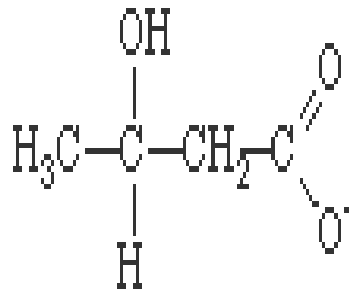
KETONE BODIES

KETONE BODIES

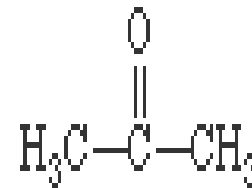
- Ketone bodies are water-soluble, that are generated from **acetyl CoA**
- There are three different types of ketone bodies.
 - **Acetoacetate**
 - **Beta hydroxybutyrate**
 - **Acetone**



Acetoacetate



D-β-Hydroxybutyrate



Acetone



CELLULAR AND TISSUE LOCATION

Ketone bodies are purely synthesized in

- Liver

- Synthesis takes place in the **mitochondria** of liver cells.
- The primary sources of ketone bodies are
 - *Fatty acids (Acetyl CoA)*
 - *Ketogenic amino acids (leu, Ile, Lys, Trp, Phe, Tyr (Acetyl CoA or AcetoacetylCoA))*



CONDITIONS WHERE KETONE BODIES ARE SYNTHESIZED

Rate of Ketone bodies synthesis increases during

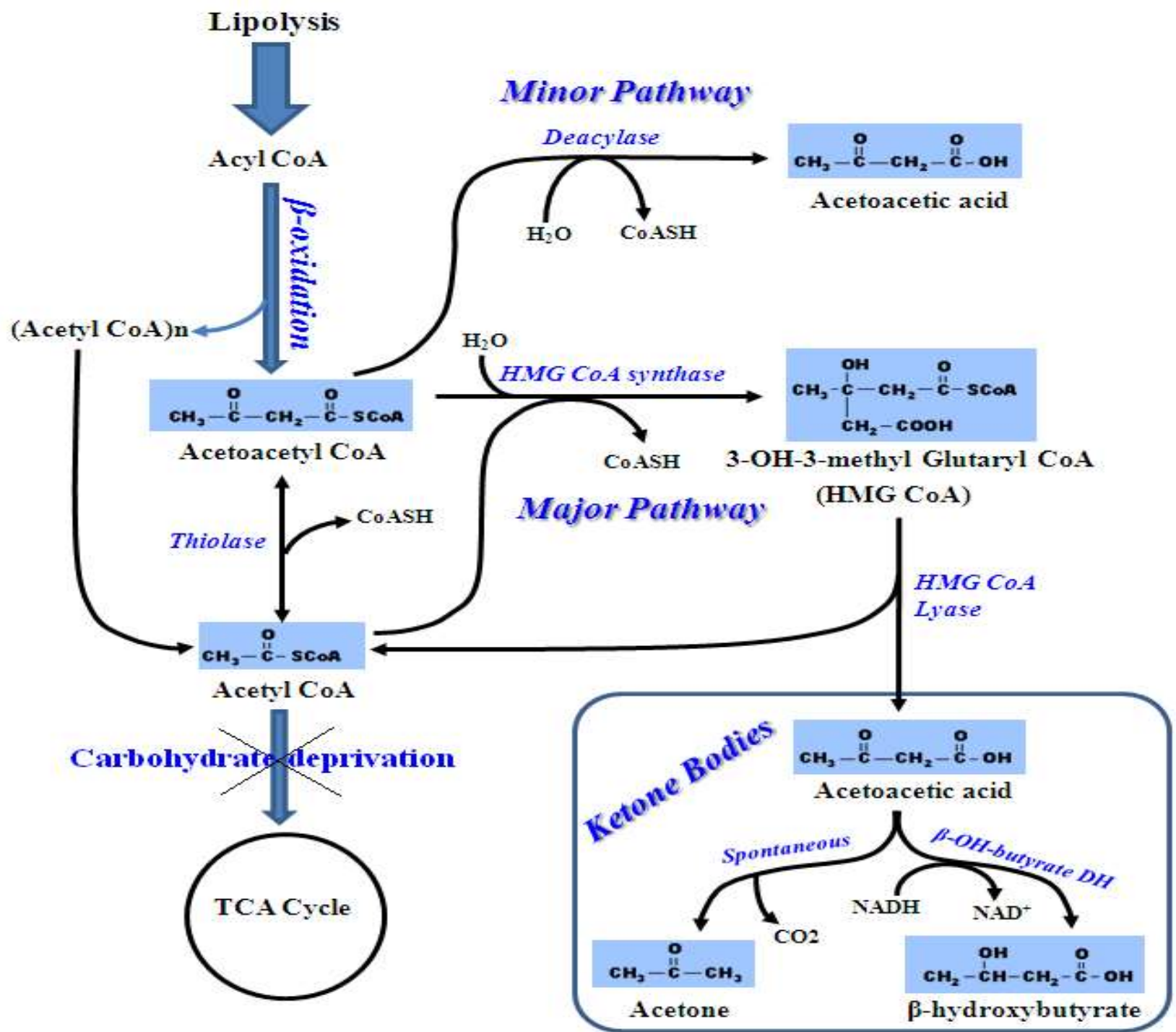
- ❑ Fasting or starvation
- ❑ Heavy or Prolonged physical exercises
- ❑ Extremely low carbohydrates diet (as with the high-protein diets).
- ❑ Uncontrolled diabetes mellitus (Type I)



KETOGENESIS

- ❑ Two molecules of acetyl CoA condense to form acetoacetyl CoA, catalyzed by **thiolase**, which is the reverse of the thiolysis step in the oxidation of fatty acids.
- ❑ Acetoacetyl CoA then reacts with acetyl CoA and water to give 3-hydroxy-3-*methyglutaryl CoA (HMG-CoA) and CoASH*.
- ❑ The reaction is catalyzed by *HMG CoA synthase*. This enzyme is exclusively present in **liver mitochondria**.
- ❑ There are two isoforms, the mitochondrial enzyme is needed for ketogenesis while the **cytosolic form is associated with cholesterol biosynthesis**.
- ❑ **Acetoacetic acid** is the primary ketone body, which will be reduced to beta-hydroxybutyrate and spontaneously decarboxylated to acetone





Peculiarity of acetone

- ☐ **Acetone is formed by decarboxylation** in the presence of decarboxylase enzyme and undergoes a slow, spontaneous decarboxylation to acetone.
- ☐ Not used as metabolic fuel
- ☐ Do not cause acidosis
- ☐ Lost via expired air
- ☐ Produce characteristic odor of acetone may be detected in the breath of a person who has a high level of acetoacetate in the blood.
- ☐ **“Acetone-breath” has been used as a crude method of diagnosing individuals with untreated Type I diabetes mellitus.**

FORMATION OF β -HYDROXY BUTYRATE

- **β -Hydroxy Butyrate** is formed by the reduction of acetoacetate in the mitochondrial matrix by **D(-)3-hydroxybutyrate dehydrogenase**.
- The **β -hydroxybutyrate dehydrogenase** reaction has two functions:
 - 1) It stores energy equivalent to an NADH in the ketone body for export to the tissues
 - 2) It produces a more stable molecule (3:1).
- The ratio of β hydroxybutyrate to acetoacetate depends on the **NADH/NAD⁺ ratio inside mitochondria**.
- If NADH concentration is high, the liver releases a higher proportion of **β -hydroxybutyrate**.



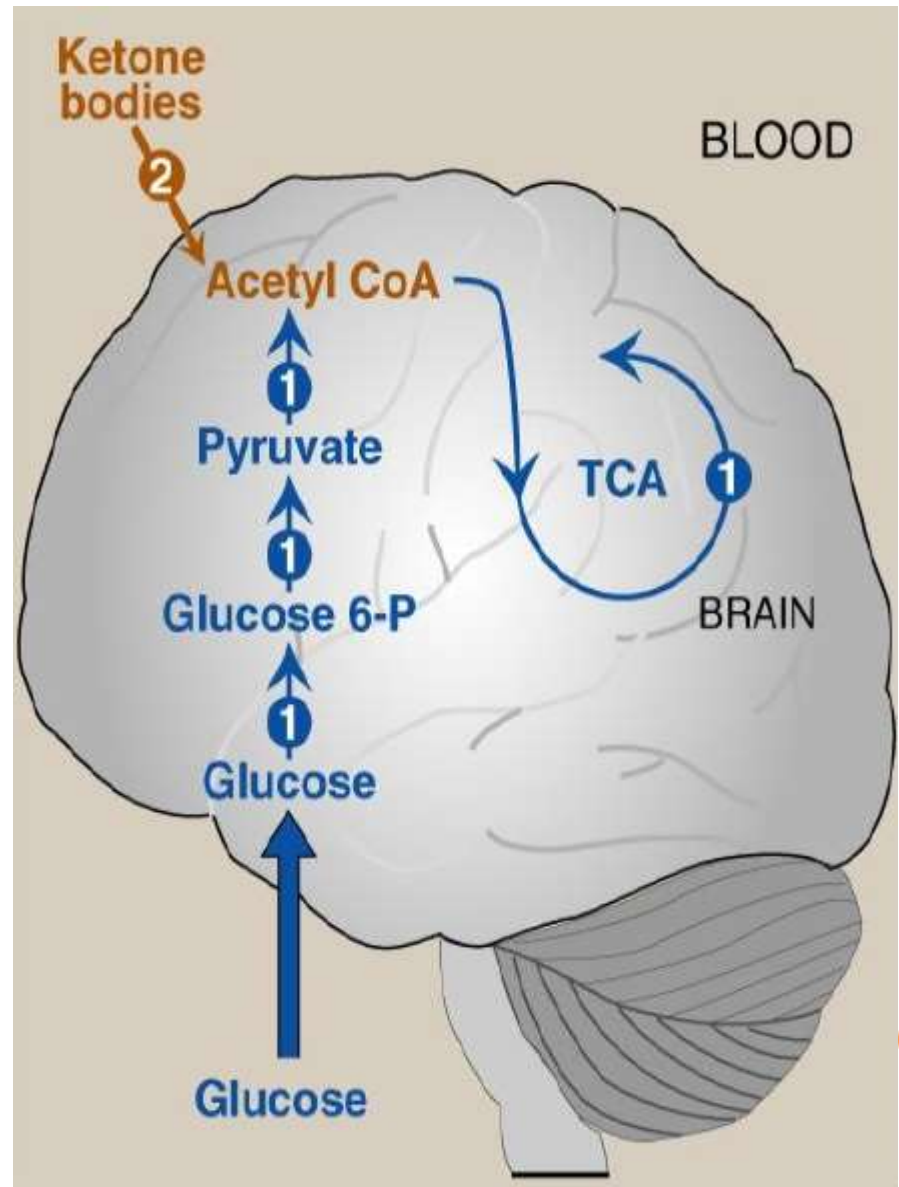
BIOLOGICAL SIGNIFICANCE OF KETONE BODIES

Ketone bodies serve as a **fuel for extra hepatic tissues**

Brain, heart, skeletal muscles

Brain:

- ❑ It is metabolically active and metabolically privileged.
- ❑ The brain generally uses 60-70% of total body glucose requirements.
- ❑ As glucose availability decreases, the brain is forced to use **either amino acids or ketone bodies for fuel**.
- ❑ In prolonged starvation, 30-40% of the fuel needs of the brain are met by ketone bodies.
- ❑ **Heart muscle** and the **renal cortex** use acetoacetate in preference to glucose



TISSUES THAT USE KETONE BODIES AS FUEL

❑ **During starvation**

- Brain
- Intestinal mucosa
- Adipocytes
- Developing fetus use ketone bodies
- Almost all tissues except liver and RBC

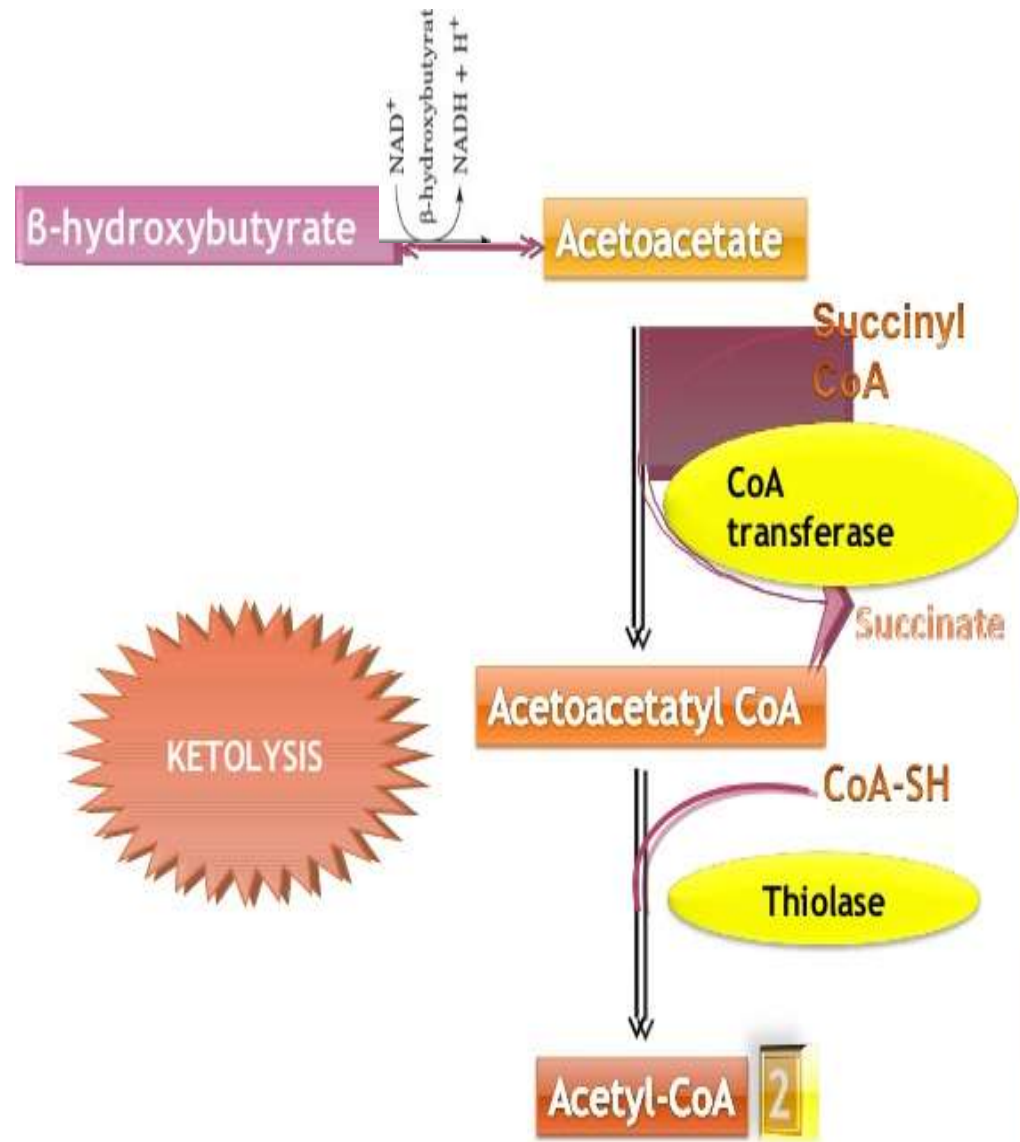


KETOLYSIS

Utilization of *Beta*-hydroxybutyrate

1. *Beta*-hydroxybutyrate, is first oxidized to acetoacetate with the production of one NADH.

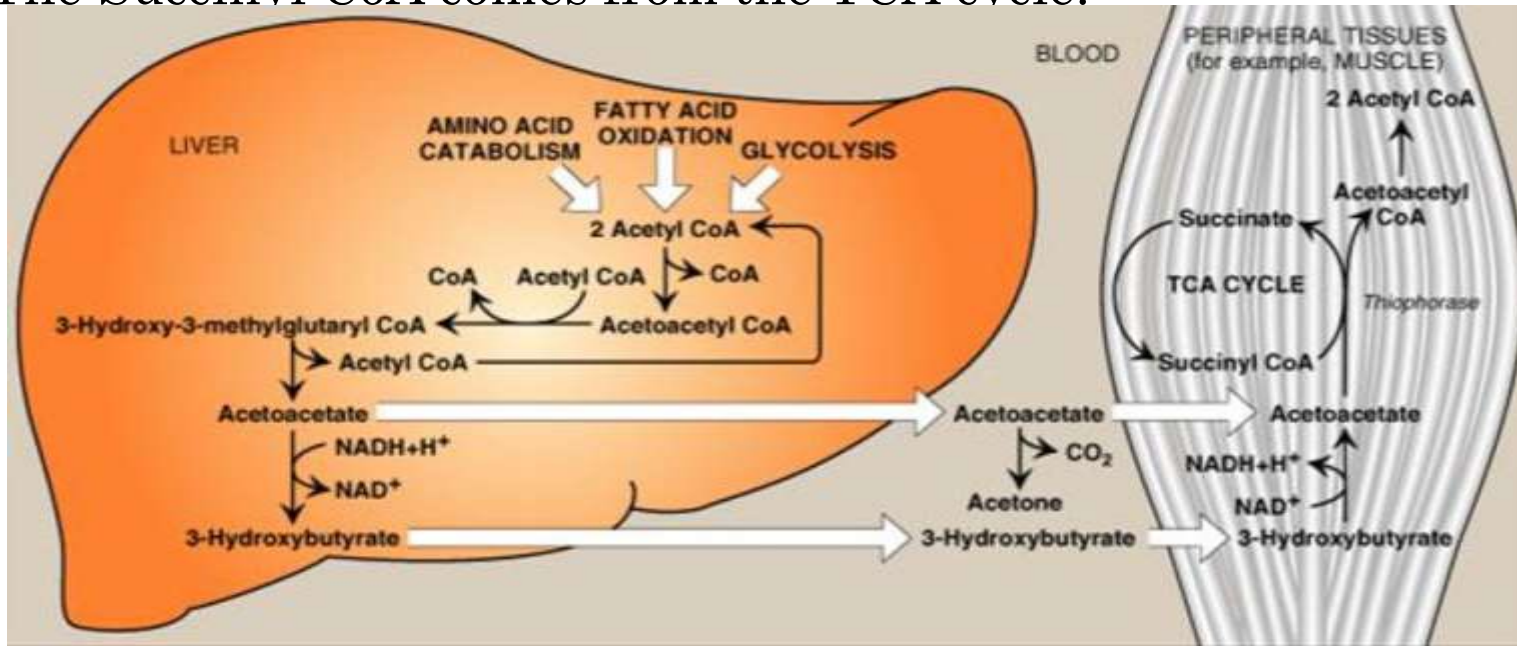
Under conditions where tissues are utilizing ketones for energy production their NAD⁺/NADH ratios are going to be relatively high, thus driving the β -hydroxybutyrate dehydrogenase catalyzed reaction in the direction of acetoacetate.



UTILIZATION OF KETONE BODIES

2) Coenzyme A must be added to the acetoacetate.

- ❑ The thioester bond is a high energy bond, so ATP equivalents must be used.
- ❑ In this case the **energy comes from a trans esterification** of the CoAS from **Succinyl CoA** to acetoacetate by Coenzyme A transferase, also called **Succinyl co A : Acetoacetate co A transferase**, also known as **Thiophorase**.
- ❑ The Succinyl CoA comes from the TCA cycle.



REGULATION OF KETOGENESIS

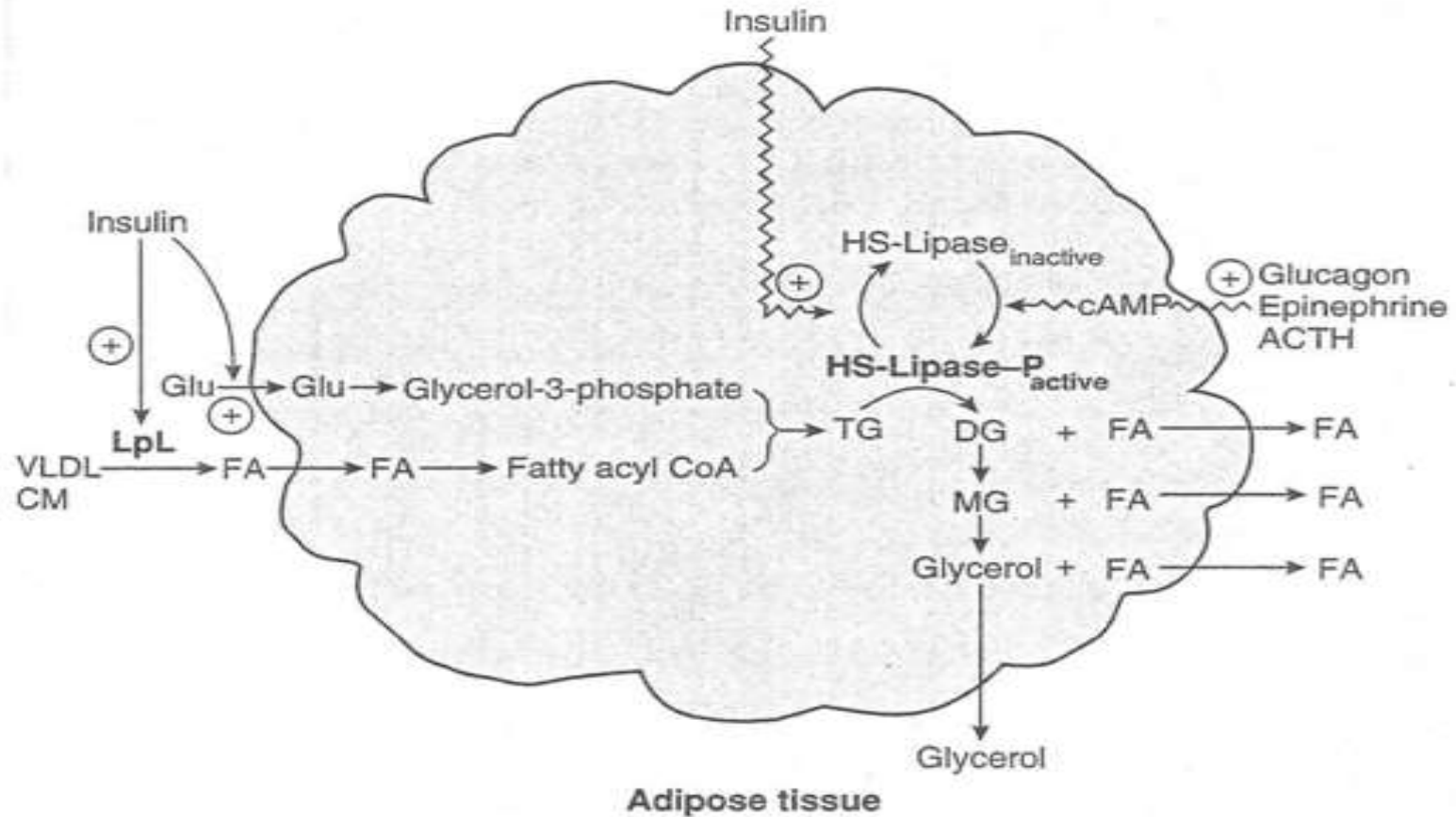
Ketogenesis is regulated at three steps-

1) Lipolysis in Adipose tissue

- ❑ Ketogenesis does not occur unless there is an  FA that arise from lipolysis of triacylglycerol in adipose tissue.
- ❑ When glucose levels fall, lipolysis induced by glucagon secretion causes increased hepatic ketogenesis due to increased substrate (free fatty acids) delivery from adipose tissue.
- ❑ Conversely, insulin, released in the well-fed state, inhibits ketogenesis via the triggering dephosphorylation and inactivation of adipose tissue hormone sensitive lipase (HSL).



LIPOLYSIS IN ADIPOSE TISSUE



Hormone sensitive lipase exists in two forms inactive dephosphorylated (brought by Insulin) and active phosphorylated form (brought by glucagon, catecholamines). Insulin promotes lipogenesis while the other hormones promote lipolysis.

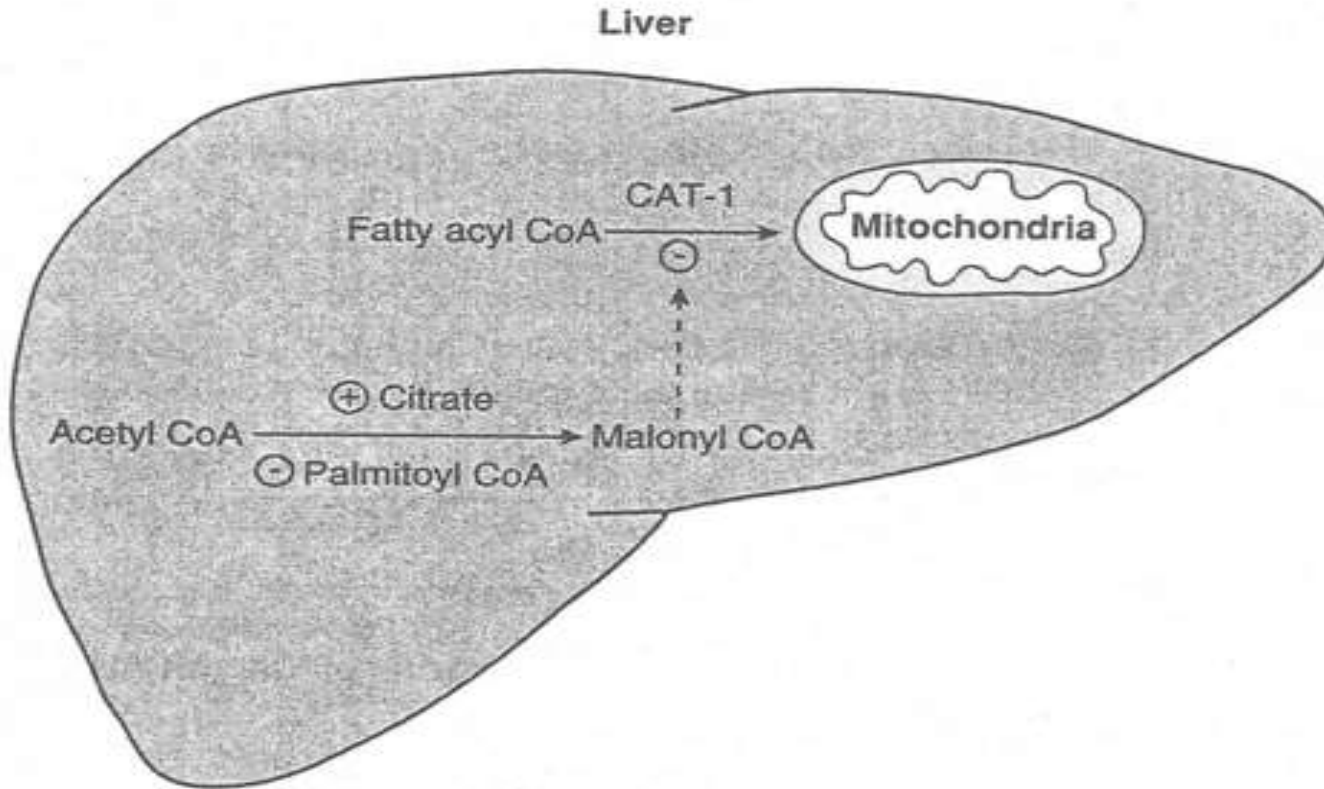
REGULATION OF KETOGENESIS

2)Transport of fatty acid-There is regulation of entry of fatty acids into the oxidative pathway by **carnitine Acyl transferase-I (CAT-I)**

- ❑ **Malonyl-CoA**, the initial intermediate in fatty acid biosynthesis formed by acetyl-CoA carboxylase in the fed state, is a potent inhibitor of CAT-I.
- ❑ Under these conditions, free fatty acids enter the liver cell in low concentrations and are nearly all esterified to acylglycerols and transported out of the liver in very low density lipoproteins (VLDL).



REGULATION OF CAT-1 ACTIVITY



CAT-I activity is low in the fed state, leading to depression of fatty acid oxidation. However, CAT-1 activity is higher in starvation, allowing fatty acid oxidation to increase.



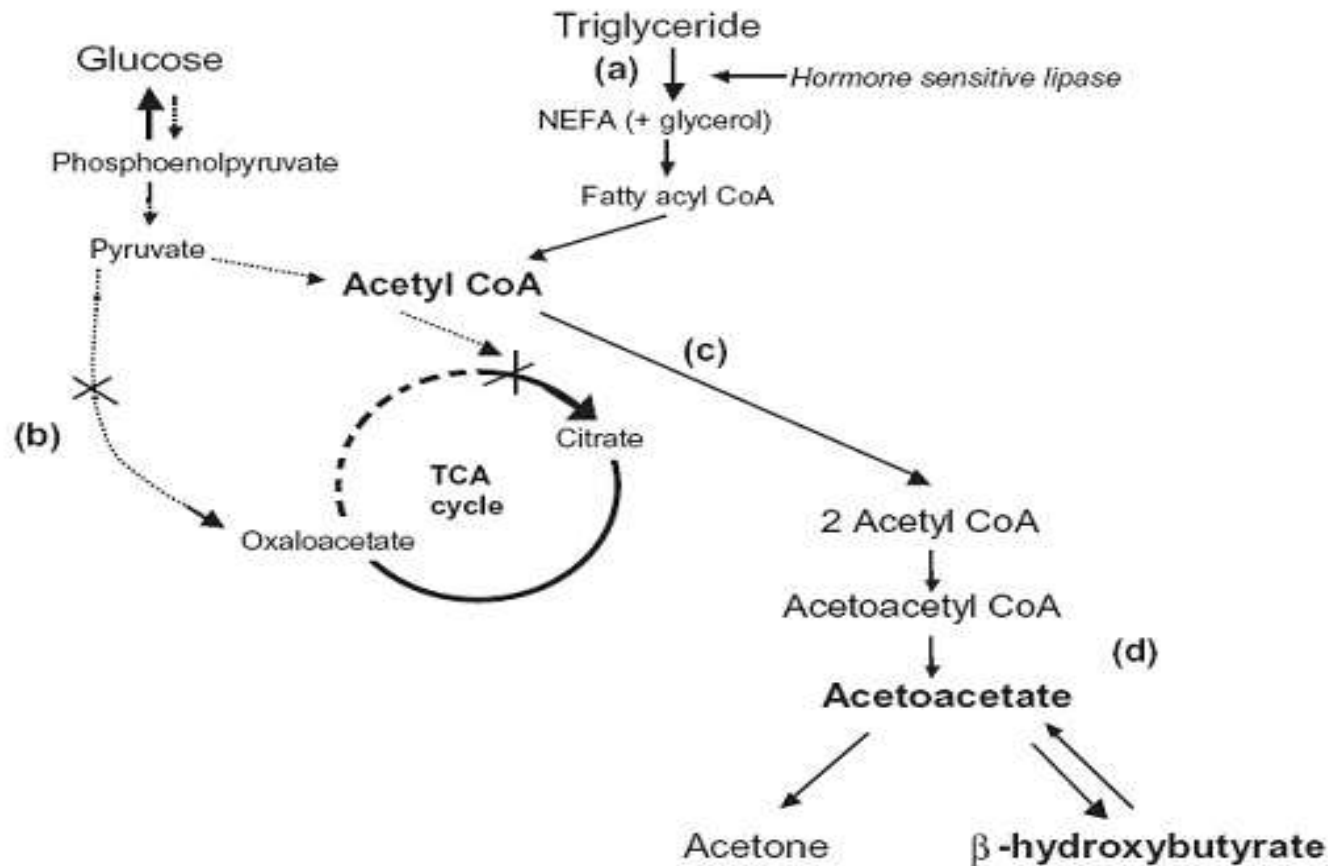
REGULATION OF KETOGENESIS

3) Fate of Acetyl co A

- ❑ The acetyl-CoA formed in beta-oxidation is oxidized in the citric acid cycle, or it enters the pathway of ketogenesis to form ketone bodies.
- ❑ As the level of serum free fatty acids is raised, proportionately more free fatty acids are converted to ketone bodies and less are oxidized via the citric acid cycle to CO_2 .
- ❑ Entry of acetyl CoA into the citric acid cycle depends on the availability of **Oxaloacetate for the formation of citrate**, but the **concentration of Oxaloacetate is lowered if carbohydrate is unavailable or improperly utilized**.



REGULATION OF KETOGENESIS



During high rates of fatty acid oxidation, primarily in the liver, large amounts of acetyl-Co A are generated. These exceed the capacity of the TCA cycle, and one result is the synthesis of ketone bodies.

NORMAL LEVELS OF KETONE BODIES

- ❑ Normal plasma concentration : 1-3 mg/dL.
- ❑ In urine ... less than 125 mg in 24 hrs.



CERTAIN TERMINOLOGIES

- ❑ **Ketosis:** Accumulation of abnormal amounts of ketone bodies in tissues and body fluids
- ❑ **Ketonemia:** level of ketone bodies in blood above normal level.
- ❑ **Ketonuria :** Urinary excretion of ketone bodies exceeds the normal amounts.
- ❑ **Ketoacidosis:** Acetoacetic acid and beta hydroxy butyric acid are moderately strong acids. When their synthesis exceeds their utilization, their amount exceeds in blood and tissues. They need to be buffered. There can be progressive loss of buffer cations and this results in ketoacidosis.



CAUSES OF KETOSIS

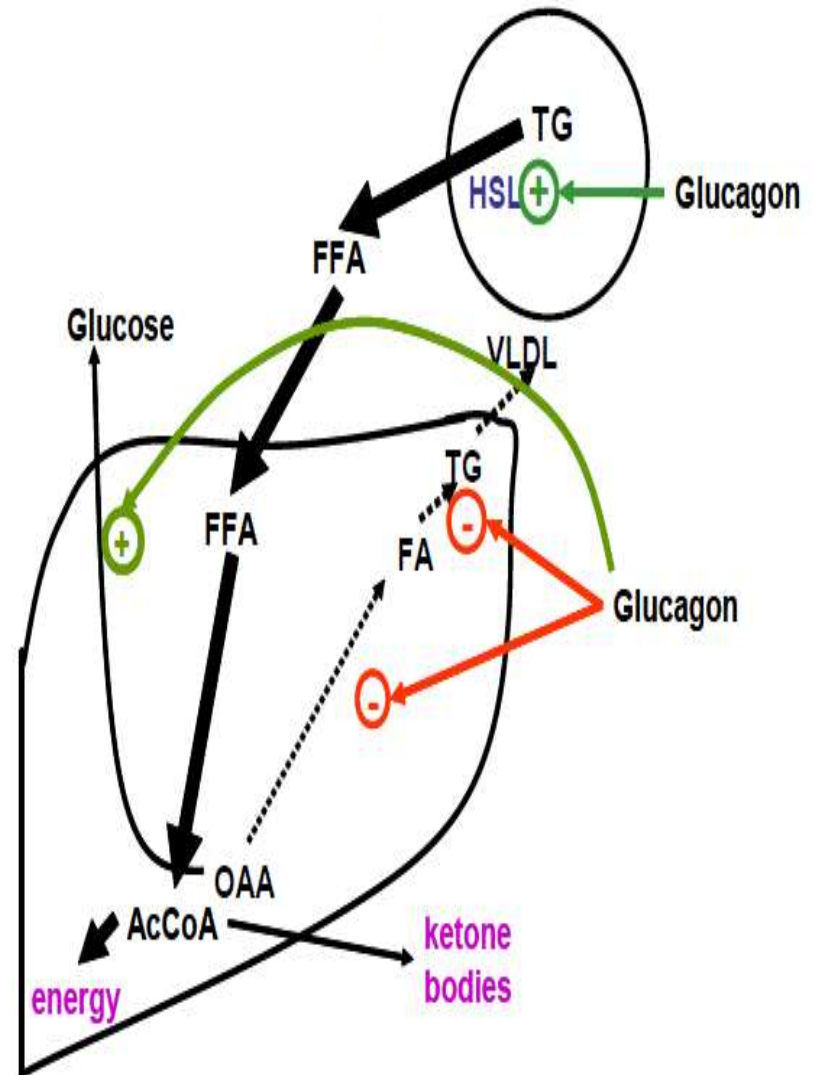
- ❑ Uncontrolled diabetes mellitus (Type I)
- ❑ Starvation
- ❑ Chronic alcoholism
- ❑ Von- Gierke's disease
- ❑ Heavy exercise
- ❑ Low carbohydrate diet- For weight loss
- ❑ Pyruvate carboxylase deficiency



STARVATION INDUCED KETOSIS

Prolonged fasting may result: From an inability to obtain food, from the desire to lose weight rapidly, or in clinical situations in which an individual cannot eat because of trauma, surgery, neoplasms, burns etc.

- In the absence of food the plasma levels of glucose, amino acids and triacylglycerols fall, triggering a decline in insulin secretion and an increase in glucagon release.



STARVATION INDUCED KETOSIS

- ❑ The decreased insulin to glucagon ratio, makes this period of nutritional deprivation a **catabolic state**, characterized by **degradation of glycogen, triacylglycerol and protein**.
- ❑ This sets in to motion an exchange of substrates between **liver, adipose tissue, muscle and brain** that is guided by two priorities-
 - (i) the need to maintain glucose level to sustain the energy metabolism of brain ,red blood cells and other glucose requiring cells and
 - (ii) to supply energy to other tissues by mobilizing fatty acids from adipose tissues and converting them to ketone bodies to supply energy to other cells of the body.



HEAVY EXERCISE INDUCED KETOSIS



LOW CARBOHYDRATES DIET INDUCE KETOSIS



CLINICAL SIGNIFICANCE-KETOACIDOSIS

❑ Both β -hydroxybutyrate and acetoacetate are **organic acids**. and are released in the protonated form, to lower the pH of the blood.

In normal individuals, other mechanisms compensate for the increased proton release.

❑ When ketone bodies are released in large quantities the normal pH-buffering mechanisms are overloaded ; the reduced pH, in combination with a number of other metabolic abnormalities results in **ketoacidosis**.

❑ In severe ketoacidosis, cells begin to lose ability to use ketone bodies also.

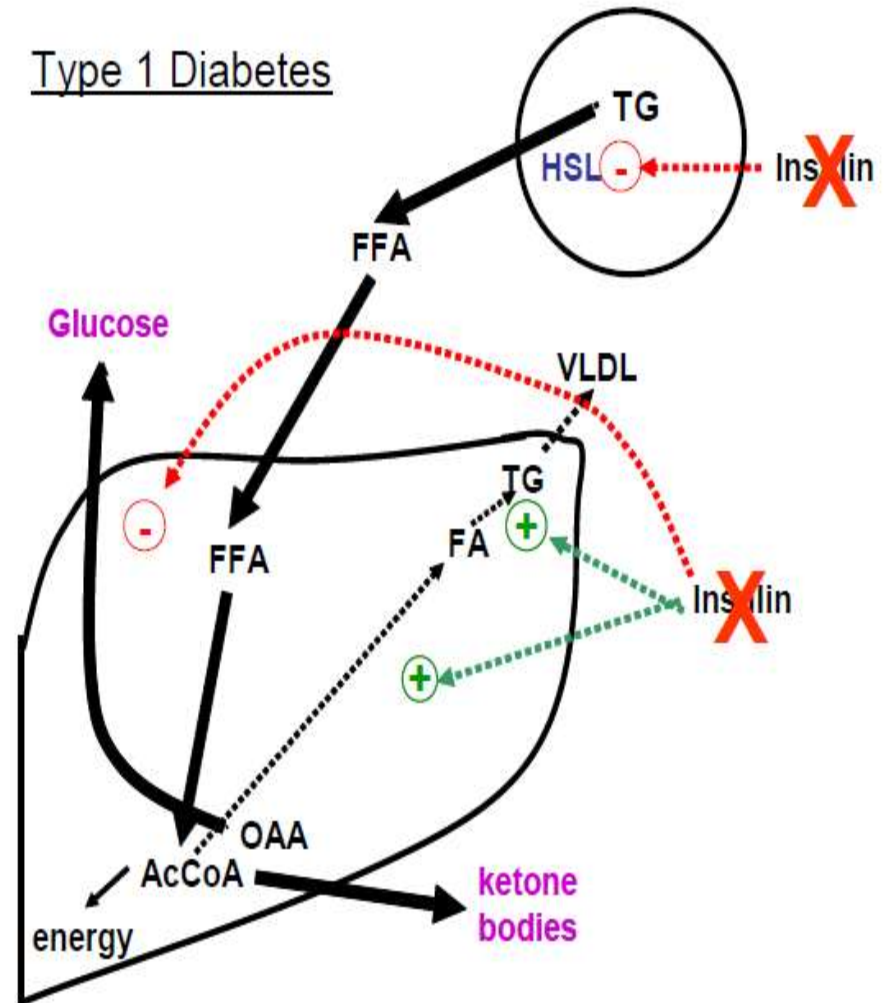


KETOGENESIS IN DIABETES

- ❑ Insufficient insulin leads to..

Increase in lipolysis ,
increase fatty acids flood
the liver Acetyl-CoA

- ❑ Decreased insulin-to-glucagon ratio trigger more glucose synthesis,
- ❑ oxaloacetate is needed to produce glucose, citric Acid Cycle is limited



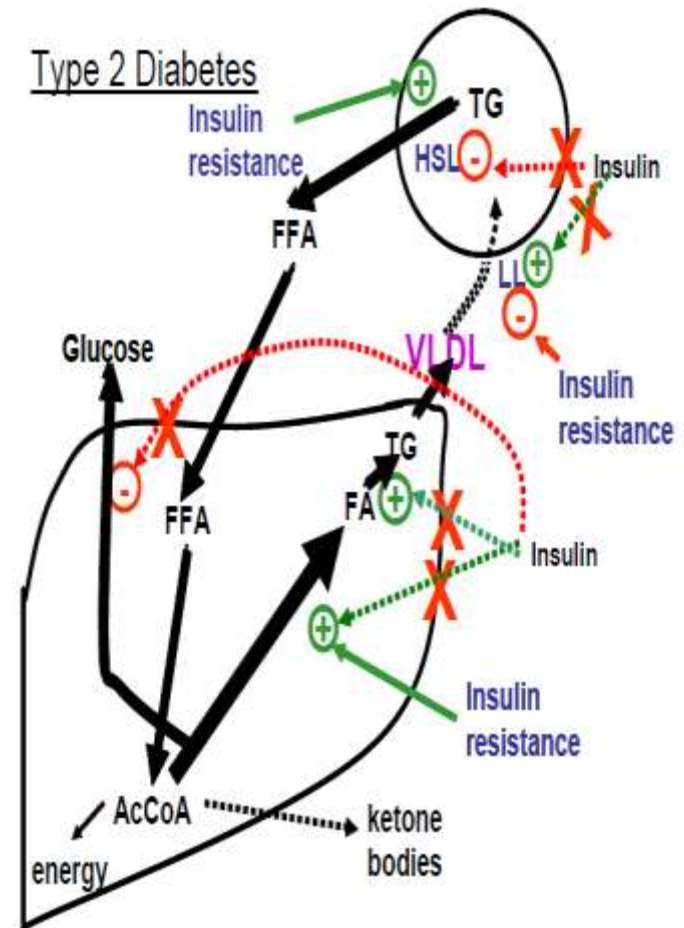
KETOACIDOSIS IN DIABETES MELLITUS

- ❑ In uncontrolled type 1 diabetes mellitus ----- severe deficiency or absence of insulin ---- lipolysis ---- very high levels of FFA ---- high levels of acetyl CoA ---- raised ketogenesis.
- ❑ In severe ketosis ---- blood level above 90 mg/dl and urine level above 5000 mg/24 hrs.
- ❑ With each ketone body , one hydrogen atom is released in blood --- lowering of pH.... Acidosis.
- ❑ The body initially buffers these with the bicarbonate buffering system, and other mechanisms to compensate for the acidosis, such as hyperventilation to lower the blood carbon dioxide levels.
- ❑ This hyperventilation, in its extreme form, may be observed as **Kussmaul respiration**.
- ❑ Ketones, too, participate in **osmotic diuresis** and lead to further electrolyte losses

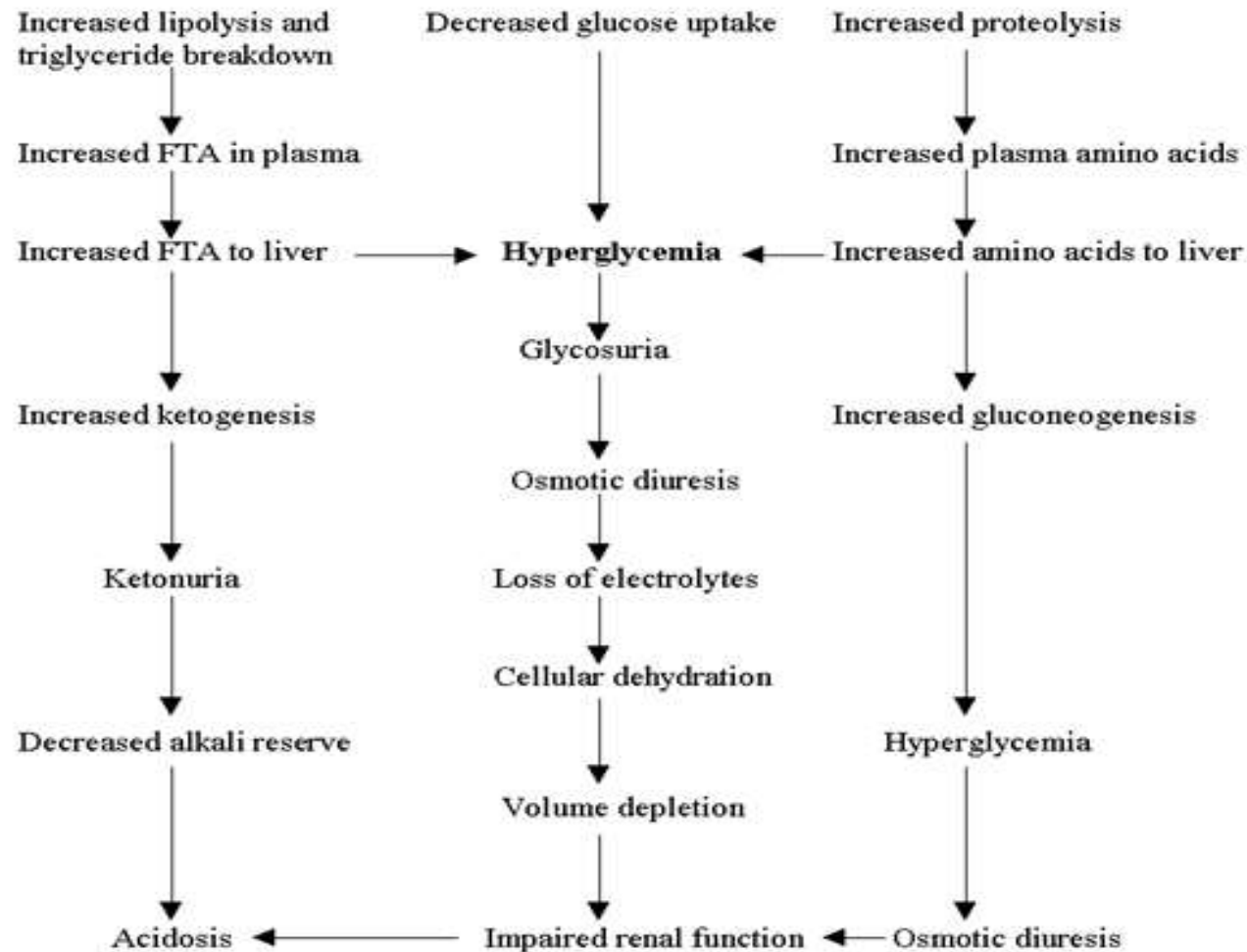


DIABETIC KETO- ACIDOSIS

- ❑ It happens predominantly in type 1 diabetes mellitus,
- ❑ But can also occur in **type 2 diabetes mellitus under certain circumstances.**
- ❑ This may be due to **inter-current illness** (pneumonia, influenza, gastroenteritis, a urinary tract infection),
- ❑ Young patients with recurrent episodes of DKA may have an **underlying eating disorder**, or may be using insufficient insulin for fear that it will cause weight gain.



DIABETIC KETO- ACIDOSIS



Diabetic Ketoacidosis may be diagnosed when the combination of **hyperglycemia (high blood sugars)**, **ketones on urinalysis** and **acidosis** are demonstrated.



ALCOHOLIC KETOACIDOSIS(AKA)

- Although the general physiological factors and mechanisms leading to AKA are understood, the precise factors have not been fully defined.

The following are the 3 main predisposing events:

- Delay and decrease in insulin secretion and excess glucagon secretion, induced by **starvation**
- Elevated ratio of (NADH)/NAD⁺) secondary to alcohol metabolism
- Volume depletion resulting from vomiting and poor oral intake of fluids



ALCOHOLIC KETOACIDOSIS(AKA)

- ❑ The metabolism of alcohol itself is a probable contributor to the ketotic state.
- ❑ Alcohol dehydrogenase metabolizes alcohol to acetaldehyde in the cytoplasm of hepatocyte.
- ❑ Acetaldehyde is metabolized further to acetic acid by aldehyde dehydrogenase mitochondria .
- ❑ Both steps require (NAD⁺) and produce (NADH).



ALCOHOLIC KETOACIDOSIS(AKA)

The decreased ratio of NAD^+/NADH has the following implications:

- ❑ Impaired conversion of lactate to pyruvate with an **increase in serum lactic acid levels**
 - ❑ Impaired gluconeogenesis because **pyruvate** is not available as a substrate for glucose production
 - ❑ A shift in the hydroxybutyrate ($\beta\text{-OH}$) to acetoacetate (AcAc) equilibrium toward $\beta\text{-OH}$ butyrate
 - ❑ In contrast to diabetic ketoacidosis, the predominant ketone body in AKA is **$\beta\text{-OH}$** . Routine clinical assays for ketonemia test for AcAc and acetone but not for $\beta\text{-OH}$.
 - ❑ Clinicians underestimate the degree of ketonemia if they rely solely on the results of laboratory testing.
- 

ALCOHOLIC KETOACIDOSIS(AKA)

- ❑ Prolonged vomiting leads to dehydration, which decreases renal perfusion, thereby limiting urinary excretion of ketoacids.
- ❑ Moreover, volume depletion increases the concentration of counter-regulatory hormones, further stimulating lipolysis and ketogenesis.
- ❑ The pivotal variable appears to be a relative deficiency of insulin.
- ❑ Individuals with higher insulin levels are more likely to present with the syndrome of alcohol-induced **hypoglycemia without ketoacidosis**

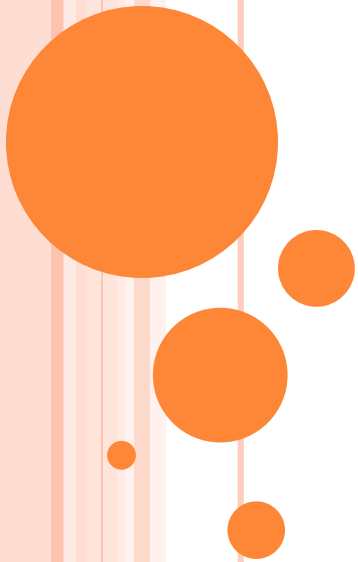


ALCOHOLIC KETOACIDOSIS(AKA)

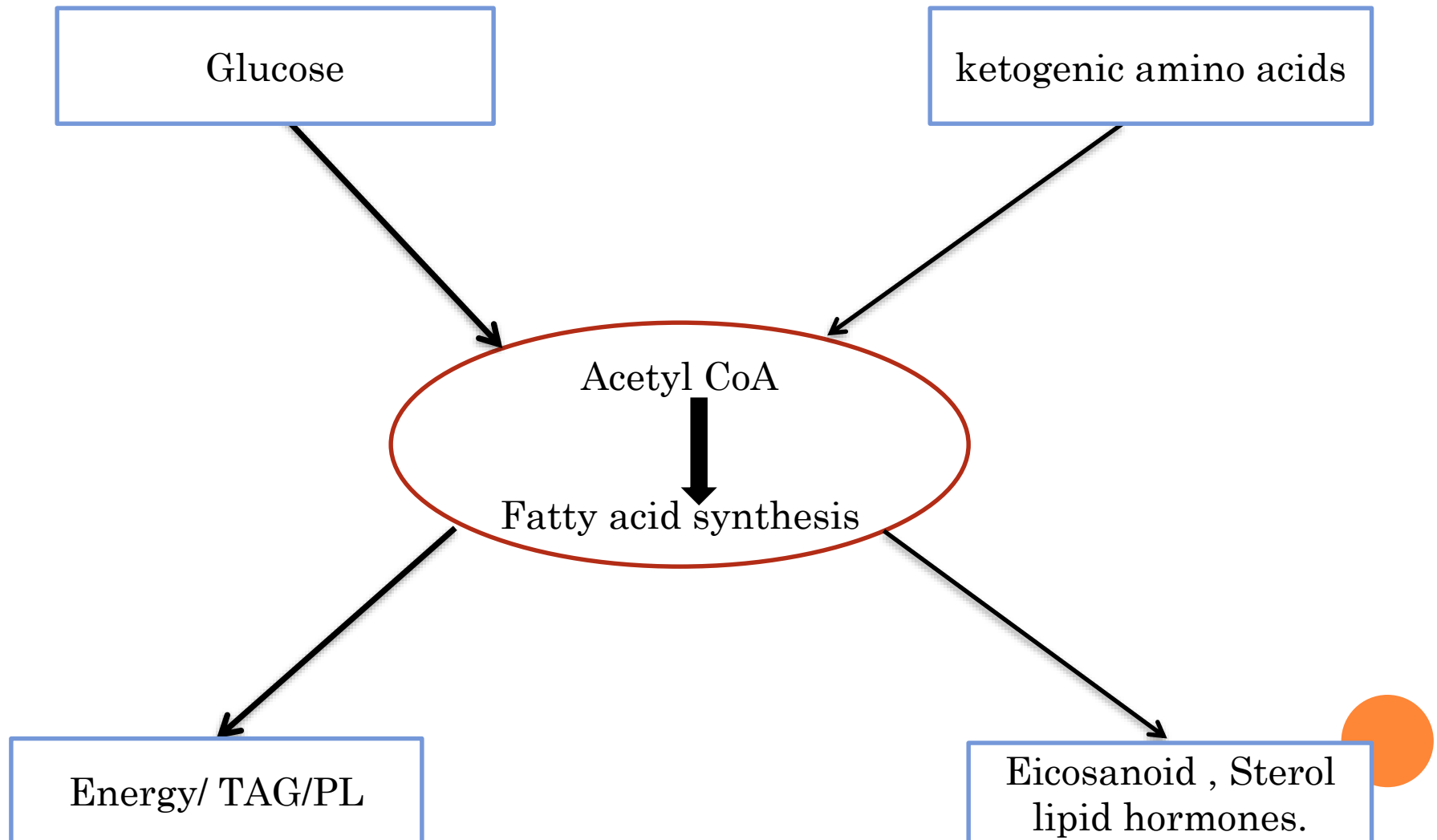
- ❑ Most cases of AKA occur when a **person with poor nutritional status due to long-standing alcohol abuse** who has been on a drinking binge suddenly **decreases energy intake because of abdominal pain, nausea, or vomiting.**
- ❑ In addition, AKA is often precipitated by another medical illness such as **infection or pancreatitis.**
- ❑ AKA results from the accumulation of the ketoacids, hydroxybutyric acid, and acetoacetic acid.
- ❑ Such accumulation is caused by the complex interaction stemming from **alcohol cessation, decreased energy intake, volume depletion, and the metabolic effects of hormonal imbalance.**



DE NOVO SYNTHESIS OF FATTY ACID



FATTY ACID SYNTHESIS SERVES TWO MAIN FUNCTIONS

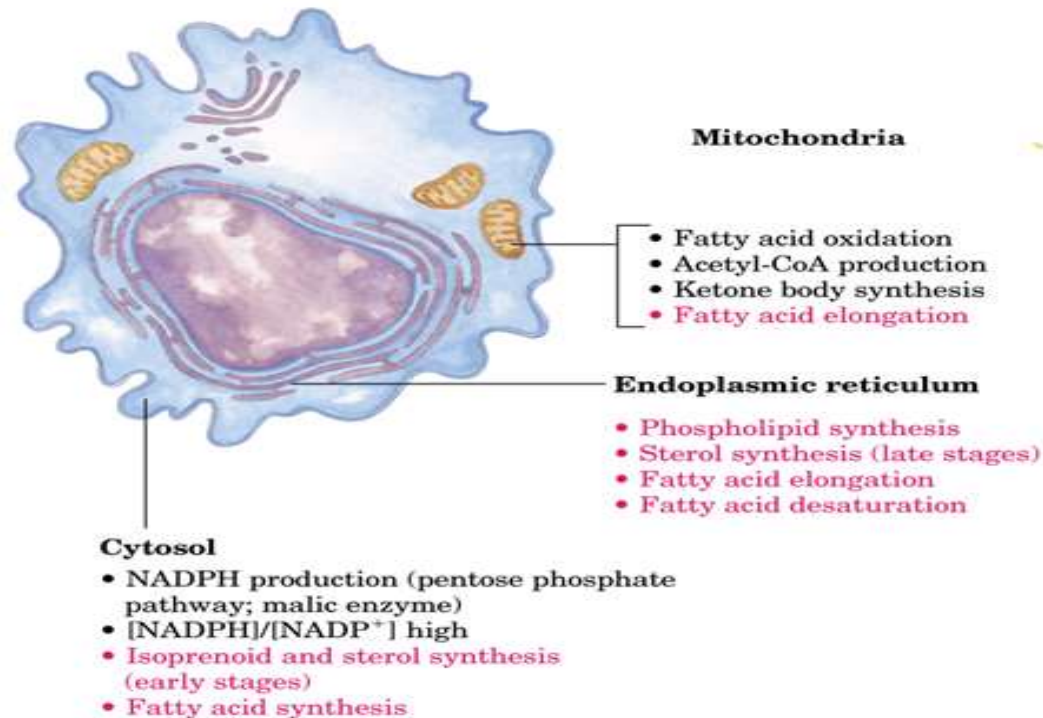


TISSUE AND CELLULAR LOCATIONS

Fatty acid synthesis occurs primarily in

- Liver
- Adipose tissue (fat)
- Lactating mammary glands

Fatty acid synthesis takes places in the cytosol of the cells



CONDITIONS AT WHICH FATTY ACID SYNTHESIS OCCURS

- ❑ Excess energy following a meal (↑insulin/glucagon ratio)
- ❑ High-carbohydrate diet.
- ❑ Low fat diet
- ❑ FA synthesis is very high during embryogenesis and in fetal lungs.
- ❑ **Fatty acid synthesis** is also greatly increased in cancer cells.



PRECURSORS FOR FATTY ACID SYNTHESIS

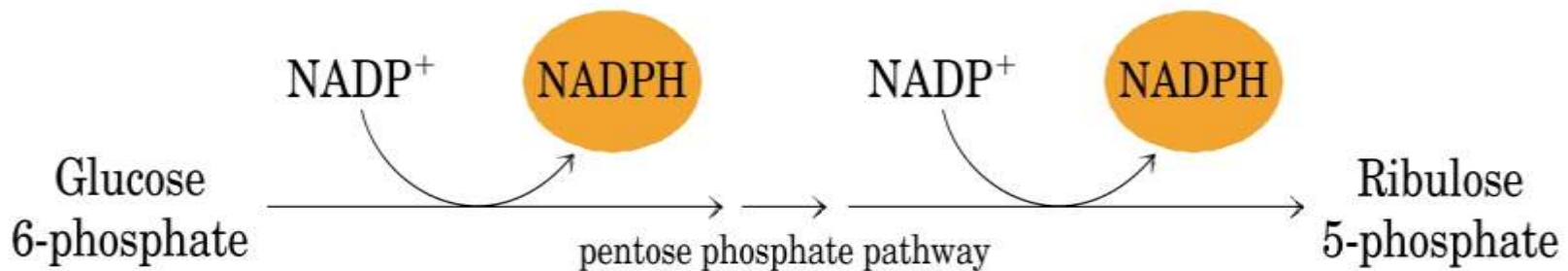
- ❑ Metabolites required
 - ❑ NADPH
 - ❑ Acetyl-CoA
 - ❑ ATP, Mn^{2+}
 - ❑ Biotin
 - ❑ HCO_3^- (as a source of CO_2).

- ❑ Enzymes required
 - ❑ Acetyl CoA Carboxylase
 - ❑ Fatty acid synthase complex



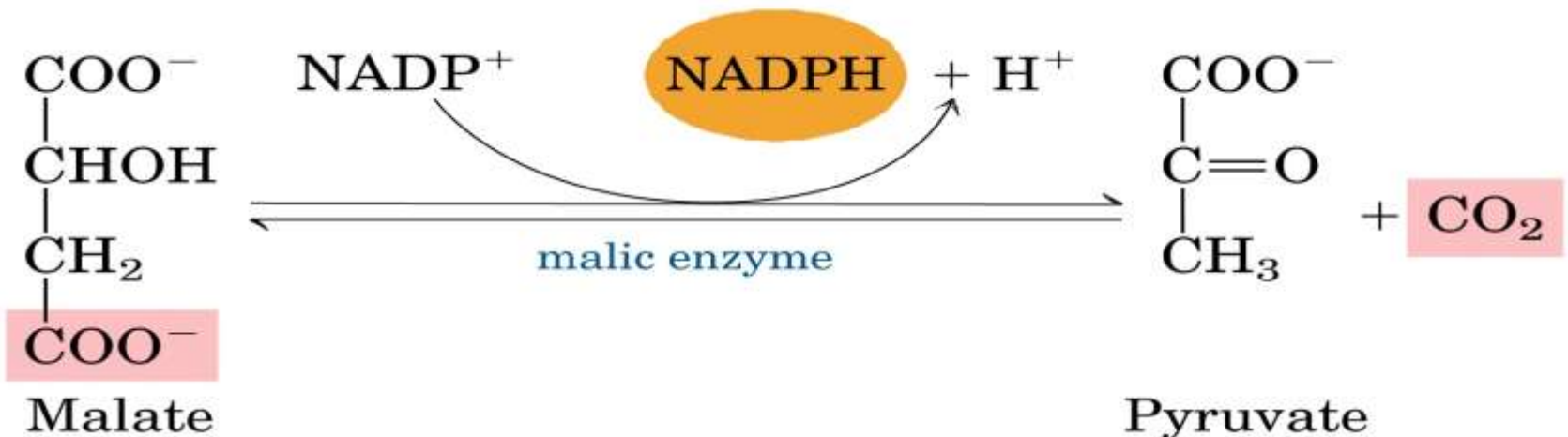
PPP -SOURCES OF NADPH

- The oxidative reactions of the **pentose phosphate pathway are the chief source** of the hydrogen required for the reductive synthesis of fatty acids.
- NADPH is involved **as donor of reducing equivalents**
- In hepatocytes, adipose tissue and the lactating mammary glands, the NADPH is supplied primarily by the pentose phosphate pathway



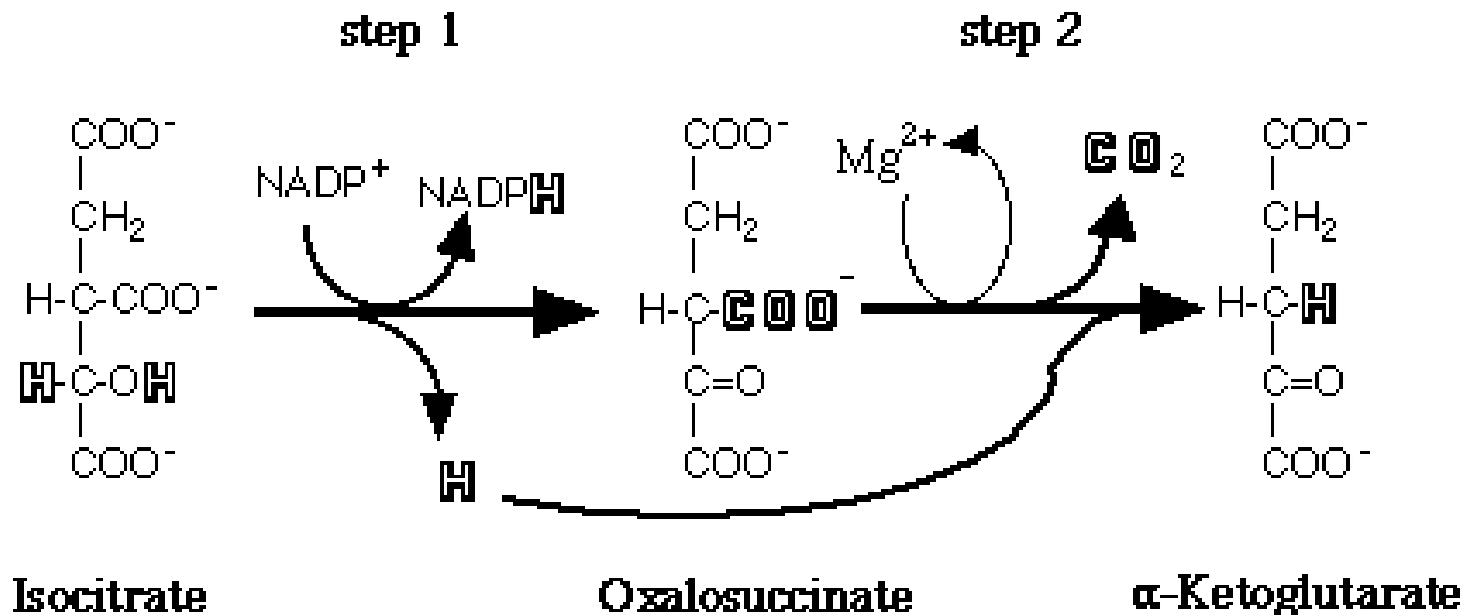
THE MALIC ENZYME- SOURCE OF NADPH

The second important NADPH source, is produced by reversible reaction of Malate to pyruvate produced in the reaction reenters the mitochondrion for further utilization

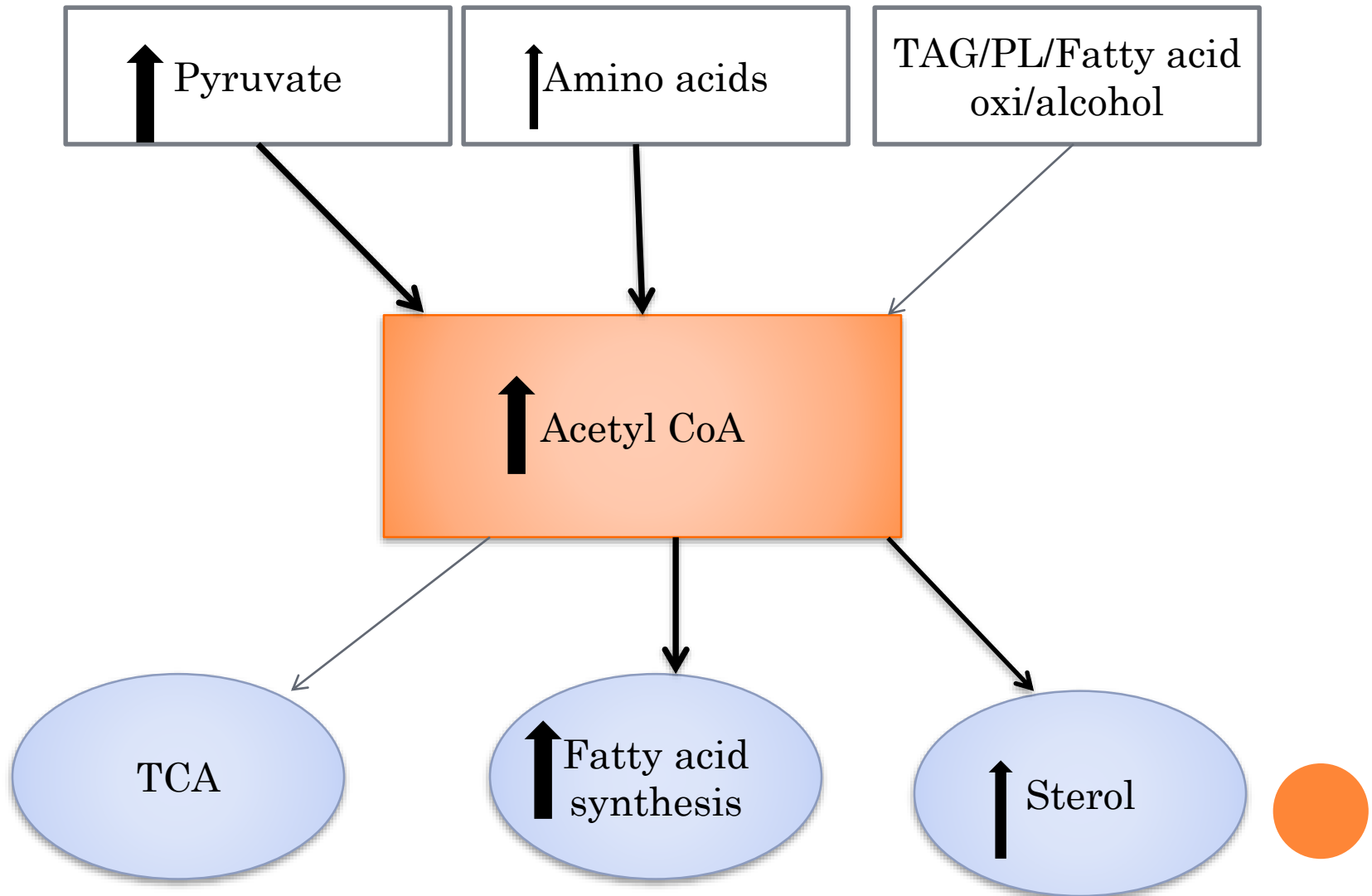


CYTOSOLIC ISOCITRATE DEHYDROGENASE- SOURCE OF NADPH

- There are three isoenzymes of **isocitrate dehydrogenase**. One, which uses NAD^+ , is found only in mitochondria. The other two use NADP^+ and are found in mitochondria and the cytosol.

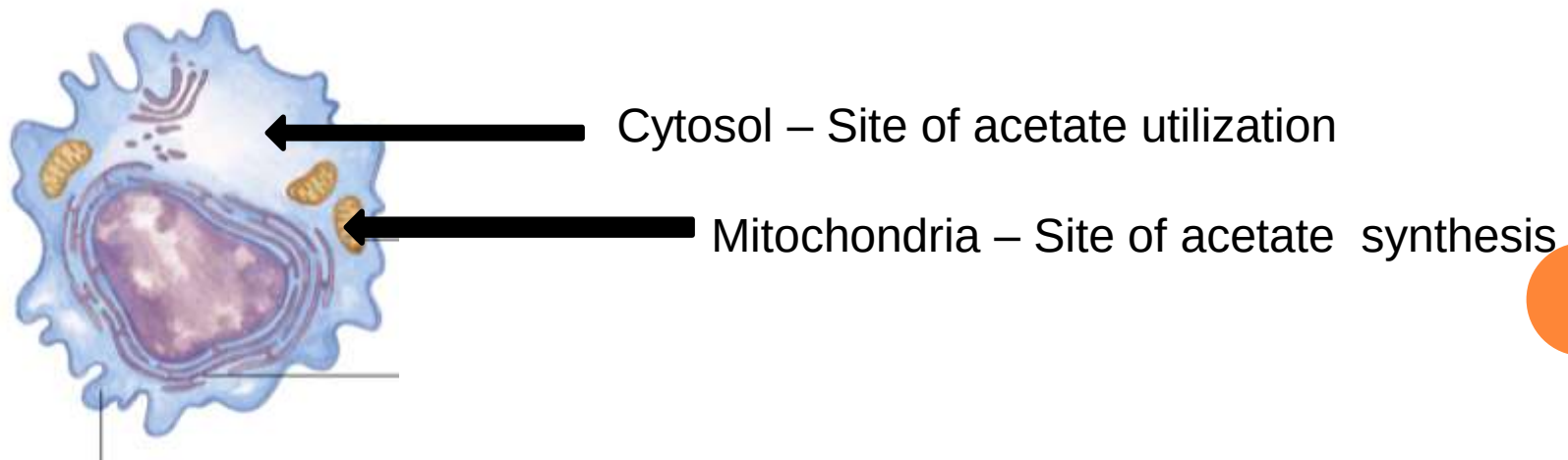


SOURCES OF ACETYL COA FOR FA SYNTHESIS



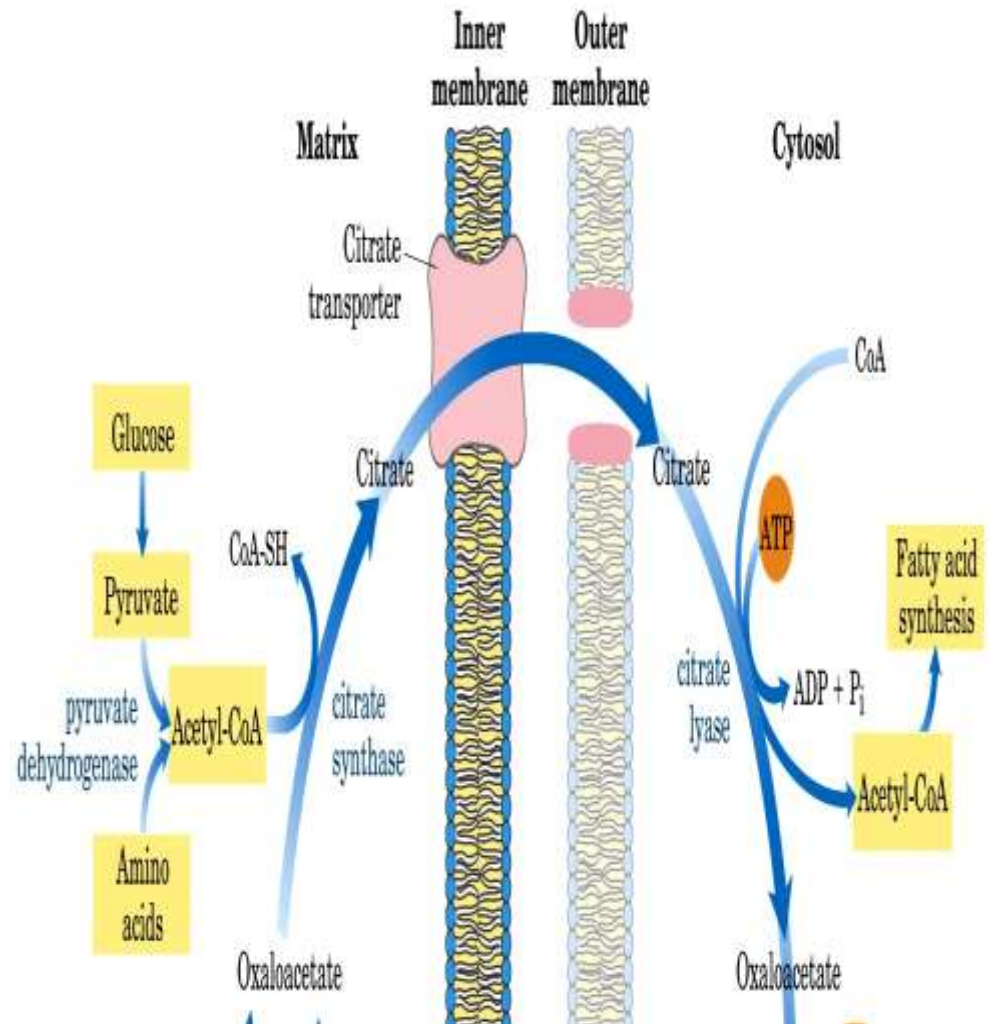
TRANSPORTATION OF ACETYL CO A

- ❑ Fatty acid synthesis requires considerable amounts of acetyl-CoA
- ❑ Nearly all acetyl-CoA used in fatty acid synthesis is formed in **mitochondria**
- ❑ Acetyl co A has to move out from the **mitochondria to the cytosol**



TRANSPORTATION OF ACETYL CO A

- ❑ **Acetate** is shuttled out of mitochondria as **citrate**
- ❑ The mitochondrial inner membrane is impermeable to acetyl-CoA
- ❑ Intra-mitochondrial acetyl-CoA first reacts with oxaloacetate to form citrate, in the TCA cycle catalyzed by citrate synthase
- ❑ Citrate then passes into the cytosol through the mitochondrial inner membrane on the **citrate transporter**.
- ❑ In the cytosol, citrate is cleaved by **citrate lyase** regenerating acetyl-CoA.



FATE OF OXALOACETATE

The product of Citrate cleavage, **oxaloacetate** can be-

- ❑ Channeled towards glucose production
- ❑ Converted to malate by malate dehydrogenase
- ❑ Converted to Pyruvate by **Malic enzyme**, producing more NADPH, that can be used for fatty acid synthesis
- ❑ Pyruvate and Malate pass through special transporters present in the inner mitochondrial membrane

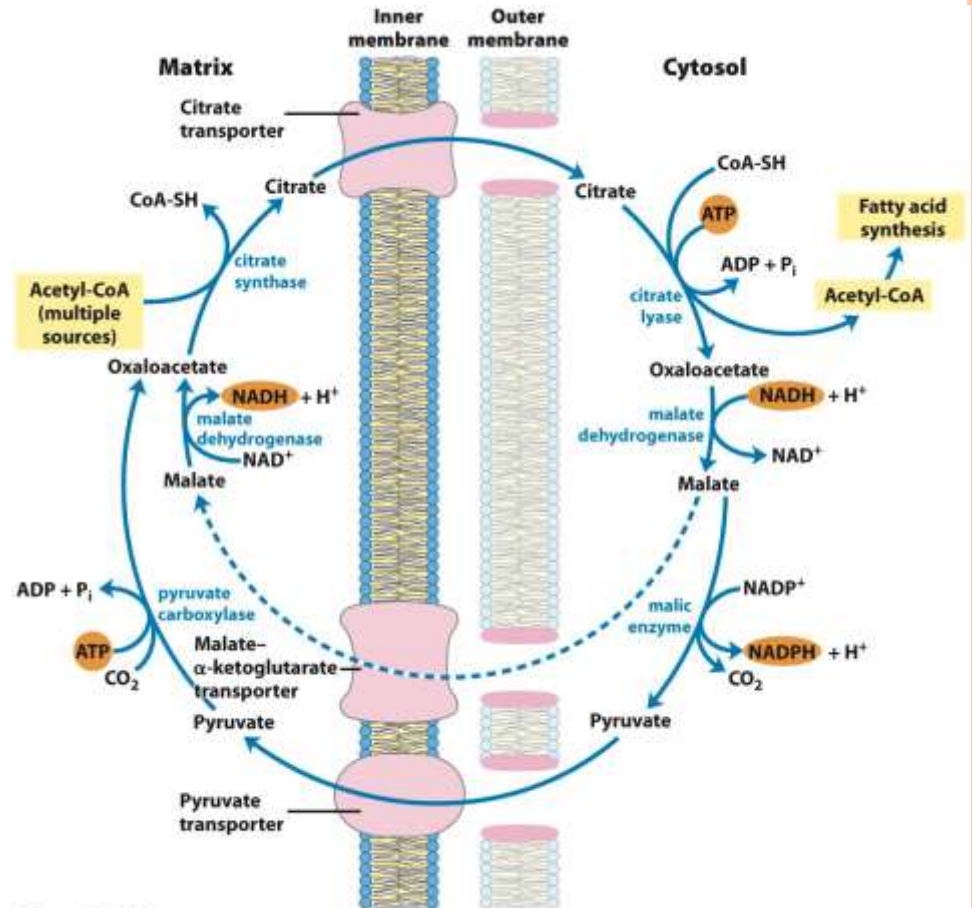


Figure 21-10
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ENZYMES INVOLVED IN FATTY ACID SYNTHESIS

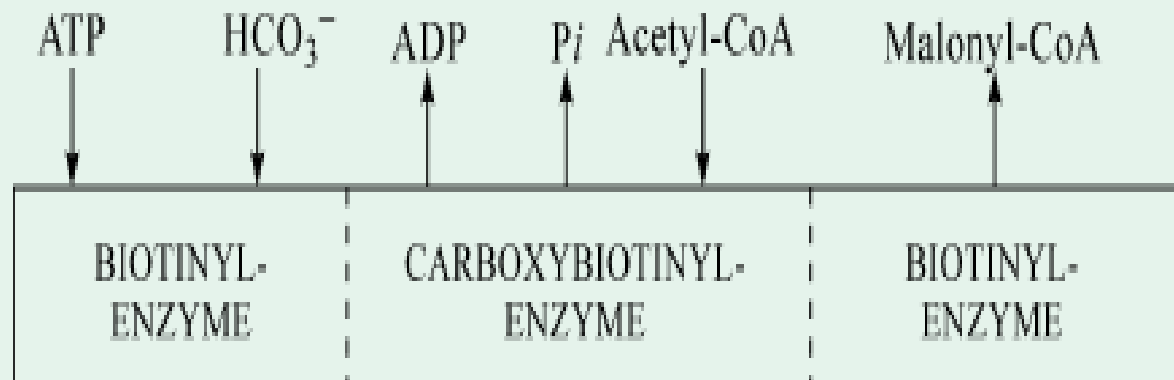
Two main enzymes-

- Acetyl co A carboxylase
 - Fatty acid Synthase
- Both the enzymes are multienzyme complexes



ACETYL-CoA CARBOXYLASE

- ❑ The production of **malonyl-CoA** is the initial committed step in the fatty acid synthesis
- ❑ The reaction is catalyzed by the enzyme, **acetyl-CoA carboxylase**.
- ❑ **AcetylCoA carboxylase** -Is the initial & rate controlling step in fatty acid synthesis
- ❑ It is a multienzyme complex containing- Biotin
 - Biotin Carboxylase
 - Biotin carboxyl carrier protein
 - Transcarboxylase



FORMATION OF MALONYL CoA

- ❑ ATP-dependent carboxylation provides energy input.
- ❑ The CO_2 is lost later during condensation with the growing fatty acid.
- ❑ The spontaneous decarboxylation drives the condensation reaction.
- ❑ As with other carboxylation reactions, the enzyme prosthetic group is biotin.
- ❑ The reaction takes place in two steps: carboxylation of biotin (involving ATP) and transfer of the carboxyl to acetyl-CoA to form malonyl-CoA.

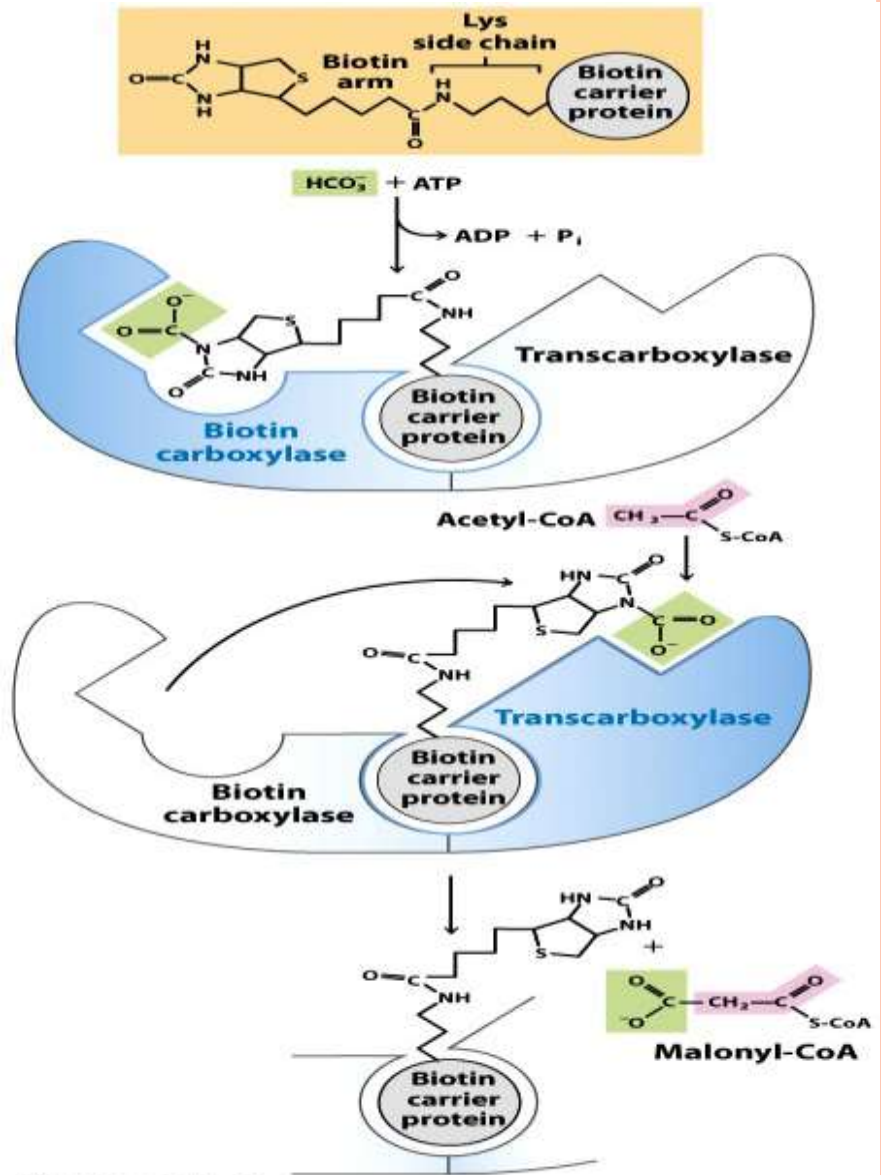


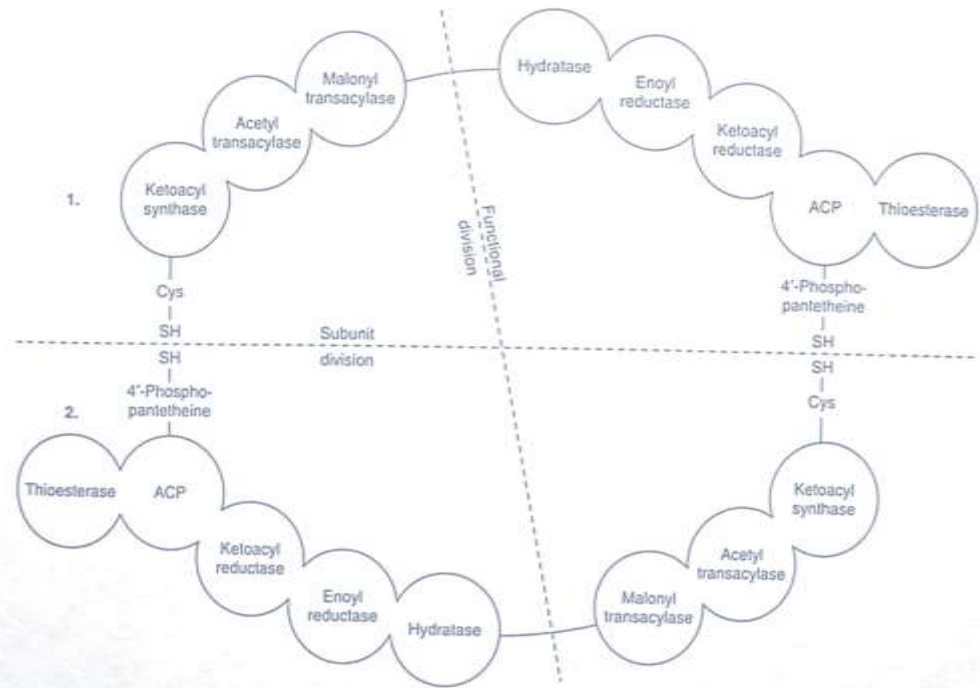
Figure 21-1

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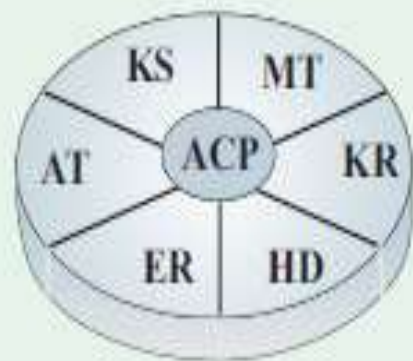
FATTY ACID SYNTHASE ENZYME COMPLEX

- ❑ The Fatty Acid Synthase Complex is a polypeptide containing seven enzyme activities
- ❑ In bacteria and plants, the individual enzymes of the fatty acid synthase system are separate, and the acyl radicals are found in combination with a protein called the acyl carrier protein (ACP).
- ❑ In yeast, mammals, and birds, the synthase system is a multienzyme polypeptide complex that incorporates ACP, which takes over the role of CoA.

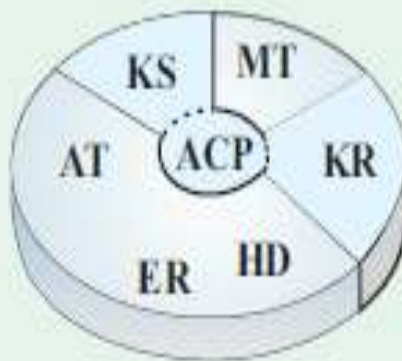


Fatty acid synthase multienzyme complex. The complex is a dimer of two identical polypeptide monomers, 1 and 2, each consisting of seven enzyme activities and the acyl carrier protein (ACP). (Cys—SH, cysteine thiol.) The —SH of the 4'-phosphopantetheine of one monomer is in close proximity to the —SH of the cysteine residue of the ketoacyl synthase of the other monomer, suggesting a "head-to-tail" arrangement of the two monomers. Though each monomer contains all the partial activities of the reaction sequence, the actual functional unit consists of one-half of one monomer interacting with the complementary half of the other. Thus, two acyl chains are produced simultaneously.

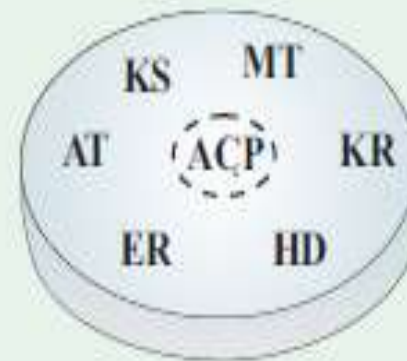
<i>Component</i>	<i>Abbreviation</i>	<i>E.C. No.</i>	<i>Role</i>
Acyl carrier protein	ACP	6.4.1.2	Carries acyl groups in thioester linkage
Acyl-CoA-ACP transacetylase	AT	2.3.1.9	Transfers acyl group from CoA to cysteine residue of KS
Malonyl-CoA-ACP transferase	MT	2.8.3.3	Transfers malonyl group from CoA to ACP
β -ketoacyl-ACP synthase	KS	2.3.1.16	Condenses acyl and malonyl groups
β -ketoacyl-ACP reductase	KR	1.1.1.36	Reduces β -keto group to β -hydroxy group
β -hydroxyacyl-ACP dehydrogenase	HD	4.2.1.17	Removes H_2O from β -hydroxyacyl-ACP, creating double bond
Enoyl-ACP reductase	ER	1.3.99.2	Reduces double bond, forming saturated acyl-ACP



Bacteria + Plants



Yeast



Vertebrates

SUBSTRATE CHANNELING

The organized structure of the fatty acid synthases of higher organisms enhances the efficiency of the overall process because of the following reasons :

1. The intermediates are directly transferred from one active site to the next.
2. The intermediates are not diluted in the cytosol.
3. The intermediates do not have to find each other by random diffusion.
4. The covalently-bound intermediates are secluded and protected from competing reactions.

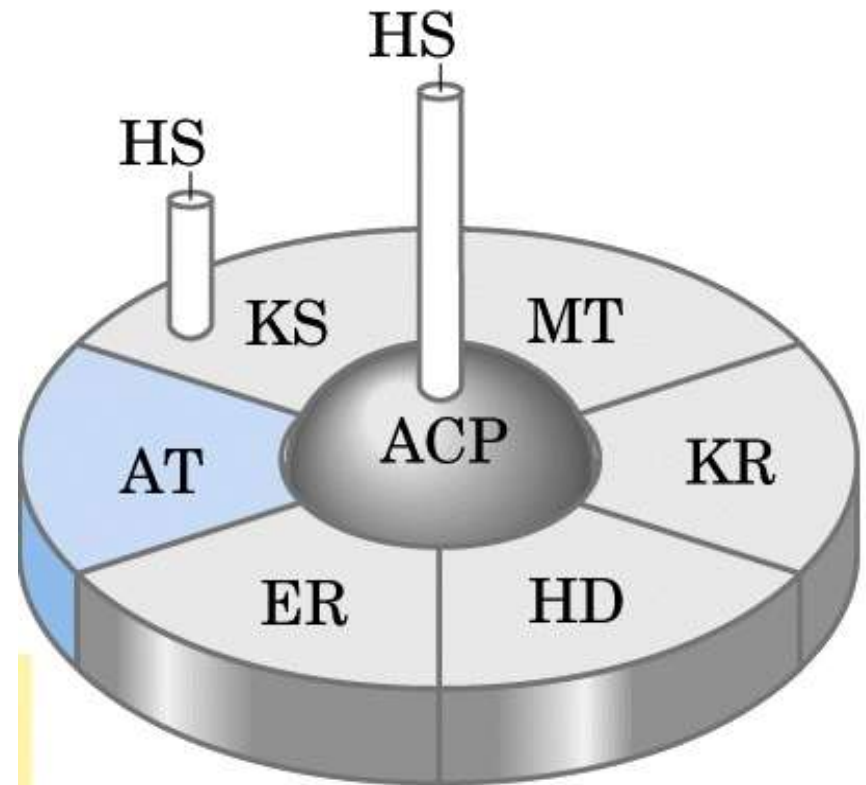


FATTY ACID SYNTHASE COMPLEX

- ❑ Each segment of the disk represents one of the six enzymatic activities of the complex
- ❑ At the center is the **ACP – acyl carrier protein** - with its phosphopantetheine arm ending in –SH.

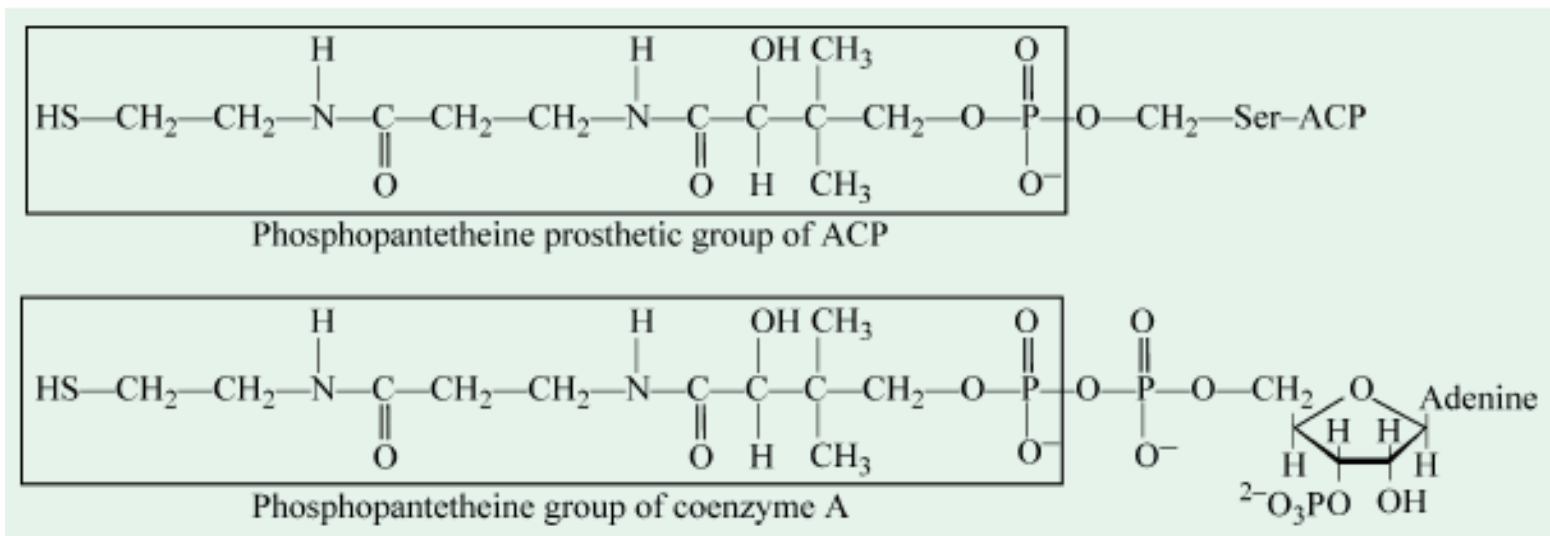
Fatty Acid Synthase prosthetic groups:

- ❑ The **thiol** (-SH) of the side-chain of a **cysteine** residue of **keto acyl synthase** enzyme (also called condensing enzyme)
- ❑ The **thiol** (-SH) of **phosphopantetheine**, equivalent in structure to part of coenzyme A. It is a component of Acyl carrier protein



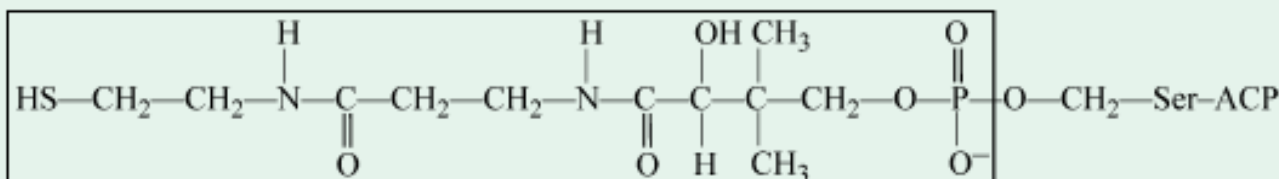
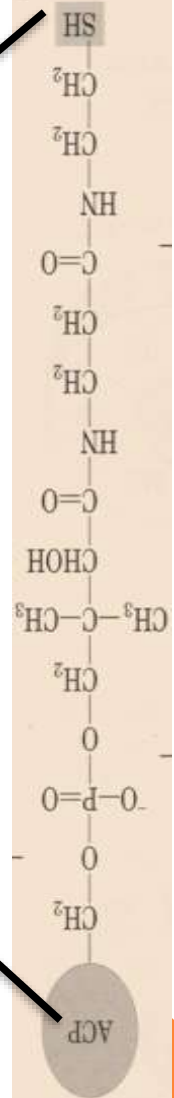
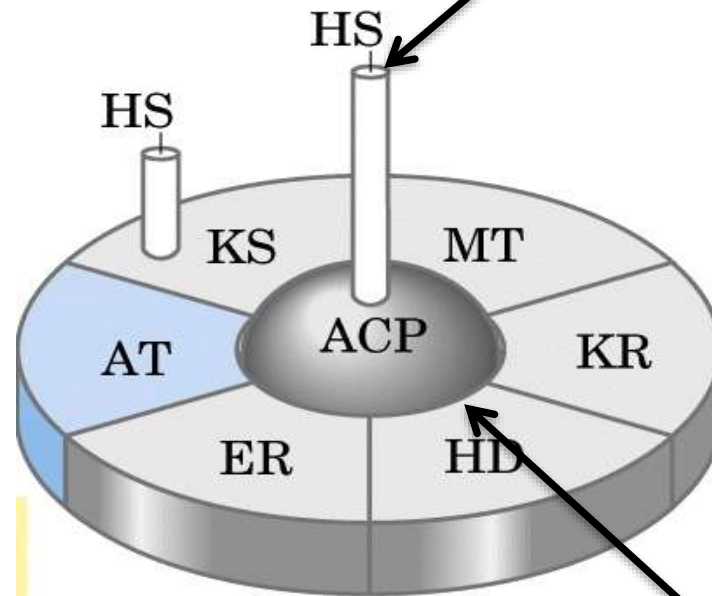
INTERMEDIATES IN FATTY ACID SYNTHESIS AND THE ACP

- ❑ *The intermediates in fatty acid synthesis are linked to an acyl carrier protein. Specifically, the intermediates are attached to the sulfhydryl(–SH) terminus of a phosphopantetheine group.*
- ❑ This single polypeptide chain of 77 residues can be regarded as a giant prosthetic group, a “*macro-CoA*”.



STRUCTURE/FUNCTION OF PHOSPHOPANTETHEINE

- ❑ **Phosphopantetheine** (Pant) is covalently linked via a phosphate ester to a serine OH of the **acyl carrier protein** domain of **Fatty Acid Synthase**.
- ❑ The **long flexible arm** of phosphopantetheine helps its thiol to move from one active site to another within the complex.
- ❑ Carrying the reaction intermediates from one enzyme active site to the next.



Phosphopantetheine prosthetic group of ACP

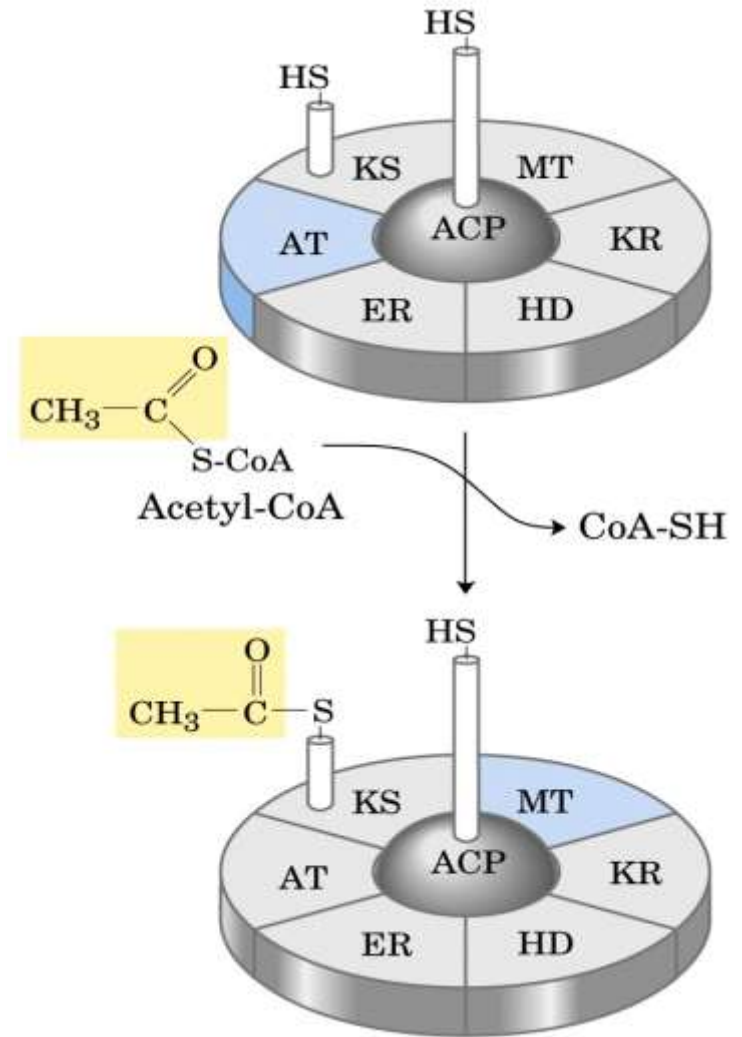
THE FIRST ROUND OF FA BIOSYNTHESIS

- ❑ To initiate FA biosynthesis, **malonyl** and **acetyl** groups are activated on to the enzyme **fatty acid synthase**.
- ❑ Initially, a priming molecule of **acetyl-CoA** combines with a cysteine —SH group catalyzed by **acetyl transacylase**
- ❑ Malonyl-CoA combines with the adjacent —SH on the 4'-phosphopantetheine of ACP of the other monomer, catalyzed by **malonyl transacylase** (to form **acetyl (acyl)-malonyl enzyme**).



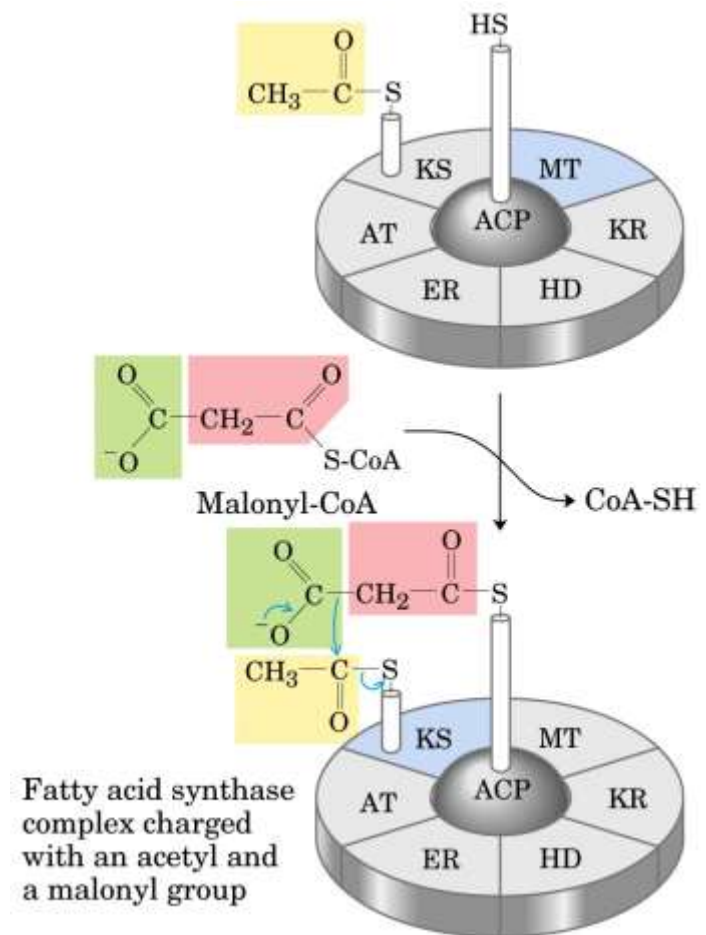
THE ACTIVATION OF THE ACETYL GROUP

- ❑ The acetyl group from acetyl-CoA is transferred to the Cys-SH group of the β -ketoacyl ACP synthase
- ❑ This reaction is catalyzed by **acetyl-CoA transacetylase**.



THE ACTIVATION OF THE MALONYL GROUP

- ❑ Transfer of the malonyl group to the –SH group of the ACP is catalyzed by **malonyl-CoA ACP transferase**.
- ❑ The charged acetyl and malonyl groups are now in close proximity to each other



SERIES OF REACTIONS

After activation, the processes involved are-

1. Condensation
2. Reduction
3. Dehydration
4. Reduction

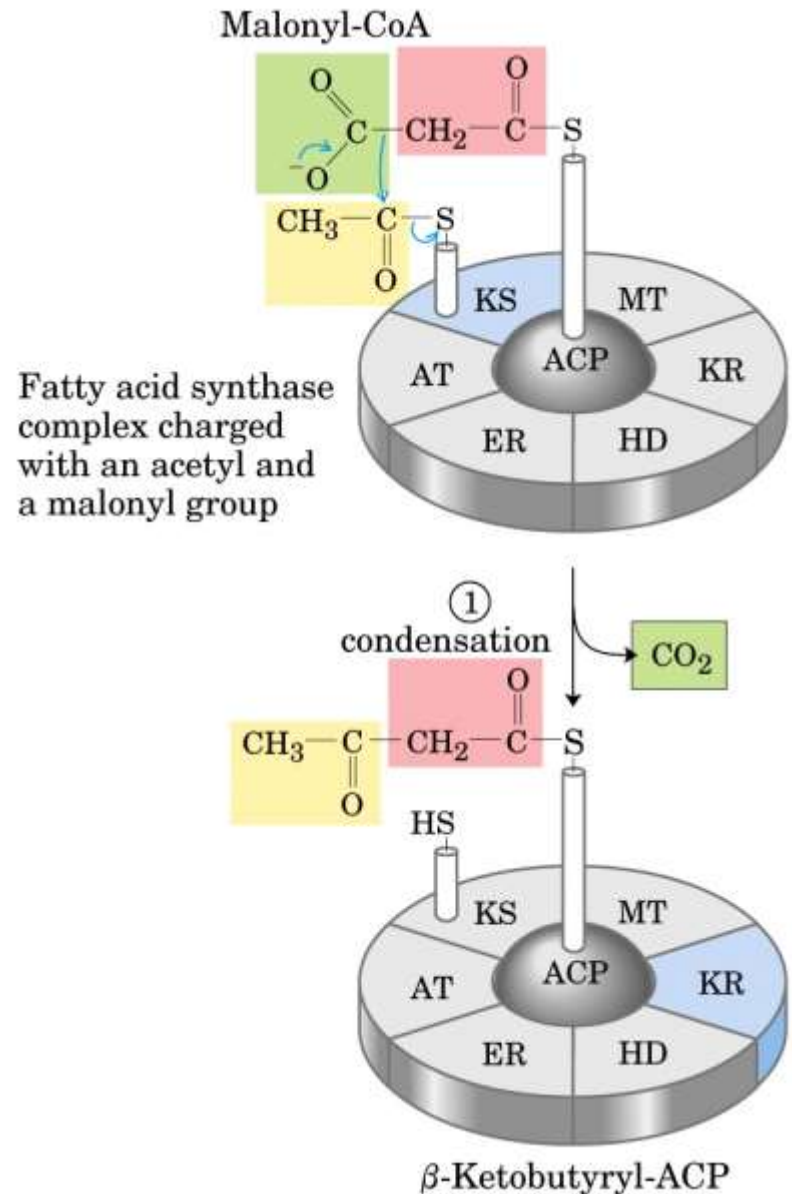
These steps are repeated till a fatty acid with 16 carbon atoms is synthesized



STEP-1(CONDENSATION)

1. Condensation –

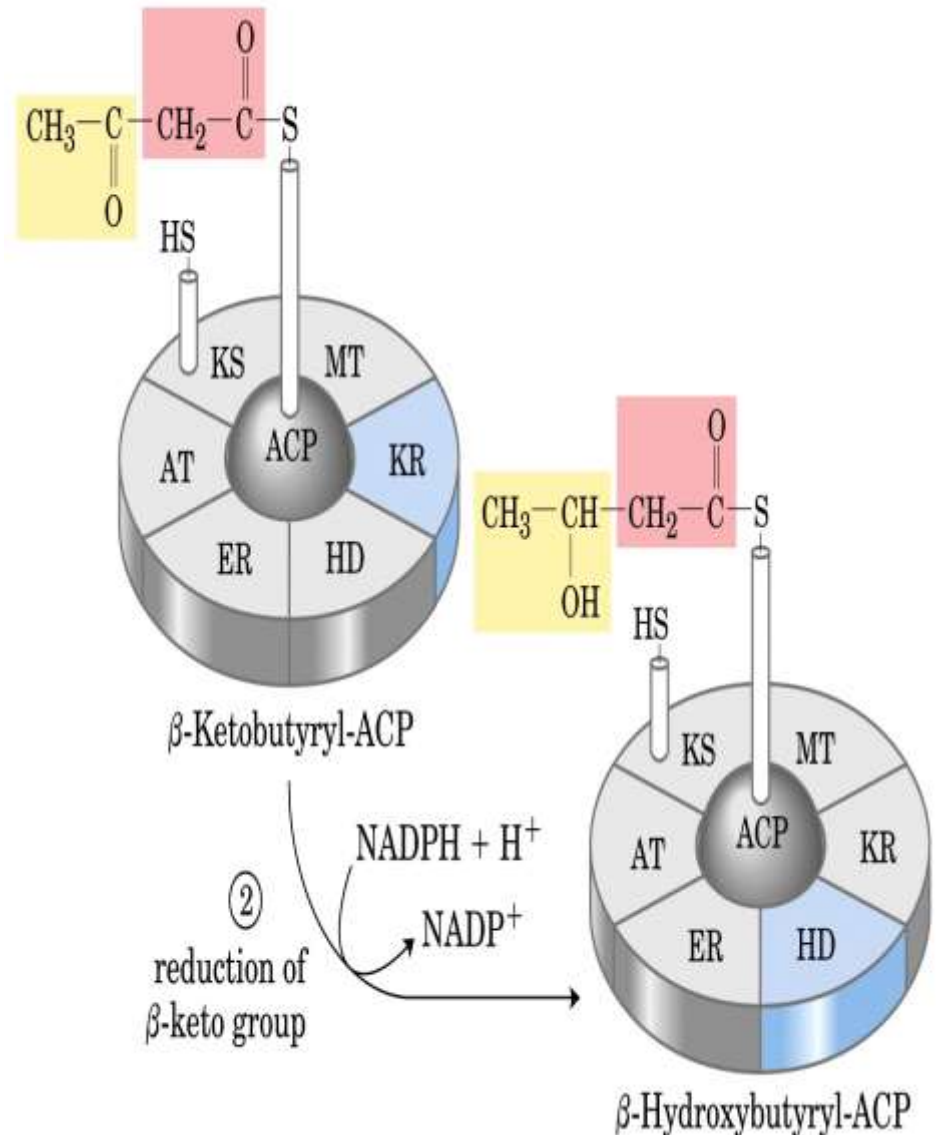
- ❑ The acetyl group attacks the methylene group of the malonyl residue to form **Acetoacetyl-ACP**
- ❑ The reaction is catalyzed by **β -ketoacyl-ACP synthase.**
- ❑ Decarboxylation allows the reaction to go to completion, pulling the whole sequence of reactions in the forward direction.



STEP-2 (REDUCTION)

2. Reduction-

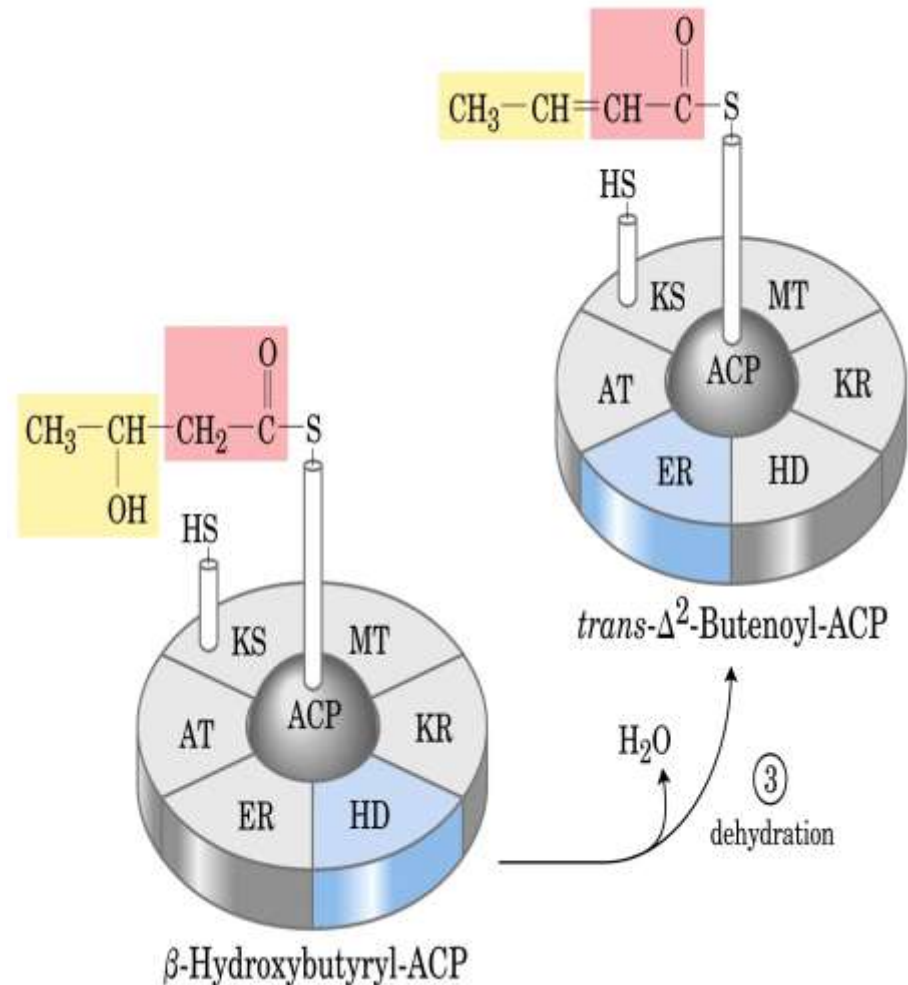
- The Acetoacetyl-ACP is reduced to β -hydroxybutyryl-ACP, catalyzed by **β -ketoacyl-ACP reductase**
- $\text{NADPH} + \text{H}^+$ are required for reduction



STEP-3 (DEHYDRATION)

3. Dehydration –

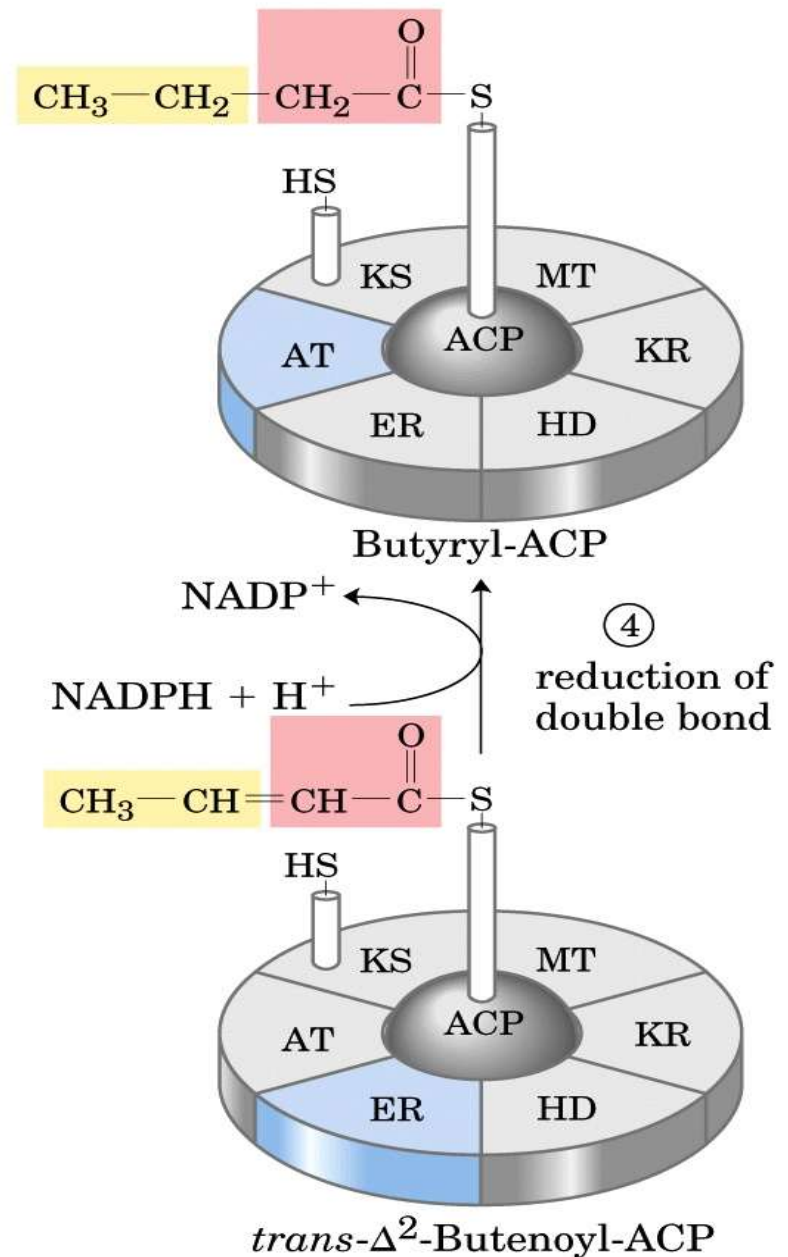
- Dehydration yields a double bond in the product, *trans*- Δ^2 -butenoyl-ACP,
- Reaction is catalyzed by β -hydroxybutyryl-ACP dehydratase.



STEP-4 (REDUCTION)

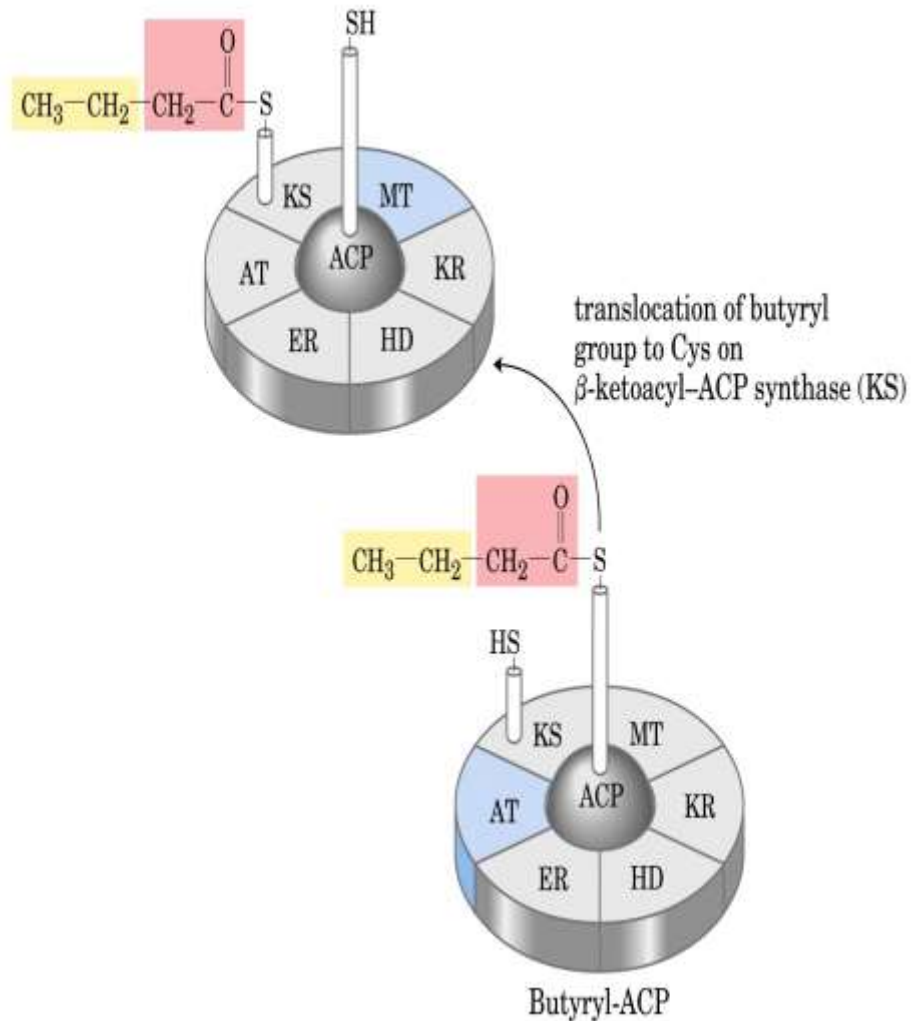
4. Reduction

- ❑ Reduction of the double bond takes place to form butyryl-ACP, Reaction is catalyzed by **enoyl-reductase**.
- ❑ Another **NADPH** dependent reaction.



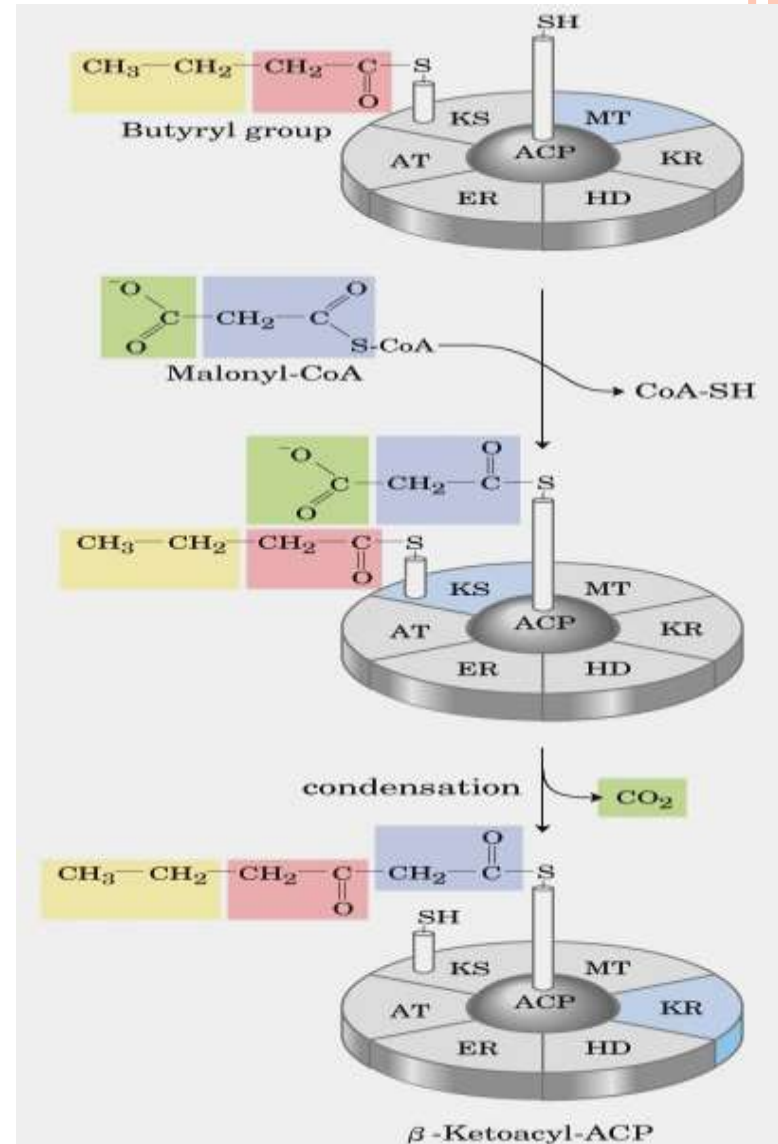
THE GROWING CHAIN IS TRANSFERRED FROM THE ACYL CARRIER PROTEIN

- ❑ This reaction makes way for the next incoming malonyl group.
- ❑ The enzyme involved is **acetyl-CoA transacetylase**



BEGINNING OF THE SECOND ROUND OF THE FA SYNTHESIS CYCLE

- ❑ The butyryl group is on the Cys-SH group
- ❑ The incoming malonyl group is first attached to ACP.
- ❑ In the condensation step, the entire butyryl group is exchanged for the carboxyl group on the malonyl residue

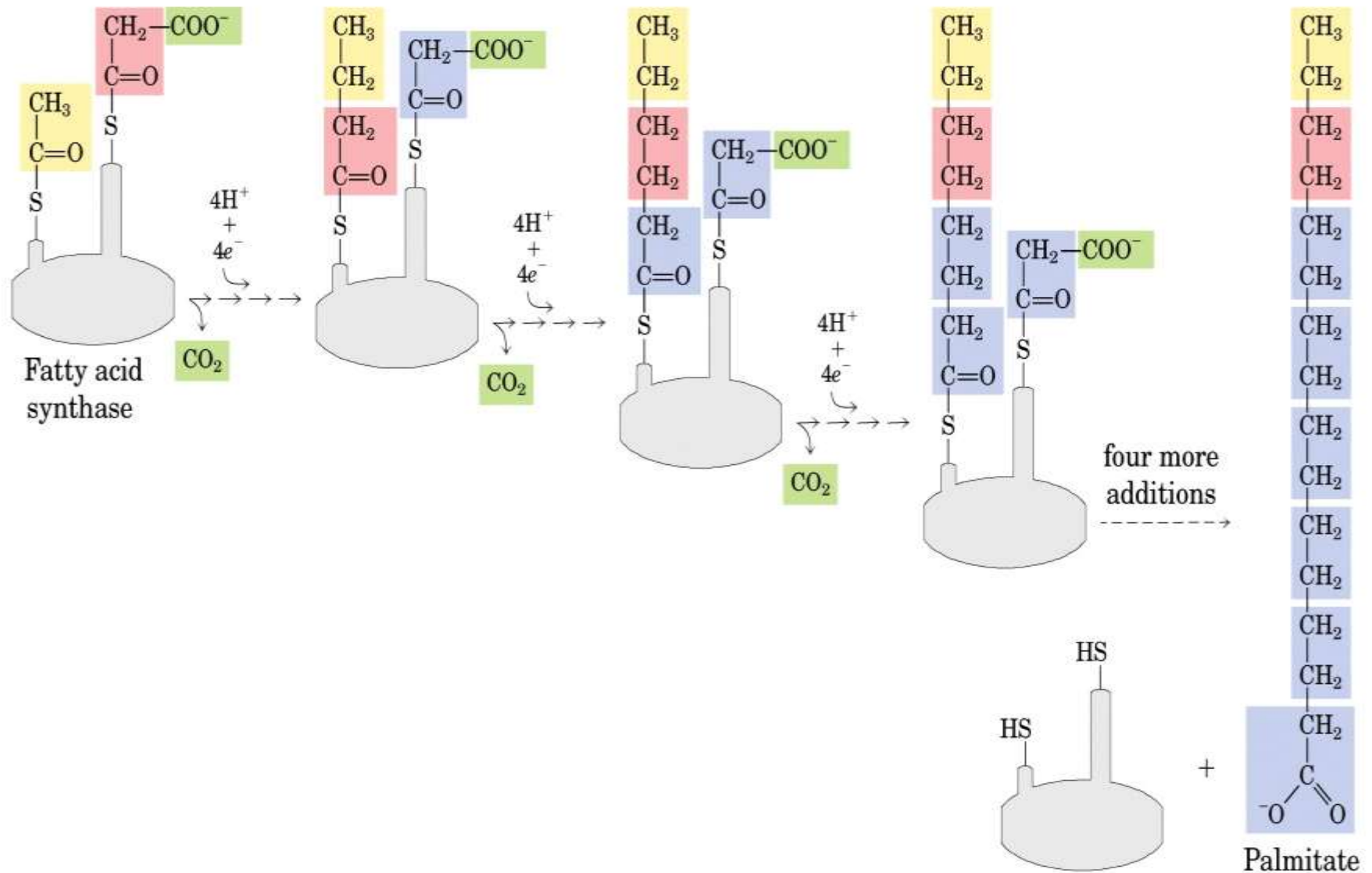


REPETITION OF THESE FOUR STEPS LEADS TO FATTY ACID SYNTHESIS

- ❑ The 3-ketoacyl group is **reduced, dehydrated, and reduced again** (reactions 2, 3, 4) to form the corresponding saturated acyl-S-enzyme.
- ❑ A new malonyl-CoA molecule combines with the —SH of 4'-phosphopantetheine, displacing the saturated acyl residue onto the free cysteine —SH group.
- ❑ The sequence of reactions are repeated until a saturated 16-carbon acyl radical (**Palmityl**) has been assembled.
- ❑ It is liberated from the enzyme complex by the activity of a seventh enzyme in the complex, **Thioesterase** (deacylase).



REPETITION OF THESE FOUR STEPS LEADS TO FATTY ACID SYNTHESIS



THE RESULT OF FATTY ACID SYNTHASE ACTIVITY

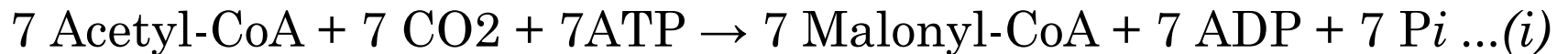
- ❑ Seven cycles of condensation and reduction produce the 16-carbon saturated palmitoyl group, still bound to ACP.
- ❑ Chain elongation usually stops at this point, and free palmitate is released from the ACP molecule by hydrolytic activity in the synthase complex.
- ❑ Smaller amounts of longer fatty acids such as stearate (18:0) are also formed
- ❑ In mammary gland, there is a separate **Thioesterase** specific for acyl residues of C₈, C₁₀, or C₁₂, which are subsequently found in milk lipids.



STOICHIOMETRY OF FATTY ACID SYNTHESIS

The overall reaction for the synthesis of palmitate from acetyl-CoA can be broken down into 2 parts :

(a) *the formation of 7 malonyl-CoA molecules*



(b) *the 7 cycles of condensation and reduction*



Hence, the overall process *for the synthesis of palmitate is :*

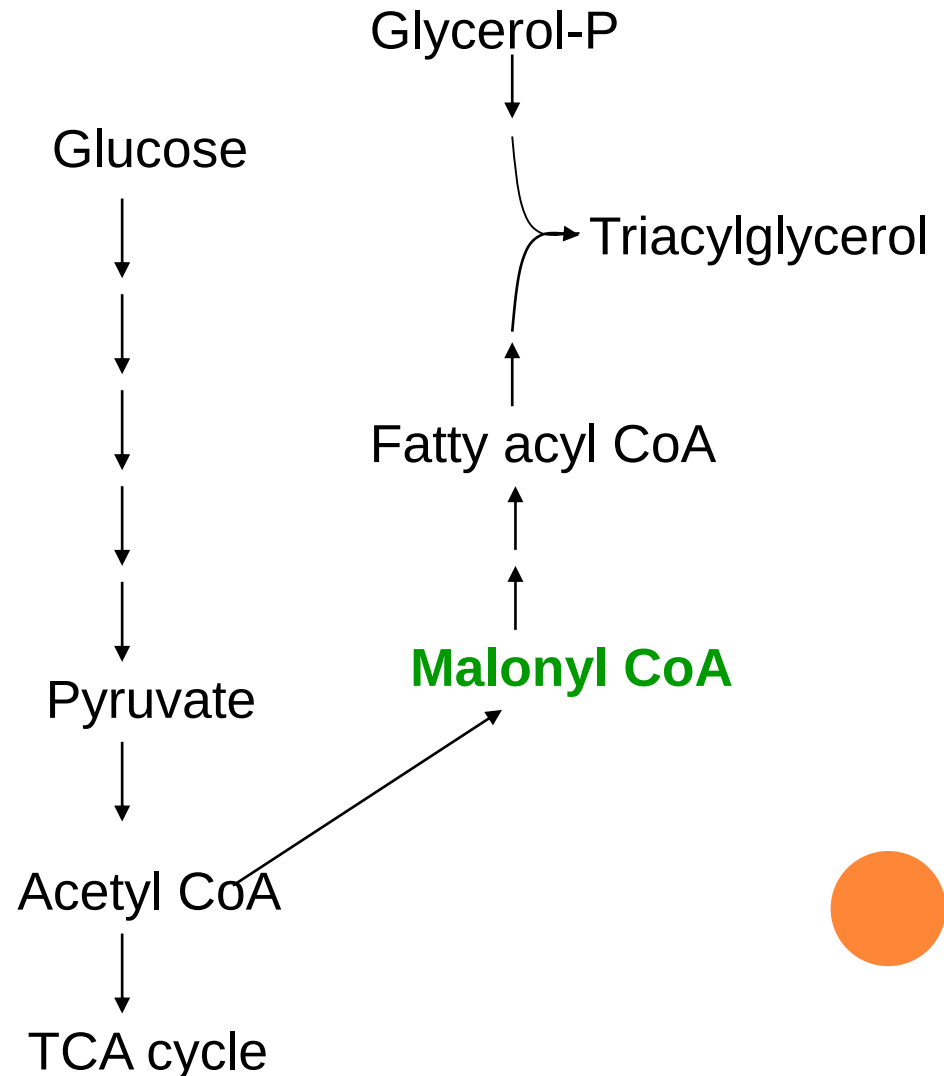


Note that the CO₂ utilized (formation of malonyl-CoA) and the CO₂ produced (condensation reaction) cancel each other when the overall stoichiometry is tabulated.



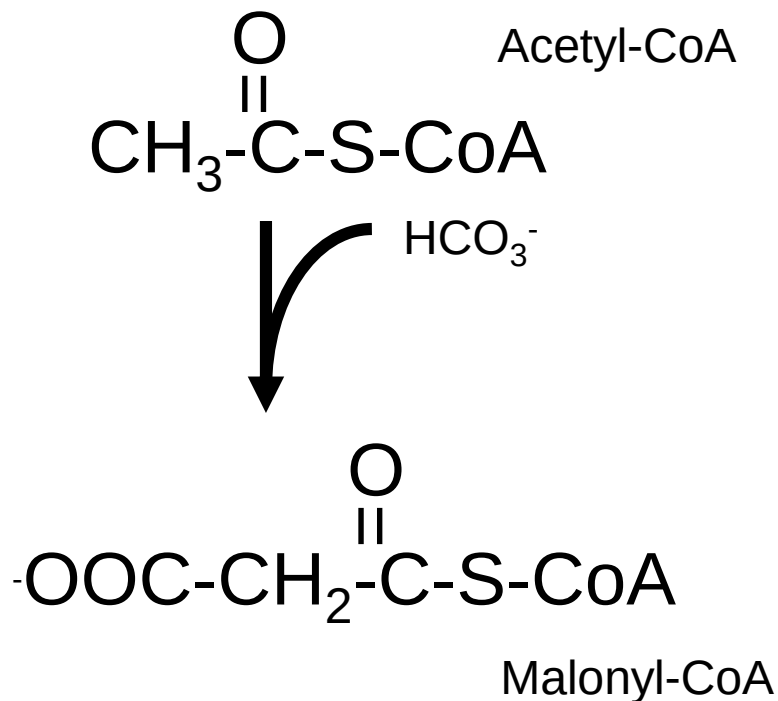
REGULATION OF FATTY ACID SYNTHESIS

When a cell has more energy, the excess is generally converted to Fatty Acids and stored as lipids such as triacylglycerol.



REGULATION OF FATTY ACID SYNTHESIS

- The reaction catalyzed by acetyl-CoA carboxylase is the rate limiting step in the biosynthesis of fatty acids.



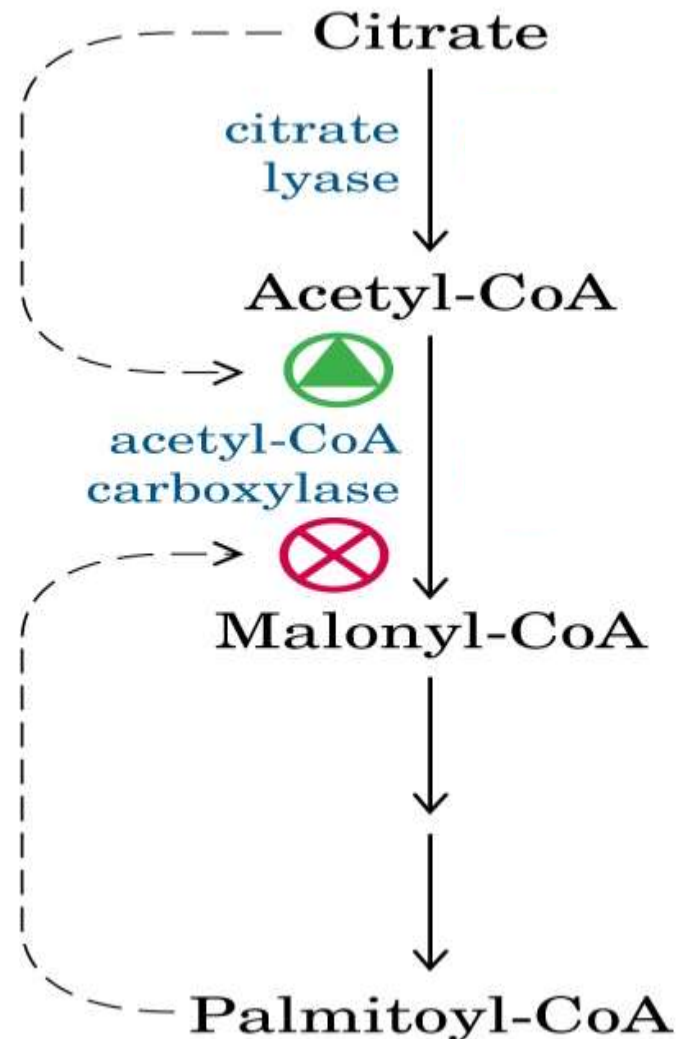
The mammalian enzyme is **regulated**, by

- ♦ Allosteric control by local metabolites
- ♦ Phosphorylation
- ♦ **Conformational changes** associated with regulation:
 - ♦ In the **active** conformation, Acetyl-CoA Carboxylase associates to form multimeric **filamentous** complexes.
 - ♦ Transition to the **inactive** conformation is associated with dissociation to yield the **monomeric** form of the enzyme (protomer).

REGULATION OF ACETYL-CoA CARBOXYLASE

Allosteric control

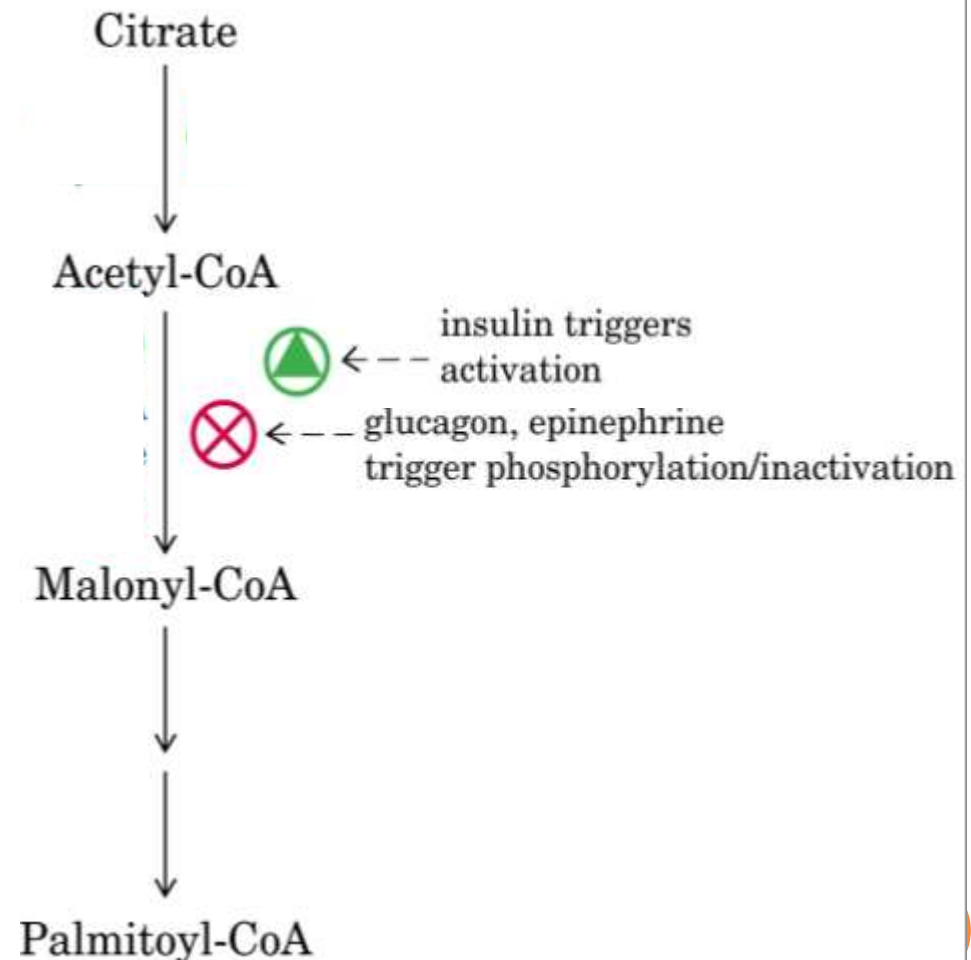
- ❑ Palmitoyl-CoA acts as a feedback inhibitor of the enzyme, and **citrate** is an activator.
- ❑ When there is an increase in mitochondrial acetyl-CoA and ATP, citrate is transported out of mitochondria
- ❑ **Citrate** becomes both the precursor of cytosolic acetyl-CoA and a signal for the activation of **acetyl-CoA carboxylase**.



REGULATION OF ACETYL-COA CARBOXYLASE

□ Phosphorylation

- Acetyl-CoA carboxylase is also regulated by hormones such as **glucagon**, **epinephrine**, and **insulin** via changes in its phosphorylation state



REGULATION OF ACETYL-COA CARBOXYLASE

- ❑ Additionally, these pathways are regulated at the level of gene expression
- ❑ Long-chain fatty acid synthesis is controlled in the **short term** by allosteric and covalent modification of enzymes and in the **long term** by changes in gene expression governing rates of synthesis of enzymes.



NUTRITIONAL STATE REGULATES LIPOGENESIS

- ❑ Excess carbohydrates is stored as fat in many animals in anticipation of periods of caloric deficiency such as **starvation, hibernation**, etc, and to provide energy for use between meals in animals, including humans, that take their food at spaced intervals.
- ❑ The nutritional state of the organism is the main factor regulating the rate of lipogenesis.



FATTY ACID SYNTHESIS DURING FED STATE

- ❑ The rate is **higher in the well-fed state** if the diet contains a high proportion of carbohydrate
- ❑ Lipogenesis converts surplus glucose and intermediates **such as** pyruvate, lactate, and acetyl-CoA to fat, assisting the anabolic phase of this feeding cycle
- ❑ Lipogenesis is **increased when sucrose is fed** instead of glucose because fructose bypasses the **phosphofructokinase** control point in glycolysis and floods the lipogenic pathway



FATTY ACID SYNTHESIS DURING FASTING

- ❑ It is depressed by restricted **caloric intake, high fat diet, or a deficiency of insulin, as in diabetes mellitus**
- ❑ These conditions are associated with increased concentrations of plasma free fatty acids
- ❑ An inverse relationship has been demonstrated between **hepatic lipogenesis and the concentration of serum-free fatty acids.**



ROLE OF INSULIN IN FATTY ACID SYNTHESIS

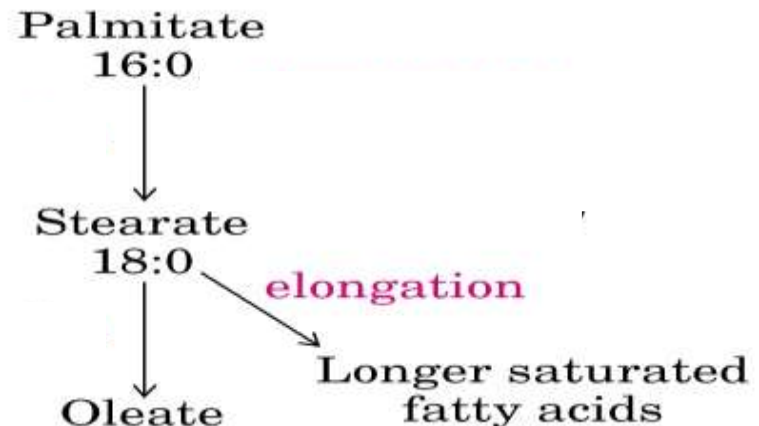
- ❑ **Insulin** stimulates lipogenesis by several other mechanisms as well as by increasing acetyl-CoA carboxylase activity.
- ❑ It **increases the transport of glucose** into the cell (eg, in adipose tissue),
- ❑ Increases the **availability of both pyruvate for fatty acid synthesis and glycerol 3-phosphate** for esterification of the newly formed fatty acids,
- ❑ Insulin converts the inactive form of pyruvate dehydrogenase to the active form in adipose tissue but not in liver, thus provides more of Acetyl co A
- ❑ Insulin also acts by **inhibiting c AMP mediated lipolysis** in adipose tissue and thereby reduces the concentration of plasma free fatty acids (long-chain fatty acids are inhibitors of lipogenesis).



- Insulin is a powerful anabolic signal, particularly in hepatocytes and adipocytes, where it induces synthesis of the lipogenic family of enzymes, which includes acetyl- CoA carboxylase, citrate lyase, the malic enzyme, glucose 6-phosphate dehydrogenase, pyruvate kinase, and the FAS complex.
 - The mechanism underlying this action by insulin involves activation of the sterol regulatory element-binding protein- 1 (SREBP- 1), a membrane-bound transcription factor that enhances transcription of the genes encoding proteins required for fatty acid synthesis. Glucagon, on the other hand, represses de novo synthesis of these enzymes in adipocytes and liver, and stimulates degradation of the lipogenic family of enzyme proteins. AMP-activated protein kinase also suppresses expression of fatty acid synthase, acetyl-CoA carboxylase, and citrate lyase.
- 

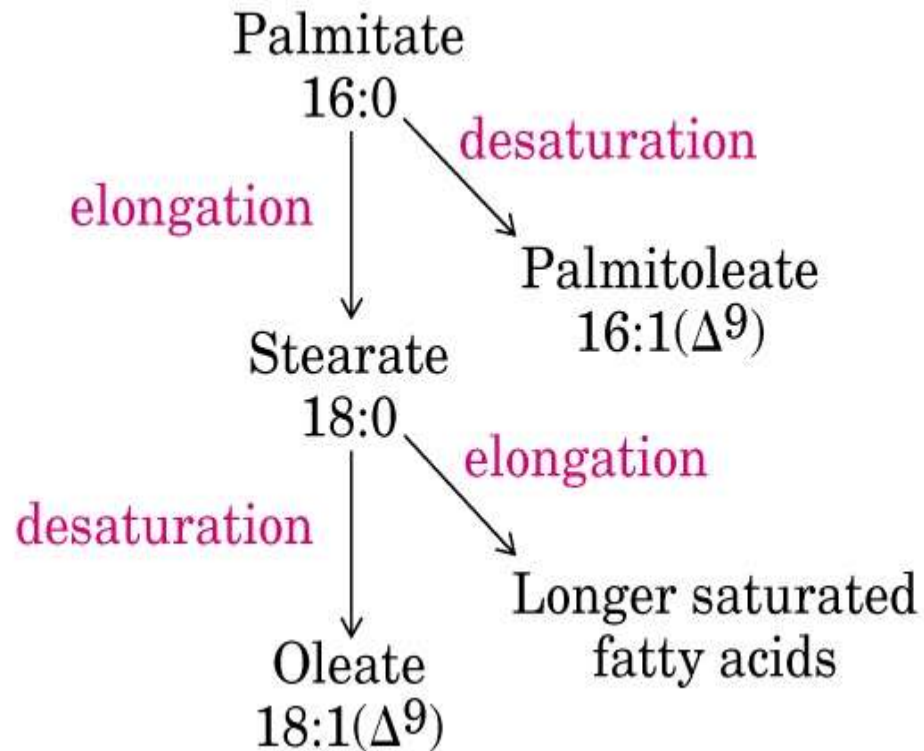
FATTY ACID ELONGATION

- ❑ **Palmitate** in animal cells is the precursor of other long-chained FAs.
- ❑ Elongation of fatty acids occurs primarily in the **endoplasmic reticulum** and utilizes **malonyl-CoA** to add two-carbon units to long-chain fatty acyl-CoAs. (microsomal chain elongation system, elongase)
- ❑ There is a minor, secondary chain elongation system (elongase) in **mitochondria** that utilizes **acetyl-CoA** as the two-carbon donor



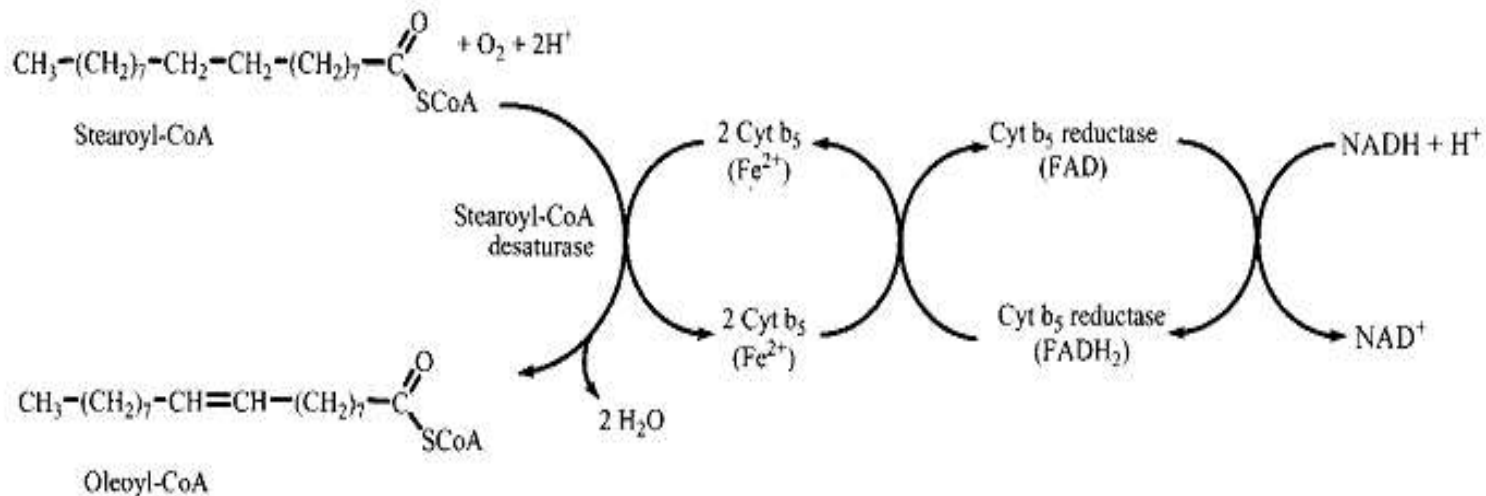
BIOSYNTHESIS OF UNSATURATED FATTY ACIDS

- **Palmitate and stearate** serve as precursors of the two most common monounsaturated fatty acids of animal cells: palmitoleate ($16:1\Delta^9$), and Oleate ($18:1\Delta^9$).
- The double bond is introduced by **fatty acyl-CoA desaturase** in the **smooth endoplasmic reticulum**.

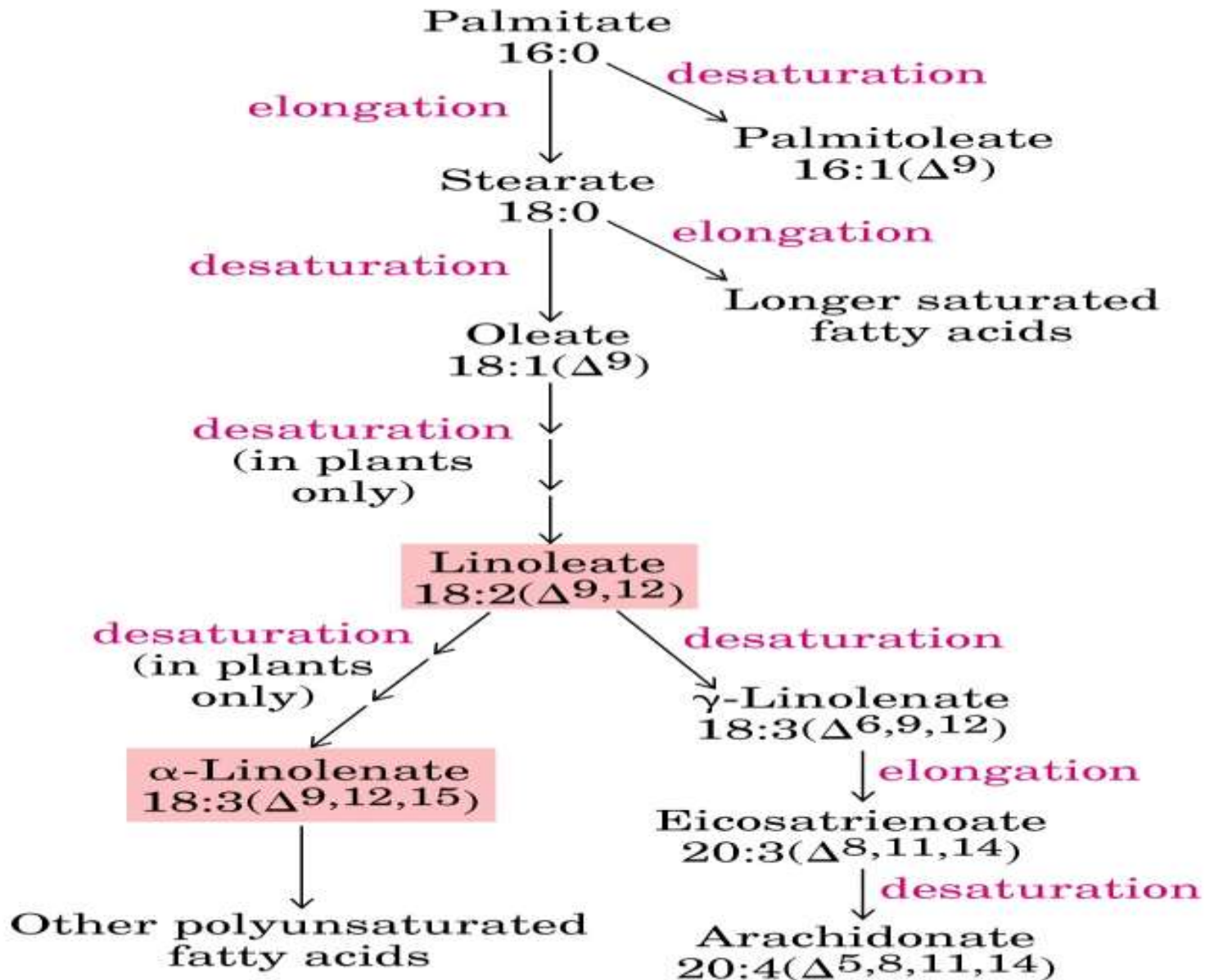


BIOSYNTHESIS OF UNSATURATED FATTY ACIDS

- ❑ **Palmitate** and **stearate** serve as precursors of the two most common monounsaturated fatty acids of animal tissues, **palmitoleate** and **oleate**.
- ❑ The double bond is introduced into the fatty acid chain by an oxidative reaction catalyzed by **fatty acyl-CoA desaturase**. The enzyme is an example of a mixed-function oxidase



ESSENTIAL FATTY ACIDS



EICOSANOIDS

Compounds containing a 20-carbon core

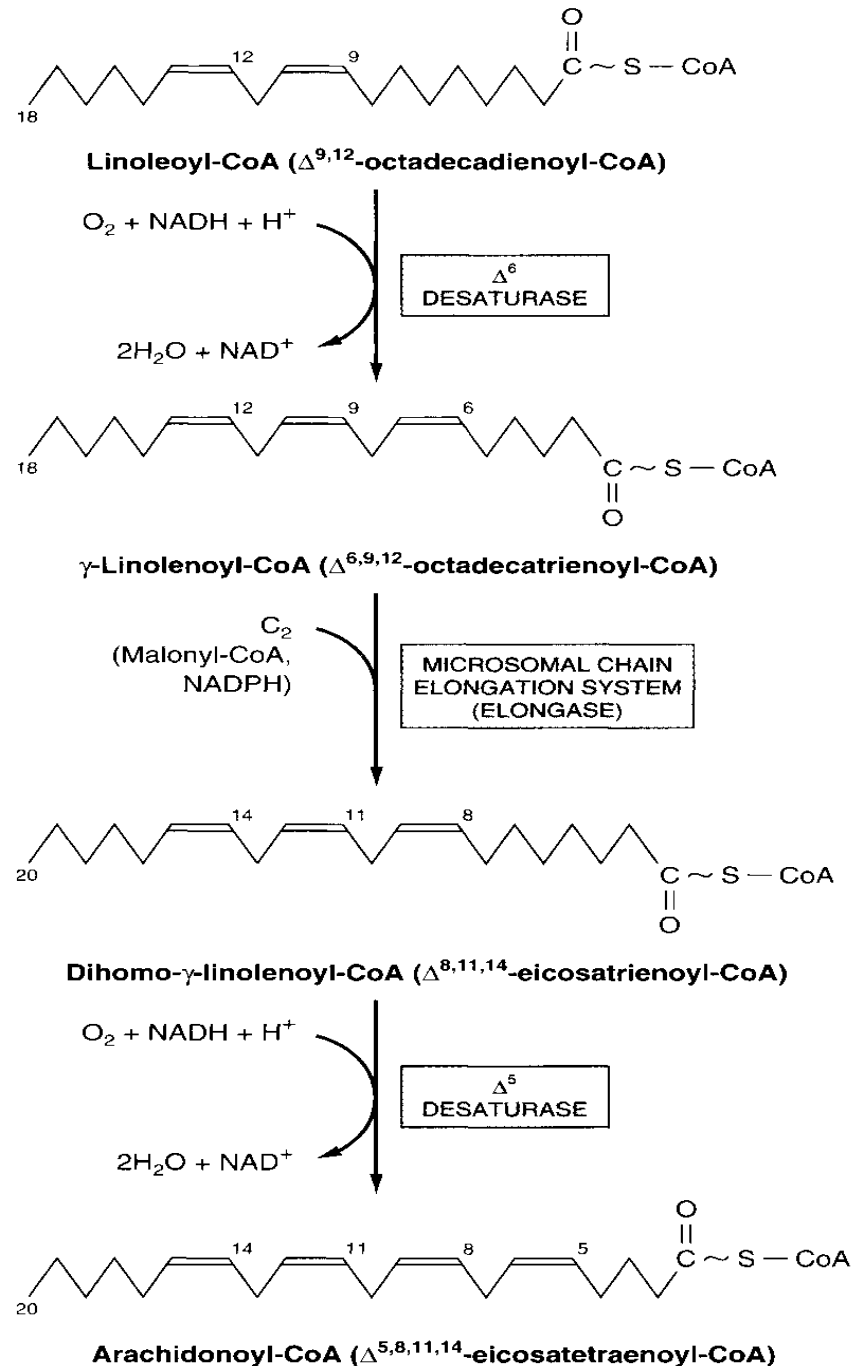
Comprise:

- prostaglandins
 - thromboxanes
 - leukotrienes
 - lipoxins
 - hydroxyeicosatetraenoic acids (HETEs)
 - hepoxilins
- } prostanoids

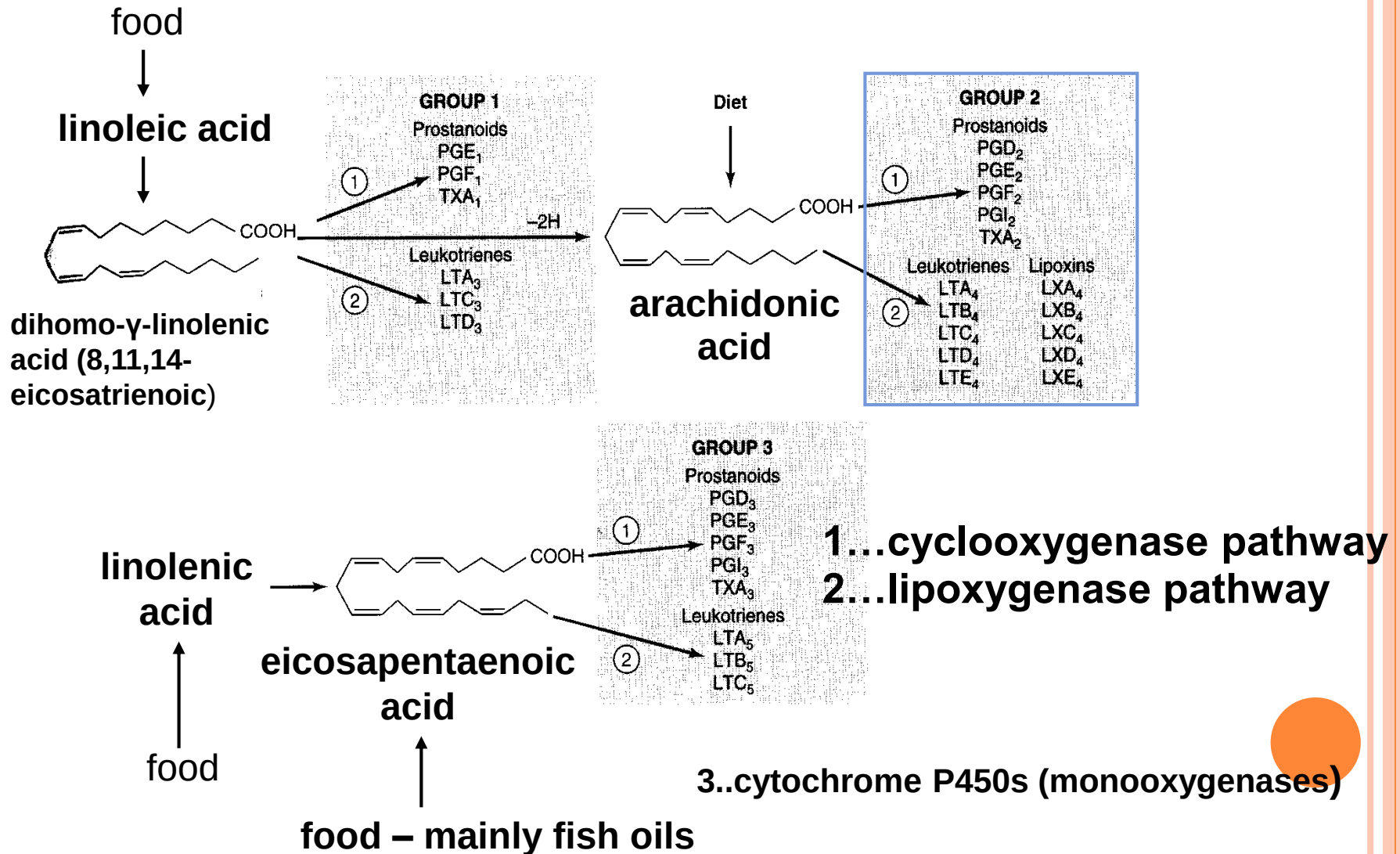


□ In humans, arachidonic acid is formed from **linoleic acid**:

□ In humans, the double bonds cannot be introduced beyond the Δ^9 position $\Rightarrow$ linoleic and linolenic acids are **essential**: must be supplied in food (plant oils, peanut, soybean, corn)



EICOSANOID PRODUCTION FROM PUFAS



MAIN SITES OF EICOSANOID BIOSYNTHESIS

- ❑ Endothelial cells
 - ❑ Leukocytes
 - ❑ Platelets
 - ❑ Kidney
-
- Eicosanoids are NOT synthesized in advance and stored in granules – when needed, they can be produced very quickly from arachidonate released from membranes



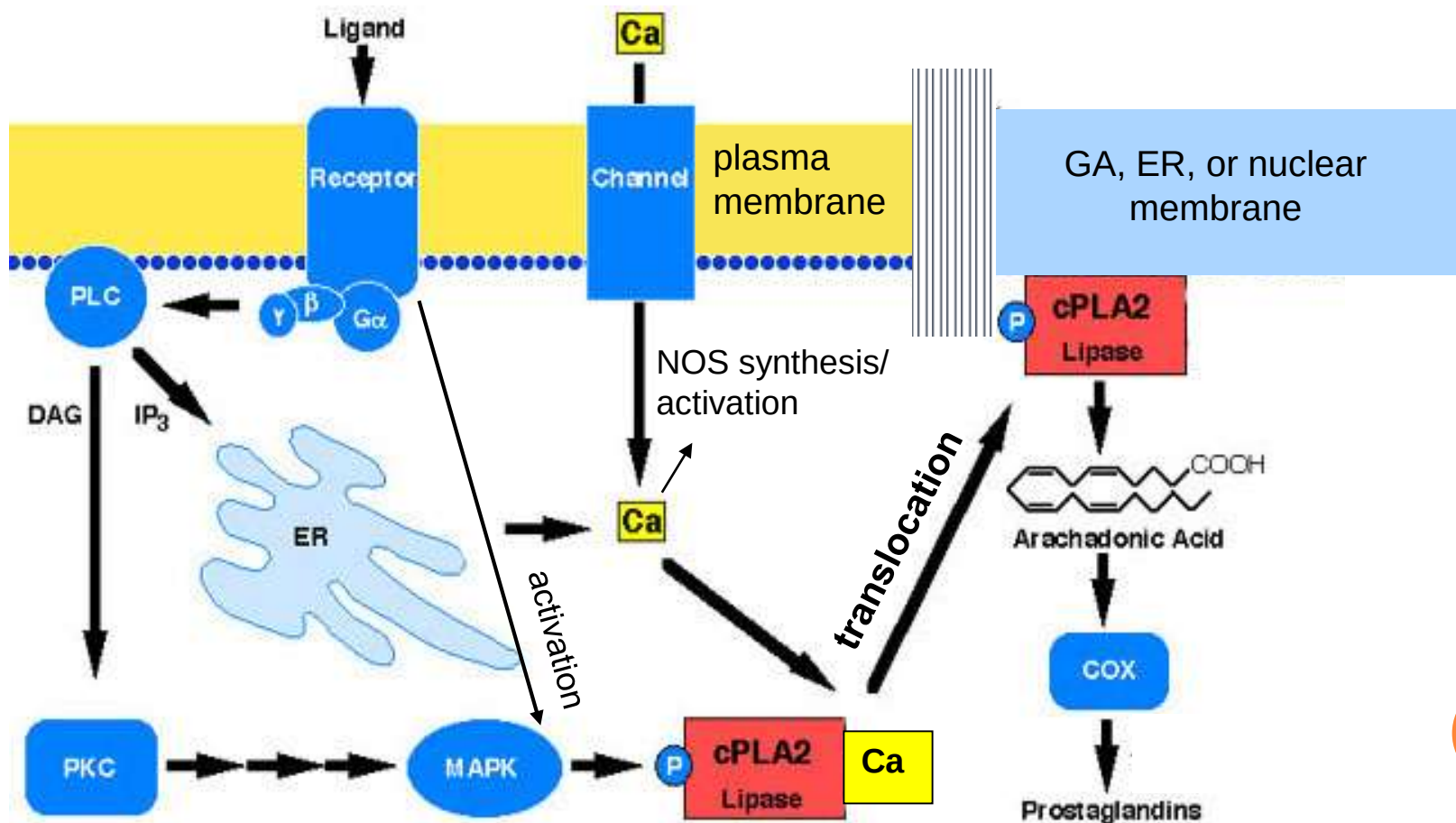
MAIN STEPS OF EICOSANOID BIOSYNTHESIS

- 1) Activation of phospholipase A₂ (PLA₂)
- 2) Release of arachidonate from membrane phospholipids by PLA₂
- 3) Eicosanoid synthesis: COX or LO pathway + subsequent cell-specific modifications by synthases / isomerases (conversion of the precursor PGH₂ to another prostanoid, conversion of LTA₄...)



1) PHOSPHOLIPASE A₂ ACTIVATION

Ligand binding to a receptor induces phospholipase C (PLC) activation → PLC cleaves PIP₂ to DAG and IP₃ that opens the Ca²⁺ channels in the ER. PLA₂, activated by Ca²⁺ and probably also by phosphorylation (MAPK), translocates to membranes of GA, ER, or nucleus from which it releases arachidonate for here residing COX/LO.



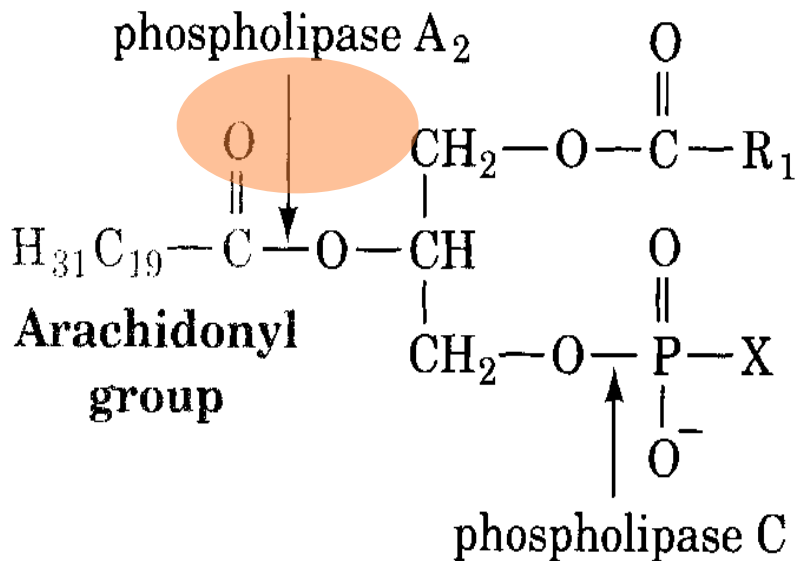
PLA₂ expression / activity is stimulated by:

- Interleukin-1
- Angiotensin II
- Bradykinin
- EGF
- Thrombin
- Epinephrine...



2) ARACHIDONATE RELEASE FOR EICOSANOID SYNTHESIS

- From membrane phospholipids – mainly by the action of phospholipase A₂:



Arachidonate release from phospholipids can be blocked by the anti-inflammatory steroids!



3) EICOSANOID BIOSYNTHESIS

In almost all cell types (except for red blood cells)

3 pathways:

A) Cyclooxygenase (COX) – produces **prostaglandins** and **thromboxanes**

B) Lipoxygenase (LO) – produces leukotrienes, **lipoxins**, **12- and 15-HETEs**, and **hepoxilins**

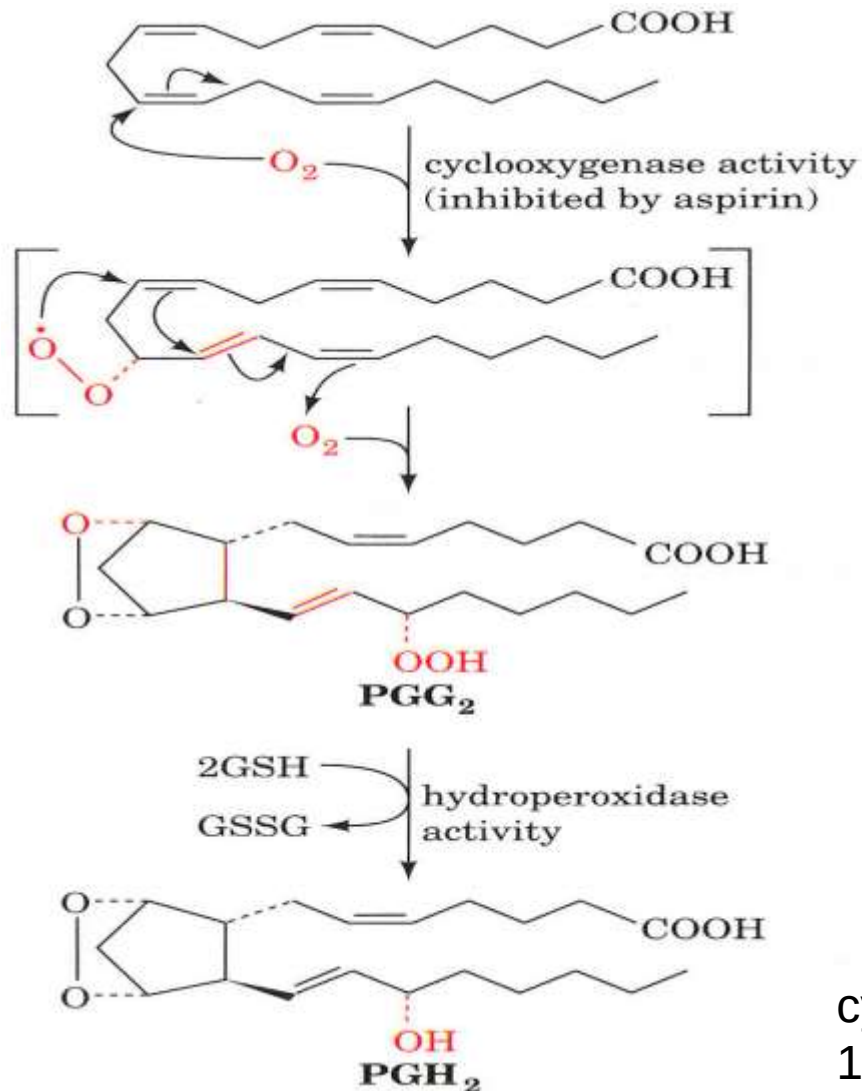
C) Cytochrome P450s (monooxygenases) – produce the other HETEs (20-HETE); principal pathway in the proximal tubules



A) CYCLOOXYGENASE (COX) PATHWAY

- **Prostaglandin H synthase**, present as two isoenzymes (PGHS-1/COX-1, PGHS-2/COX-2), each possessing two activities:
 - **cyclooxygenase** – catalyzes addition of two molecules of O_2 to the arachidonic acid molecule, forming PGG_2
 - **hydroperoxidase** – converts the hydroperoxy function of PGG_2 to an OH group (of PGH_2)
- Mostly, a given cell type produces 1 type of prostanoids: platelets produce almost exclusively thromboxanes, vascular endothelial cells prostacyclins, heart muscle makes PGI_2 , PGE_2 , $PGF_{2\alpha}$

PROSTAGLANDIN H SYNTHASE

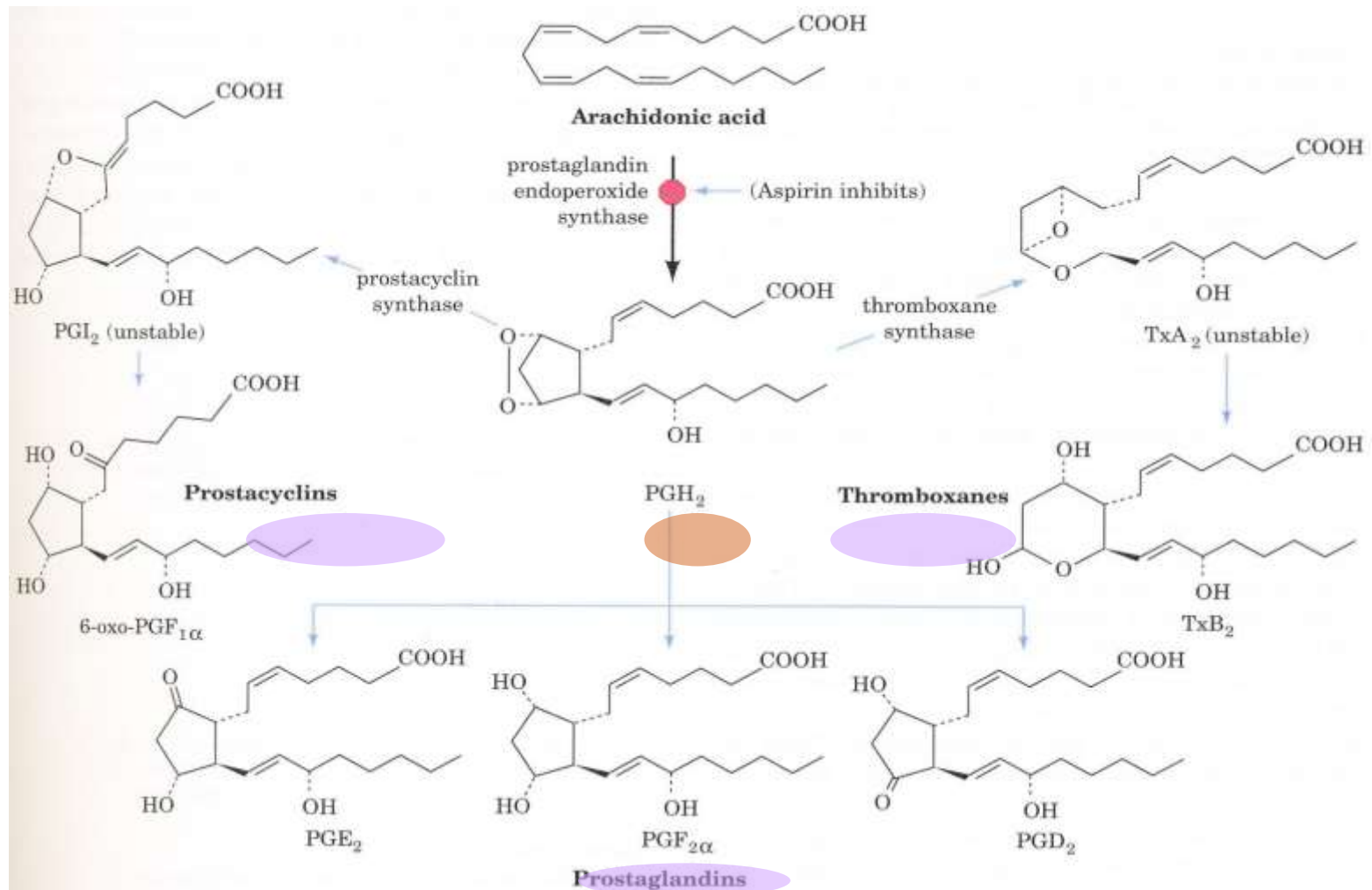


- PGH₂ = precursor of all series 2 prostaglandins and thromboxanes

cyclic 9,11-endoperoxide,
15-hydroperoxide is formed

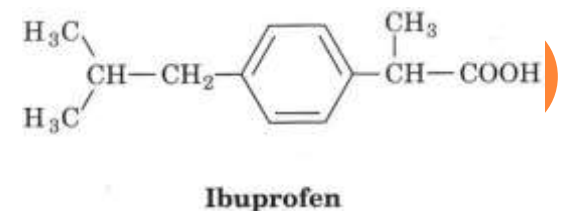
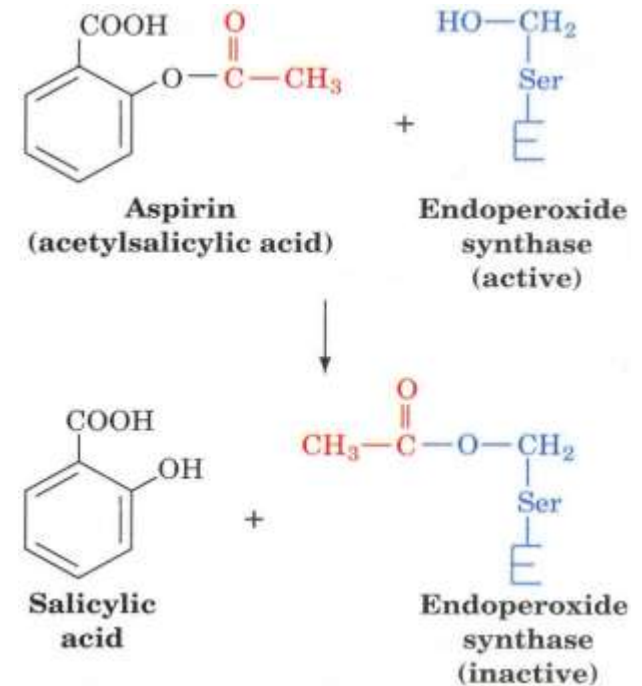
PRODUCTS OF THE COX PATHWAY

- ❑ Platelets contain thromboxane synthase producing TXA_2 , TXB_2
- ❑ Vascular endothelial cells contain prostacyclin synthase which converts PGH_2 to prostacyclin PGI_2

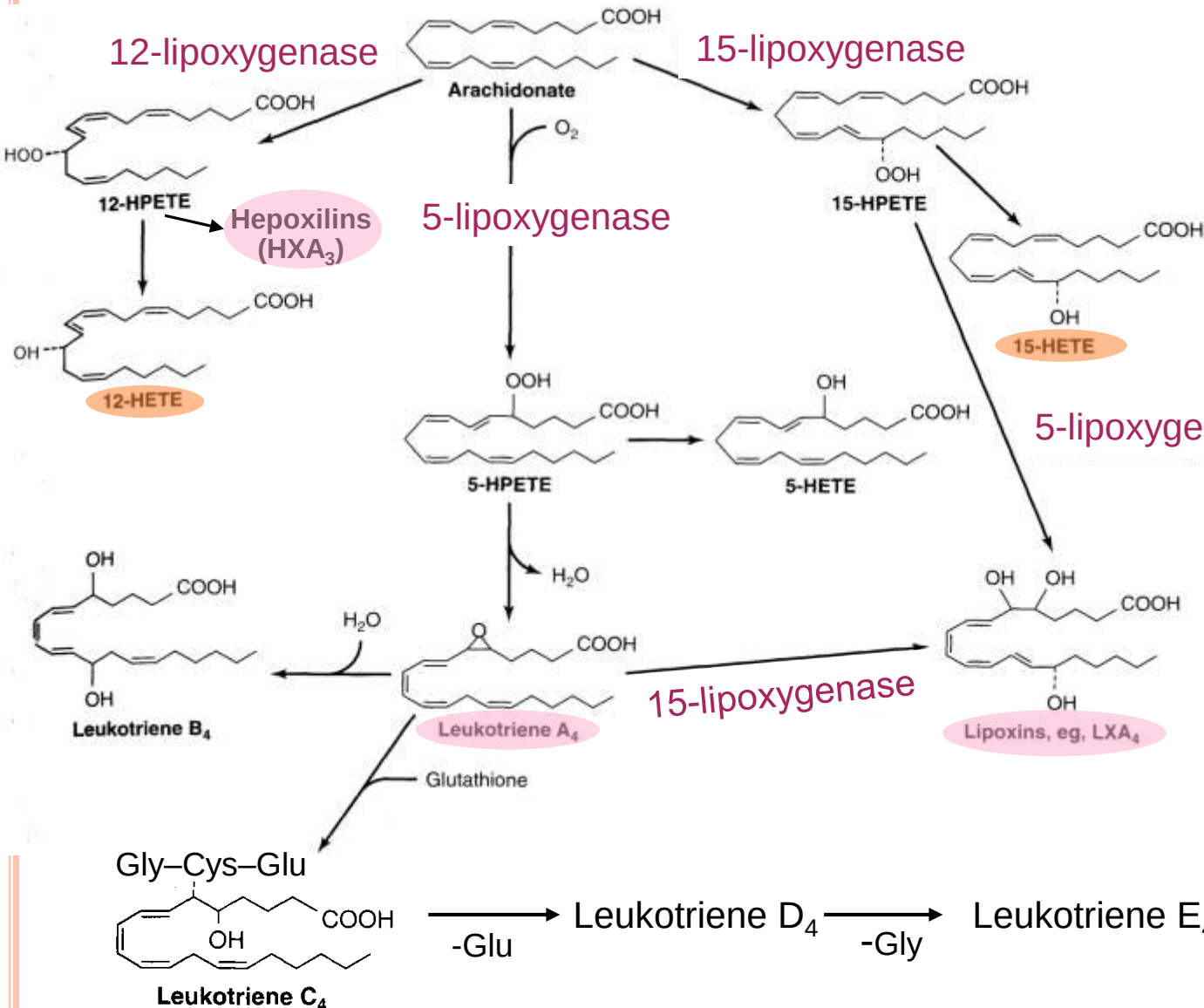


INHIBITION OF THE COX PATHWAY

- Aspirin inhibits the COX activity of both PGHS-1 and PGHS-2 (by acetylation of a distinct Ser of the enzyme)
- Other nonsteroidal anti-inflammatory drugs (NSAIDs) also inhibit the COX activity (ibuprofen competes with arachidonate)
- Transcription of PGHS-2 can be blocked by anti-inflammatory corticosteroids



B) LIPOXYGENASE (LO) PATHWAY



- 3 different lipoxygenases insert oxygen into the 5, 12, or 15 position of arachidonate; the first product is the hydroperoxy-eicosatetraenoic acid (HPETE)

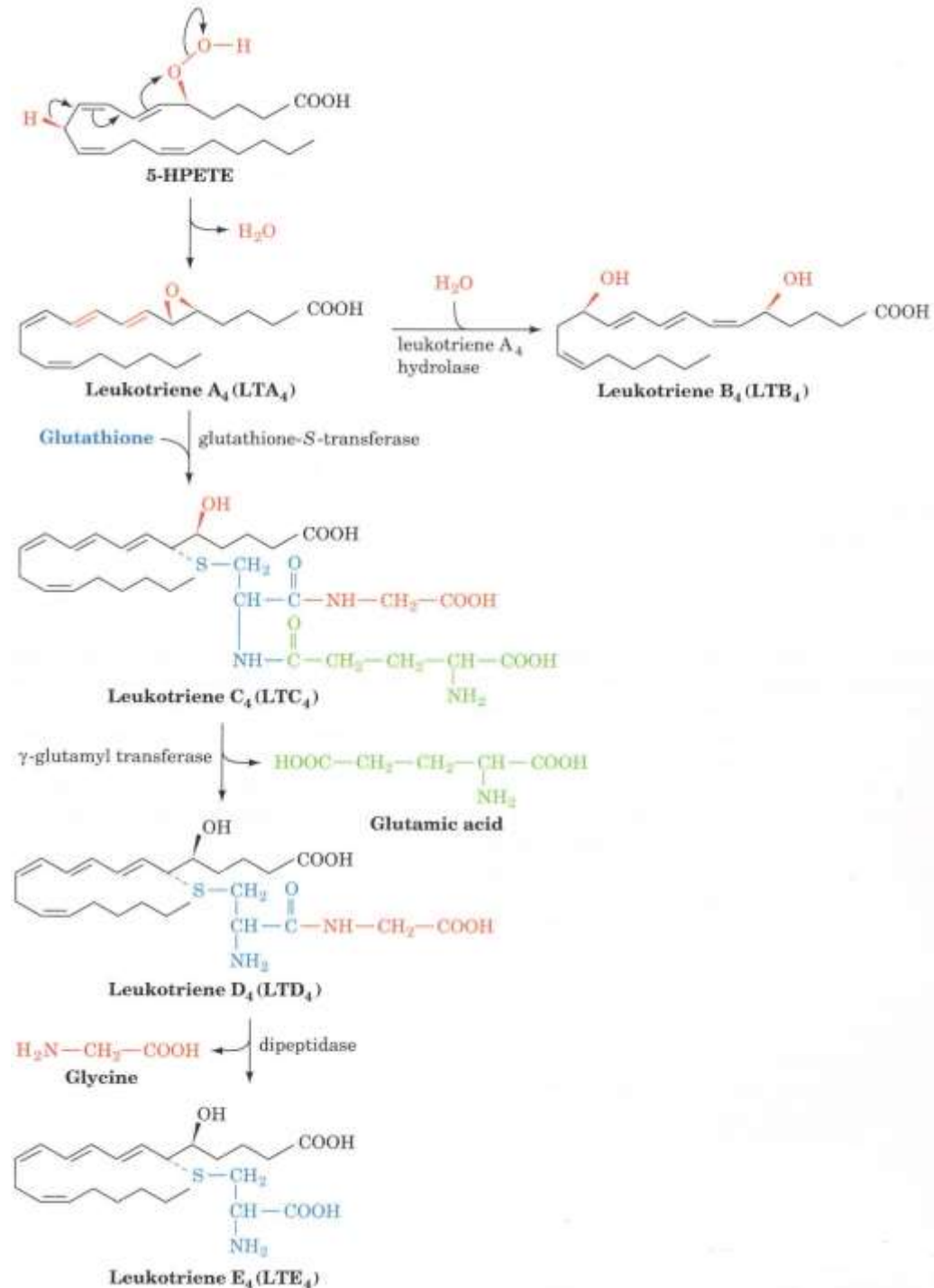
- Only 5-lipoxygenase produces leukotrienes; requires protein FLAP

peptidoleukotrienes

PEPTIDOLEUKOTRIENE BIOSYNTHESIS:

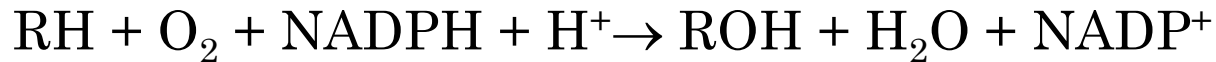
Requires glutathione!!!

Peptido-leukotrienes are short-lived organic molecules known to have potent biological effects as mediators of inflammation, hypersensitivity and respiratory



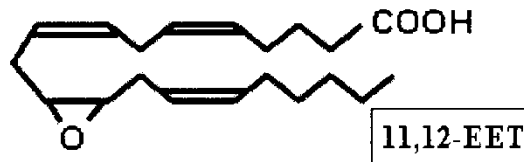
C) EICOSANOID SYNTHESIS BY CYP450s

- Cytochrome P450s – monooxygenases:

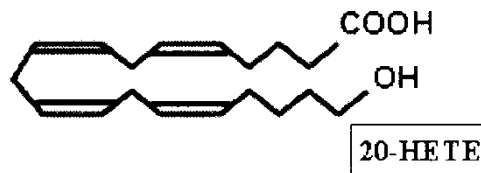


- Two main classes of compounds are formed:

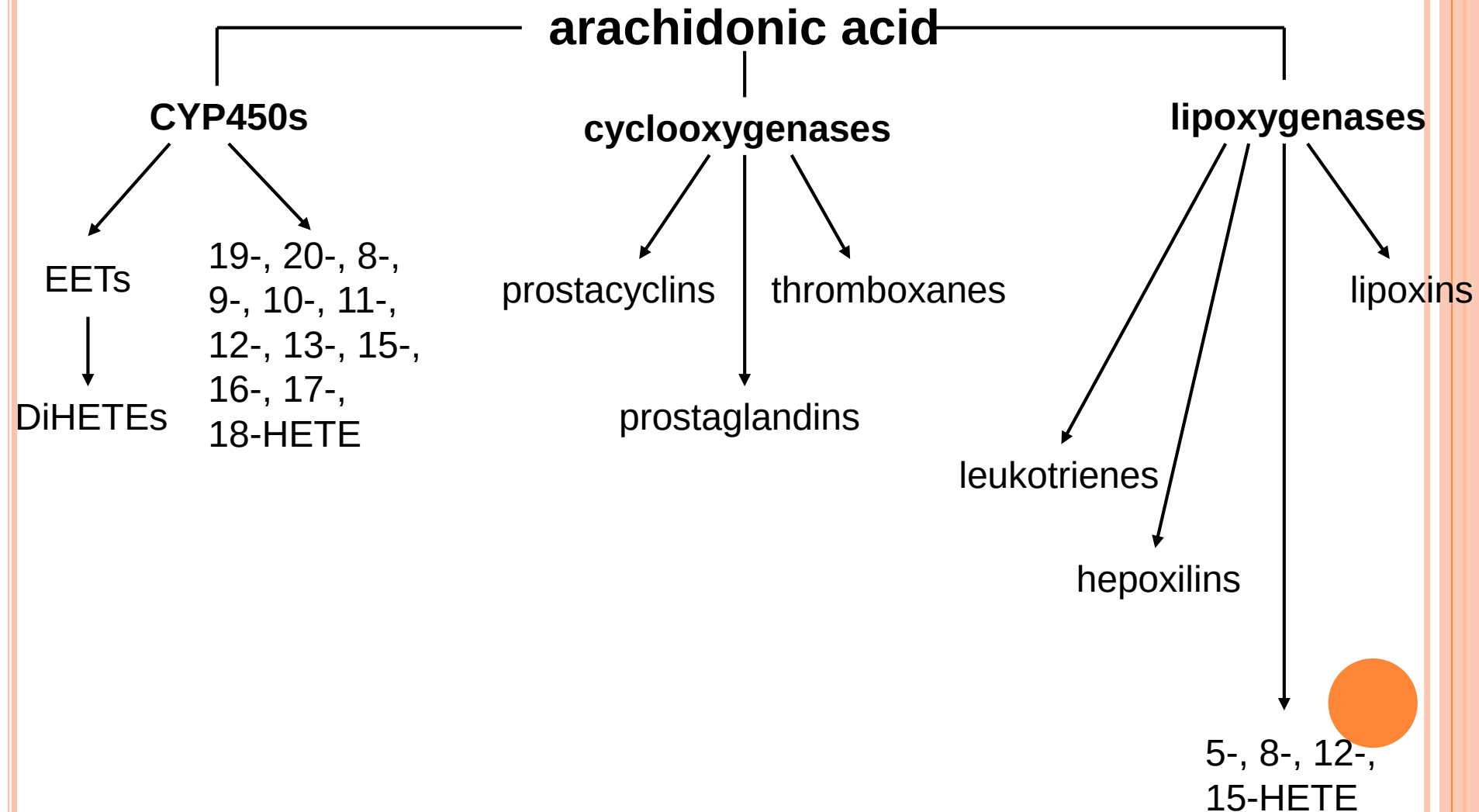
- Epoxygenases catalyze the formation of **epoxyeicosatrienoic acids (EETs)** that are further metabolized by epoxide hydrolases to dihydroxyeicosatrienoic acids (DiHETEs) which are almost inactive:



- Hydroxylases catalyze the formation of HETEs (20-HETE, 13-HETE...)



SUMMARY OF THE PRODUCTS



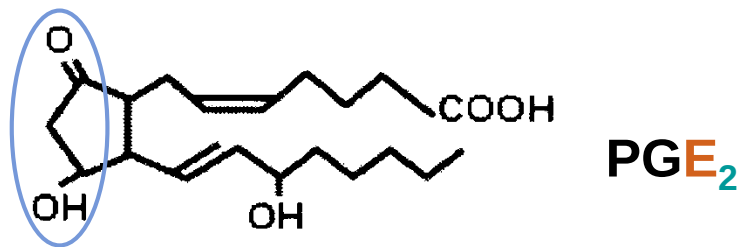
STRUCTURAL FEATURES

- ❑ Prostaglandins – cyclopentane ring
- ❑ Thromboxanes – six-membered oxygen-containing ring
- ❑ Leukotrienes – 3 conjugated double bonds + one more unconjugated
- ❑ Lipoxins – conjugated trihydroxytetraenes



PROSTAGLANDIN NOMENCLATURE

- ❑ The three classes A, E, F (third letter) are distinguished on the basis of the functional groups about the cyclopentane ring
- ❑ The subscript numerals refer to the number of double bonds in the side chains
- ❑ The subscript α refers to the configuration of the 9-OH group (projects down from the plane of the ring)



E... β -hydroxyketone
2 double bonds





BIOLOGICAL EFFECTS OF EICOSANOIDS



- ❑ Eicosanoids, like hormones, display profound effects at extremely low concentrations (10^{-9} TO 10^{-14})
- ❑ They have a very short half-life; thus, they act in an **autocrine or paracrine** manner (unlike hormones)
- ❑ Biological effects depend not only on the particular eicosanoid but also on the local availability of receptors that it can bind to



IN GENERAL, EICOSANOIDS MEDiate

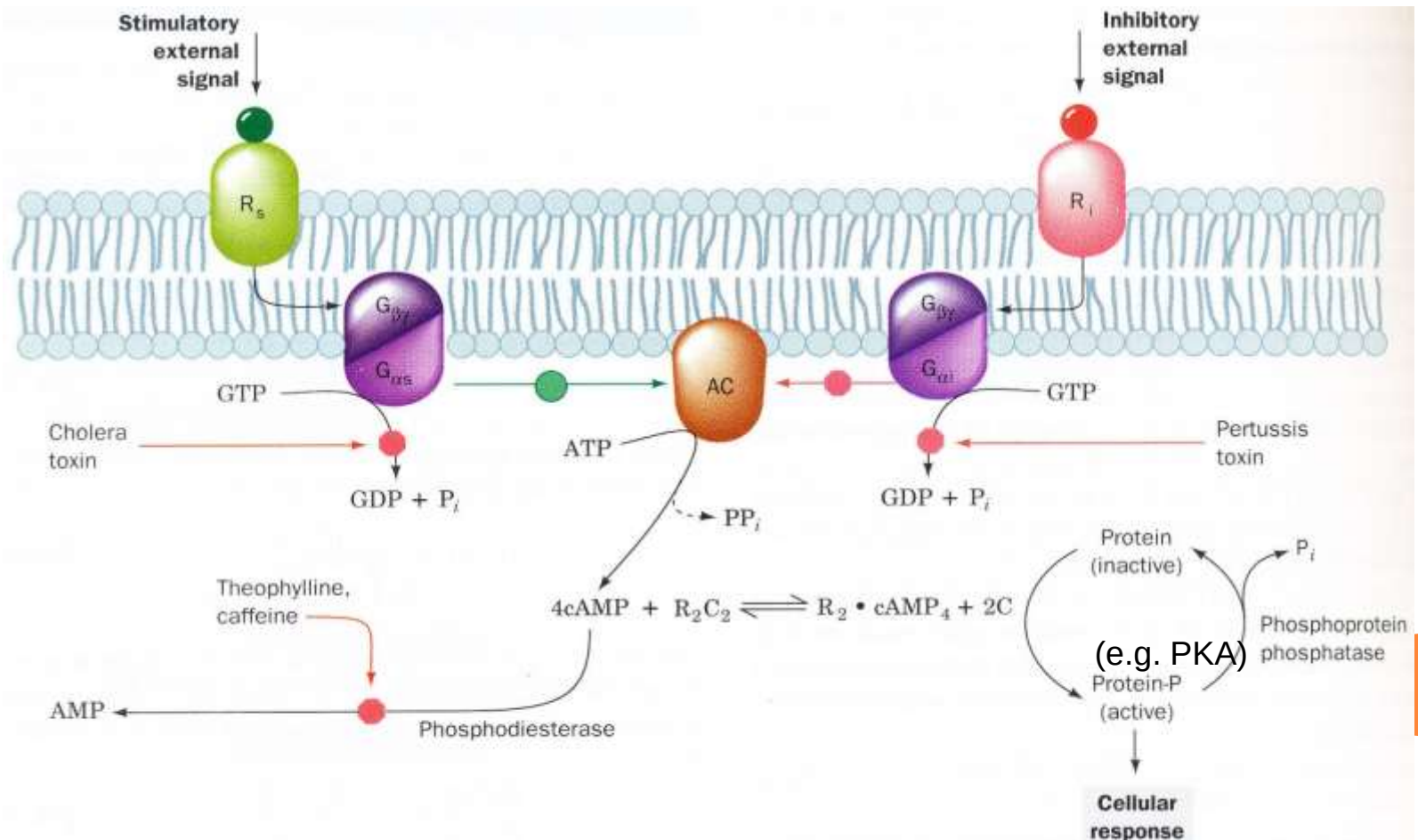
- ❑ Inflammatory response, notably as it involves the joints (rheumatoid arthritis), skin (psoriasis), and eyes
- ❑ Production of pain and fever
- ❑ Regulation of blood pressure
- ❑ Regulation of blood clotting
- ❑ Regulation of renal function
- ❑ Control of several reproductive functions, such as the induction of labor



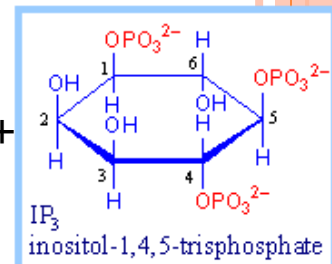
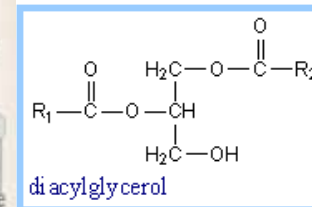
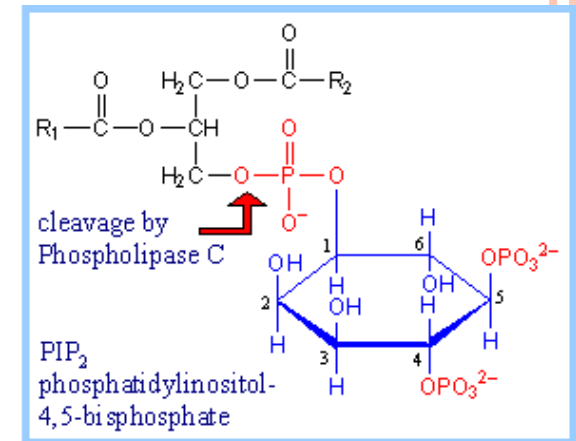
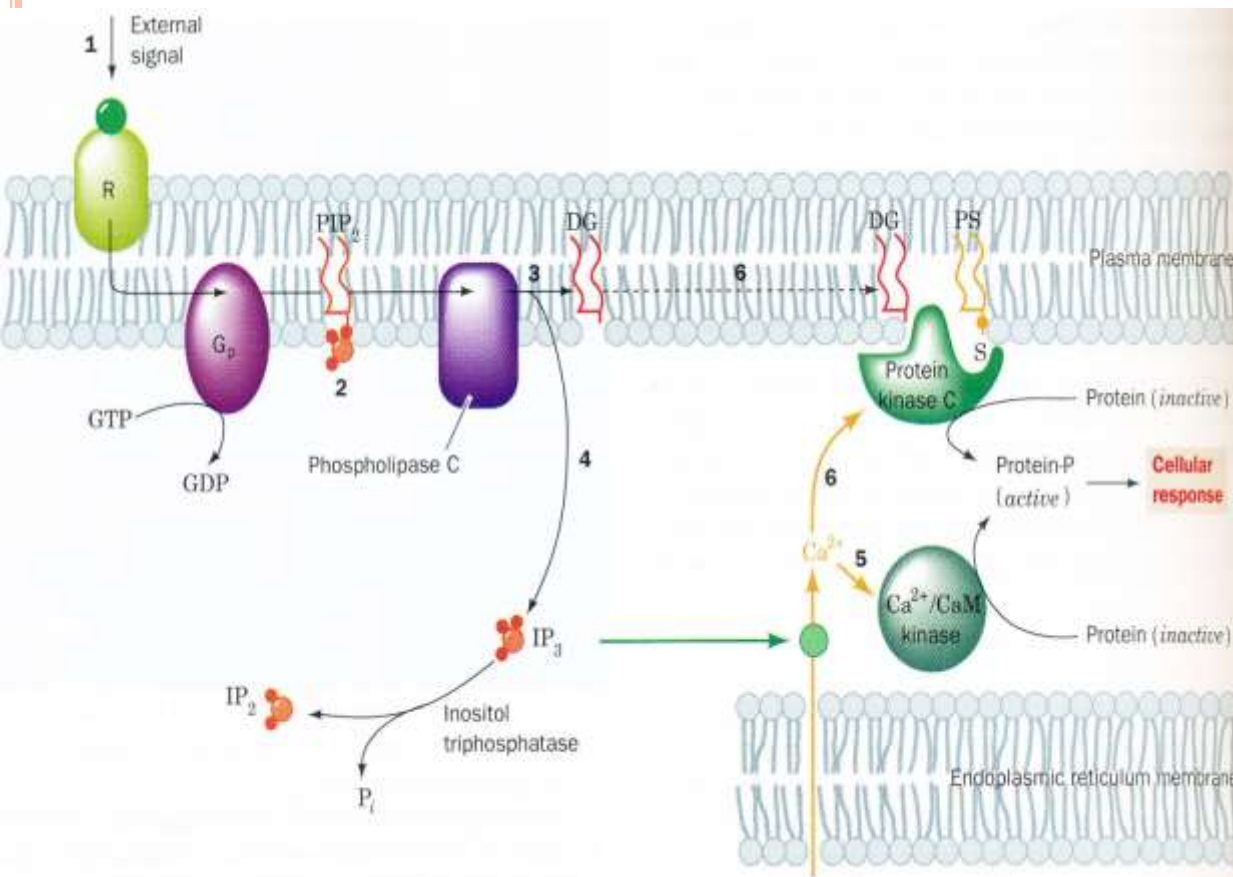
MECHANISMS OF ACTION

Via the G protein-coupled receptors:

- a) $G_{\alpha s}$ stimulate adenylate cyclase (AC)
- b) $G_{\alpha i}$ inhibit adenylate cyclase



- c) G_q activates **phospholipase C** that cleaves phosphatidylinositol-4,5-bisphosphate (PIP_2) to inositol-1,4,5-trisphosphate (IP_3) and diacylglycerol (DAG); DAG together with Ca^{2+} activates protein kinase C, IP_3 opens Ca^{2+} channels of the ER



EFFECTS OF PROSTAGLANDINS

- ❑ Mediate **inflammation**:
 - cause vasodilation $\Rightarrow$ redness, heat (PGE_1 , PGE_2 , PGD_2 , PGI_2)
 - increase vascular permeability $\Rightarrow$ swelling (PGE_2 , PGD_2 , PGI_2)
- ❑ Regulate **pain and fever** (PGE_2)
- ❑ PGE_2 , PGF_2 stimulate uterine muscle contractions during labor
- ❑ Prostaglandins of the PGE series **inhibit gastric acid secretions** (synthetic analogs are used to treat gastric ulcers)
- ❑ Regulate **blood pressure**: vasodilator prostaglandins PGE , PGA , and PGI_2 lower systemic arterial pressure
- ❑ Regulate **platelet aggregation**: PGI_2 = potent inhibitor of platelet aggregation
- ❑ PGE_2 inhibits reabsorption of Na^+ and water in the collecting duct.
 PGI_2 : vasodilatation and regulation of glomerular filtration rate.

BIOLOGICAL ROLE OF THROMBOXANES

- ❑ Thromboxanes are synthesized by platelets and, in general, cause **vasoconstriction and platelet aggregation**
- ❑ TXA_2 is also produced in the kidney (by podocytes and other cells) where it causes vasoconstriction and mediates the response to ANGII
- ❑ Thus, **both thromboxanes and prostaglandins (PGI_2) regulate coagulation**
 - ❑ In Eskimos, higher intake of eicosapentaenoic acid and group 3 prostanoids may be responsible for low incidence of heart diseases and prolonged clotting times since TXA_3 is a weaker aggregator than TXA_2 and both PG_3 and TXA_3 inhibit arachidonate release and TXA_2 formation



BIOLOGICAL ROLE OF LEUKOTRIENES

- ❑ LTs are produced mainly in leukocytes that also express receptors for LTs
- ❑ Leukotrienes are very potent **constrictors of the bronchial airway muscles**: (LTC_4 , LTD_4 , and LTE_4 = *the slow-reacting substance of anaphylaxis*)
- ❑ They **increase vascular permeability**
- ❑ They cause **attraction** (LTB_4) and **activation of leukocytes** (primarily eosinophils and monocytes), promote **diapedesis** (increase expression of integrins on the leukocyte surface), enhance **phagocytosis**
- ❑ They regulate vasoconstriction



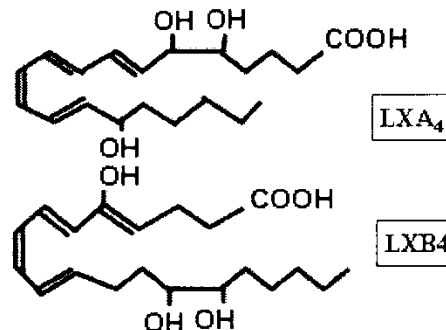
they regulate inflammatory reactions, host defense against infections as well as hyperreactivity (asthma...)

LIPOXINS

- Lipoxins are produced mainly by **leukocytes and platelets** stimulated by cytokines (IL-4, TGF- β):
 - a) 5-lipoxygenase (5-LO) of neutrophils produces leukotriene LTA₄ which enters platelets where it is converted by 15-LO to LXA₄ or LXB₄
 - b) 15-LO of epithelial cells and monocytes forms 15-HPETE which becomes a substrate of 5-LO and epoxid hydrolase of leukocytes

...transcellular biosynthesis

- Main products: LXA₄, LXB₄



BIOLOGICAL ROLES OF LIPOXINS

- Unlike pro-inflammatory eicosanoids, lipoxins **attenuate the inflammation** and appear to facilitate the resolution of the acute inflammatory response
- Hypothesis: in the first phase of the inflammatory response, leukotrienes are produced (e.g. LTB_4) → then, the level of PGs rises and PGs „switch“ the synthesis from leukotriene production to the pathway which, in the 2nd phase, produces lipoxins promoting the resolution of inflammation
- Therefore, potential therapeutic use of LXs in the treatment of inflammatory diseases (glomerulonephritis, asthma) is being extensively studied

EFFECTS OF LXs MEDIATING THE RESOLUTION OF INFLAMMATION

- ❑ LXs inhibit chemotaxis of neutrophils and eosinophils and diapedesis
 - ❑ Inhibit formation of ROS (neutrophils, lymphocytes) and ONOO⁻ (neutrophils)
 - ❑ Inhibit production of specific cytokines by leukocytes
 - ❑ Stimulate non-inflammatory phagocytosis (of apoptotic neutrophils...)
 - ❑ Antagonize LT receptors
 - ❑ Affect not only the cells of the myeloid line:
 - inhibit the contraction of the bronchial smooth muscle
 - inhibit production of cytokines by the cells of colon, fibroblasts...
 - inhibit the interaction between leukocytes and endothelial cells
- 

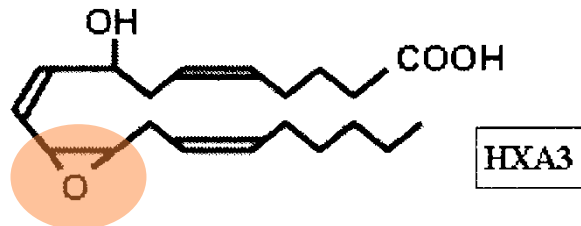
BIOLOGICAL EFFECTS OF HETES

- ❑ 5-HETE participates in **host defense** against bacterial infection (chemotaxis and degranulation of neutrophils and eosinophils)
- ❑ 20-HETE causes **vasoconstriction** (by its effect on the smooth muscle of vessels); in the kidney, it regulates Na^+ excretion, diuresis, and blood pressure
- ❑ 12- and 15-HETE are produced in the kidney and participate in the **regulation of the renin-angiotensin system** (12-HETE also mediates secretion of **aldosterone induced by ANGII**)

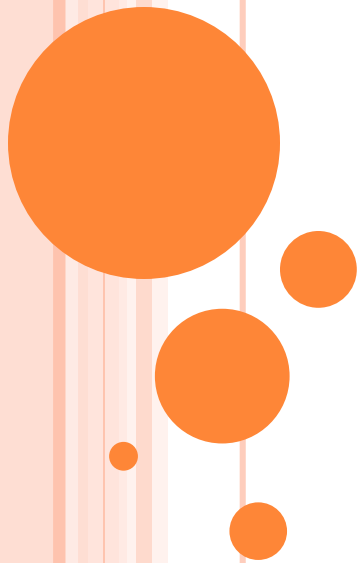


BIOLOGICAL ROLES OF HEPOXILINS

- ❑ HXA₃ stimulates glucose-induced **insulin secretion** by pancreatic β cells
- ❑ Under oxidative stress, HXA₃ formation is stimulated and HXA₃ **upregulates** the expression of **glutathione peroxidase**...compensatory defense response to protect cell viability?



BIOSYNTHESIS OF TRIACYLGLYCEROL



SYNTHESIS OF TAG

- ❑ The pathway for triacylglycerol synthesis in most tissues, including **liver and adipocytes**, utilizes **glycerol 3-phosphate and fatty acyl-CoA**.
- ❑ Synthesis of TAG takes place in endoplasmic reticulum.
- ❑ The activated fatty acids (i.e., fatty acids attached to coenzyme A) are derived either from **endogenous, de novo fatty acid synthesis or from dietary fats**.
- ❑ By contrast, triacylglycerol synthesis in the small intestine begins with **2-monoacylglycerol**.



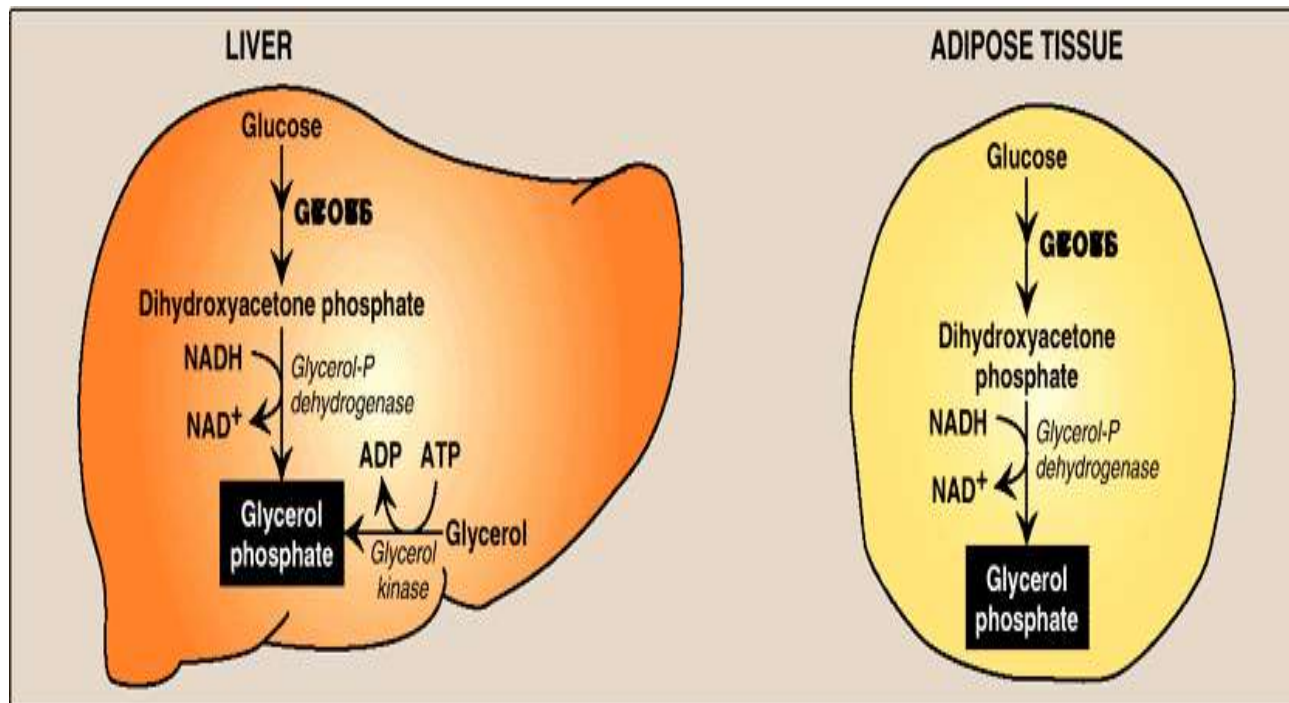
SYNTHESIS OF TRIACYLGLYCEROLS

- 1- Synthesis of glycerol phosphate
- 2- Formation of fatty acyl CoA
- 3- Formation of a molecule of TAG



SYNTHESIS OF GLYCEROL PHOSPHATE

- ❑ **In liver;** this process depends on supply of glucose or free glycerol coming to liver is converted to glycerol phosphate by enzyme Glycerol kinase.
- ❑ In adipose tissue glucose uptake is insulin dependent as it has GLUT-4 receptors . Low glucose--- low insulin --- -- no synthesis of TAG in adipocytes.



FORMATION OF ACTIVATED FREE FATTY ACID

- ❑ Long chain fatty acids are converted to fatty acyl CoA .

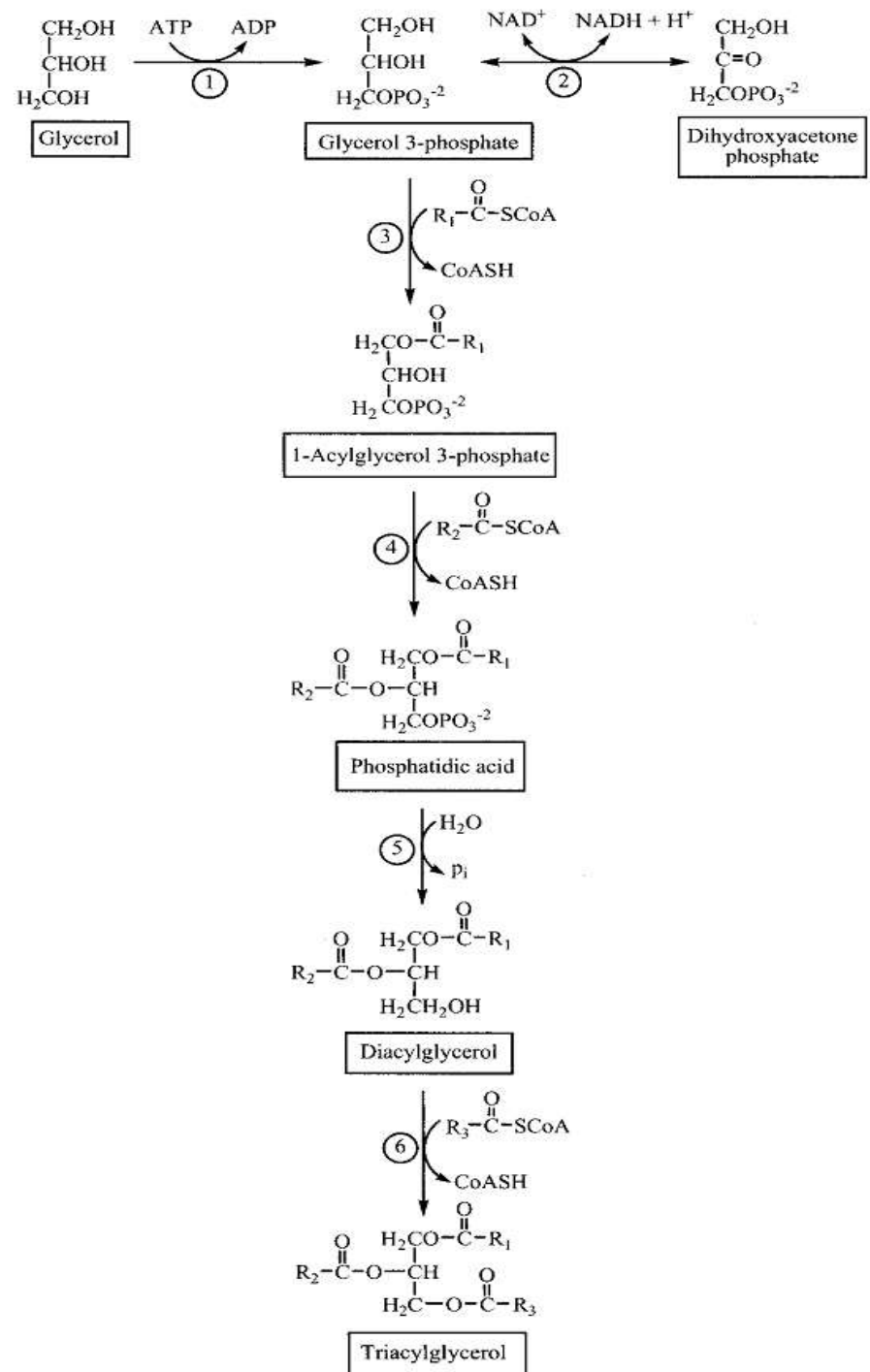
Enzyme required is Fatty acyl CoA synthase.

- ❑ Fatty acyl CoA participates in TAG synthesis.



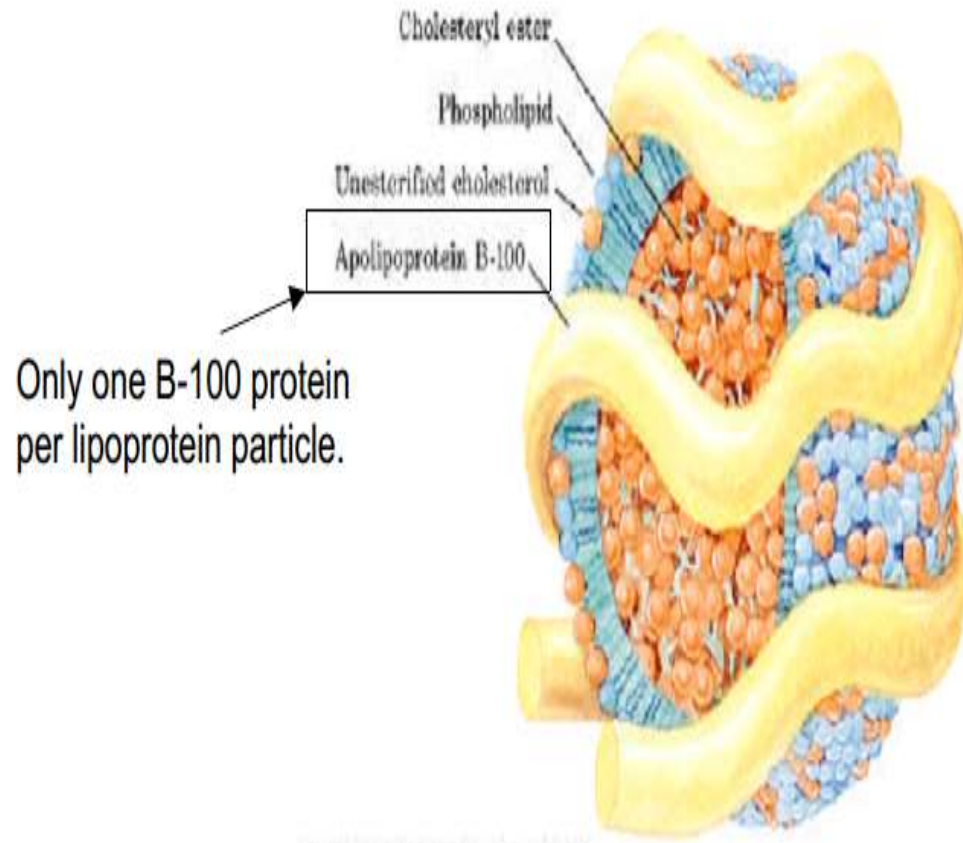
SYNTHESIS OF A MOLECULE OF TAG

- Glycerol phosphate combines with a fatty acyl CoA and forms Lysophosphatidic acid. Enzyme is **Acyltransferase** which removes CoA.
- Lysophosphatidic acid combines with the second fatty acyl CoA to form DAG phosphate. Enzyme is **Acyltransferase**.
- Phosphatase** removes phosphate and forms DAG. DAG combines with the third fatty acyl CoA and forms TAG.



FATES OF TAG IN LIVER AND ADIPOSE TISSUE

- Adipose tissue ---- TAG stored in cytosol
- Liver --- very little stored. Exported out of liver in VLDL , which exports endogenous lipids to peripheral tissues.



SYNTHESIS OF PHOSPHOLIPIDS

- All cells, with the possible exception of mature red blood cells, are capable of synthesizing one or more glycerophospholipids.
- Most of the reactions involved in phospholipid synthesis occur on the **cytosolic face of the endoplasmic reticulum and Golgi complex.**
- **The liver is a major site of phospholipid synthesis**
- Two other tissues with a high capacity for phospholipid synthesis are intestinal enterocytes, **type I1 cells of the lung, which synthesize pulmonary surfactant**



SYNTHESIS OF GLYCEROPHOSPHOLIPIDS

- **Phosphatidic acid** is the basic component for glycerophospholipid synthesis which then combines with an alcohol.

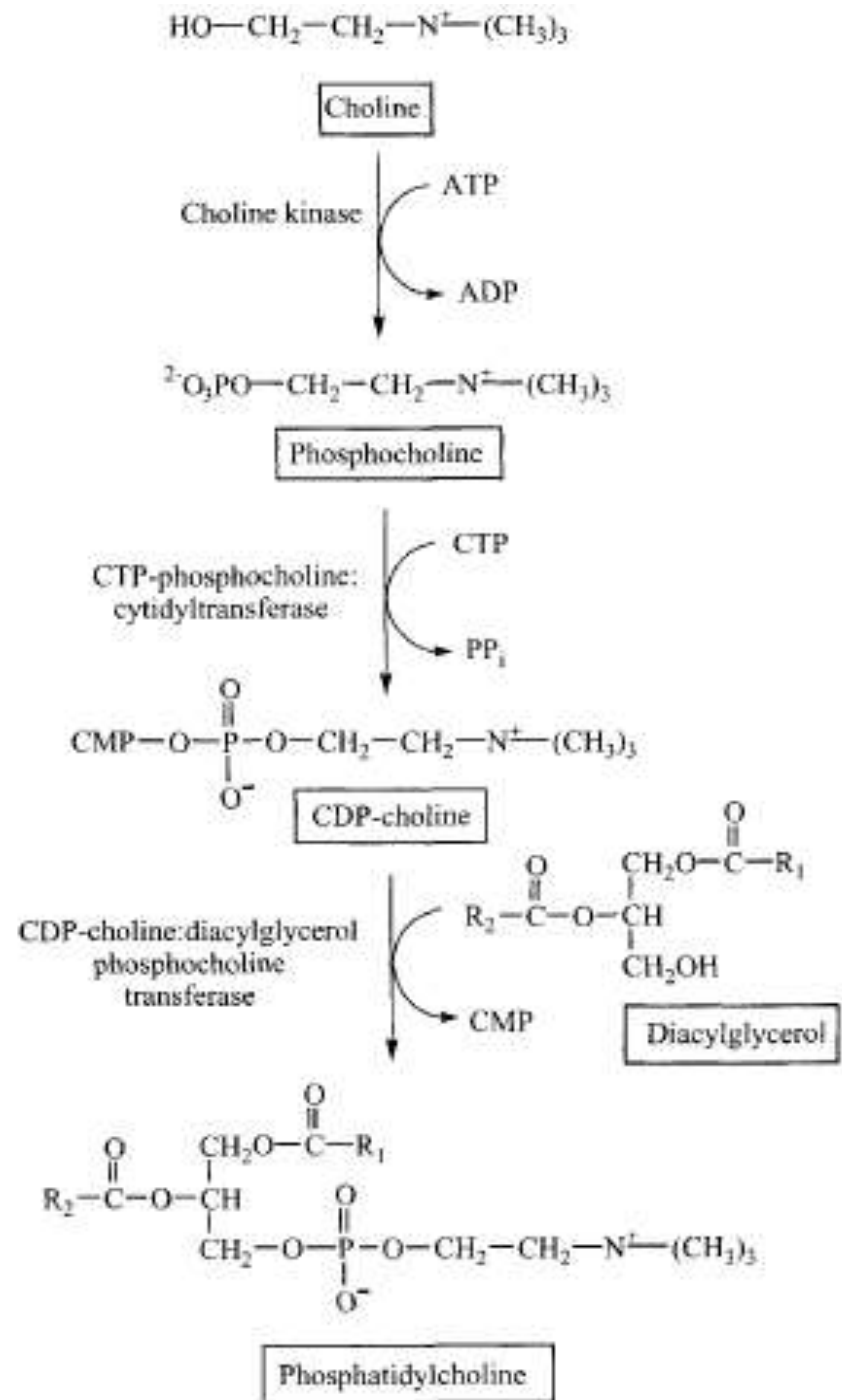
This may involve two processes, i.e

1. Phosphatidic acid may be donated from CDP-diacylglycerol to an alcohol
2. CDP- alcohol may donate its phosphomonoester to diacylglycerol.



SYNTHESIS OF PC (LECITHIN)

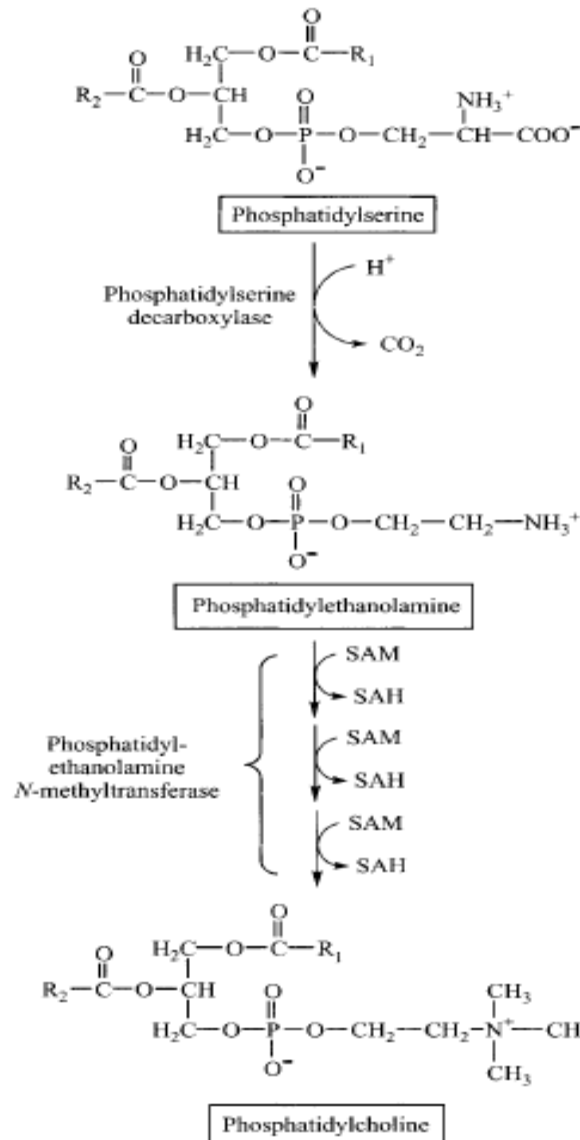
- One of the most abundant PL in cells.
- Substrates required are :-
 - Choline;** preexisting obtained from diet or from turnover of PL.
 - Diacylglycerol;** formed by removal of phosphate from phosphatidic acid. Enzyme is phosphatidate hydrolase.
- PC can also be formed from PS in the liver.



PC CAN ALSO BE FORMED FROM PS

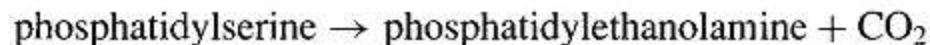
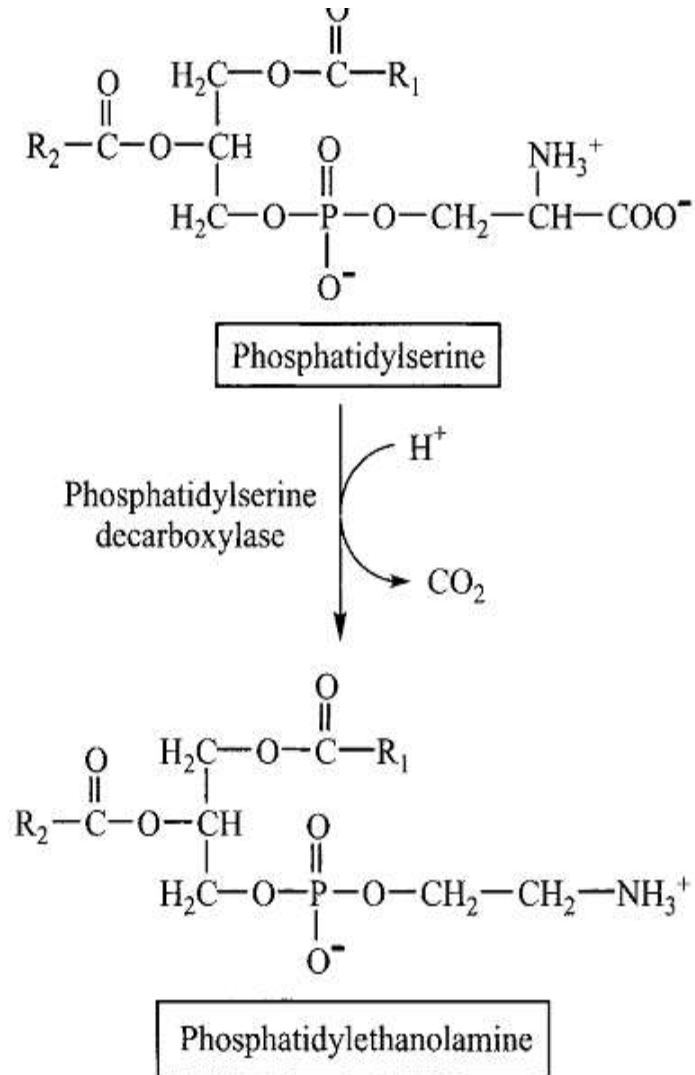
- Takes place only in liver.
- Liver can make PC by this process even when free choline levels are low.
- PS is converted to PE.
- Enzyme is PS decarboxylase.
- PE undergoes methylation.
- Enzyme is methyltransferase.
- Result is the formation of phosphatidylcholine.

Choline is now recognized as an essential nutrient, especially for people with low dietary intakes of protein whose methionine pool may be inadequate. Choline deficiency compromises hepatic VLDL synthesis and secretion and can therefore result in fatty liver



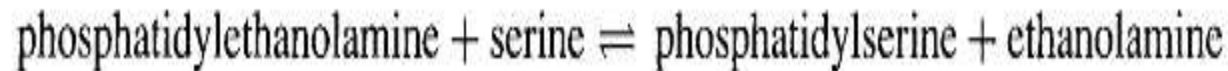
SYNTHESIS OF PE

- Formed from preexisting Ethanolamine.
- Phosphorylation of ethanolamine by Kinase.
- Formation of CDP-Ethanolamine.
- Transfer of Ethanolamine Phosphate from CDP to Diacylglycerol forms PE.
- PS can be converted to PE by reversal of decarboxylation.



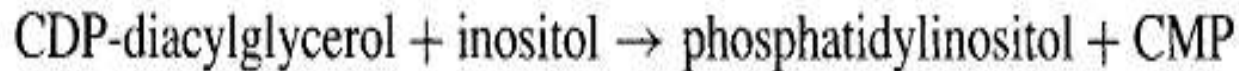
SYNTHESIS OF PS

- Formed from PE.
- PE reacts with serine to form PS. Enzyme is PE-Serine transferase.
- This is a base exchange reaction in which ethanolamine of PE is exchanged for free serine.



SYNTHESIS OF PI

- ❑ Substrates required are free inositol and CDP-diacylglycerol.
- ❑ Diacylglycerol 3 phosphate (PA) reacts with CTP to form CDP-diacylglycerol. Enzyme is Diacylglycerol-CTP synthase.



- ❑ CDP-Diacylglycerol reacts with inositol and forms Phosphatidyl inositol. Enzyme is PI Synthase.



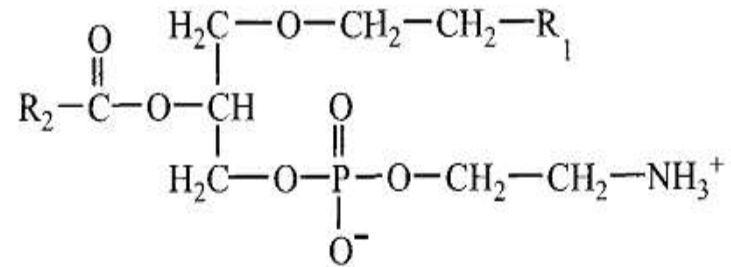
SYNTHESIS OF CARDIOLIPIN

- ❑ Cardiolipin is di-phosphatidyl glycerol in nature.
- ❑ It is composed of two molecules of phosphatidic acid connected by a molecule of glycerol.
- ❑ CDP-diacylglycerol transfers diacylglycerophosphate to phosphatidylglycerol to form cardiolipin.

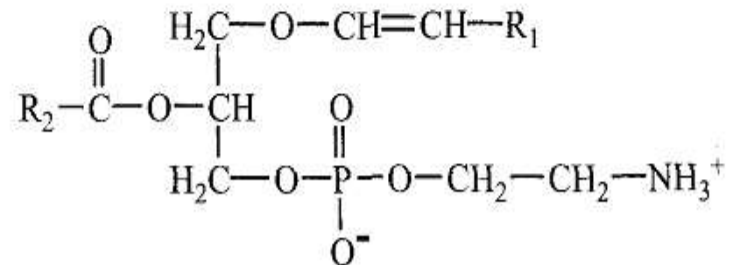
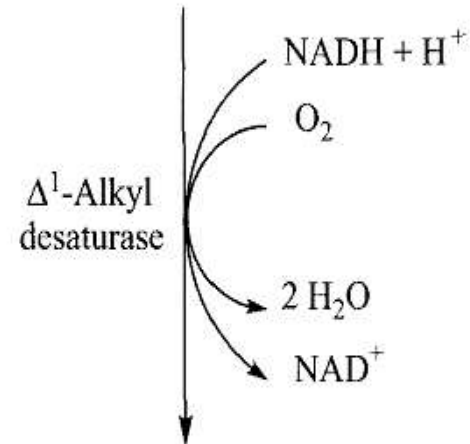


SYNTHESIS OF PLASMALOGENS

- Plasmalogens are the PL in which F.A at C1 of glycerol is attached by an ether linkage.
- Substrates required are di-hydroxy acetone phosphate and acyl CoA.



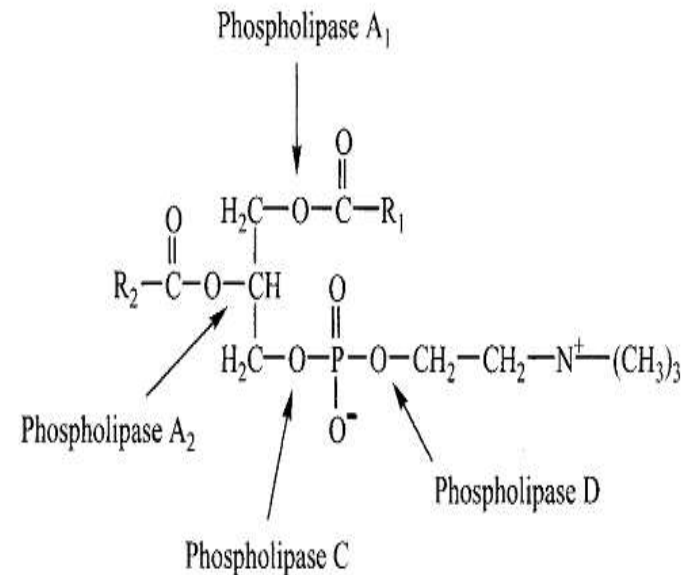
Alkylacylglycerophosphoethanolamine



Ethanolamine plasmalogen

DEGRADATION OF PHOSPHOLIPIDS

- Phospholipase A₁:
 - found in many mammalian tissues.
 - removes fatty acid from C1
- Phospholipase A₂:
 - found in many tissues and pancreatic juice
 - removes F.A at C2
 - when acts on PI, releases arachidonic acid
 - inhibited by glucocorticoids



DEGRADATION OF PHOSPHOLIPIDS

○ Phospholipase C:-

- cleaves phosphate group at C3
- found in liver lysosomes and some bacteria
- role in producing second messengers.

○ Phospholipase D:-

- found primarily in plant tissues.
- removes the compound with alcohol group on C3



DEGRADATION OF SPHINGOMYELIN

- Enzyme is **Sphingomyelinase**, a lysosomal enzyme.
- It removes **phosphorylcholine** hydrolytically and ceramide is produced.
- **Ceramide** is cleaved by **ceramidase** and leaves behind **sphingosine** and a **free fatty acid**.
- Sphingosine and ceramide act as intracellular messengers.



DEGRADATION OF GLYCOSPHINGOLIPIDS

- Done by lysosomal enzymes
- Different enzymes act on specific bonds hydrolytically ---- the groups added last are acted first.



SPHINGOLIPIDOSES

- Lipid storage diseases
- Accumulation of sphingolipids in lysosomes
- Partial or total absence of a specific hydrolase
- Autosomal recessive disorders



TYPES OF SPHINGOLOIPIDOSES

Gaucher disease:-

- most common lysosomal storage disease
- absence of glucocerebrosidase enzyme
- accumulation of glucocerebrosides
- enlargement of liver and spleen
- osteopetrosis of long bones
- CNS involvement in infants



○ **Krebbe disease:-**

- accumulation of galactocerebrosides
- absence of galactosylceramidase
- mental and motor function defect
- blindness and deafness
- loss of myelin

○ **Farber disease:-**

- accumulation of ceramide
- absence of ceramidase
- joints and skin involvement



○ **Niemann pick disease:-**

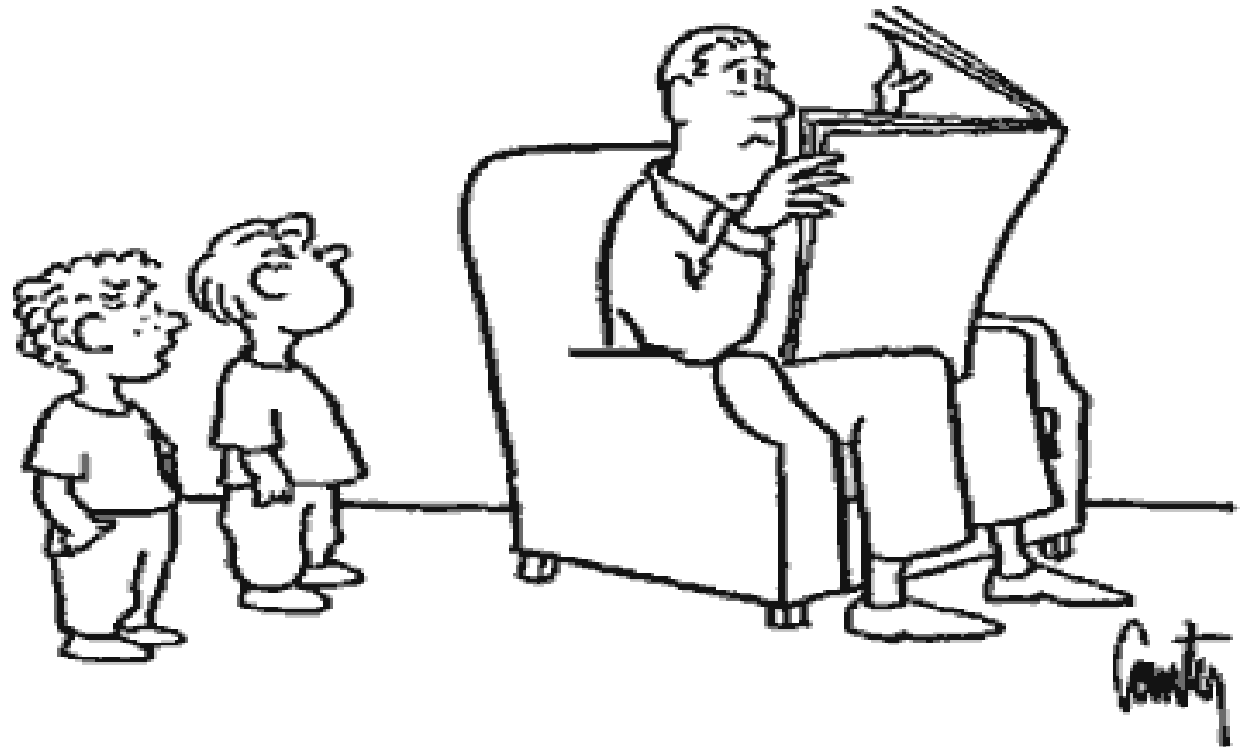
- accumulation of sphingomyelin
- absence of sphingomyelinase
- liver and spleen enlargement
- neuronal degeneration

○ **Fabry disease:-**

- accumulation of globosides
- absence of alpha galactosidase
- skin rash
- kidney and heart failure
- burning pain in legs



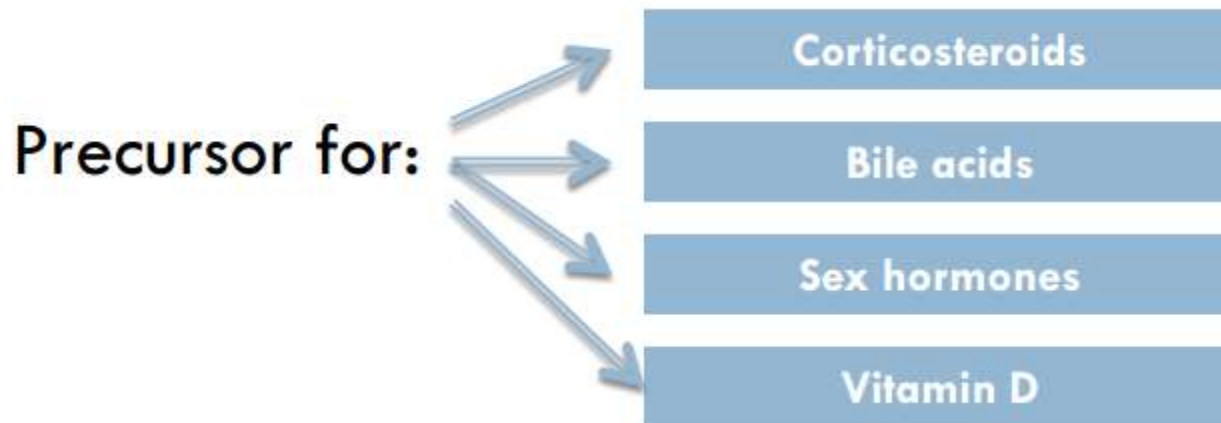
METABOLISM OF CHOLESTEROL



"Don't get too close to my dad - his cholesterol is 290."

CHOLESTEROL BIOSYNTHESIS

- ❑ Principal sterol synthesized by animals.
- ❑ In vertebrates, hepatic cells typically produce more cholesterol than other cells.
- ❑ Absent among prokaryotes.



CHOLESTEROL BIOSYNTHESIS

- Cholesterol required by the body is derived from diet (200-300mg/day) and synthesis in body (700 mg/day is synthesized).

Liver and Intestine accounts for larger proportion.

- NOTE: Virtually all cells containing nucleated cells are capable of synthesizing cholesterol which occurs in endoplasmic reticulum (ER) and cytosol of cells.
- Acetyl-CoA is the source of all carbon atoms in cholesterol.

Coronary atherosclerosis correlates with high plasma LDL/HDL cholesterol ratio.

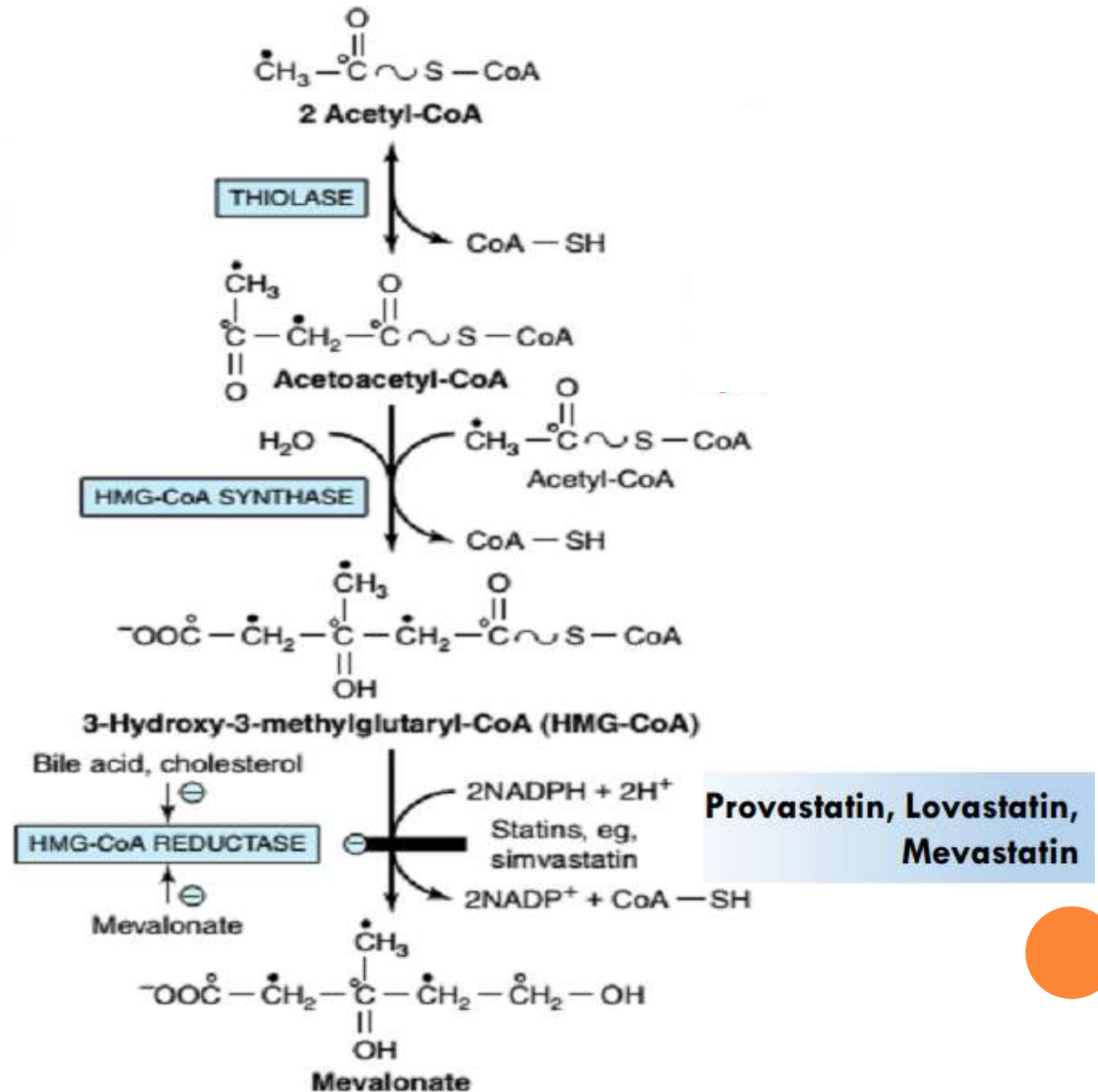


DIVIDED INTO 5 STEPS

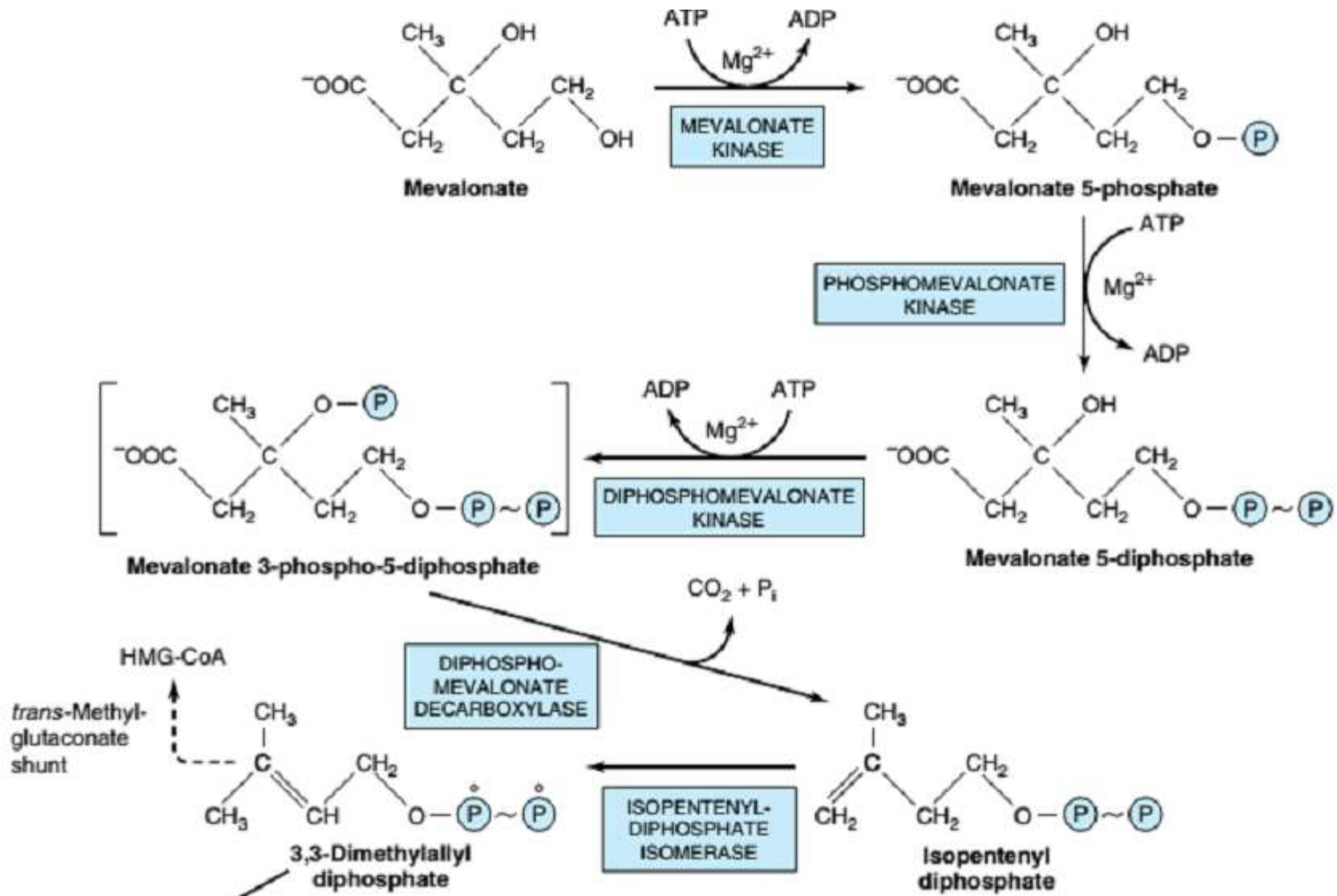
1. Synthesis of mevalonate occurs from Acetyl-CoA
2. Isoprenoid units are formed from MEVALONATE by loss of CO₂
3. Six isoprenoid units condense to form SQUALENE
4. SQUALENE is cyclized to give rise to parent steroid LANOSTEROL
5. Cholesterol is formed from LANOSTEROL



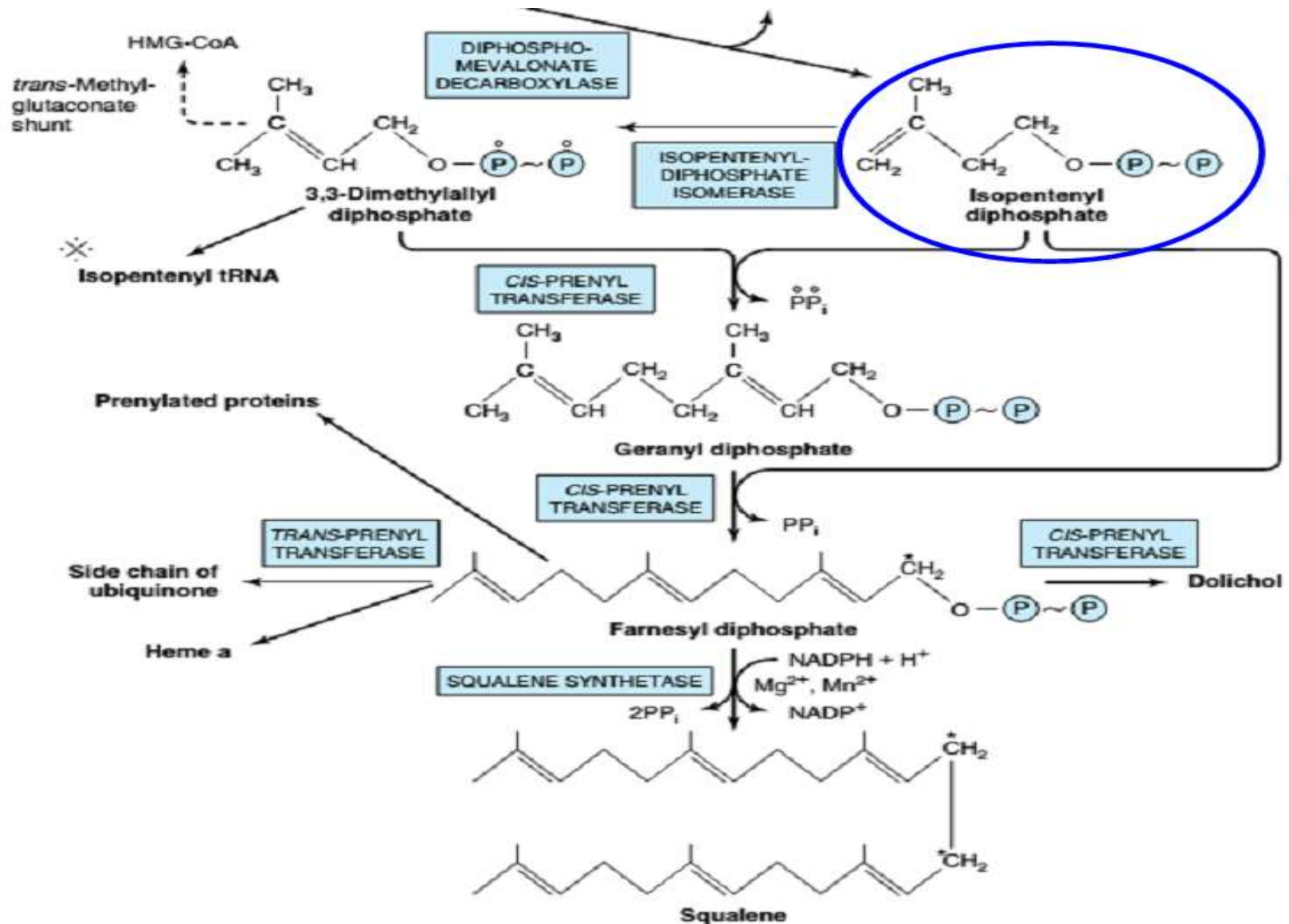
STEP 1



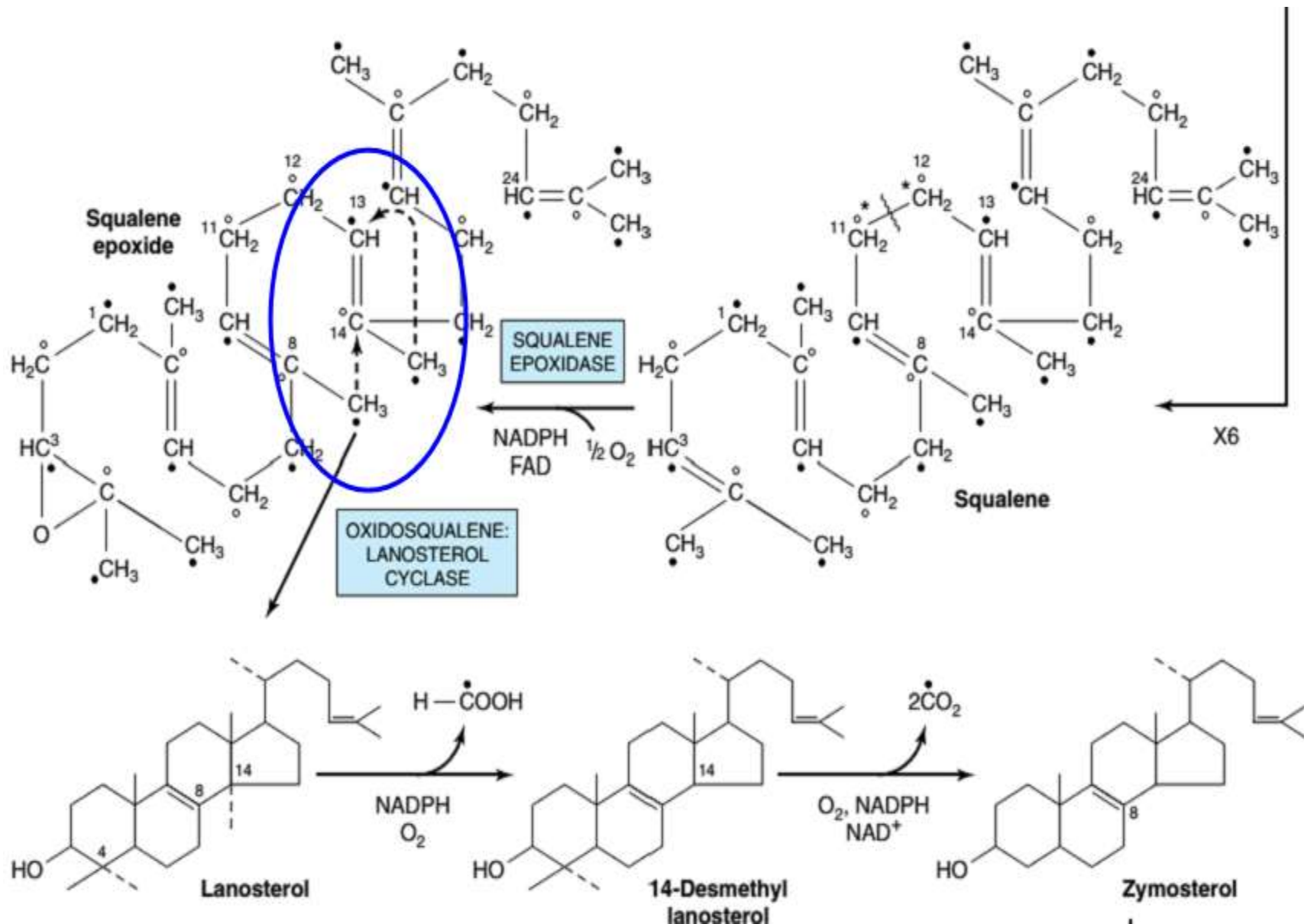
2. FORMATION OF ISOPRENOID UNITS



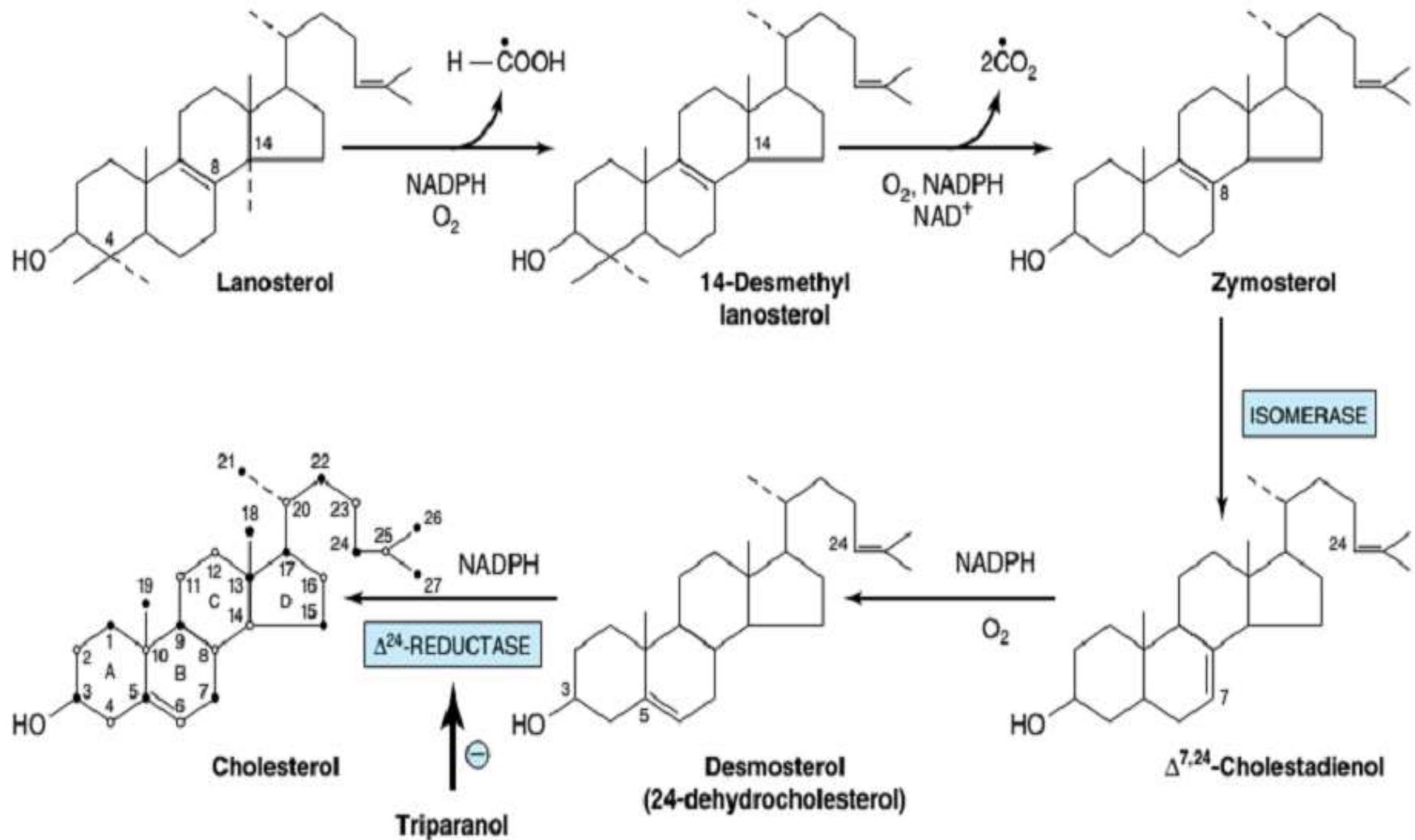
3. FORMATION OF SQUALENE



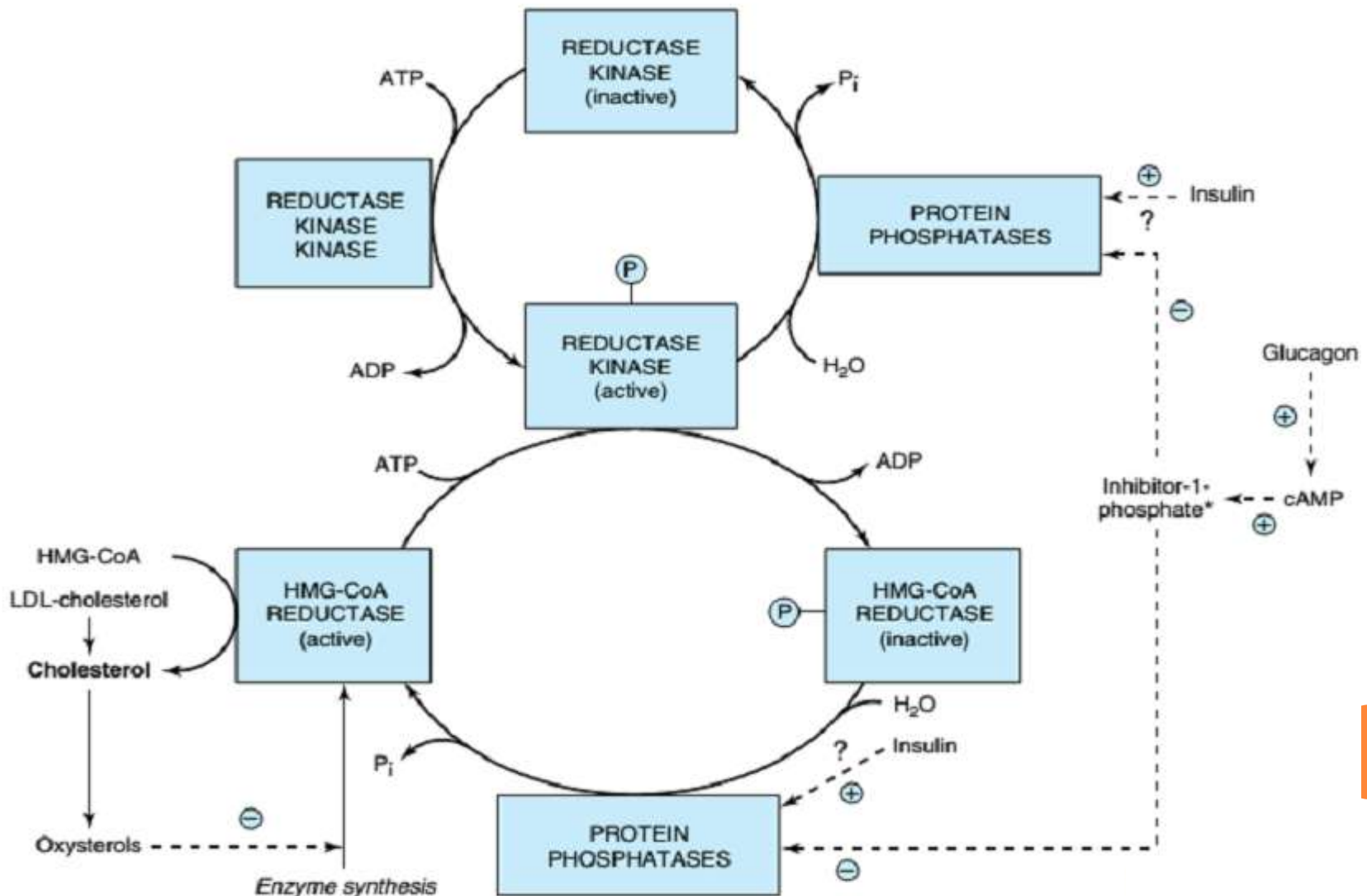
4. FORMATION OF LANOSTEROL



5. FORMATION OF CHOLESTEROL



REGULATION OF CHOLESTEROL SYNTHESIS IS CONTROLLED BY HMG-CoA REDUCTASE



REGULATION OF CHOLESTEROL SYNTHESIS IS CONTROLLED BY HMG-CoA REDUCTASE

1. Decreased synthesis of cholesterol in starving animals is accompanied by reduced activity of the enzyme.
2. Hepatic synthesis is inhibited by dietary cholesterol.
3. HMG-CoA reductase in liver is inhibited by Cholesterol.
4. Cholesterol and metabolites represses transcription of HMG-CoA reductase transcription by Sterol Regulatory Element Binding Protein (SREBP).
5. Insulin or thyroid hormone increases HMG-CoA reductase activity, While, Glucagon or glucocorticoids decrease it.



INCREASE AND DECREASE IN CHOLESTEROL

Increased by

1. Uptake of cholesterol-containing lipoproteins by receptors.
2. Uptake of free cholesterol from cholesterol-rich lipoproteins to the cell membrane
3. Cholesterol synthesis
4. Hydrolysis of cholesterol ester by the enzyme cholesterylester hydrolase.



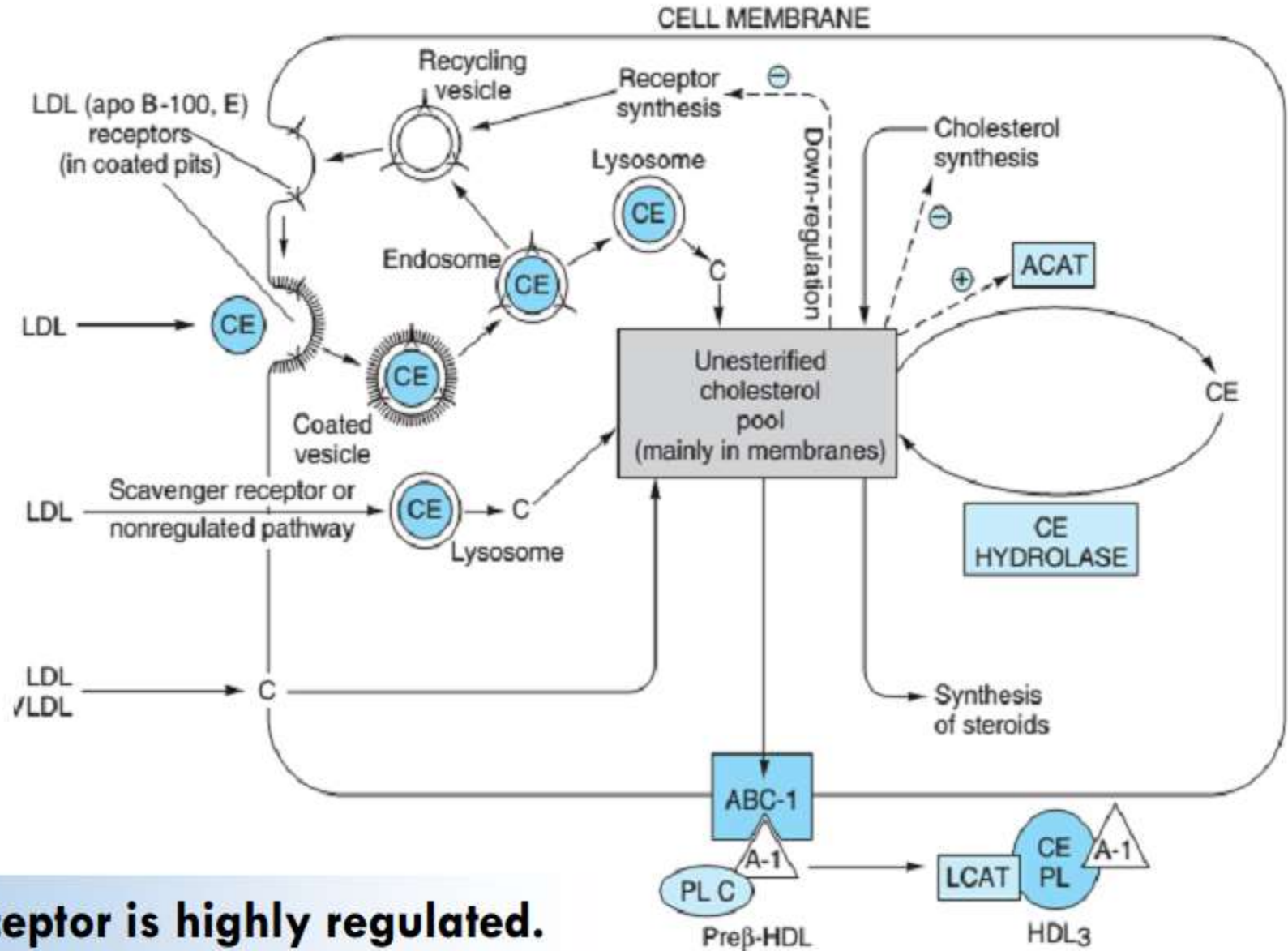
INCREASE AND DECREASE IN CHOLESTEROL

Decreased by

1. Efflux of cholesterol from the membrane to lipoproteins of low cholesterol potential
2. Esterification of cholesterol by ACAT (acyl-CoA: cholesterol acyltransferase)
3. Utilization of cholesterol for synthesis of other steroids such as hormones or bile acids in liver



WHAT ARE THE FACTORS THAT INFLUENCE CHOLESTEROL BALANCE IN TISSUES?



LDL receptor is highly regulated.

DECREASE OF CHOLESTEROL IN CELLS

- ❑ Efflux of cholesterol from cells to HDL
- ❑ Esterification of cholesterol by the enzyme “Acyl-CoA-cholesterol acyl transferase” (ACAT).
- ❑ Utilization of cholesterol for synthesis of steroid hormones, viz. glucocorticoids, mineralocorticoids, gonadal hormones
- ❑ In liver cells: formation of cholic acid
- ❑ Formation of Vitamin D3



CHOLESTEROL BALANCE IN TISSUES

- ☐ Increase of cholesterol in cells
- ☐ Increased synthesis of cholesterol
- ☐ Hydrolysis of cholesterol ester by the enzyme “cholesterol ester hydrolyase”.
- ☐ Uptake and delivery of cholesterol in cells by circulating LDL
- ☐ Uptake of cholesterol-containing lipoproteins by ‘non-receptor’
- ☐ Uptake of free cholesterol by cell membranes



CONSIDERATION OF OTHER FACTORS THAT INFLUENCE CHOLESTEROL LEVEL IN BLOOD

Dietary fats (saturated FA vs Unsaturated FA)

- ☐ Dietary cholesterol
- ☐ Dietary carbohydrates
- ☐ Heredity
- ☐ Blood groups
- ☐ Calorie Intake
- ☐ Vitamin B-complex
- ☐ Minerals
- ☐ Dietary fibers
- ☐ Physical exercise
- ☐ Life style of an individual





METABOLISM OF LIPOPROTEINS

CLASSES OF LIPOPROTEINS

- ❑ On the basis of density ;-ultracentrifugation
 - chylomicrons
 - VLDL
 - LDL
 - HDL
- ❑ On electrophoresis—
 - chylomicrons at origin
 - LDL (beta lipoproteins)
 - VLDL (pre-beta lipoproteins)
 - HDL (alpha lipoproteins)



APOLIPOPROTEINS

- **Serve very important functions**
 - ❑ recognition site for cell-surface receptors
 - ❑ activators or coenzymes for enzymes of lipoprotein metabolism
 - ❑ some are essential structural component of the lipoprotein particle
 - ❑ transfer between different types of lipoproteins and bring about changes.
 - ❑ Classes from A to E. Some have sub classes.

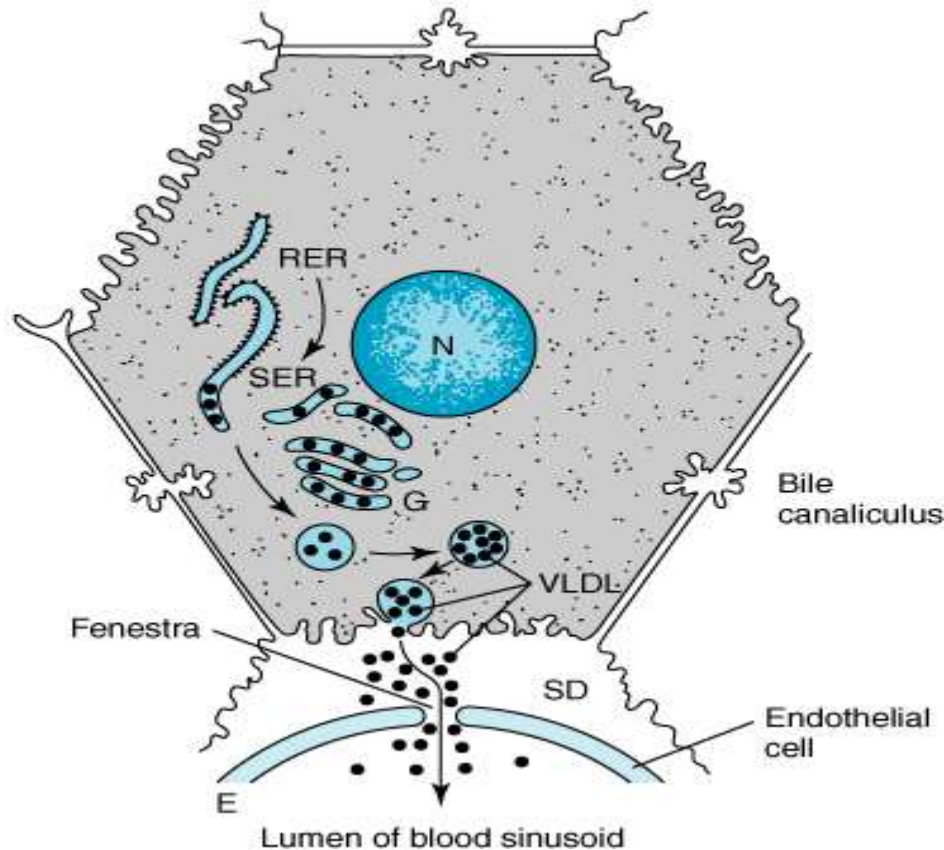


METABOLISM OF LIPOPROTEINS

- ❑ There are striking similarities in the mechanisms of formation of chylomicrons by intestinal cells and of VLDL by hepatic parenchymal cells
- ❑ Apart from the mammary gland, the intestine and liver are the only tissues from which particulate lipid is secreted.
- ❑ Newly secreted or "nascent" VLDL contain only a small amount of apolipoproteins C and E, and the full complement is acquired from HDL in the circulation
- ❑ Apo B 100 is essential for VLDL formation



METABOLISM OF VLDL



VLDL are secreted into the space of Disse and then into the hepatic sinusoids through fenestrae in the endothelial lining.

CATABOLISM OF VLDL

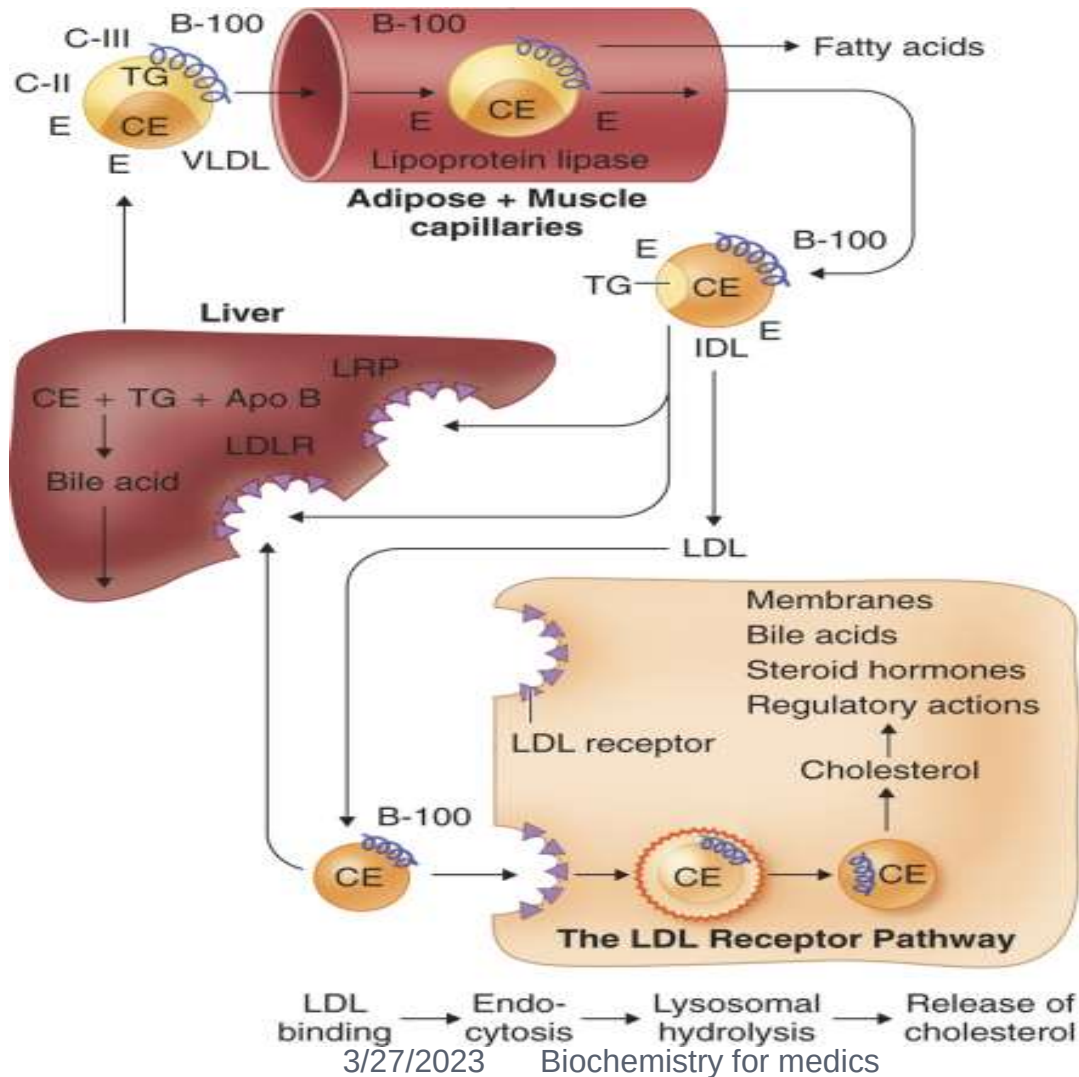
- ❑ Catabolism of VLDL is similar to chylomicrons
- ❑ Both **phospholipids** and **apo C-II** are required as cofactors for **lipoprotein lipase activity**, while **apo A-II** and apo **C-III** act as inhibitors.
- ❑ Hydrolysis takes place while the VLDLs are attached to the enzyme on the endothelium.
- ❑ Triacylglycerol is hydrolyzed progressively through a diacyl glycerol to a monoacylglycerol and finally to free fatty acids plus glycerol.
- ❑ Some of the released free fatty acids return to the circulation, attached to albumin, but the bulk is transported into the tissue .



CATABOLISM OF VLDL

- ❑ Reaction with lipoprotein lipase results in the loss of approximately 90% of the triacylglycerol of VLDLs and in the loss of apo C (which returns to HDL) but not apo E, which is retained.
 - ❑ These changes occurring to VLDL, lead to the formation of VLDL remnants or IDL (intermediate-density lipoprotein)
 - ❑ After metabolism to IDL, VLDL may be taken up by the liver directly via the LDL (apo B-100, E) receptor, or it may be converted to LDL.
 - ❑ Only one molecule of apo B-100 is present in each of these lipoprotein particles, and this is conserved during the transformations.
 - ❑ Thus, each LDL particle is derived from a single precursor VLDL particle.
 - ❑ In humans, a relatively large proportion of IDL forms LDL, accounting for the increased concentrations of LDL in humans compared with many other mammals.
- 

METABOLISM OF VLDL



- ❑ The liver and many extrahepatic tissues express the **LDL (apo B-100, E) receptor**.
- ❑ It is so designated because it is specific for apo B-100
- ❑ but not B-48, which lacks the carboxyl terminal domain of B-100 containing the LDL receptor ligand,
- ❑ and it also takes up lipoproteins rich in apo E.

METABOLISM OF LDL

- The liver and many extra hepatic tissues express the **LDL (apo B-100, E) receptor**.
- It is so designated because it is specific for apo B-100 but not B-48, which lacks the carboxyl terminal domain of B-100 containing the LDL receptor ligand, and it also takes up lipoproteins rich in apo E.
- This receptor is defective in **familial hypercholesterolemia**.
- Approximately 30% of LDL is degraded in extra-hepatic tissues and 70% in the liver.
- A positive correlation exists between the incidence of **coronary atherosclerosis** and the plasma concentration of LDL cholesterol.



METABOLISM OF HDL

Synthesis of HDL

- HDL is synthesized and secreted from both **liver and intestine** .
- However, apo C and apo E are synthesized in the liver and transferred from liver HDL to intestinal HDL when the latter enters the plasma.
- A major function of HDL is to act as a repository for the **apo C and apo E** required in the metabolism of chylomicrons and VLDL.
- Nascent HDL consists of **discoid phospholipid bilayer containing apo A and free cholesterol**.



METABOLISM OF HDL

- ❑ LCAT and the LCAT activator apo A-I—bind to the discoidal particles, and the surface phospholipid and free cholesterol are converted into cholesteryl esters and lysolecithin .
- ❑ The nonpolar cholesteryl esters move into the hydrophobic interior of the bilayer, whereas lysolecithin is transferred to plasma albumin.
- ❑ Thus, a nonpolar core is generated, forming a spherical, pseudomicellar HDL covered by a surface film of polar lipids and apolipoproteins.
- ❑ This aids the removal of excess unesterified cholesterol from lipoproteins and tissues .



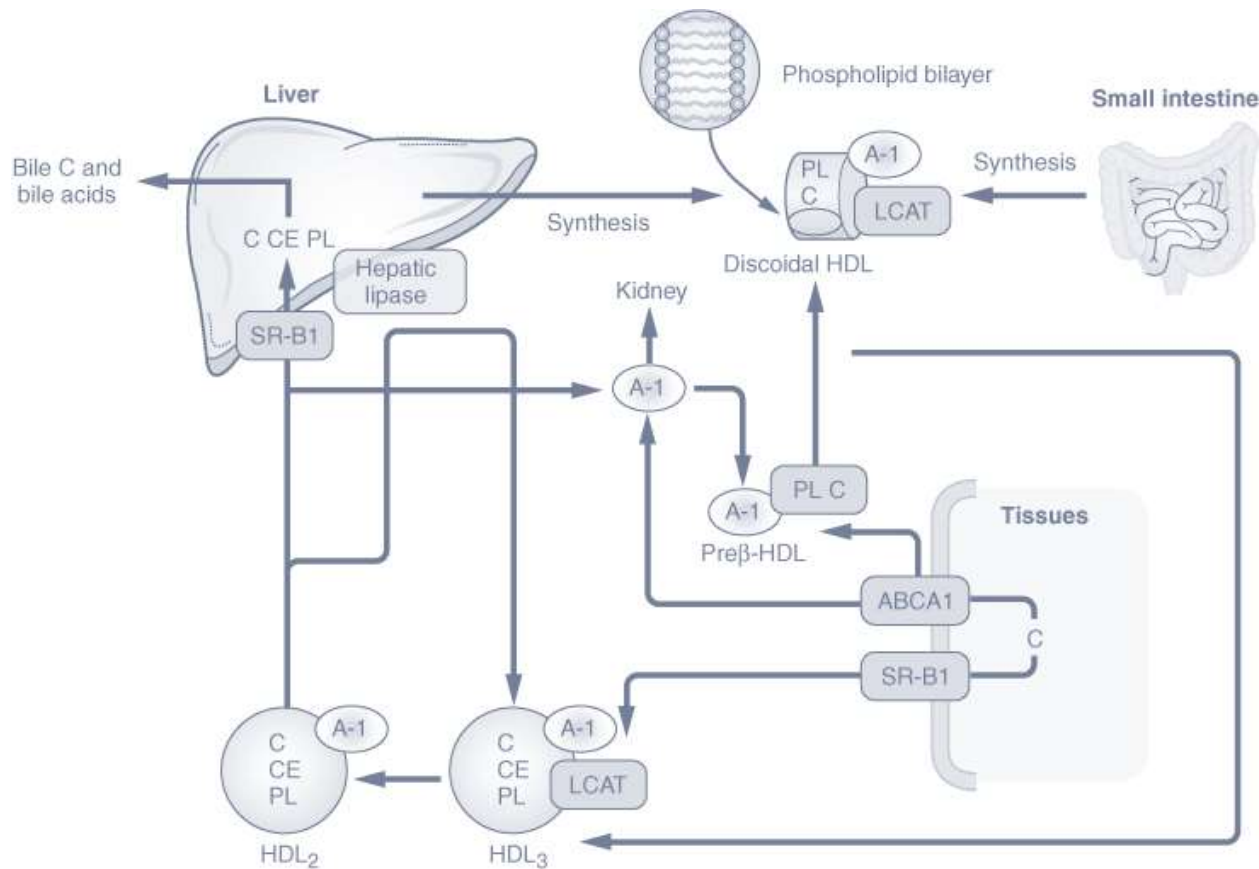
METABOLISM OF HDL

Role of LCAT

- LCAT(Lecithin Cholesterol Acyl Transferase) enzyme catalyzes the esterification of cholesterol to form Cholesteryl ester. The reaction can be represented as follows-
- $\text{Lecithin} + \text{Cholesterol} \longrightarrow \text{Lysolecithin} + \text{Cholesteryl Ester}$



REVERSE CHOLESTEROL TRANSPORT



- ❑ The cholesterol efflux is brought about by esterification of cholesterol under the effect of LCAT.
- ❑ The cholesteryl ester rich HDL (HDL₂) gains entry through Scavenger receptor (SR-B1)



METABOLISM OF HDL

- The **class B scavenger receptor B1 (SR-B1)** has been identified as an **HDL receptor with a dual role in HDL metabolism.**
- In the liver and in steroidogenic tissues, it binds HDL via **apo A-I**, and **cholesteryl ester** is selectively delivered to the cells, although the particle itself, including apo A-I, is not taken up.
- In the tissues, on the other hand, SR-B1 mediates the acceptance of cholesterol from the cells by HDL, which then transports it to the liver for excretion via the bile in the process known as **reverse cholesterol transport**



METABOLISM OF HDL

- ❑ A second important mechanism for reverse cholesterol transport involves the **ATP-binding cassette transporter A1 (ABCA1)**.
- ❑ ABCA1 is a member of a family of transporter proteins that couple the hydrolysis of ATP to the binding of a substrate, enabling it to be transported across the membrane.
- ❑ ABCA1 preferentially transfer cholesterol from cells to poorly lipidated particles such as pre -HDL or apo A-1, which are then converted to HDL₃ via discoidal HDL
- ❑ Pre -HDL is the most potent form of HDL inducing cholesterol efflux from the tissues.

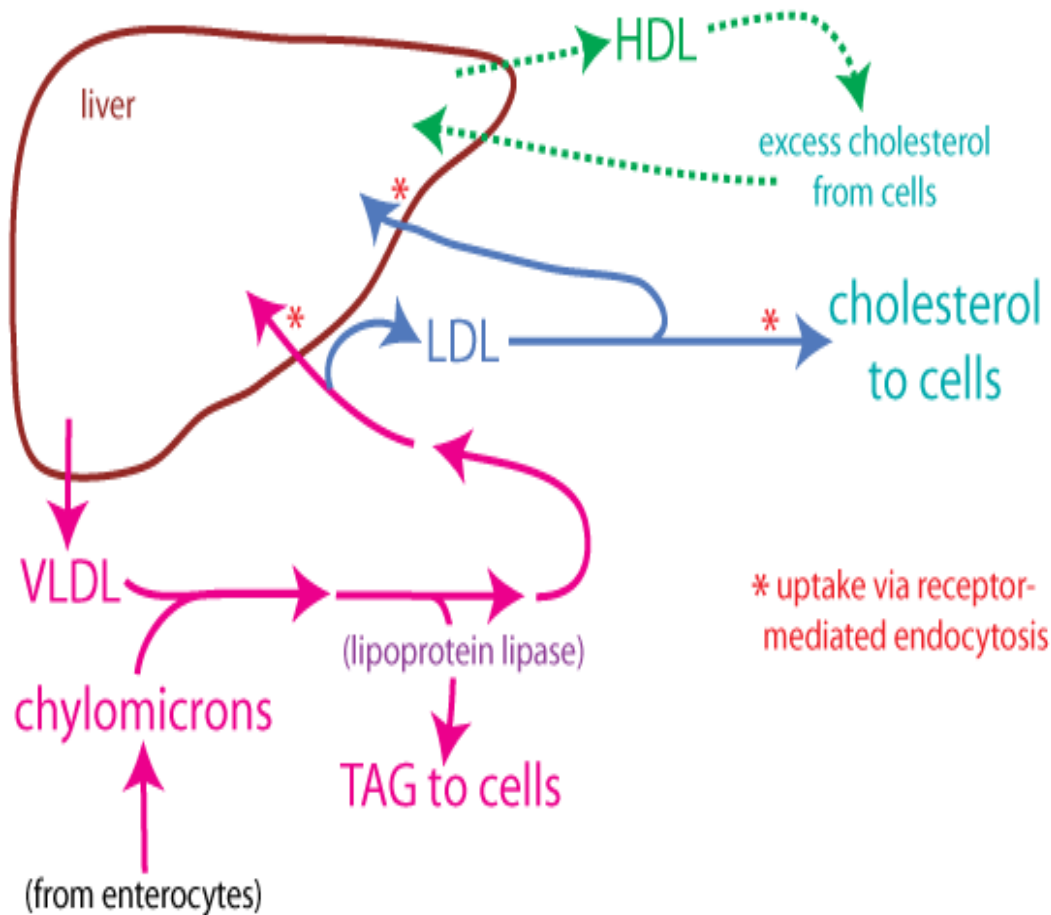


FUNCTIONS OF HDL

- ❑ Scavenging action- HDL scavenges extra cholesterol from peripheral tissues by reverse cholesterol transport
- ❑ HDL, with the help of apo E competes with LDL for binding sites on the membranes and prevents internalization of LDL cholesterol in the smooth cells of the arterial walls
- ❑ HDL contributes its apo C and E to nascent VLDL and chylomicrons for receptor mediated endocytosis
- ❑ HDL stimulates prostacyclin synthesis by the endothelial cells, which prevent thrombus formation
- ❑ HDL also helps in the removal of macrophages from the arterial walls.
- ❑ HDL prevents LDL from getting oxidized



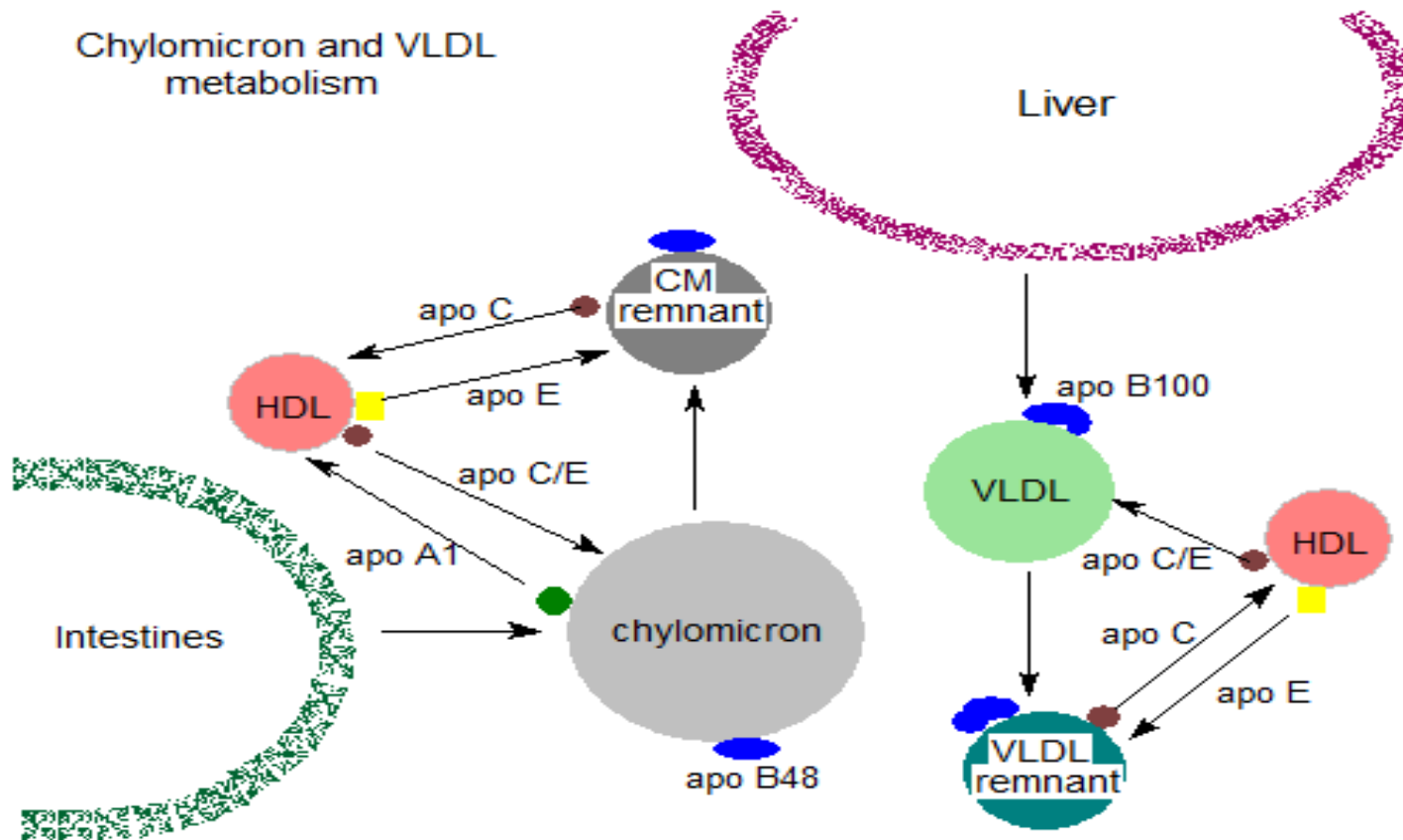
SUMMARY OF FORMATION AND FATE OF LIPOPROTEINS



- ❑ Chylomicrons is a transporter of dietary lipids whereas VLDL is a transporter of endogenous lipids (mainly TGs).
- ❑ LDL transports cholesterol to peripheral cells while HDL transports cholesterol from peripheral cells back to liver.



ROLE OF HDL IN RECEPTOR MEDIATED ENDOCYTOSIS



HDL contributes its apo C and E to nascent VLDL and chylomicrons for receptor mediated endocytosis



CLINICAL SIGNIFICANCE OF LIPOPROTEIN METABOLISM

Fatty Liver

- ❑ is an abnormal accumulation of fats (triglycerides) inside liver cells.
- ❑ Hepatic triacylglycerol synthesis provides the immediate stimulus for the formation and secretion of VLDL.
- ❑ Impaired VLDL formation or secretion leads to nonmobilization of lipid components from the liver, results in fatty liver.



FATTY LIVER (CONTD.)

Fatty livers fall into two main categories-

A) More synthesis of Triglycerides

- ❑ High carbohydrate diet
- ❑ High fat feeding
- ❑ Diabetes mellitus



FATTY LIVER (CONTD.)

B) Defective VLDL synthesis -The second type of fatty liver is usually due to a **metabolic block in the production of plasma lipoproteins**, thus allowing triacylglycerol to accumulate.

The lesion may be due to –

1). A block in apolipoproteins synthesis

- a) Protein energy Malnutrition
- b) Impaired absorption
- c) Presence of inhibitors of endogenous protein synthesis e.g.- Carbon tetra chloride, Puromycin, Ethionine , Heavy metals etc.
- d) Hypobetalipoproteinemia- Defective apo B gene can cause impaired synthesis of apo B protein.



FATTY LIVER (CONTD.)

(2) A failure in provision of phospholipids that are found in lipoproteins

- ❑ A deficiency of **choline**, a **lipotropic factor** can cause impaired formation of phosphatidyl choline (Lecithin), a glycerophospholipid.
- ❑ Methionine deficiency can also cause impaired choline synthesis
- ❑ Inositol deficiency
- ❑ Deficiency of essential fatty acids can also cause impaired PL synthesis



FATTY LIVER (CONTD.)

- (3) Impaired Glycosylation-** Orotic acid also causes fatty liver; it interferes with glycosylation of the lipoprotein, thus inhibiting release, and may also impair the recruitment of triacylglycerol to the particles. In conditions of orotic aciduria(disorder of pyrimidine nucleotide biosynthesis), fatty liver can be observed.
- 4) Impaired secretion of VLDL-** oxidative stress is a common cause for membrane disruption of lipoproteins.

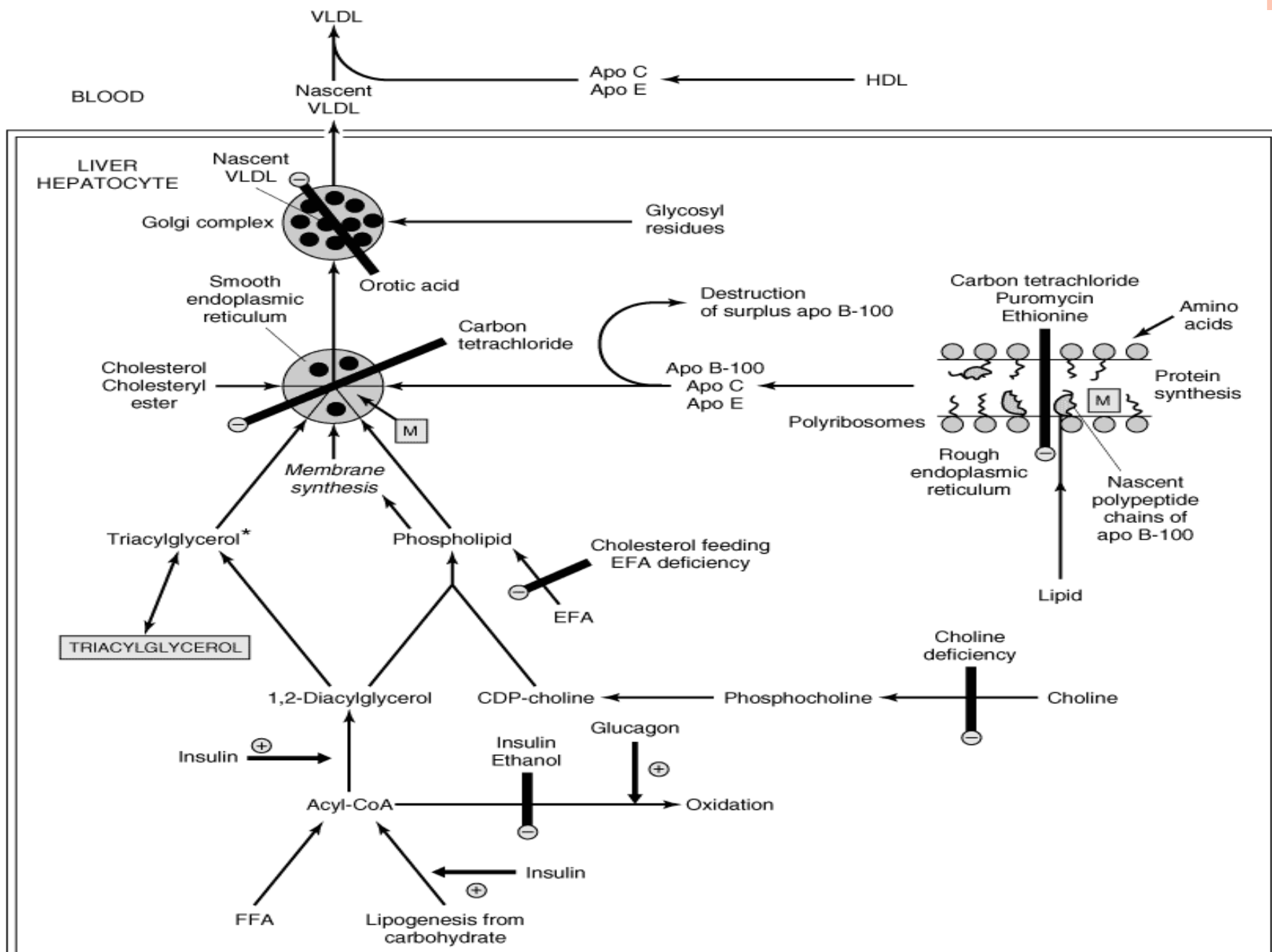


FATTY LIVER (CONTD.)

2) Alcoholic fatty liver

- ❑ **Alcoholism** leads to fat accumulation in the liver, hyperlipidemia, and ultimately **cirrhosis**.
- ❑ The fatty liver is caused by a combination of impaired fatty acid oxidation and increased lipogenesis, which is thought to be due to changes in the $[NADH]/[NAD^+]$ redox potential in the liver,
- ❑ and also to interference with the action of transcription factors regulating the expression of the enzymes involved in the pathways.





FATTY LIVER AND LIPOTROPIC AGENTS

Lipotropic agents- Agents such as-

- ❑ Choline
- ❑ Inositol
- ❑ Methionine and other essential amino acids,
- ❑ Essential fatty acids,
- ❑ Anti oxidant vitamins,
- ❑ Vitamin B12, folic acid and
- ❑ Synthetic antioxidants which have the apparent effect of removal of fats from the liver cells, and thus prevent the formation of fatty liver are called lipotropic agents.

